
ISSCC 2022 Tutorials

Analog Circuit Design in Bipolar-CMOS-DMOS (BCD) Technologies

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Marco Berkhout

- ❑ 1992/1996 MSc/PhD Electrical Engineering, Twente University, The Netherlands
- ❑ 1996-2020: Senior Principal Design Engineer Philips / NXP Semiconductors, The Netherlands
- ❑ 2020–present: Fellow Goodix Technology, The Netherlands

- ❑ Expert Class-D Audio Power Amplifiers
 - 25+ patents
 - 20+ papers at IEEE conferences / journals

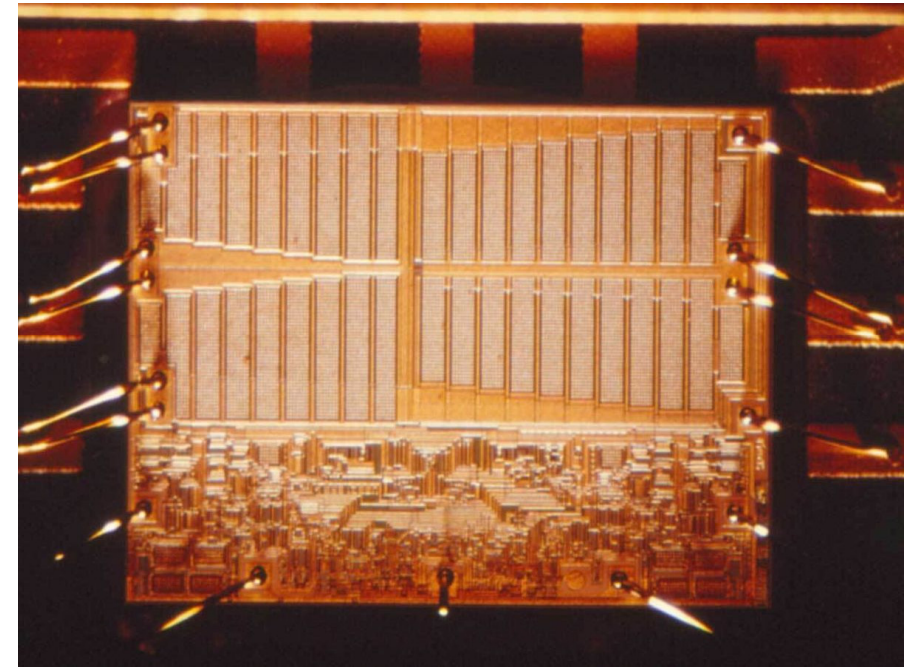
- ❑ Technical Program Committees
 - ESSCIRC 2008–2018
 - ISSCC 2013-2016, 2020-present



IEEE Milestone

- May 18, 2021, the invention of BCD technology was commemorated with an IEEE Milestone
 - <https://spectrum.ieee.org/three-chips-in-one-the-history-of-the-bcd-integrated-circuit>

SGS (now STMicroelectronics) pioneered the super-integrated silicon-gate process combining bipolar, CMOS, and DMOS (BCD) transistors in single chips for complex, power-demanding applications. The first BCD super-integrated circuit, named L6202, was capable of controlling up to 60V-5A at 300 kHz. Subsequent automotive, computer, and industrial applications extensively adopted this process technology, which enabled chip designers flexibly and reliably to combine power, analog, and digital signal processing.



BCD Applications



battery chargers



induction cooking



display drivers



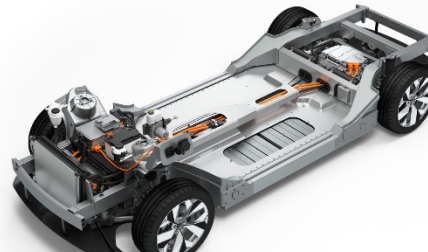
home lighting



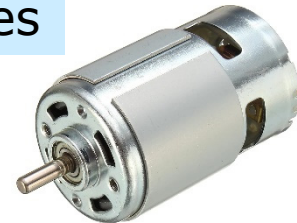
wireless charging



solar inverters



electrical vehicles



motor drivers



audio amplifiers



LED matrix lighting

BCD Technology

- Analog
 - high speed, high precision
 - feedback control, references, signal processing

- Digital
 - high density CMOS
 - control logic, interfacing, digital signal processing
 - BCD follows CMOS roadmap with about 10 years lag

- Power
 - high voltage
 - high current

What is High Voltage ?

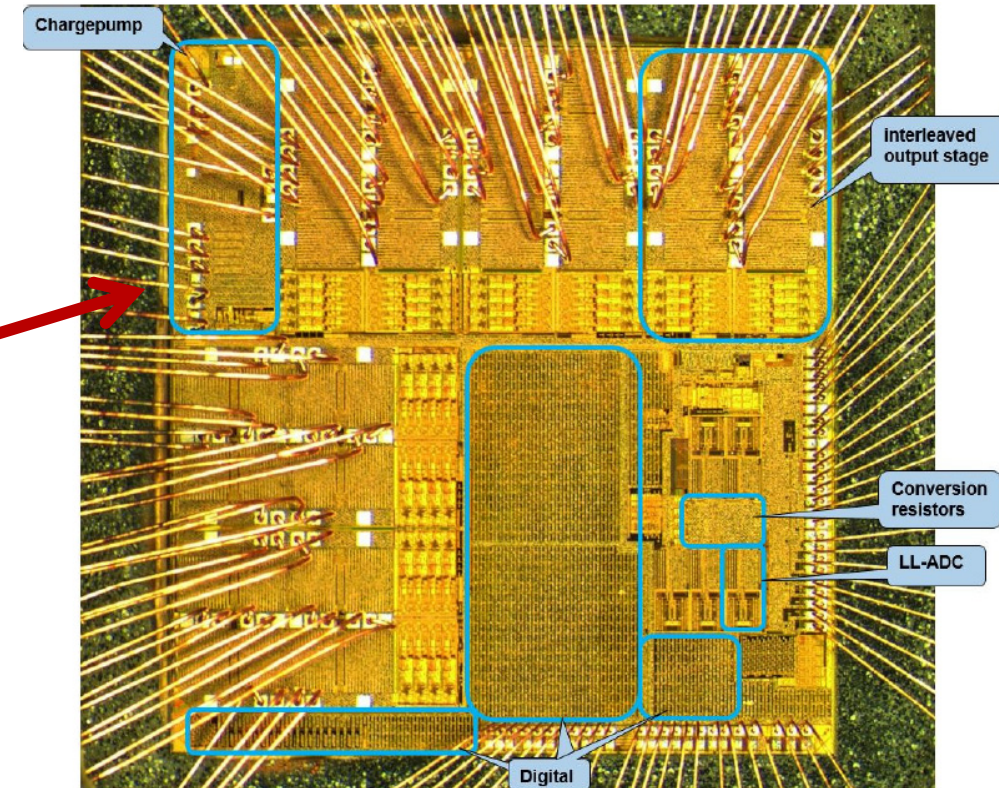
- circuit design: any voltage high enough to damage gate-oxide

- application optimized BCD process nodes
 - 8V - 20V mobile / laptop
 - 40V - 80V automotive / home
 - 400V – 1200V mains
 - 4kV – 6kV galvanic isolation

- high voltage comes at the price of silicon area
 - critical electric field
 - device terminal distance
 - isolation spacing

What is High Current ?

- circuit design: any current that causes significant conduction loss
- high current requires low-ohmic connections
 - 1A - 5A battery chargers, PMICs
 - 10A - 30A audio amplifiers, motor drivers
- process/package features
 - ultra thick Cu metallization
 - thick (multiple) bondwires ^[19]
 - bonding over active areas



Outline

- Introduction & Motivation
- BCD Process Technology & Devices
- Device Characteristics & Figures-of-Merit
- Applications
- Circuit Design

Outline

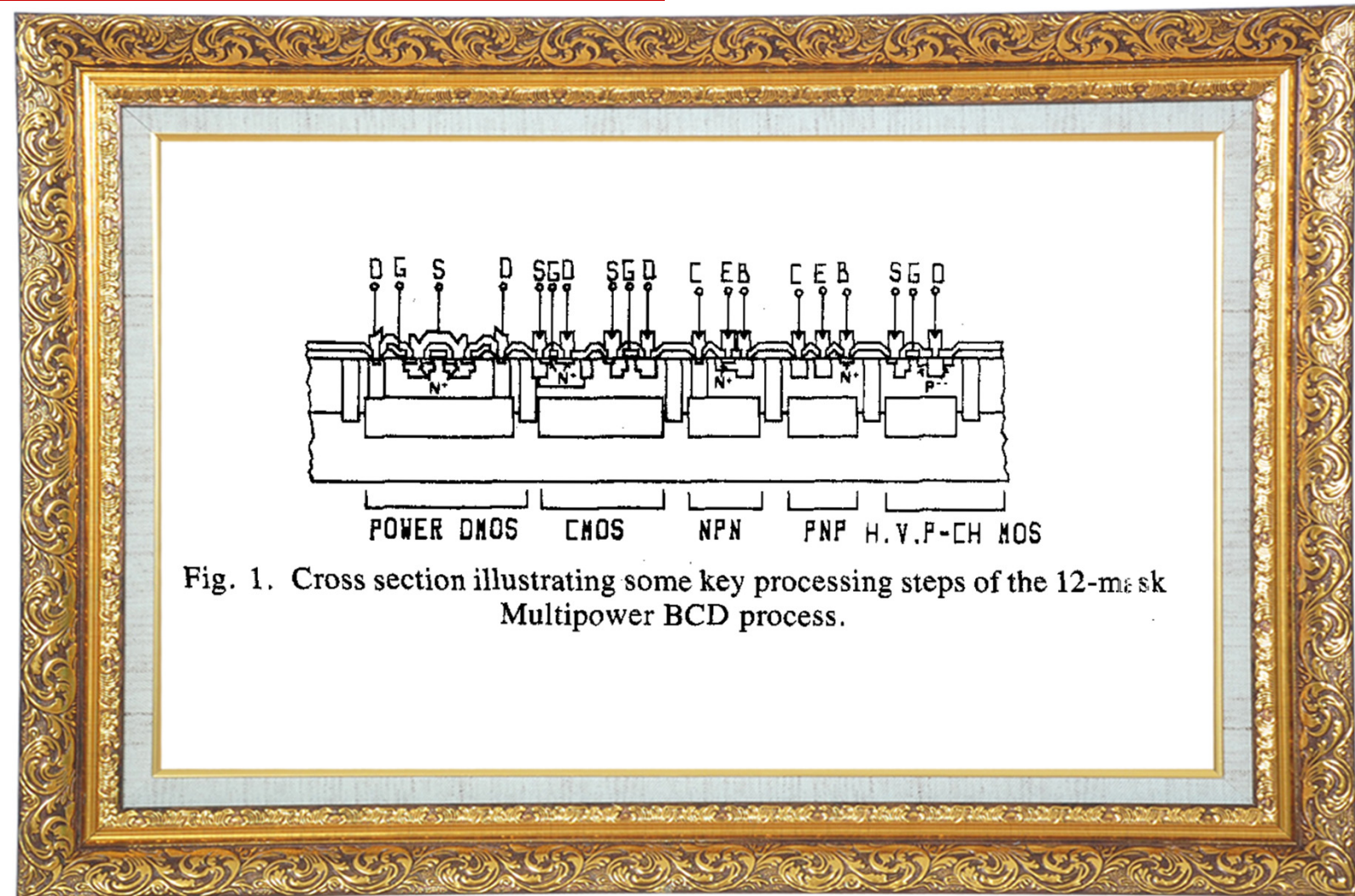
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- Circuit Design

What is BCD Technology^[1] ?

☐ **B**ipolar

☐ **C**MOS

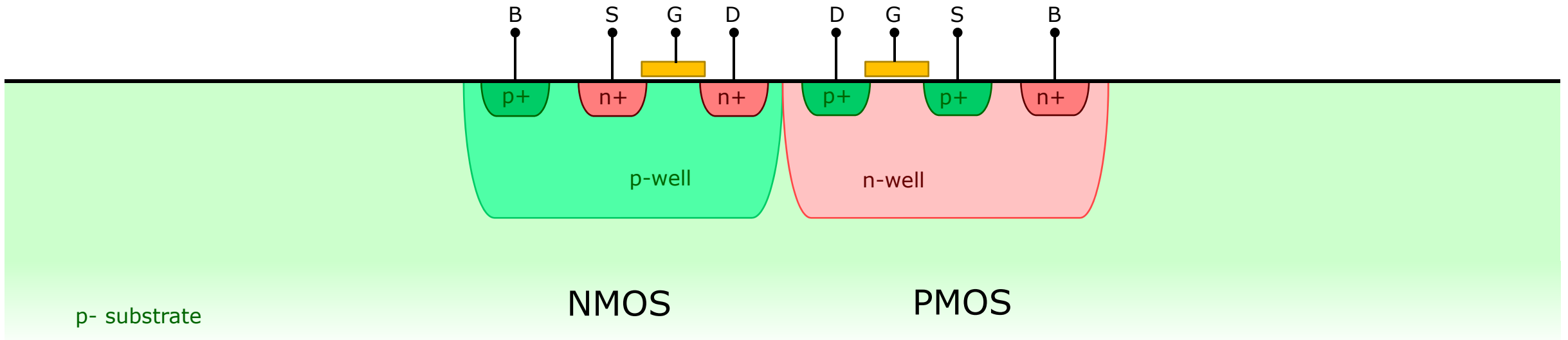
☐ **D**MOS



CMOS

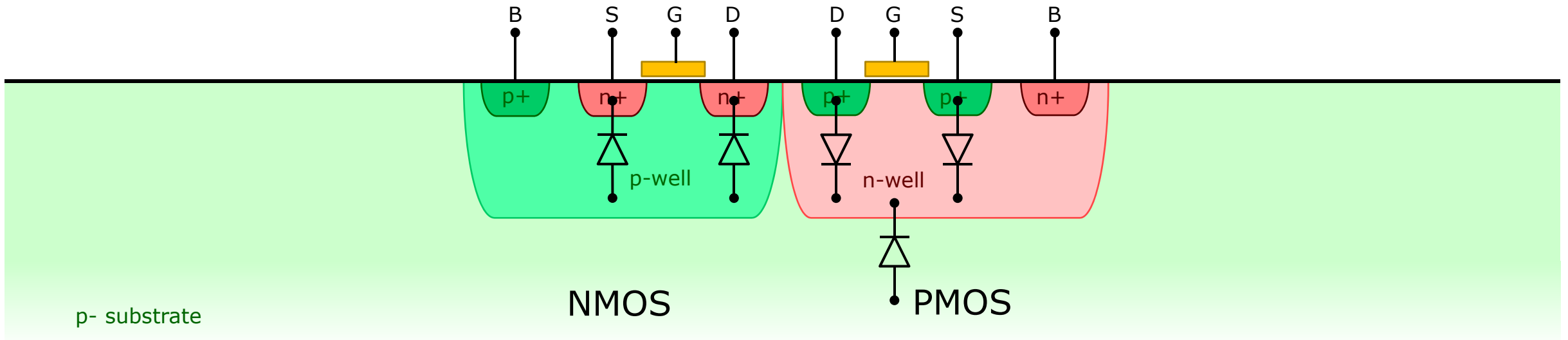
CMOS

- symmetrical devices
 - interchangeable source and drain
- well contacts
 - shared by multiple devices



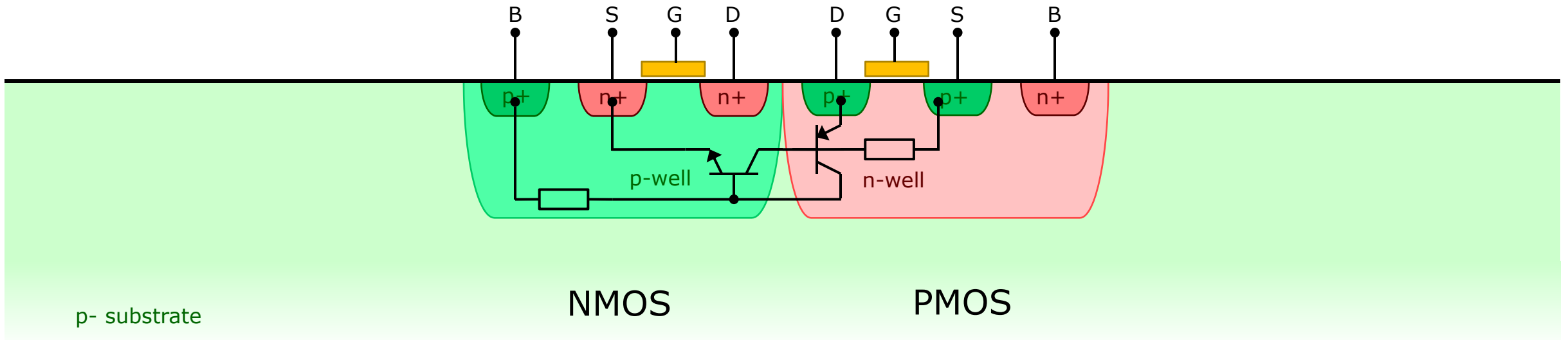
CMOS

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- well contacts
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- p-n junctions
 - shorted
 - reverse biased



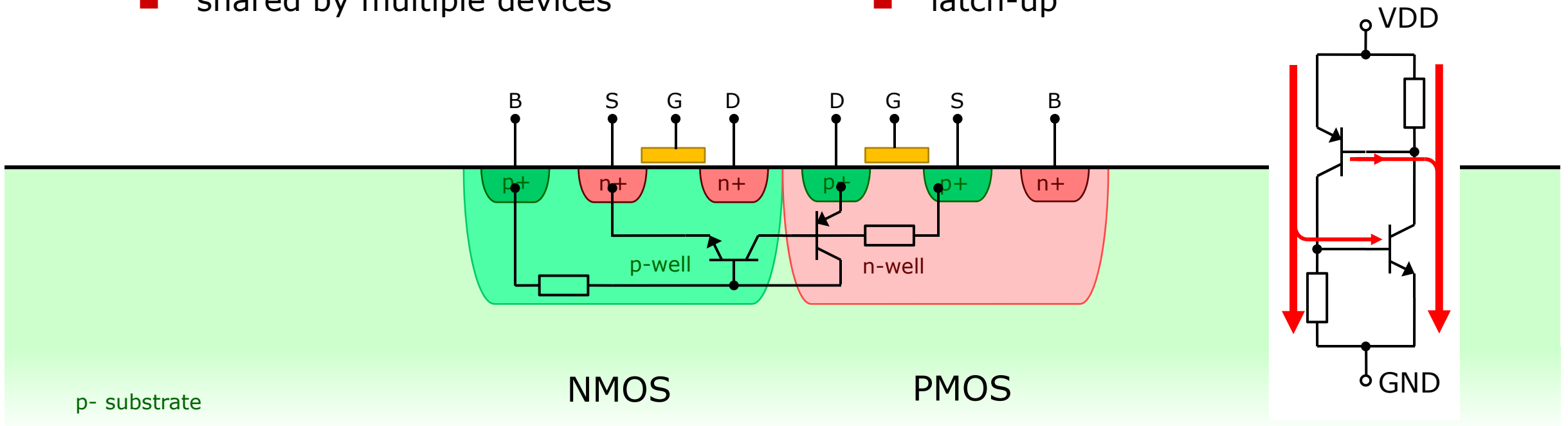
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- bipolar parasitics
 - latch-up



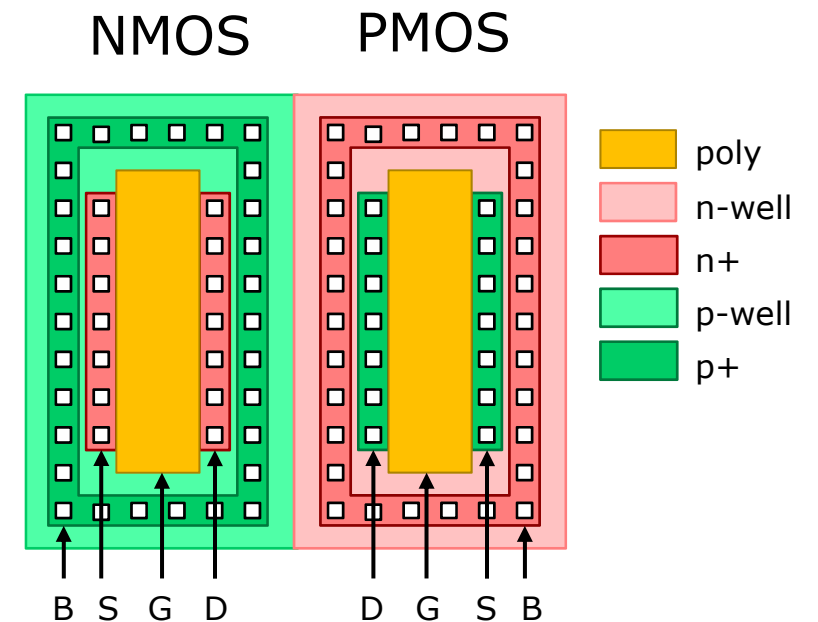
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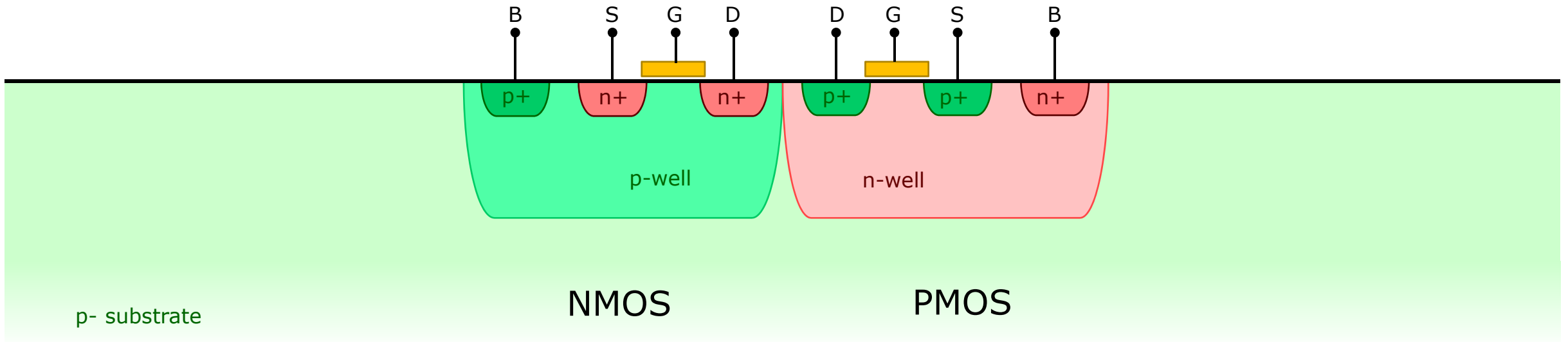


CMOS

- ❑ latchup design rules
 - part of standard design rule check
 - checks distance to nearest well connection
- ❑ digital layout
 - standard cells (NANDs, NORs, etc)
 - well contacts part of fixed layouts
 - well contacts are inserted by layout tool
 - correct by construction
- ❑ analog layout
 - well contact rings around devices

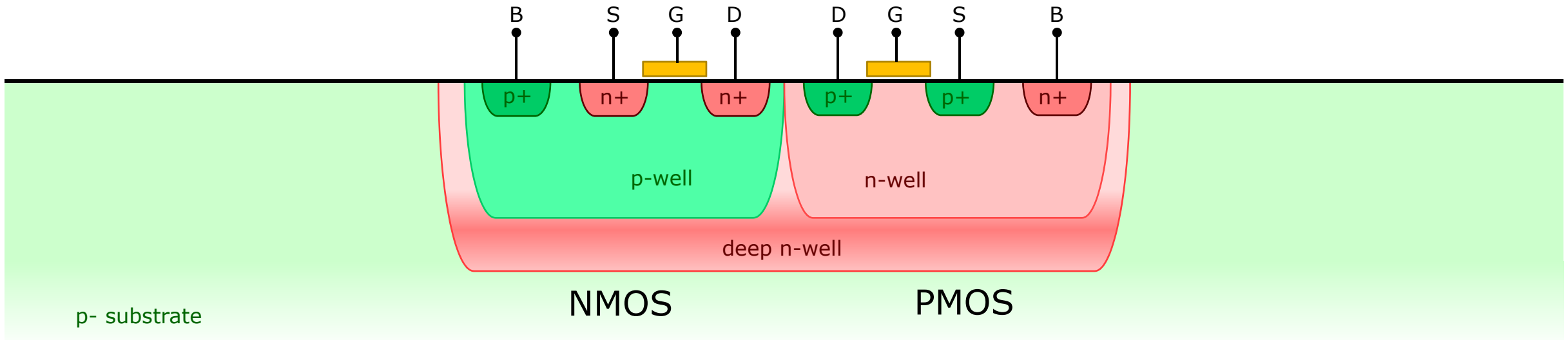


Isolated CMOS



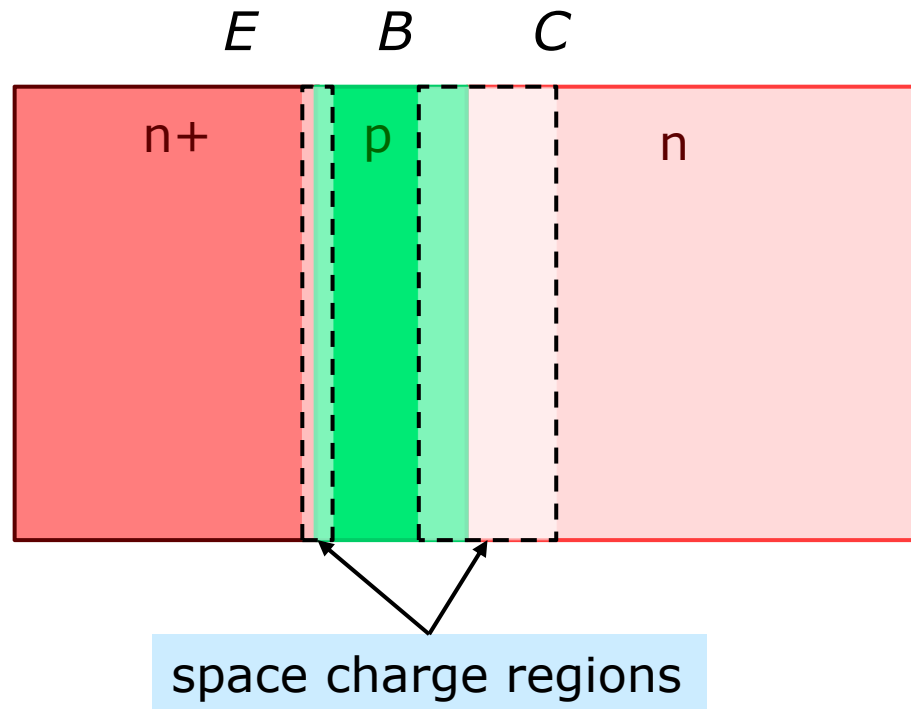
Isolated CMOS

- deep n-well
 - 'triple well'
 - high energy ion implantation
- inverse doping gradient
 - highest doping below n/p-well
 - does not affect surface behaviour
- isolation from substrate
 - NMOS body can be at different voltage than substrate



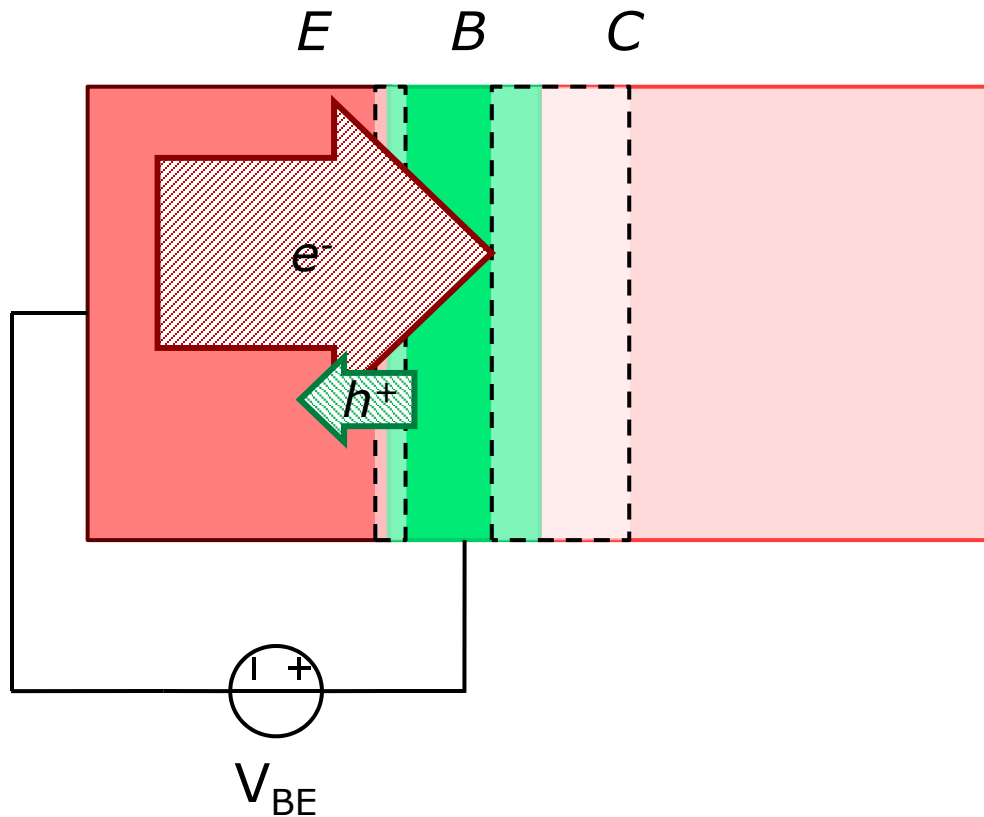
Bipolar Junction Transistor

Bipolar Junction Transistor (BJT)



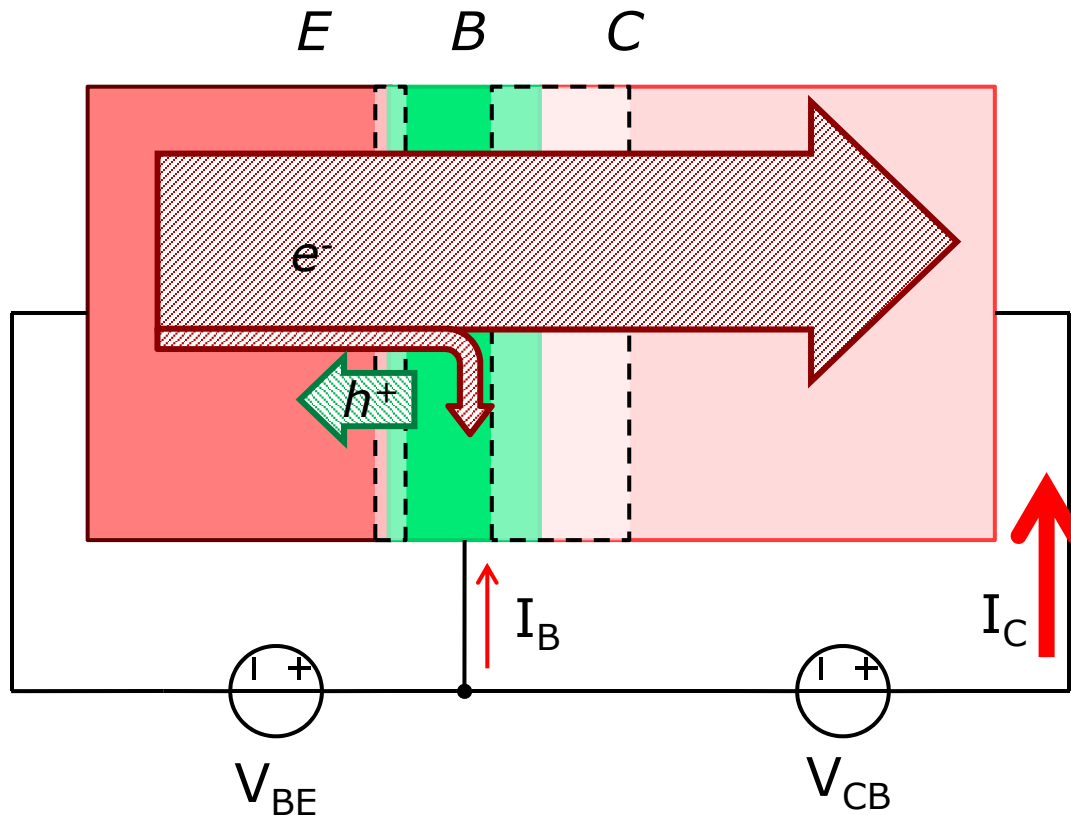
- bipolar
 - both electrons and holes are involved in transistor operation
- three regions with different dopant
 - heavily doped **emitter**
 - lightly doped **collector**
 - narrow, moderately doped **base**
- two pn-junctions
- NPN or PNP

Bipolar Junction Transistor (BJT)



- forward biased base-emitter junction
 - large electron current flowing from emitter to base
 - small hole current flowing from base to emitter

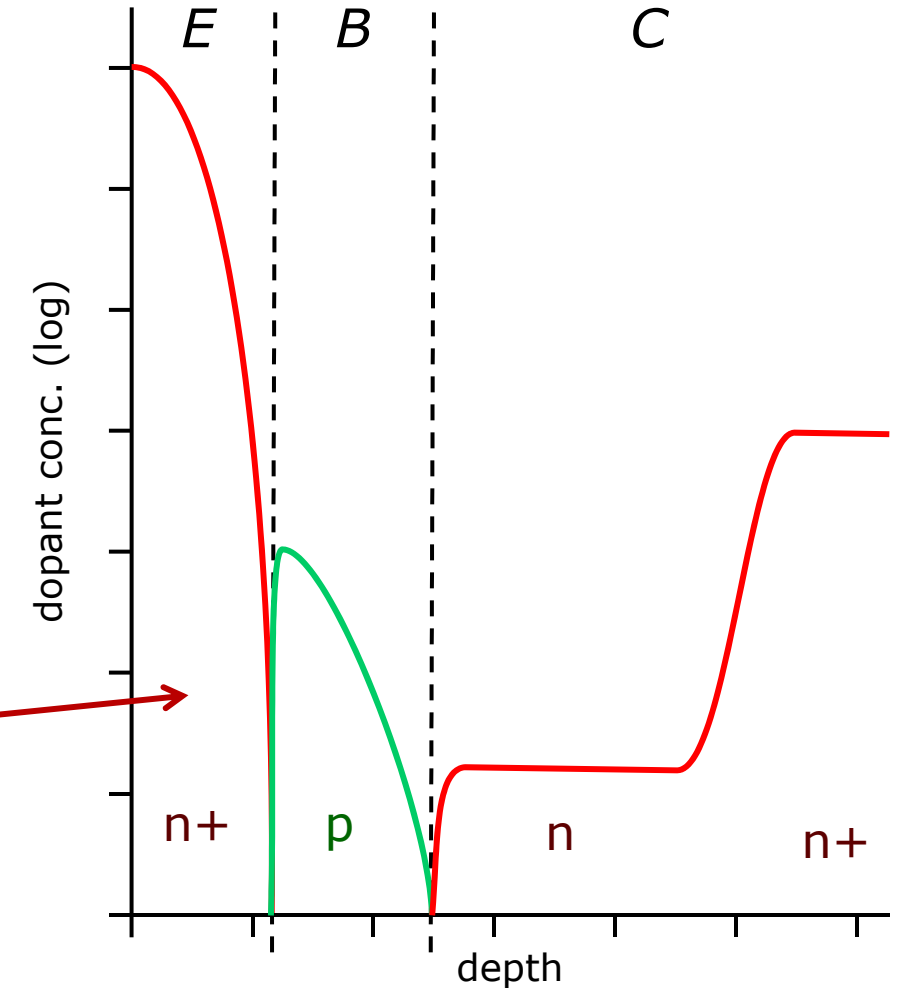
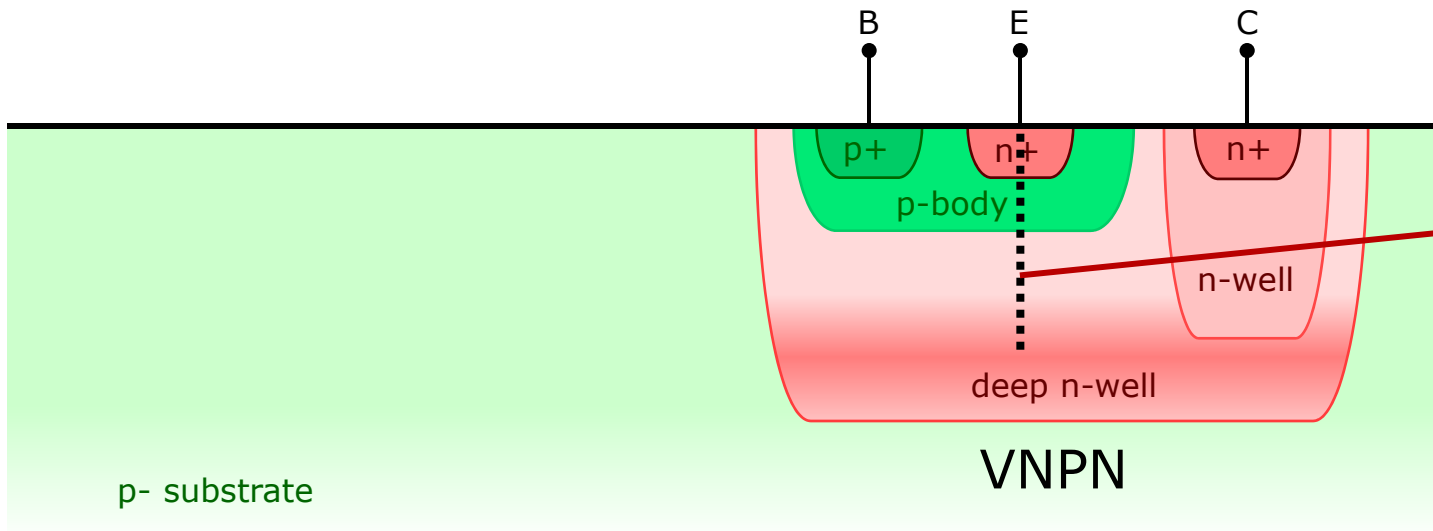
Bipolar Junction Transistor (BJT)



- forward biased base-emitter junction
 - large electron current flowing from emitter to base
 - small hole current flowing from base to emitter
- reverse biased base-collector junction
 - few electrons recombine in base
 - most electrons swept up by space-charge region base-collector junction
- high current gain
 - $I_C/I_B = <50,200>$

Bipolar Junction Transistor (BJT)

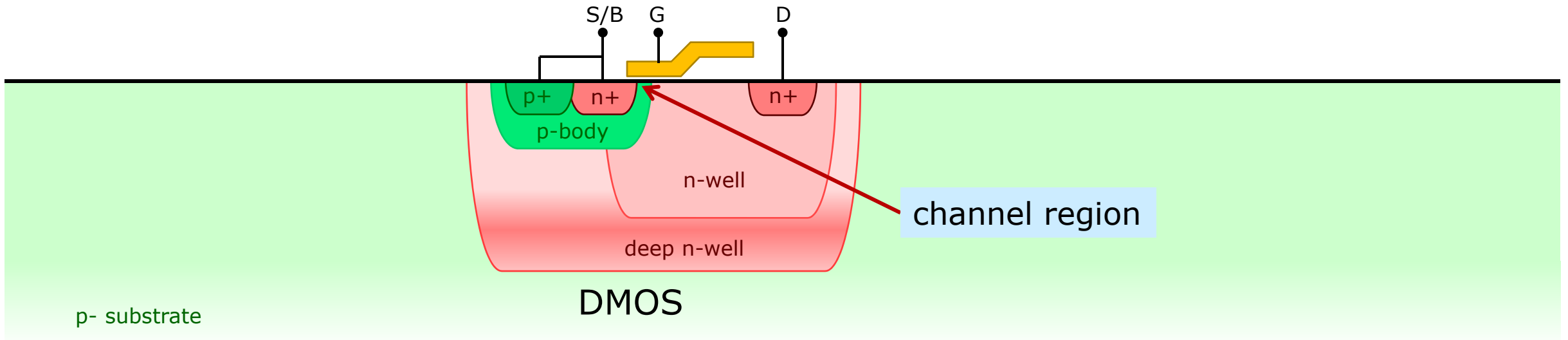
- additional p-body diffusion
 - base width determined by difference in **vertical** diffusion p-body and n+ emitter
 - **vertical** diffusion profile yields NPN with **high current gain**
 - high doping in bottom deep n-well reduces collector contact resistance



DMOS

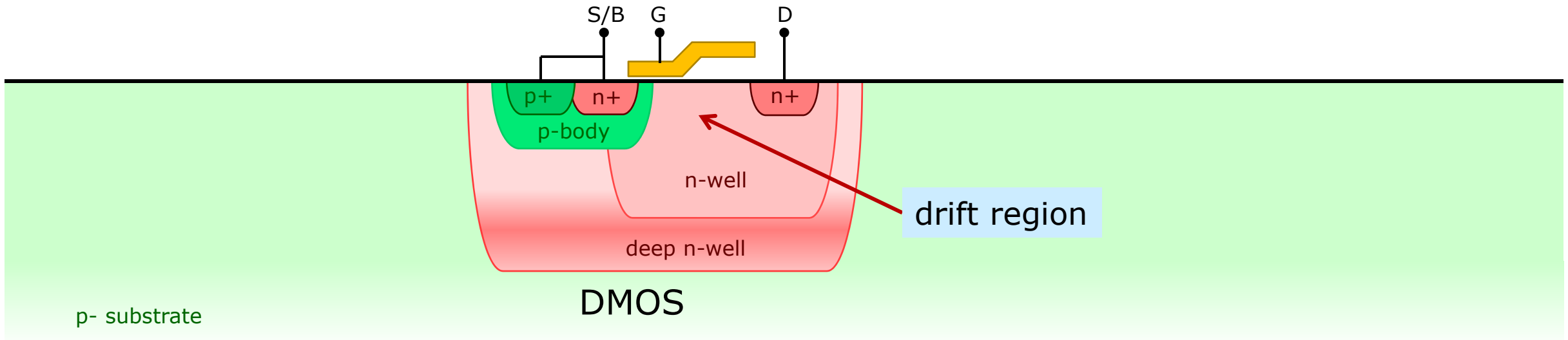
DMOS

- 'D' => double-diffused
 - refers to n+ source and p-body diffusions
 - self-alignment: poly gate
 - fixed channel length
 - difference in lateral diffusion n+ source / p-body
 - only channel width can be scaled



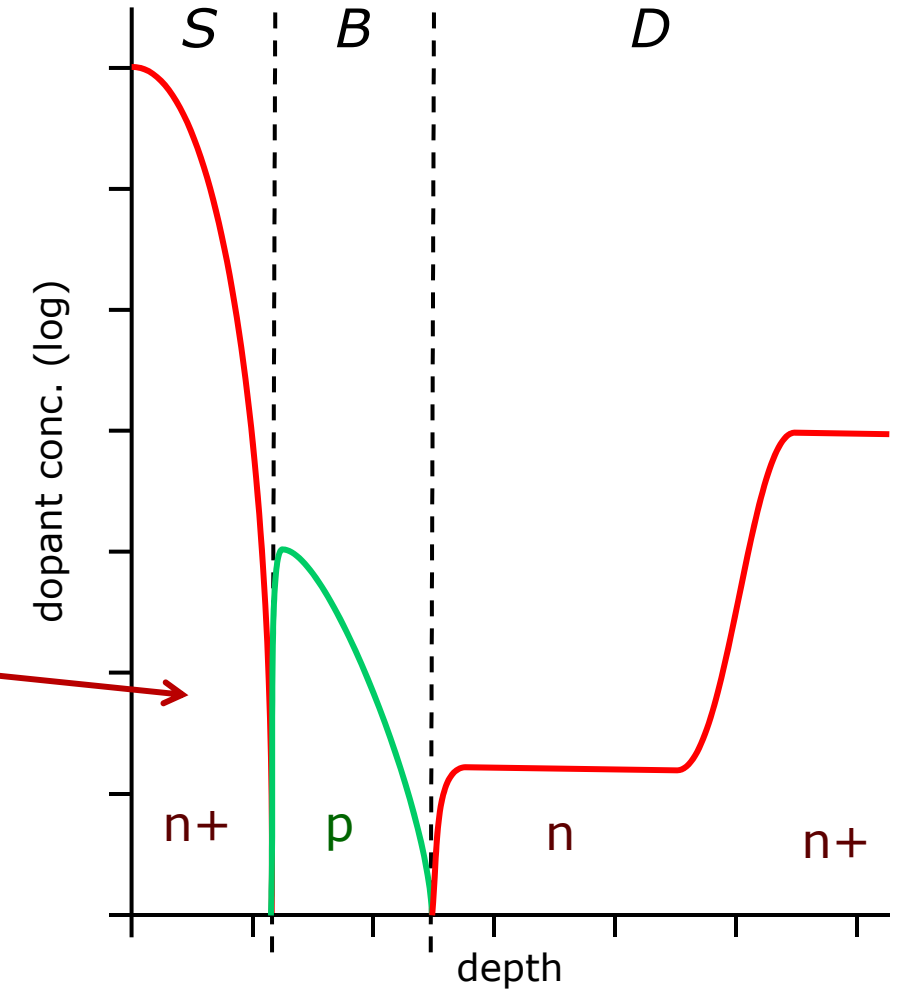
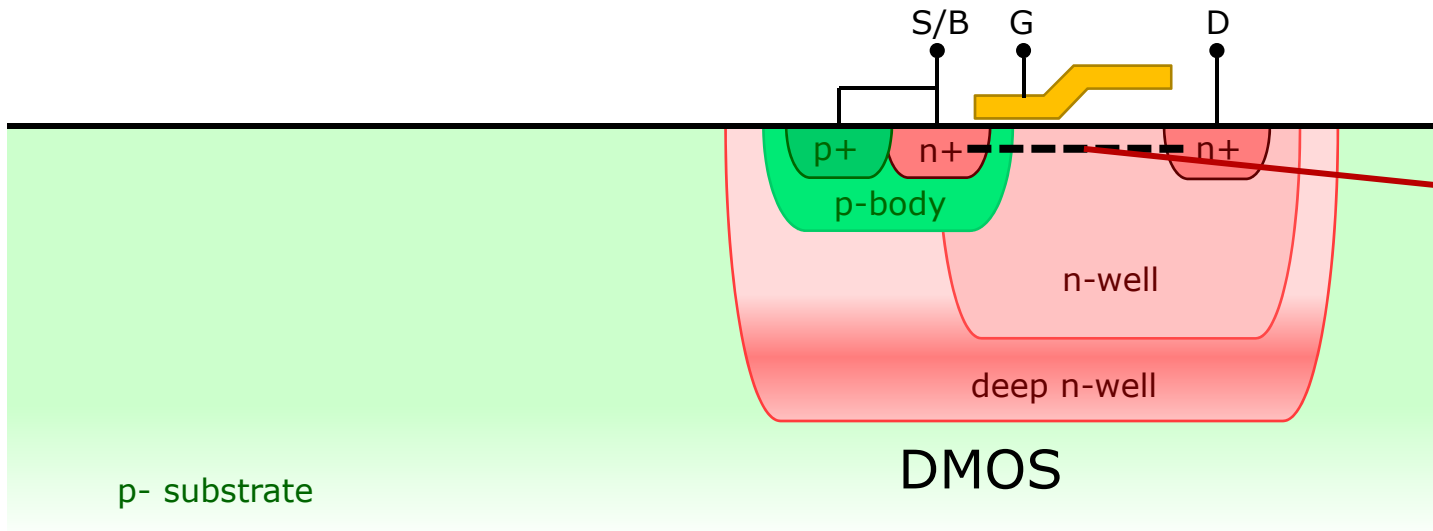
DMOS

- $V_{DSmax} \gg V_{GSmax}$
- drift region
 - lightly doped region between channel and drain contact
 - determines on-resistance
- field plate
 - step in poly-gate
 - reduced surface field (RESURF)
- deep n-well
 - isolate source/body from substrate



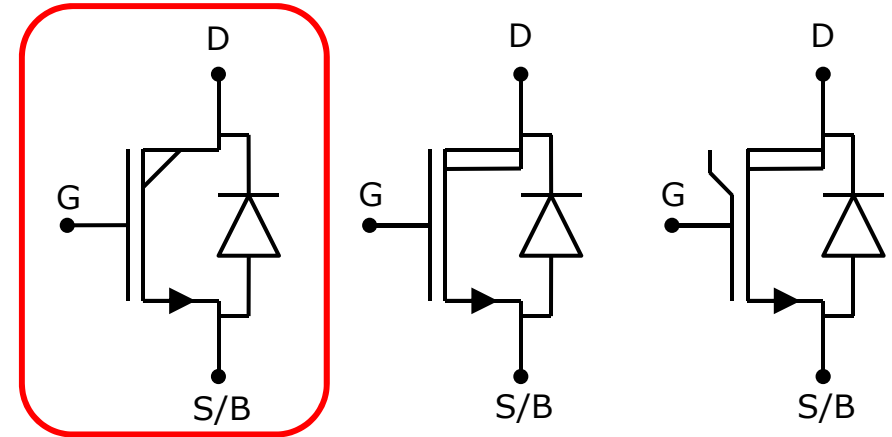
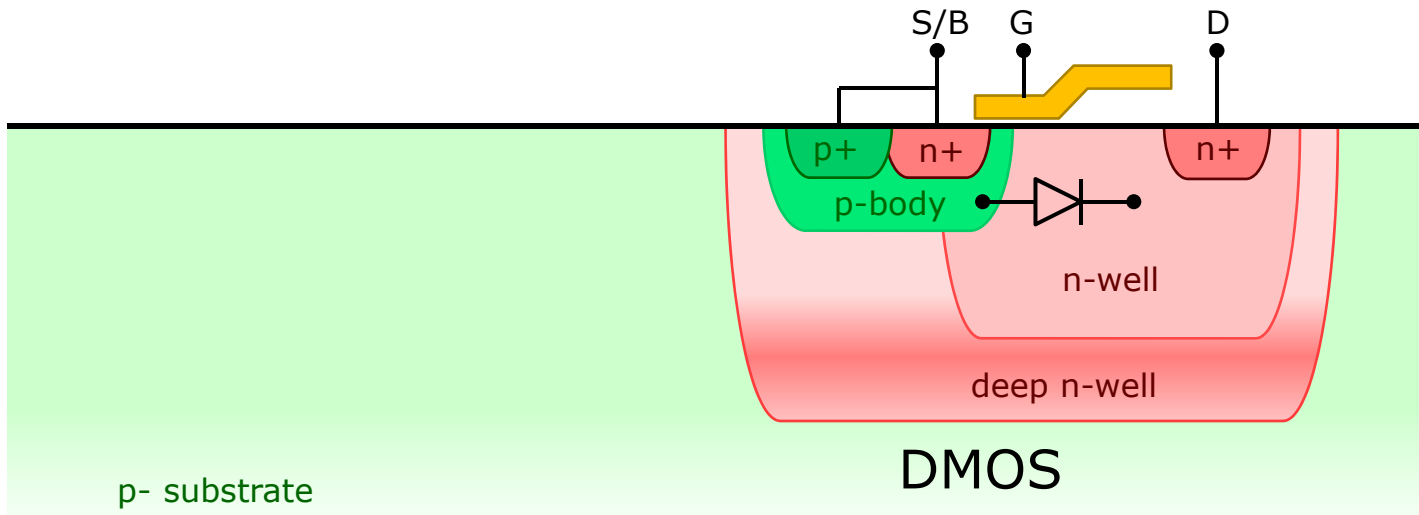
DMOS

- asymmetrical
 - drain and source not interchangeable
- very effective bipolar parasitic
 - lateral diffusion profile similar to NPN



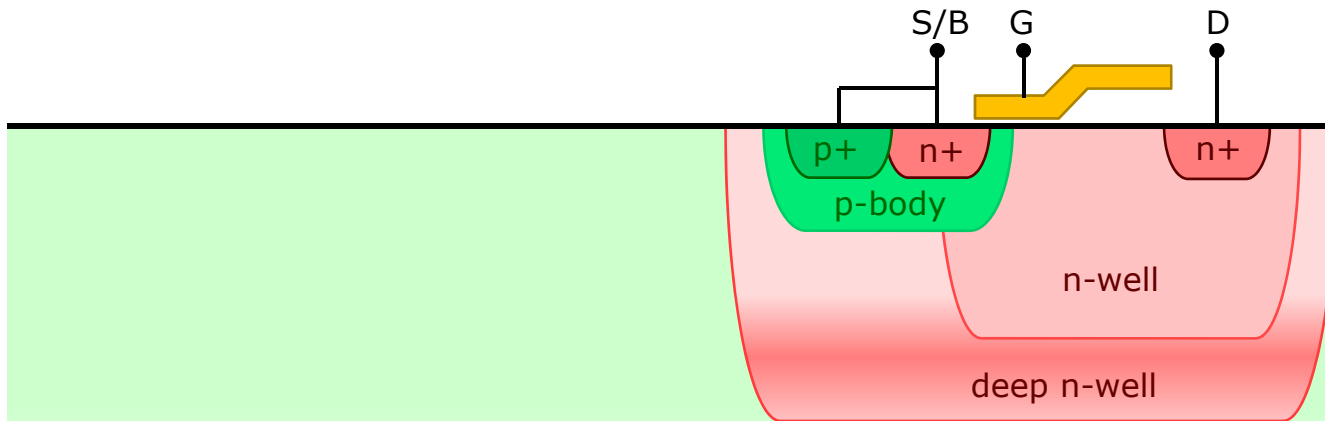
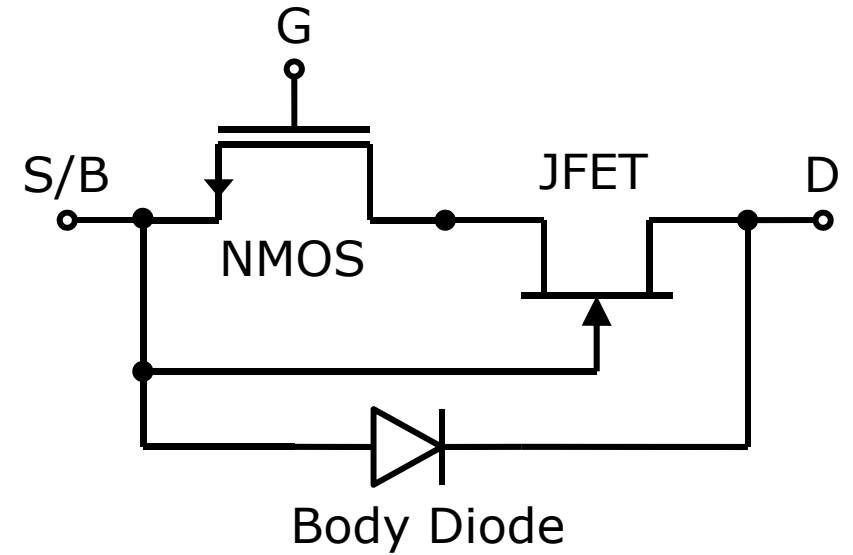
DMOS

- asymmetrical
 - drain and source not interchangeable
- very effective bipolar parasite
 - lateral diffusion profile similar to NPN
- body always shorted to source to prevent turn-on parasitic NPN
 - diode between source and drain
 - explicitly drawn in device symbol



DMOS

- device model
 - NMOS with JFET cascode
 - JFET pinch-off limits internal NMOS drain voltage
- drain source diode
 - forward conduction characterized

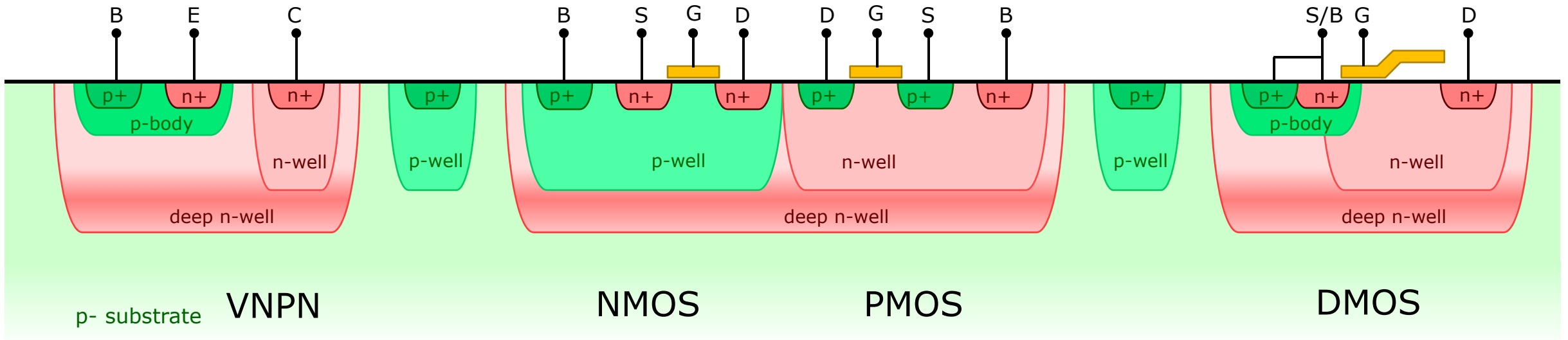


DMOS

p- substrate

BJT + CMOS + DMOS

- combining BJT, CMOS and DMOS on same wafer yields BCD technology
- lateral isolation with p-well barriers
 - lateral spacing scales with breakdown voltage



HV-PMOS ?

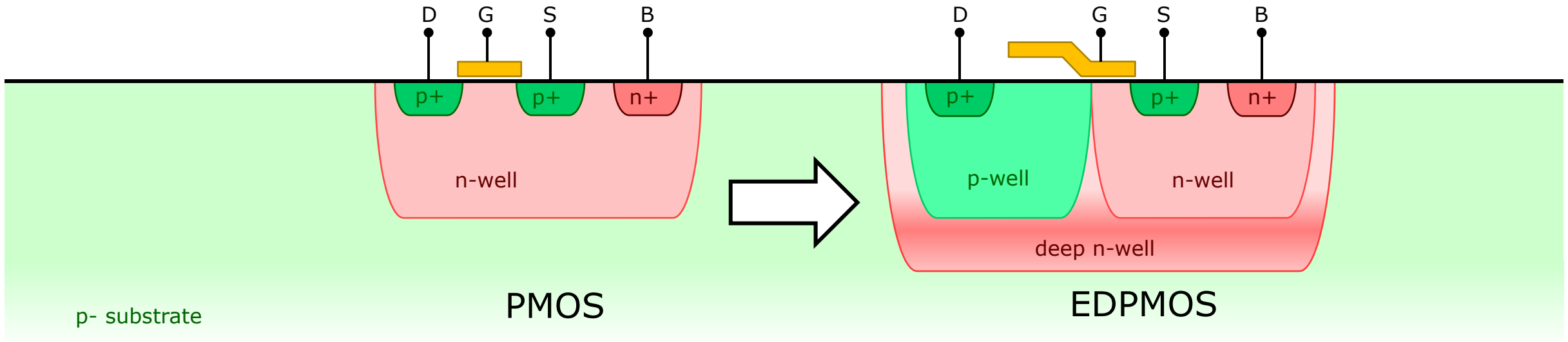
HV-PMOS

□ p-type DMOS

- possible but unusual due to higher on-resistance (hole mobility)
- requires additional diffusions

□ extended drain PMOS (EDPMOS)

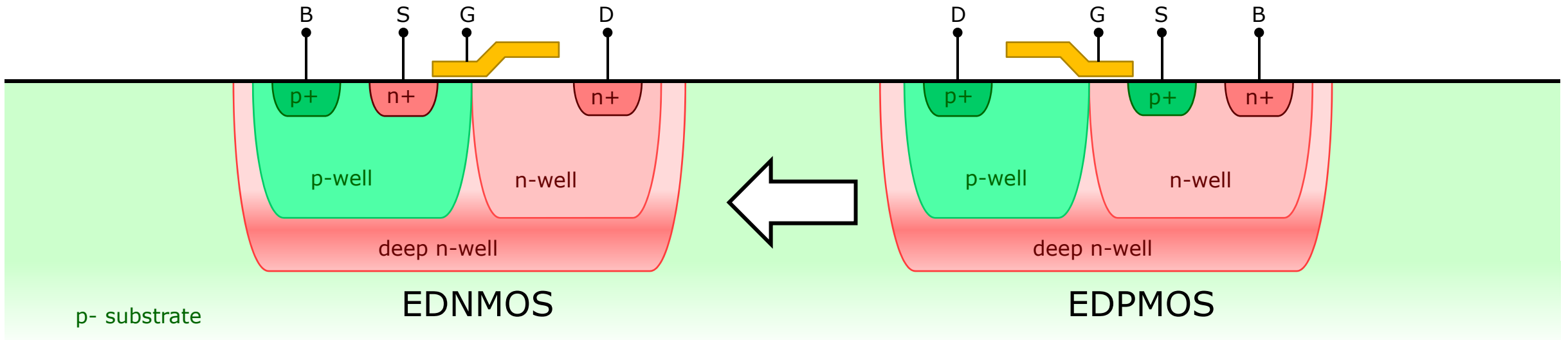
- p-well drift region
- field plate
- deep n-well isolation



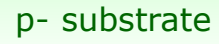
HV-NMOS

- extended drain NMOS (EDNMOS)
 - HV derivative of standard CMOS

- cheaper alternative to BCD
 - no p-body diffusion needed
 - no vertical NPN
 - higher on-resistance than DMOS



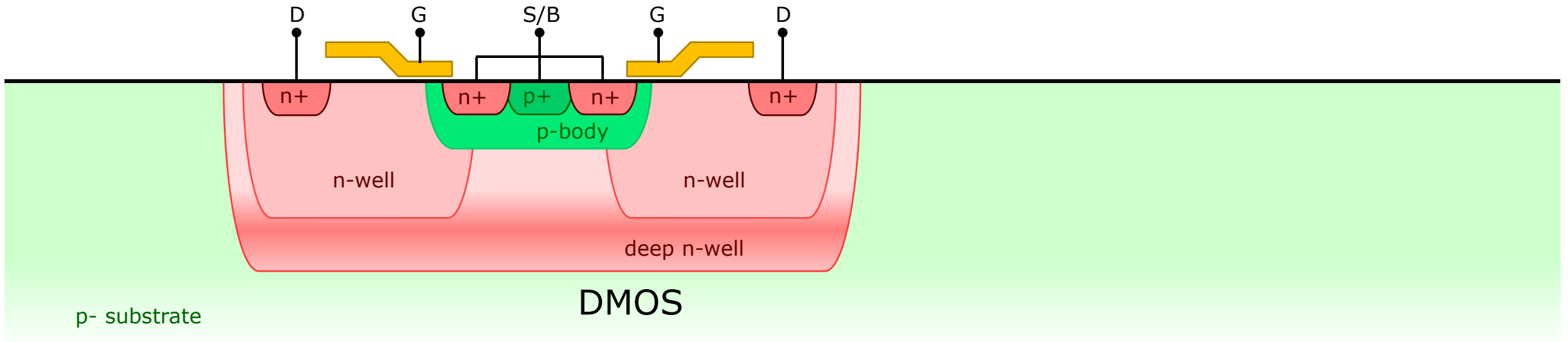
Folding





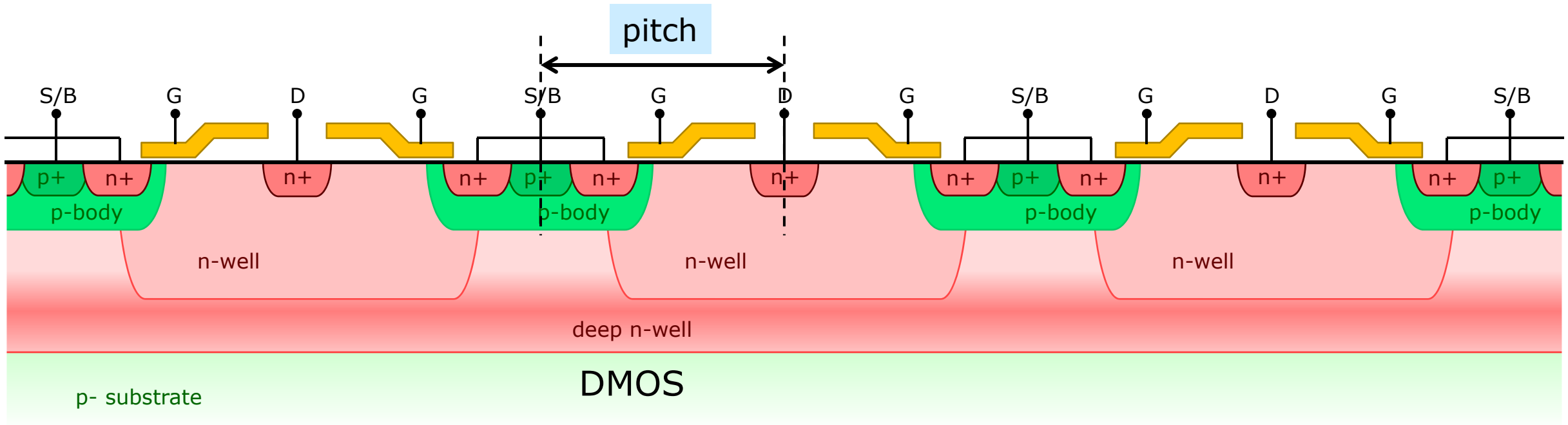
DMOS

□ Folding: Source Centered



DMOS

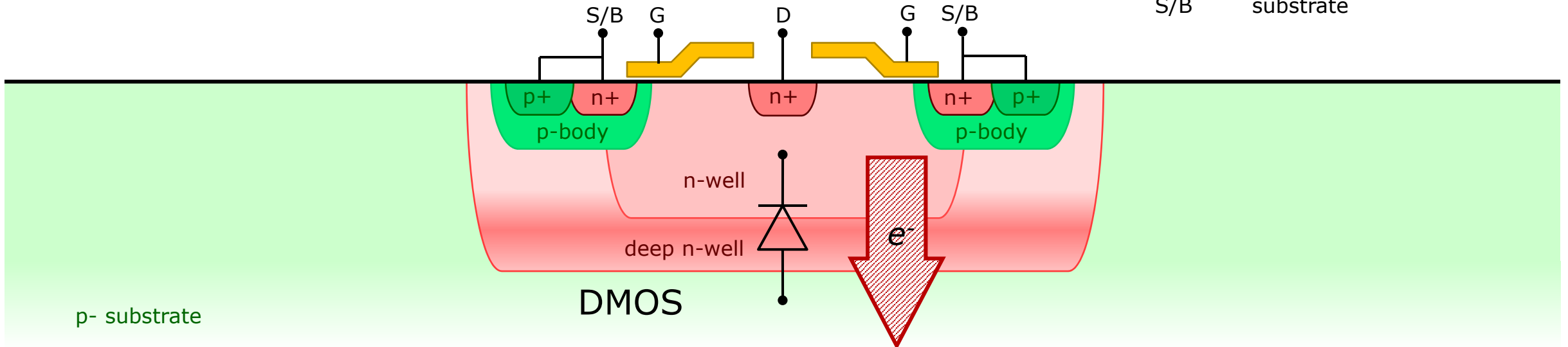
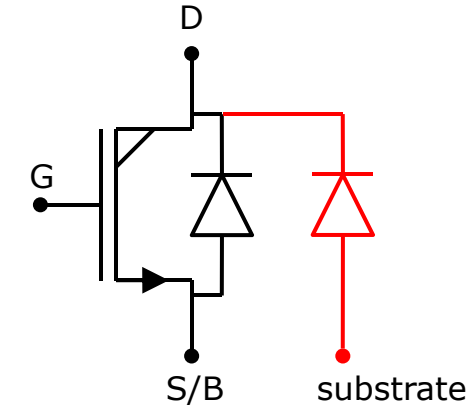
- power device
 - folded many times (e.g. >100)
- pitch
 - centre source to centre drain
- net device area
 - pitch x gate width
 - independent of folding
 - excluding isolation area / guard rings



Isolation

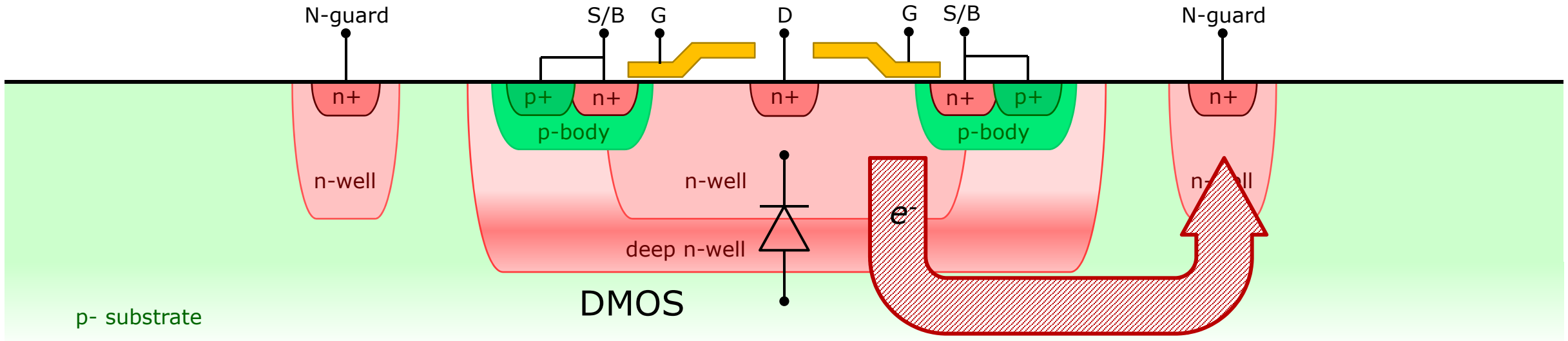
Isolated DMOS

- parasitic drain to substrate diode
 - forward biased when drain below substrate
- minority carrier injection
 - latch-up risk



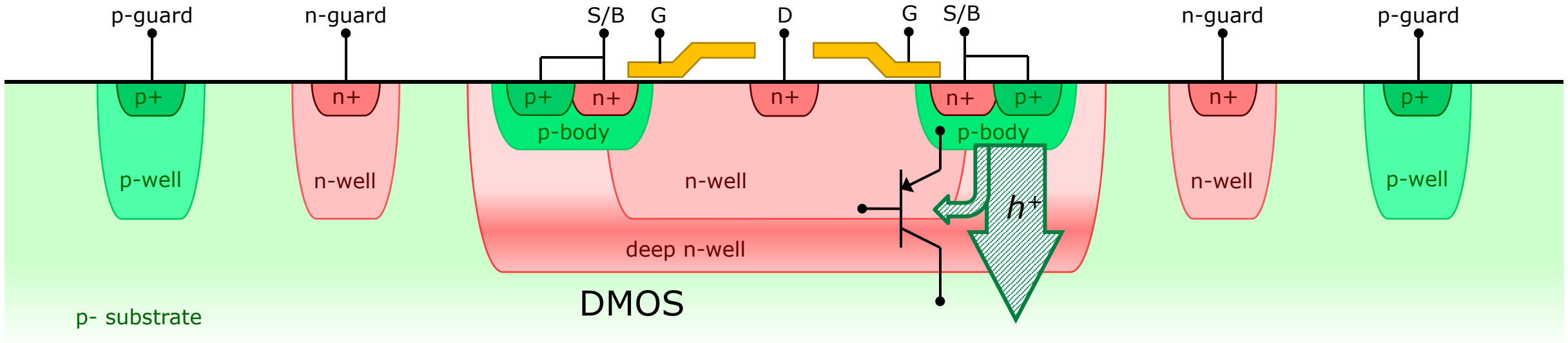
Isolated DMOS

- parasitic drain to substrate diode
 - forward biased when drain below substrate
- minority carrier injection
 - latch-up risk
- n-guard ring
 - collect minority carriers
 - connect to positive voltage



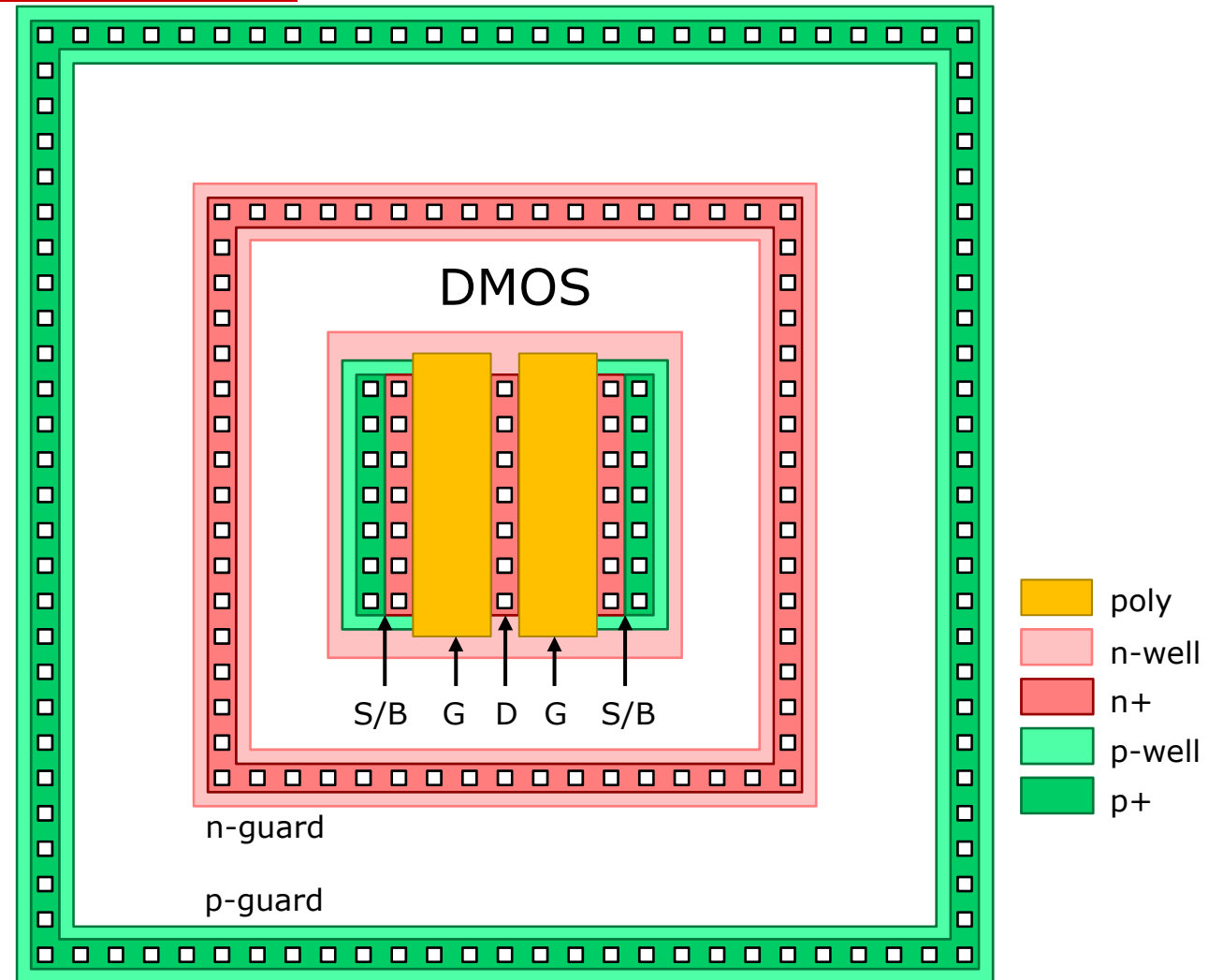
Isolated DMOS

- parasitic substrate PNP
 - active if source above drain
 - current flows to substrate i.s.o. drain
- majority carrier injection
 - high dissipation
- p-guard ring
 - fix substrate potential



Isolation Area

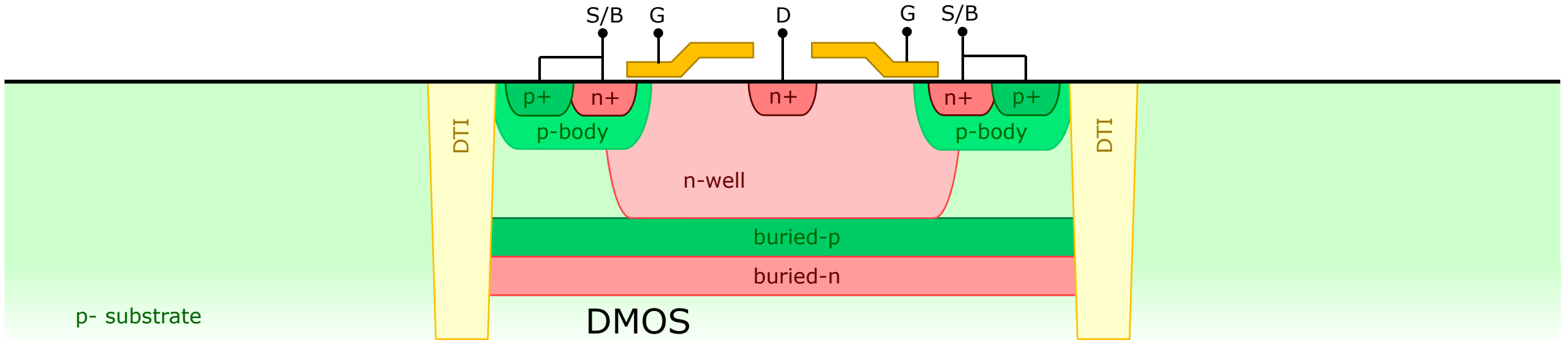
- guard rings increase device area
 - lateral spacing increases with breakdown voltage
- relatively expensive for small HV devices
- design economy
 - only use HV devices when absolutely necessary



[illegible]

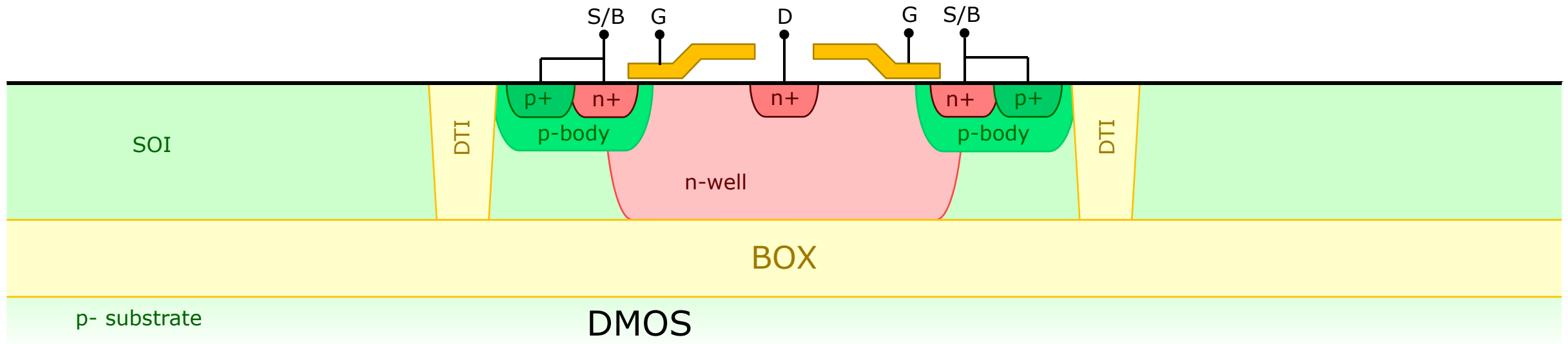
Deep Trench Isolation (DTI)

- lateral dielectric isolation
 - deep trenches (e.g. $30\mu\text{m}$)
- very expensive
 - area / waferprice trade-off
- area reduction
 - replaces guard rings



Silicon-On-Insulator (SOI)^[3]

- ❑ buried oxide (BOX)
- ❑ full dielectric isolation
 - no latch-up
 - no leakage between components
 - high temperature operation $> 175^{\circ}\text{C}$
- ❑ very, very expensive
 - requires special wafers
 - warping limits wafer size to 8"

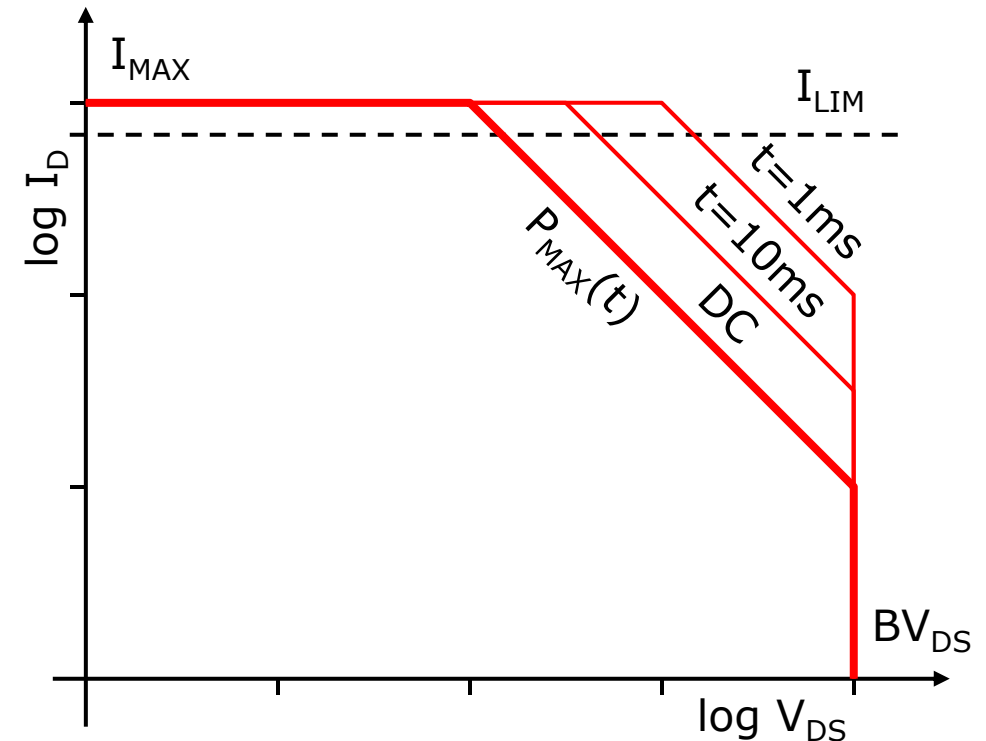


Outline

- Introduction & Motivation
- BCD Process Technology & Devices
- Device Characteristics & Figures-of-Merit
- Applications
- Circuit Design

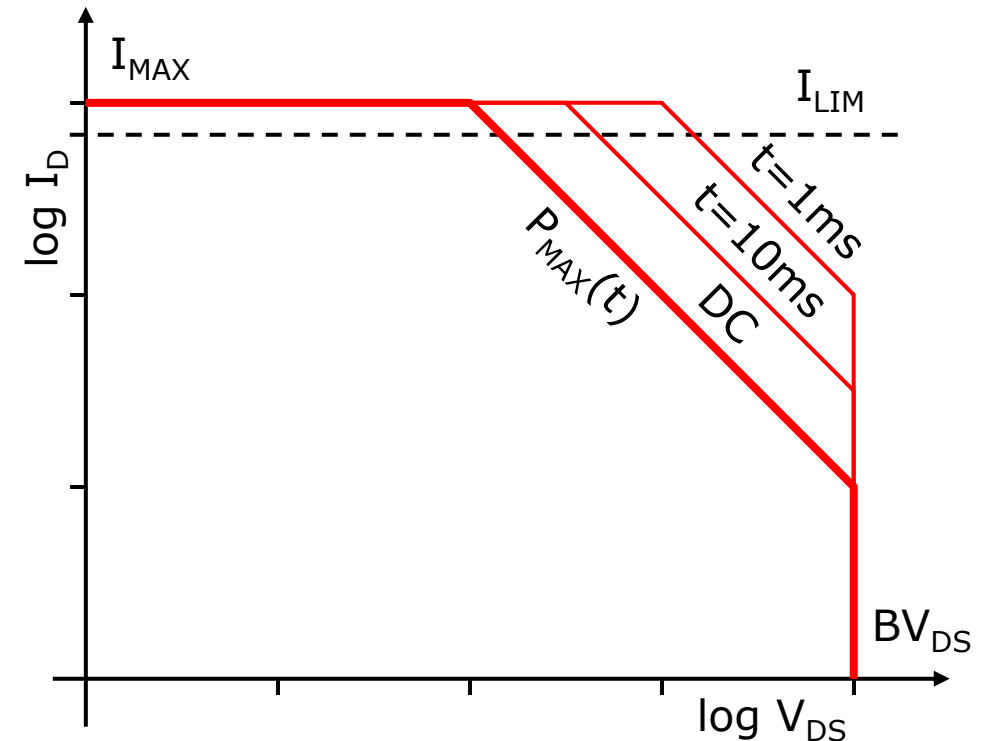
Safe Operating Area (SOA(R))

- voltage limit BV_{DS}
 - breakdown voltage
 - operating voltage $< BV_{DS}$
- current limit I_{MAX}
 - electromigration
 - bondwires
 - protection circuits limit to $I_{LIM} < I_{MAX}$
- dissipation limit $P_{MAX}(t)$
 - thermal
 - time dependent



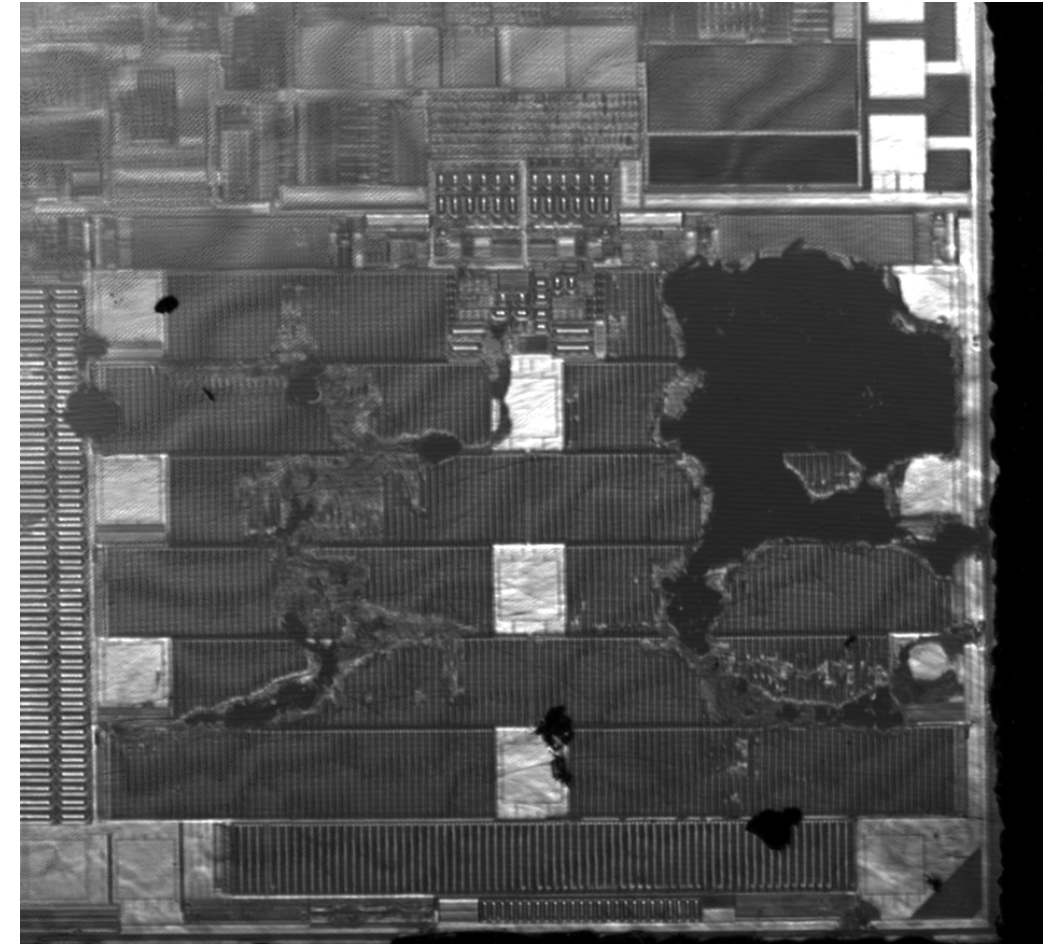
Temperature Range

- -40°C to 150°C
 - normal operating range (DC)
- 150°C to 200°C
 - thermal protection kicks in (msec)
 - e.g. sustained short circuit
- 200°C to 350°C
 - short thermal overshoot (<μsec)
 - e.g. hard switching transitions
- beyond 350°C
 - catastrophic failure



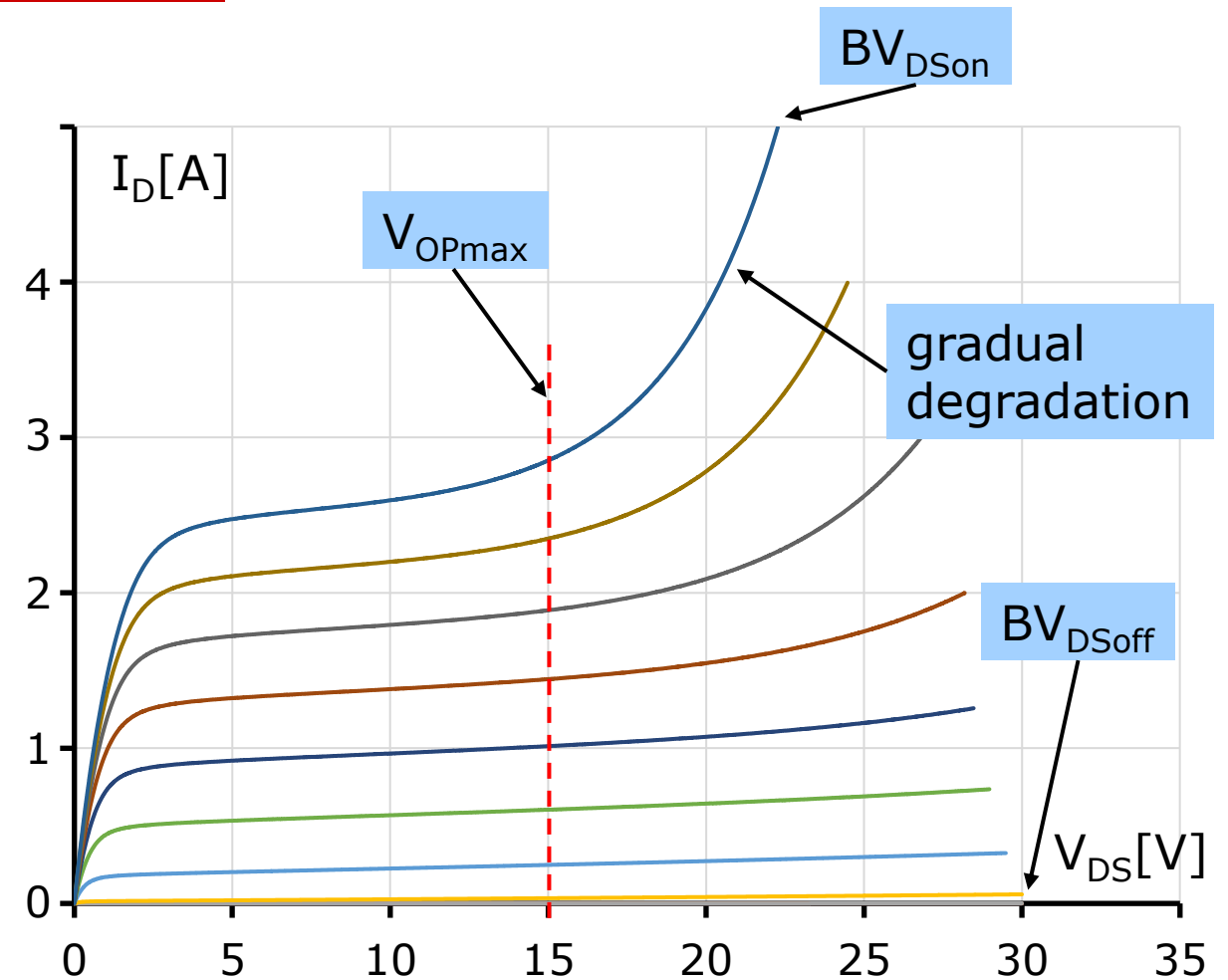
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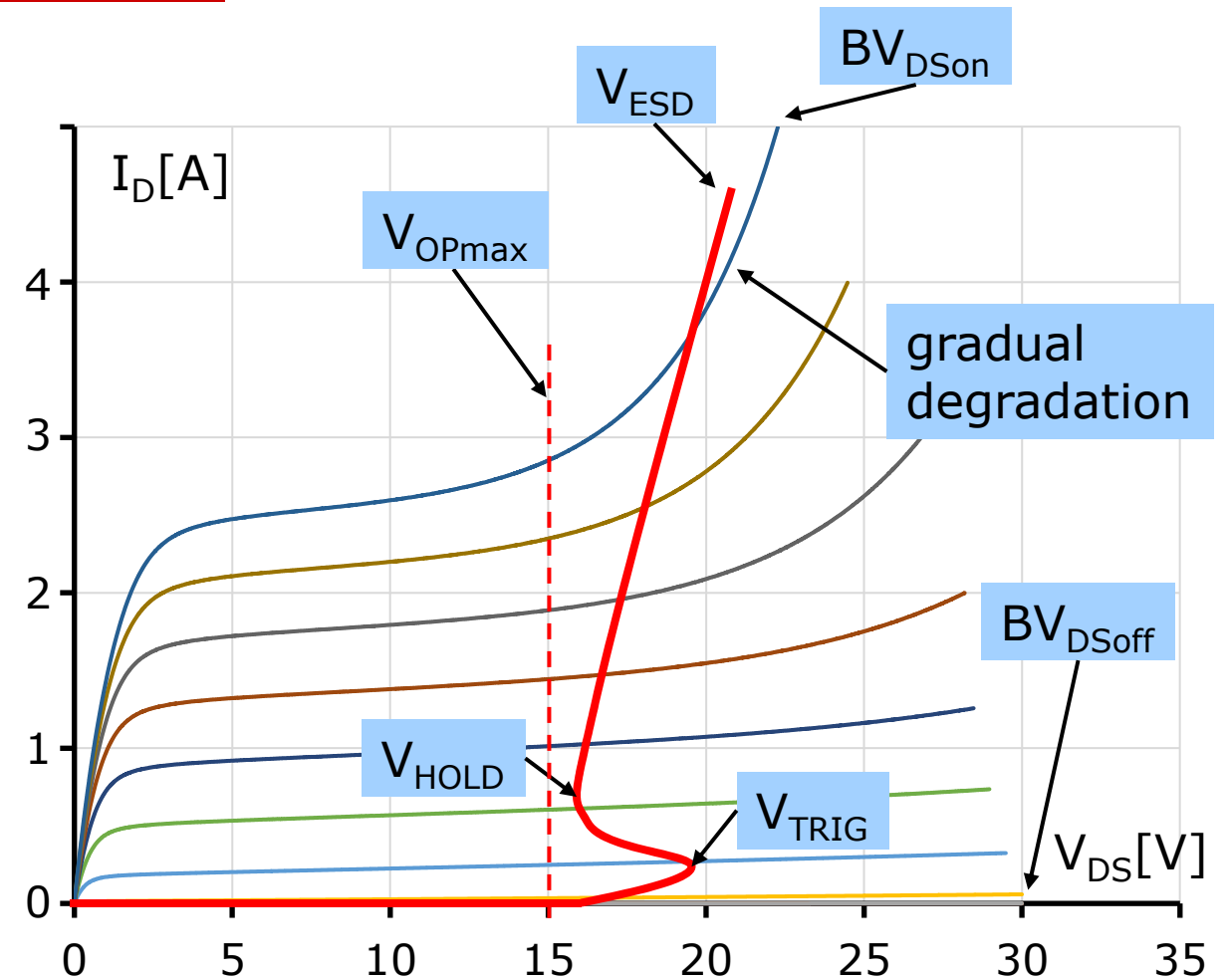
Breakdown Voltage BV_{DS}

- bias dependent
 - on-state $BV_{DSon} < \text{off-state } BV_{DSoff}$
- maximum operating voltage V_{OPmax}
 - no lifetime reduction



Breakdown Voltage BV_{DS}

- bias dependent
 - on-state $BV_{DSon} < \text{off-state } BV_{DSoff}$
- maximum operating voltage V_{OPmax}
 - no lifetime reduction
- ESD protection
 - trigger voltage V_{TRIG}
 - hold voltage V_{HOLD}
 - max ESD voltage V_{ESD}
- supply voltage V_{SUP}
 - $V_{SUP} < \min(V_{HOLD}, V_{OPmax})$
 - $\max(V_{TRIG}, V_{ESD}) < BV_{DSon}$

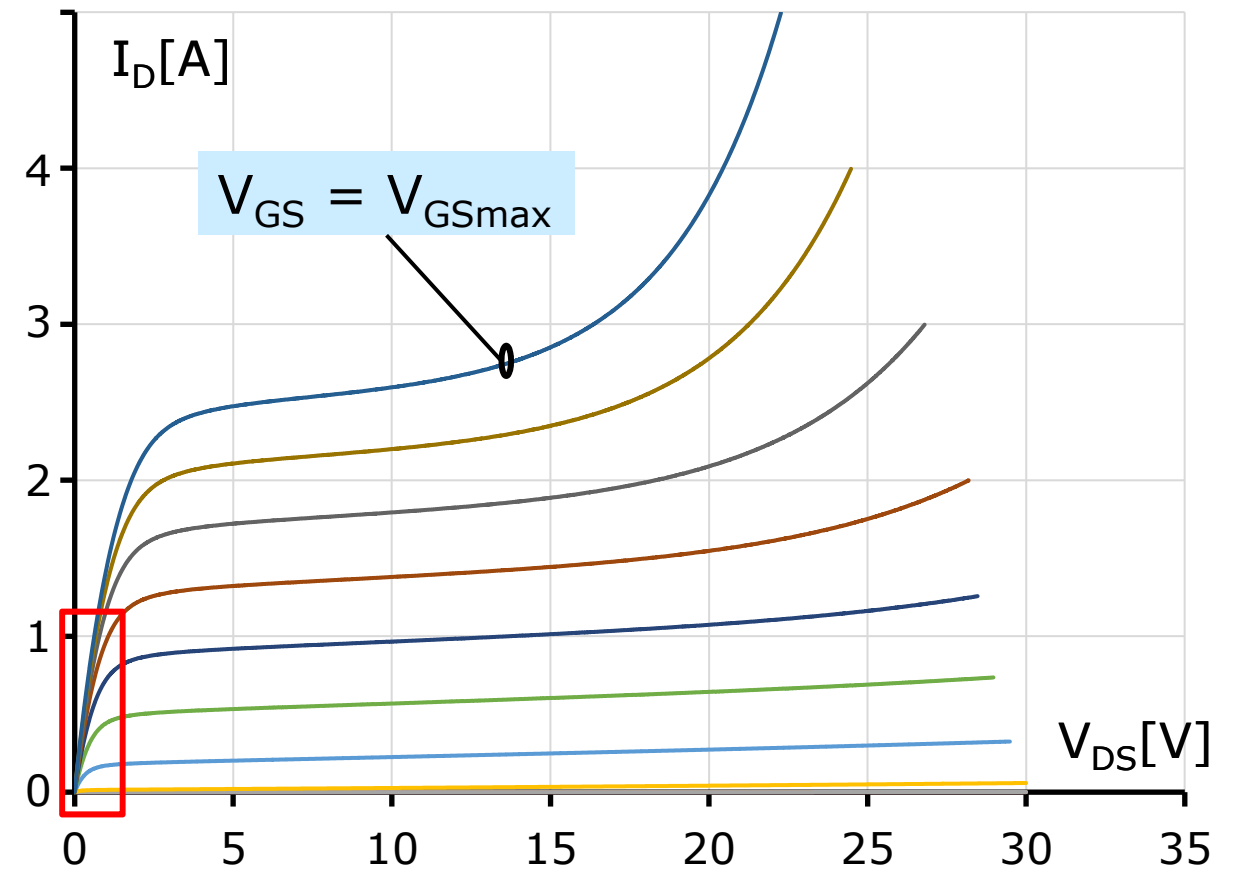


On-Resistance

□ linear region:

■ $V_{DS} < V_{GS} - V_{TH}$

■ $I_D = \beta \cdot \left(V_{GS} - V_{TH} - \frac{V_{DS}}{2} \right) \cdot V_{DS}$



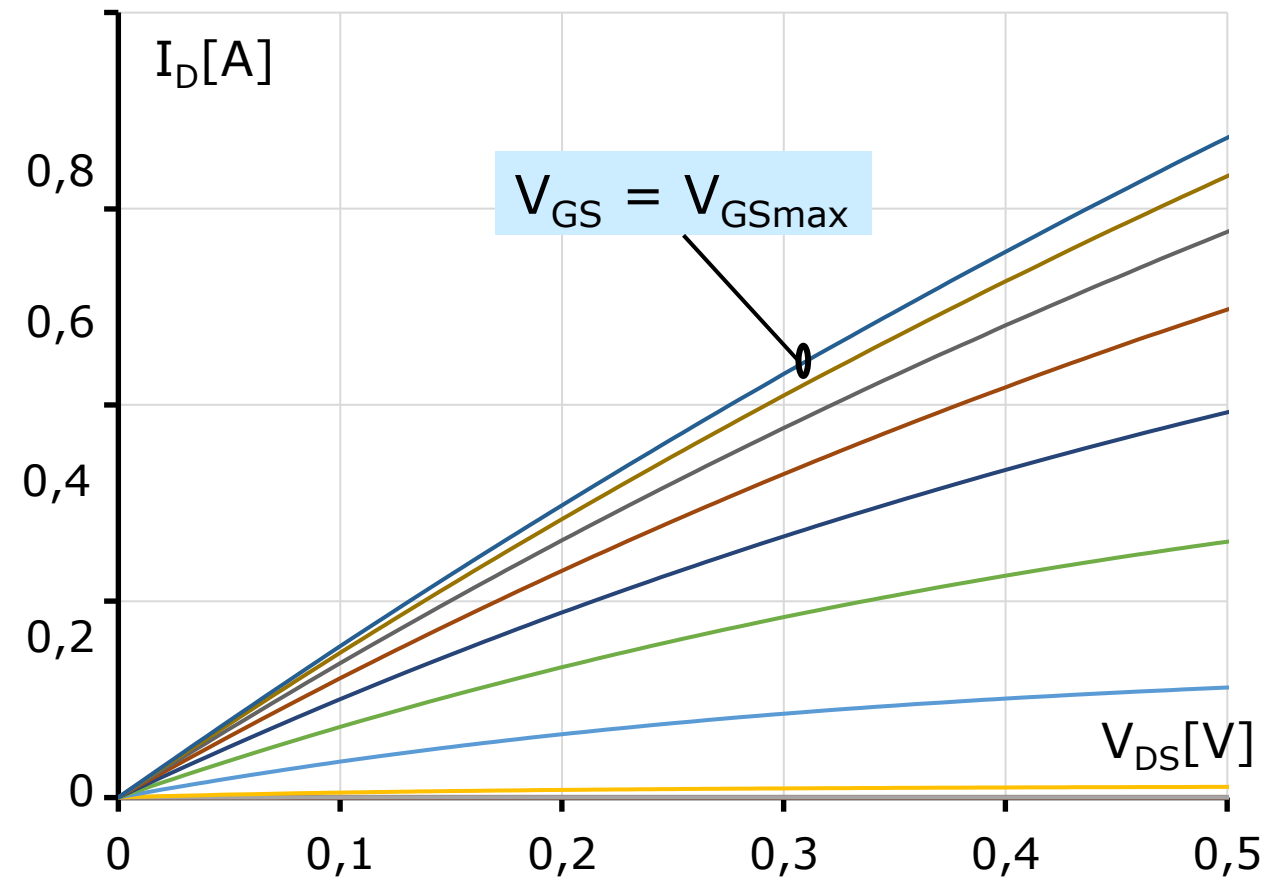
On-Resistance

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- $I_D = \beta \cdot \left(V_{GS} - V_{TH} - \frac{V_{DS}}{2} \right) \cdot V_{DS}$

□ how to determine R_{DSon} ?

- $\frac{1}{R_{DSon}} = \left. \frac{dI_D}{dV_{DS}} \right|_{V_{DS}=0, V_{GS}=V_{GSmax}}$



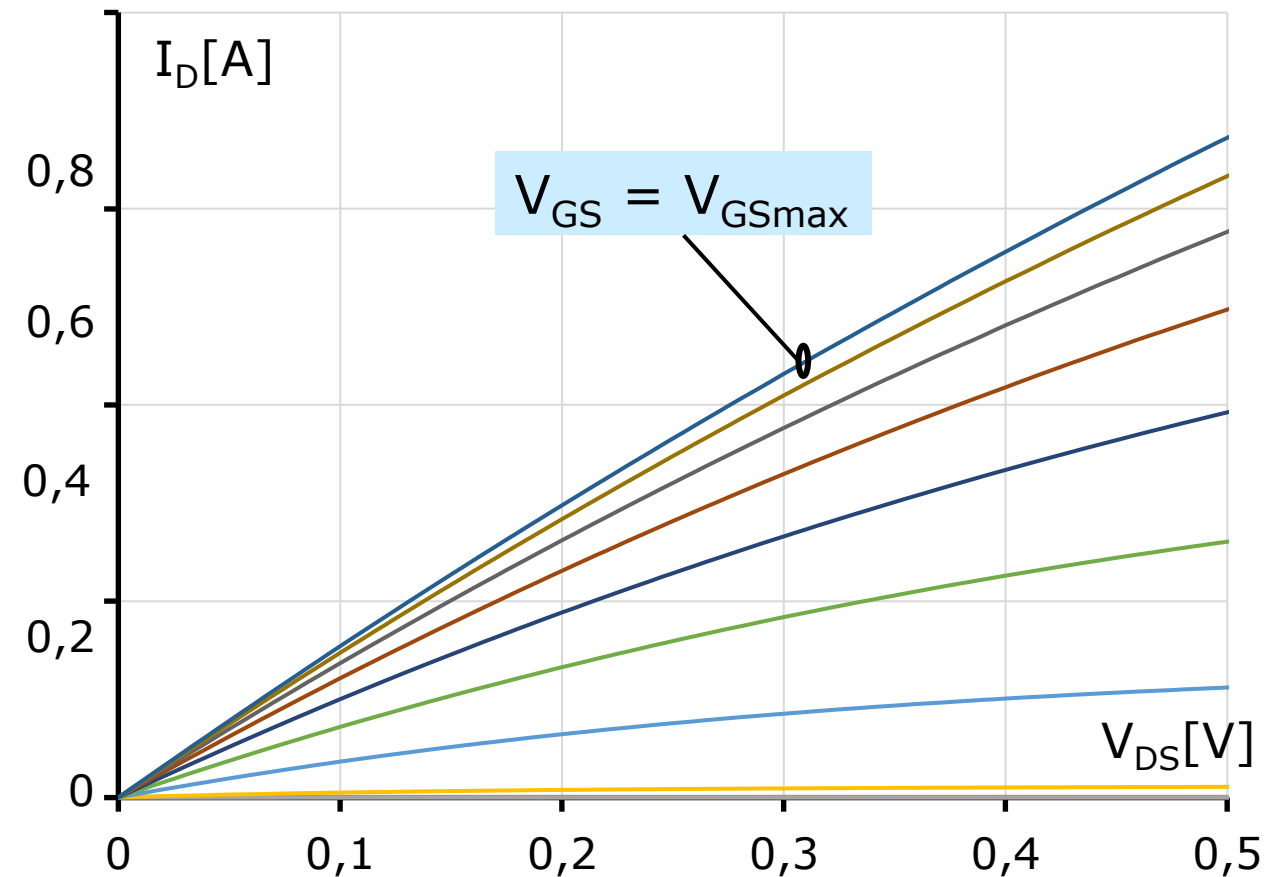
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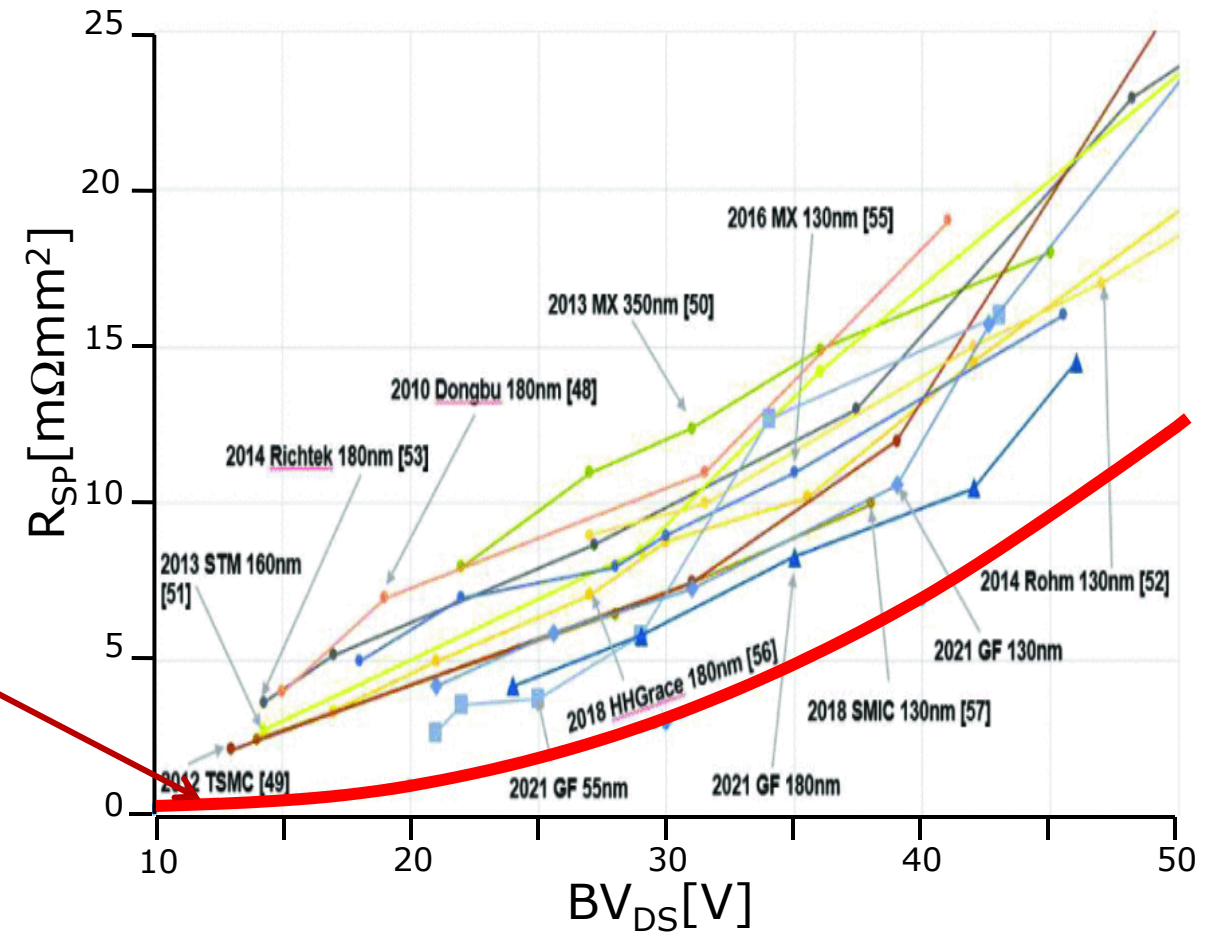
□ how to determine R_{DSon} ?

- ~~$\frac{1}{R_{DSon}} = \left. \frac{dI_D}{dV_{DS}} \right|_{V_{DS}=0, V_{GS}=V_{GSmax}}$~~
- equation disregards drift region
- measure I_D @ $V_{DS}=100\text{mV}$
- $R_{DSon} = \left. \frac{V_{DS}}{I_D} \right|_{V_{DS}=100\text{mV}, V_{GS}=V_{GSmax}}$



Specific On-Resistance $R_{SP}^{[4]}$

- inversely proportional to device area
 - $R_{DSon} \times Area$ product [$m\Omega mm^2$]
 - net area = pitch x gatewidth
- most important FoM in BCD process comparisons
- scales with BV_{DS}
 - 1-D silicon limit: $R_{SP} \approx BV_{DS}^{2.5}$
- determined at 25°C
 - R_{DSon} temperature dependent



Specific Gate Charge Q_{GSP}

- gate charge needed to turn from full OFF to full ON

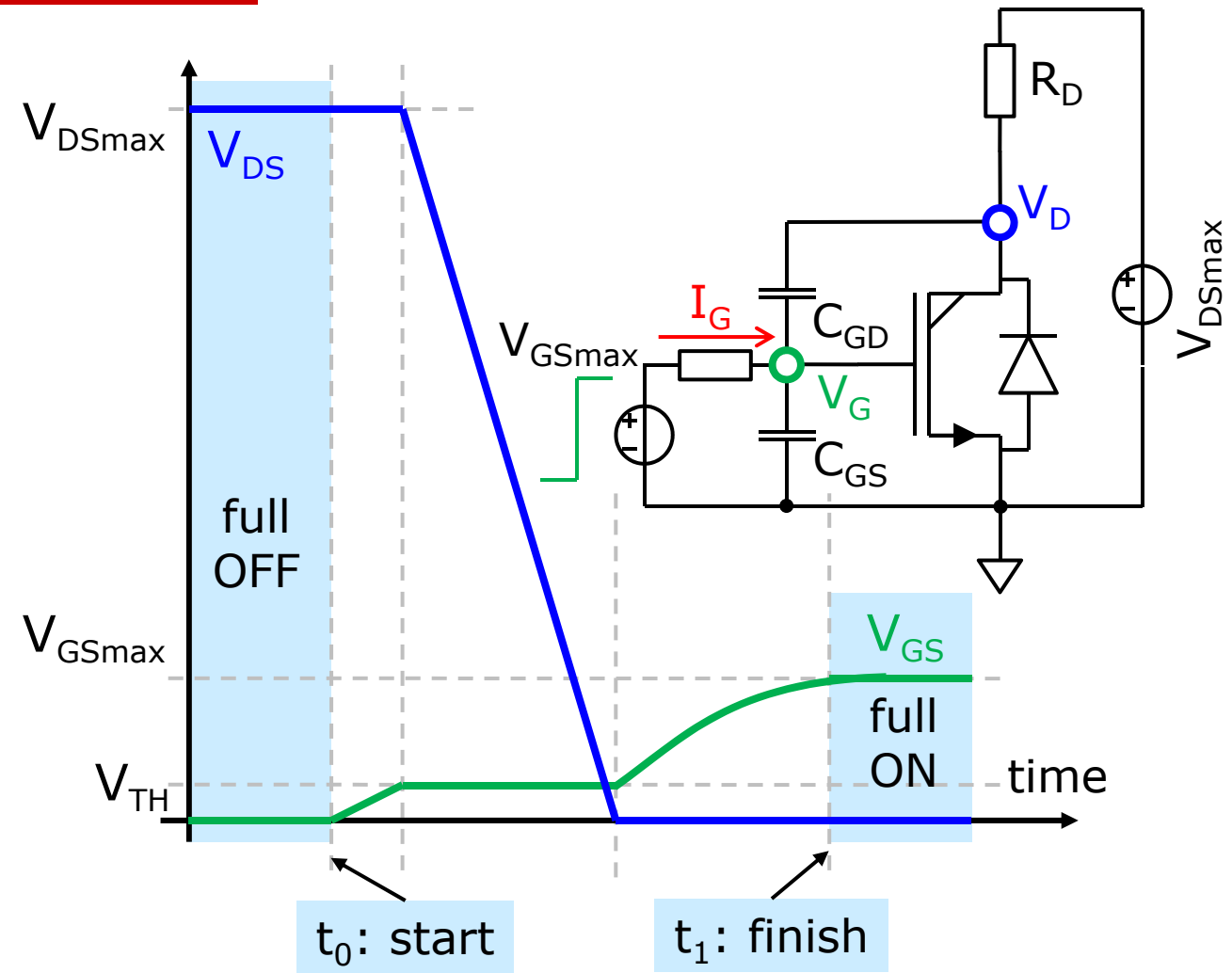
- full OFF: $V_{GS} = 0$; $V_{DS} = V_{DSmax}$

- full ON: $V_{GS} = V_{GSmax}$; $V_{DS} = 0$

$$Q_G = \int_{t_0}^{t_1} I_G \cdot dt$$

- proportional to device area

- **Q_G / Area** ratio [nC/mm²]



MOSFET FoMs

- specific on-resistance
 - **$R_{DSon} \times Area$** product [$m\Omega mm^2$]
 - conduction loss

- specific gate charge
 - **$Q_G / Area$** ratio [nC/mm^2]
 - switching loss

- area independent product
 - **$R_{DSon} \times Q_G$** [$m\Omega nC$]
 - most used FoM for discrete FETs and technologies (e.g. Si, SiC, GaN)

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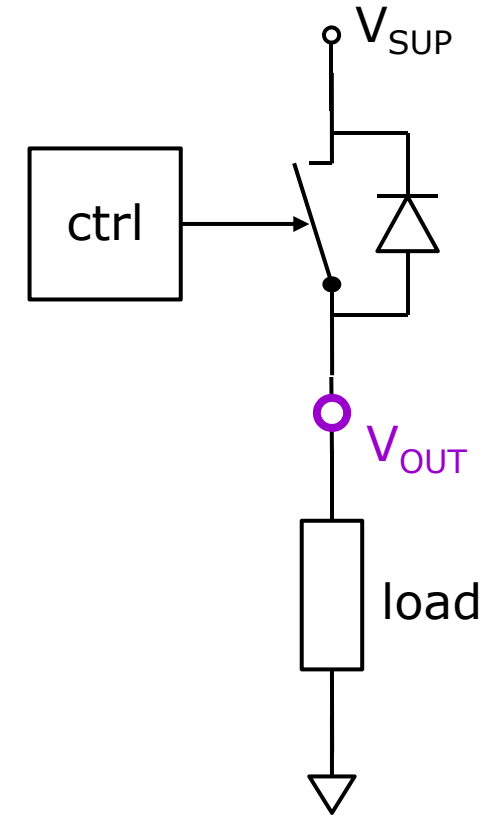
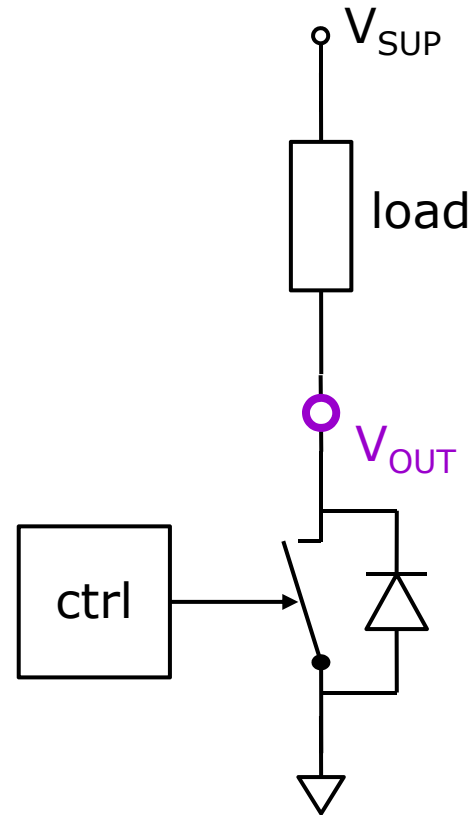
Applications of BCD Technology

- ☐ Power Conversion [5]
- ☐ Motor Drivers [6]
- ☐ Industrial [7]
- ☐ Class-D Audio Amplifiers [8-11,19]
- ☐ Automotive CAN/LIN [12]
- ☐ Biomedical [13]
- ☐ Wireless Charging [14]
- ☐ SSL LED drivers [15,16]
- ☐ Energy Harvesting [17]
- ☐ Display Drivers [18]

- ☐ **Common Denominator: Power Switches**

Power Switches

- lowside switch
 - load ground $\neq$ ground
 - heating / lighting
 - simple ctrl
- higside switch
 - load (dis)connected
 - safety (automotive)
 - load ground = ground
 - complex ctrl
- half bridge

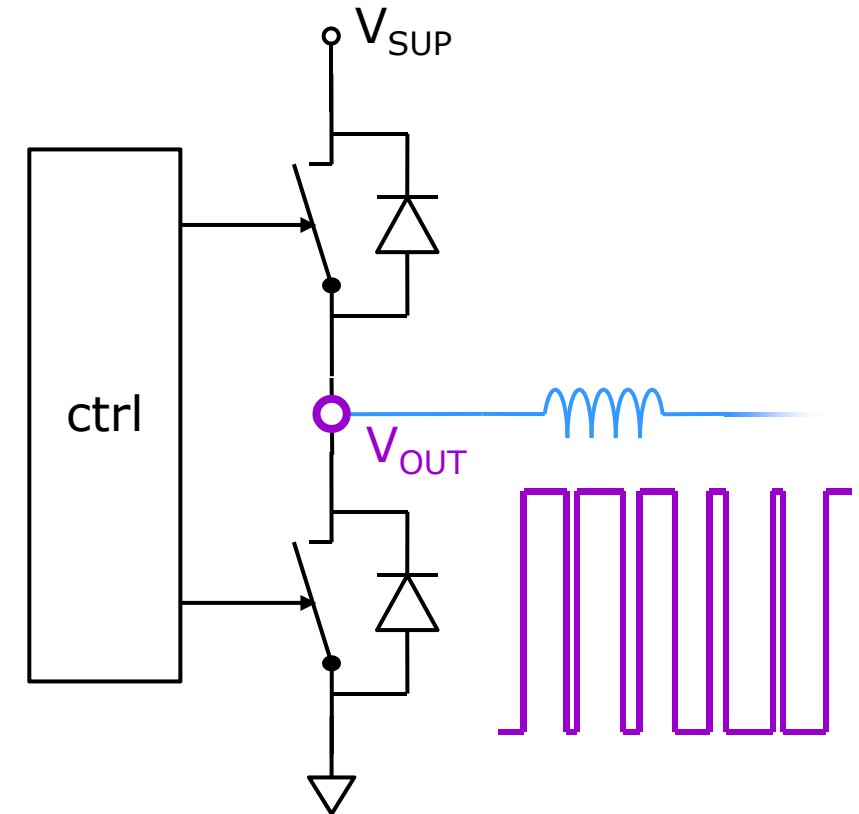


Half Bridge ?



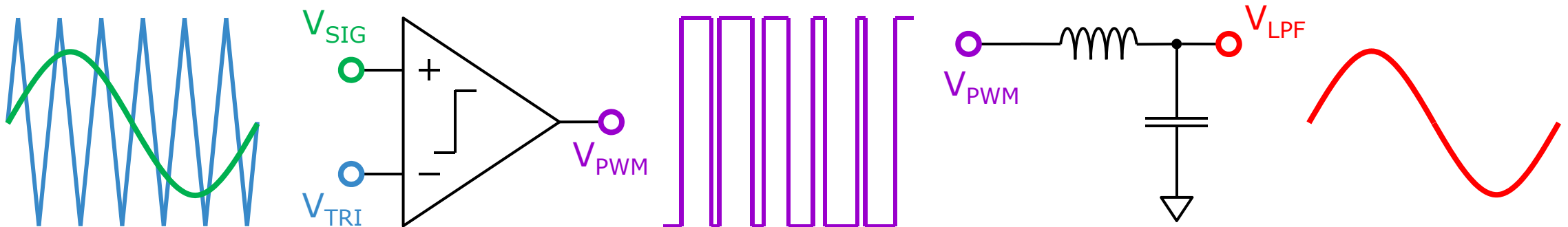
Half Bridge

- stack of lowside & highside switch
 - inductor, loudspeaker, motor windings, etc..
- break-before-make
 - (flyback) diodes
 - conductive path to supply / ground
- class-D = switch mode
 - class-D (audio) amplifiers
 - switch mode power supplies (SMPS)
 - Pulse Width Modulation (PWM)

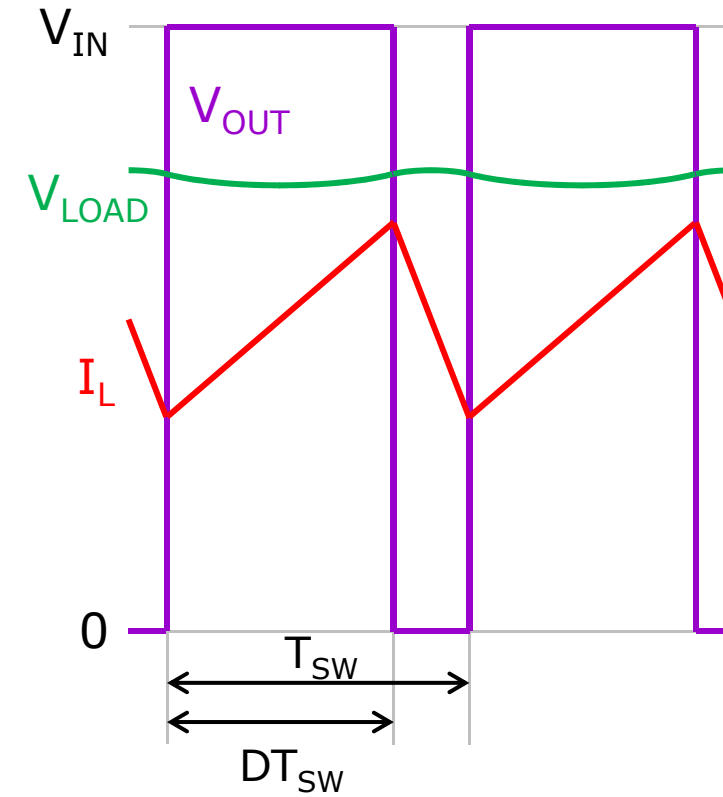
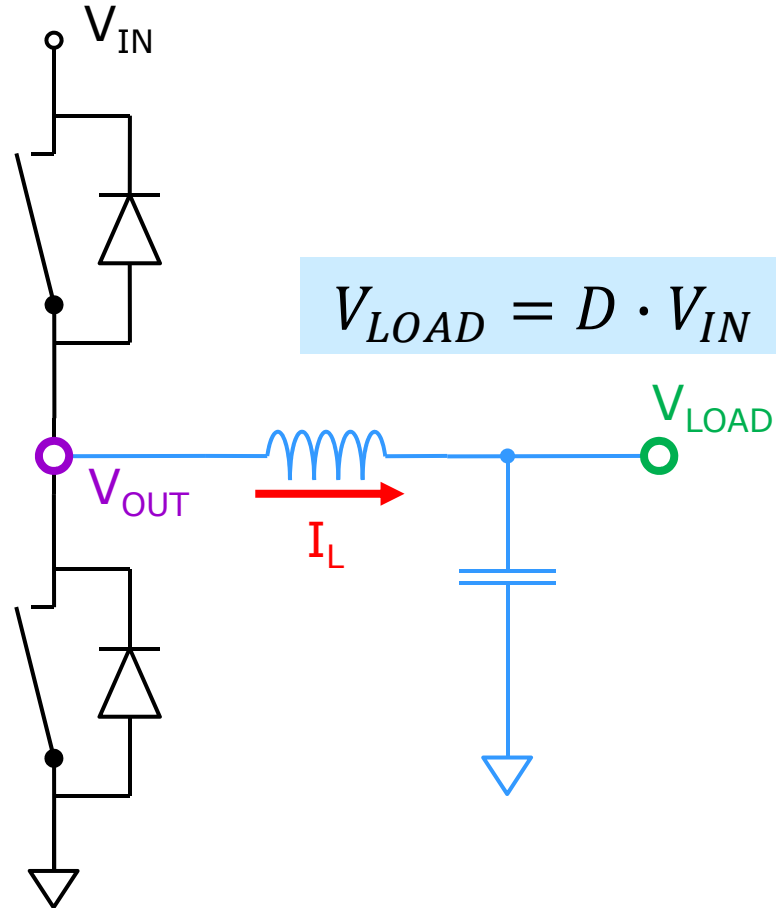


Pulse Width Modulation

- input signal V_{SIG}
- high frequency reference V_{TRI}
 - triangular or sawtooth
 - $F_{PWM} > 10 \times F_{SIG}$
- PWM signal V_{PWM}
 - duty-cycle proportional to V_{SIG}
- after low pass filter V_{LPF}
 - input signal V_{SIG}
 - plus a (small) high frequency ripple

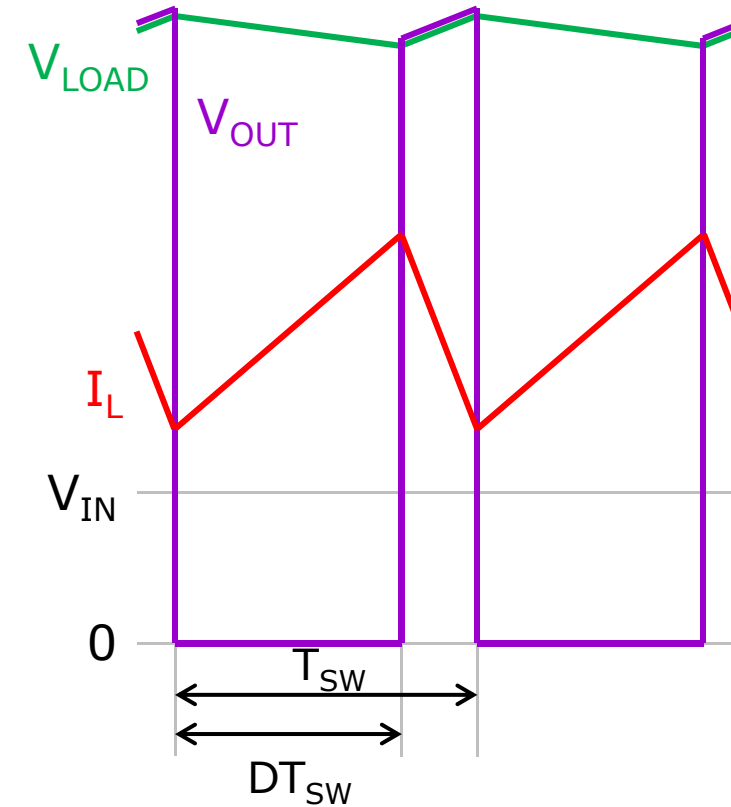
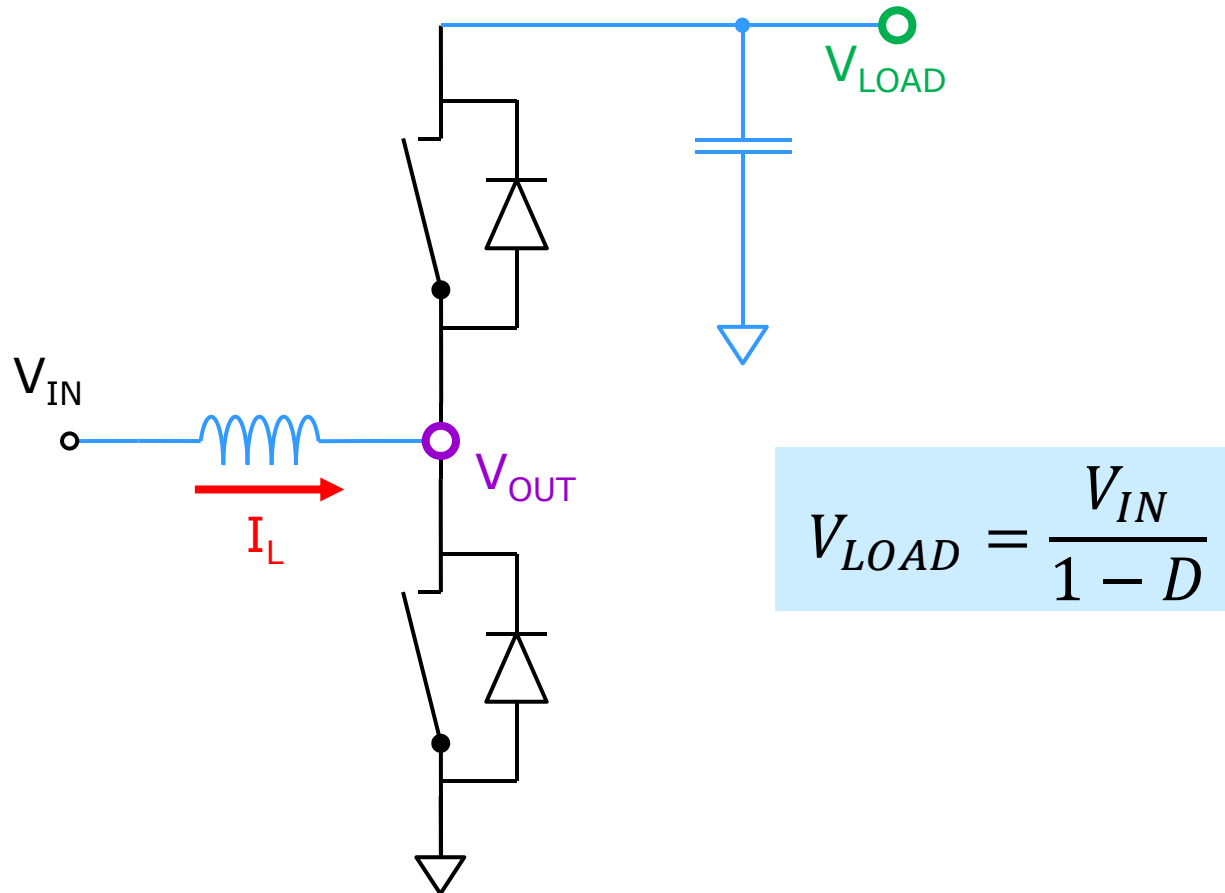


Single Half-Bridge



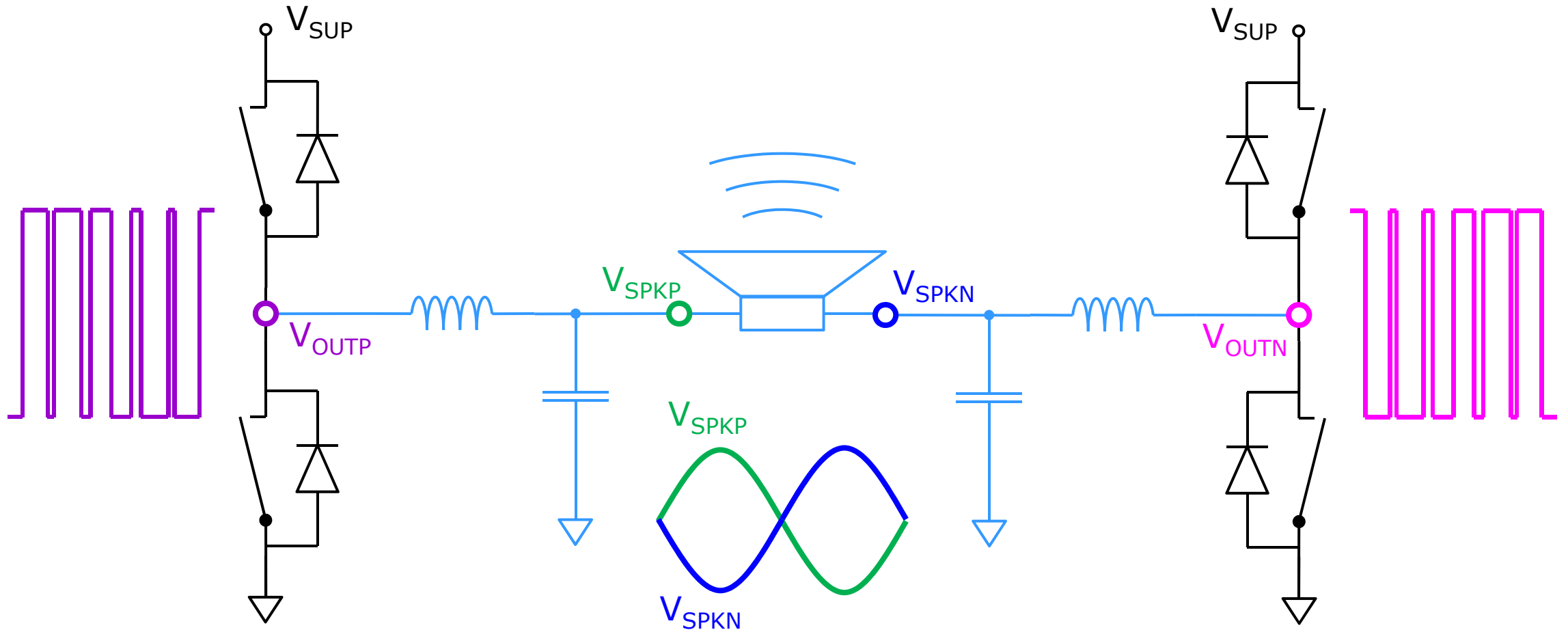
DC-DC Buck Converter

Single Half-Bridge



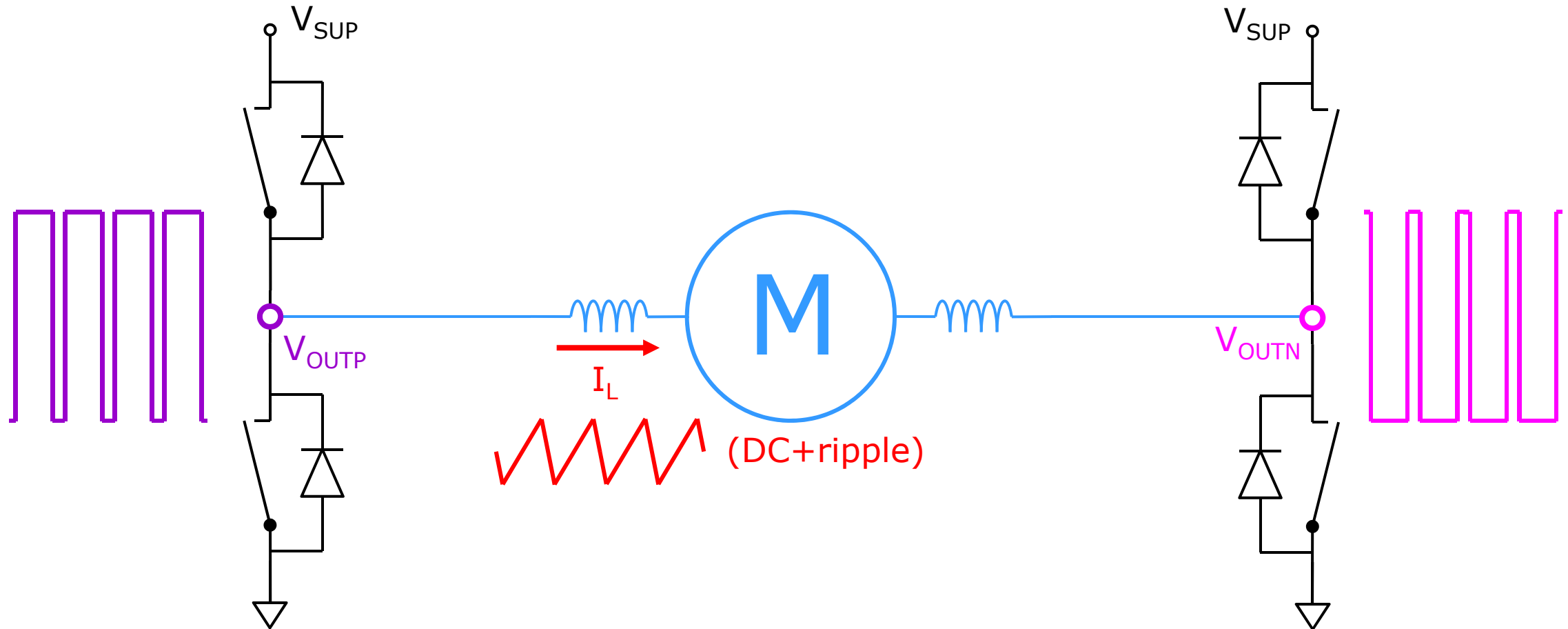
DC-DC Boost Converter

2 Half Bridges make a Full Bridge



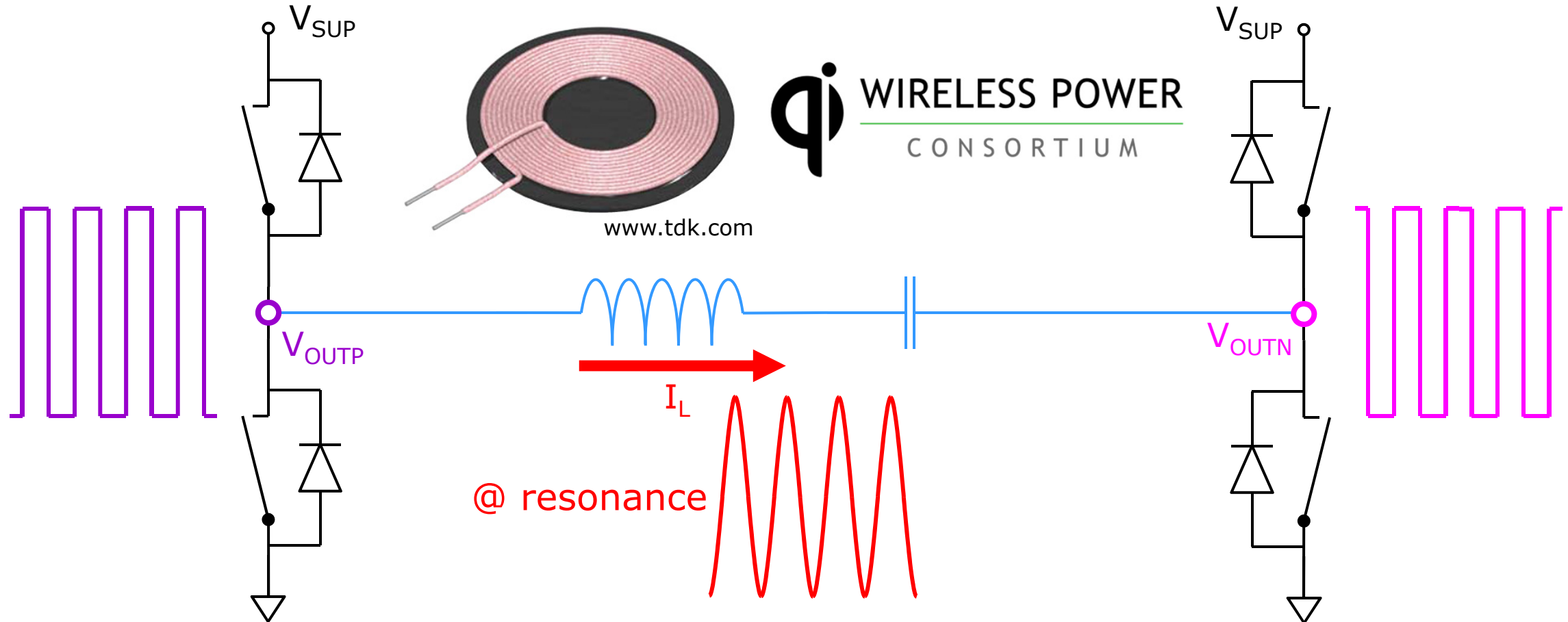
Class-D Audio Amplifier [8]

2 Half Bridges make a Full Bridge



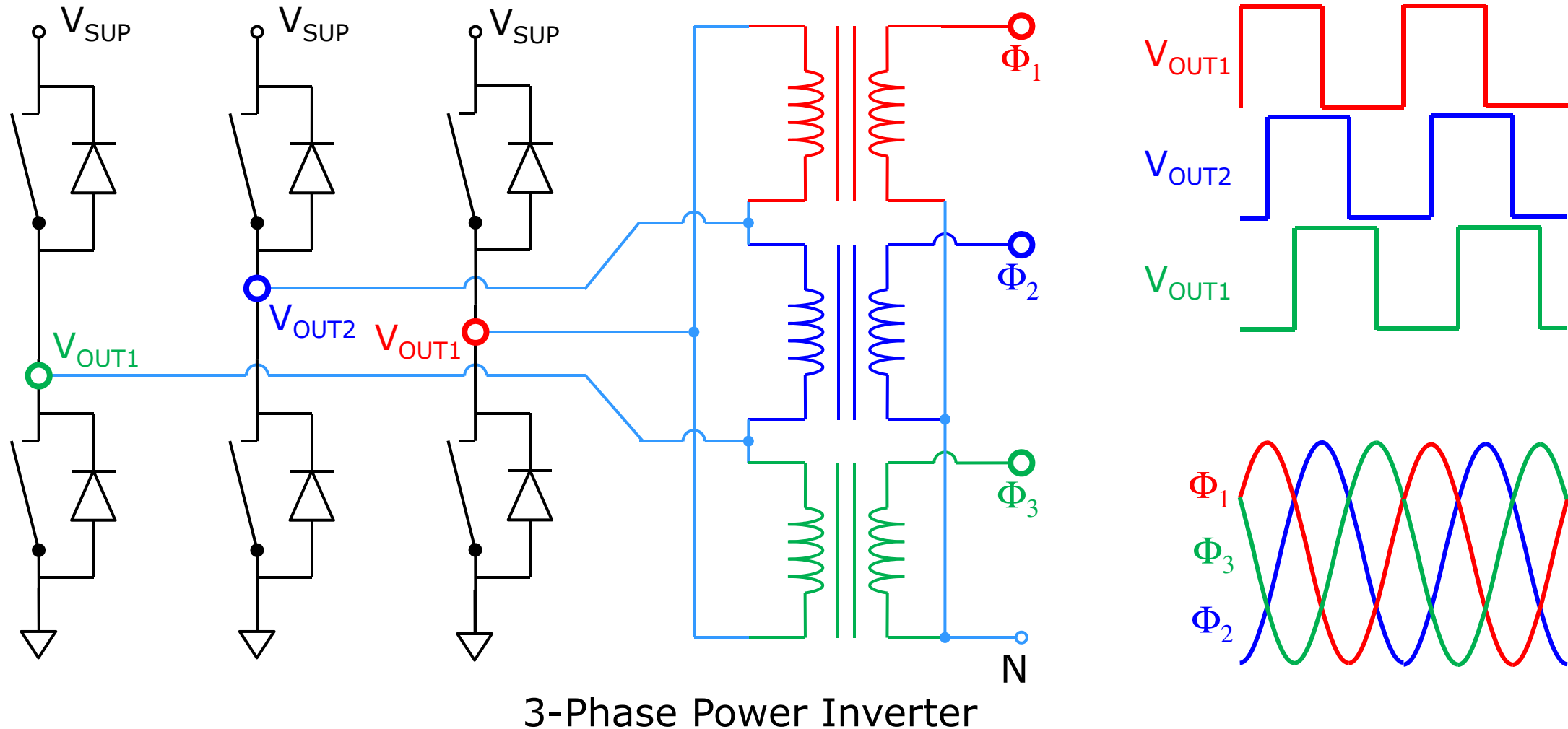
DC-Motor Driver

2 Half Bridges make a Full Bridge

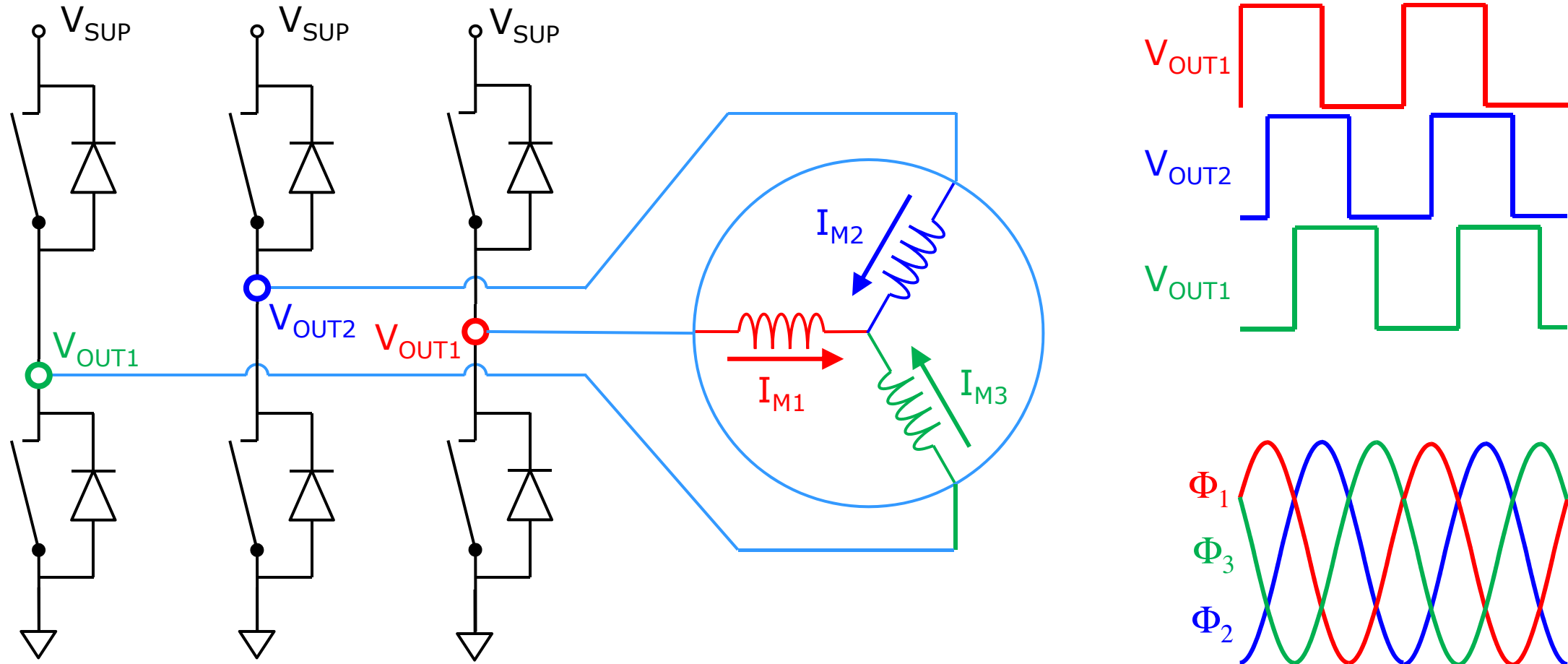


Qi Wireless Charging Transmitter

3 Half Bridges make a 3-Phase Full Bridge

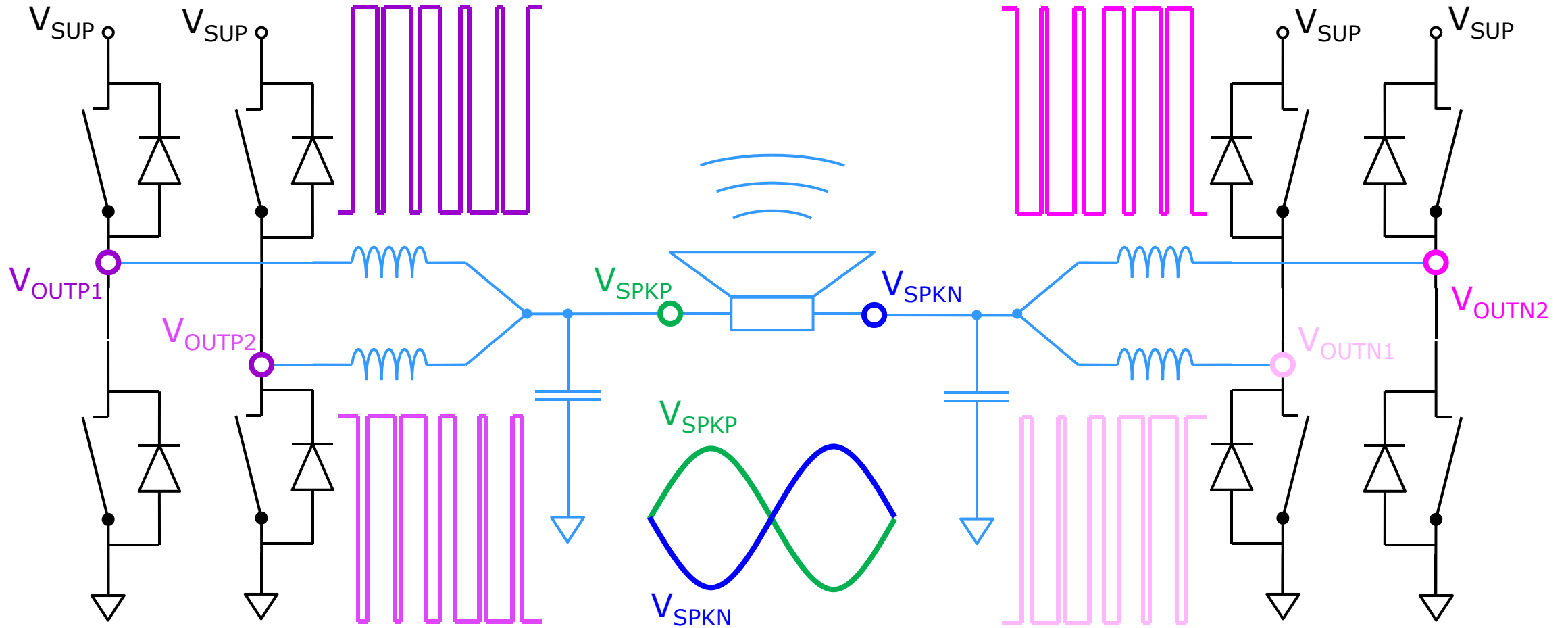


3 Half Bridges make a 3-Phase Full Bridge



Brushless DC-Motor Driver [6]

More Legs ?



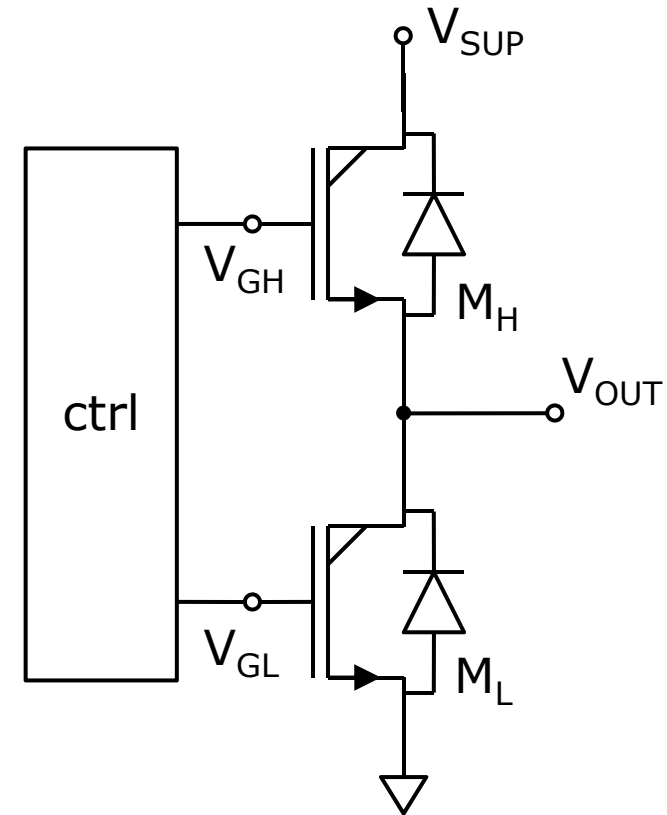
Multi-Phase PWM Class-D Audio Amplifier [19]

Outline

- Introduction & Motivation
- BCD Process Technology & Devices
- Device Characteristics & Figures-of-Merit
- Applications
- Circuit Design

Half Bridge Switch Implementation

- external (discrete) powerFETs
 - voltage x current > 1000VA
 - GaN / SiC
 - ctrl IC in BCD technology
- integrated powerFETs
 - voltage x current < 400VA
 - DMOS



Challenges of High Current

- dissipation
 - conduction loss $I^2 \times R_{DSon}$
 - thermal design

- high current switching
 - high dI/dt in ground / supply lines
 - large voltage excursions

- parasitic inductance
 - bondwires, lead fingers, PCB tracks

- separate power ground

Challenges of High Voltage

- for circuit design **Voltage** becomes **High Voltage** when $V_{SUP} > V_{GSmax}$
 - design measures needed to prevent damage

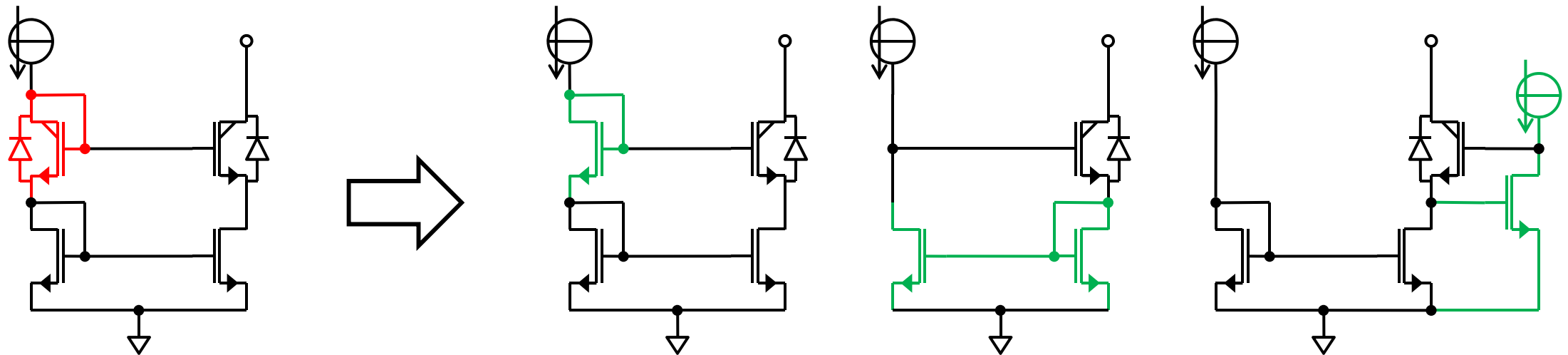
- limit voltage swings
 - voltage clamps
 - high voltage cascodes

- use local (low) voltage supply domains
 - beware voltage domain crossings
 - ESD

- beware floating nodes !!!!!
 - standby / powerdown modes

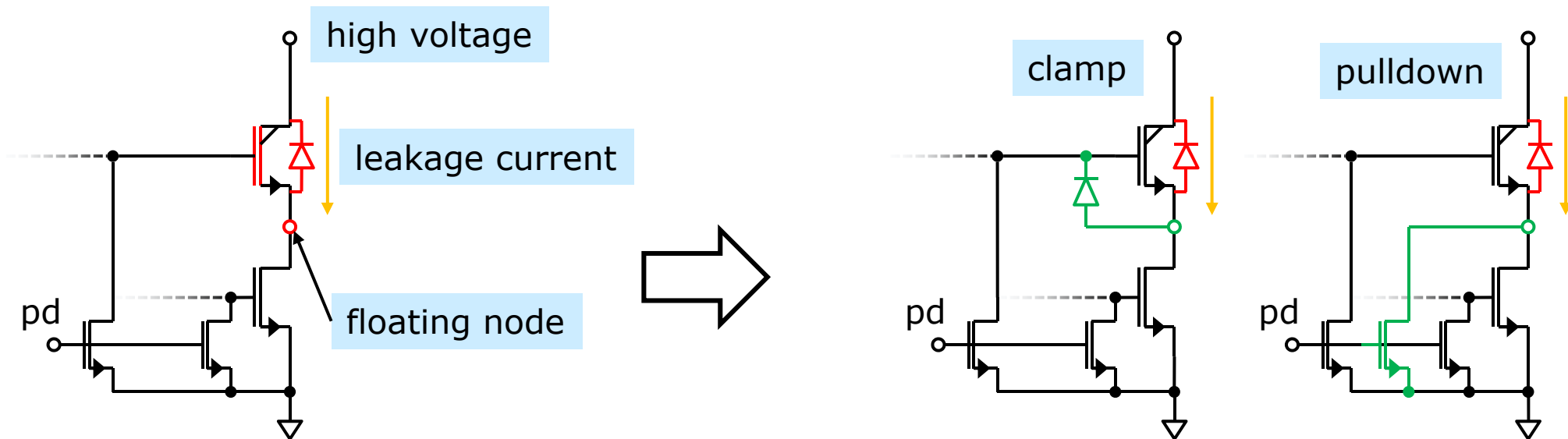
Challenges of High Voltage

- design economy
 - only use HV devices when absolutely necessary
 - usually, more area efficient solutions are possible



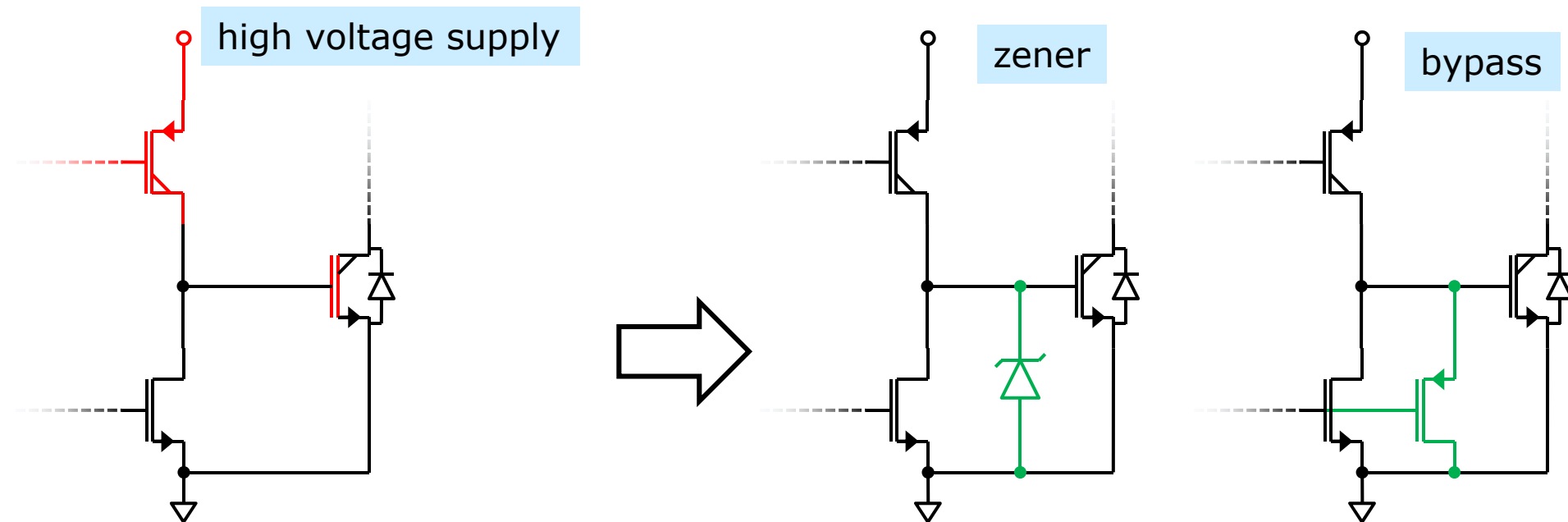
Challenges of High Voltage

- beware of floating nodes
 - biasing switched off in standby / powerdown modes => floating nodes
 - leakage through body diode => oxide damage



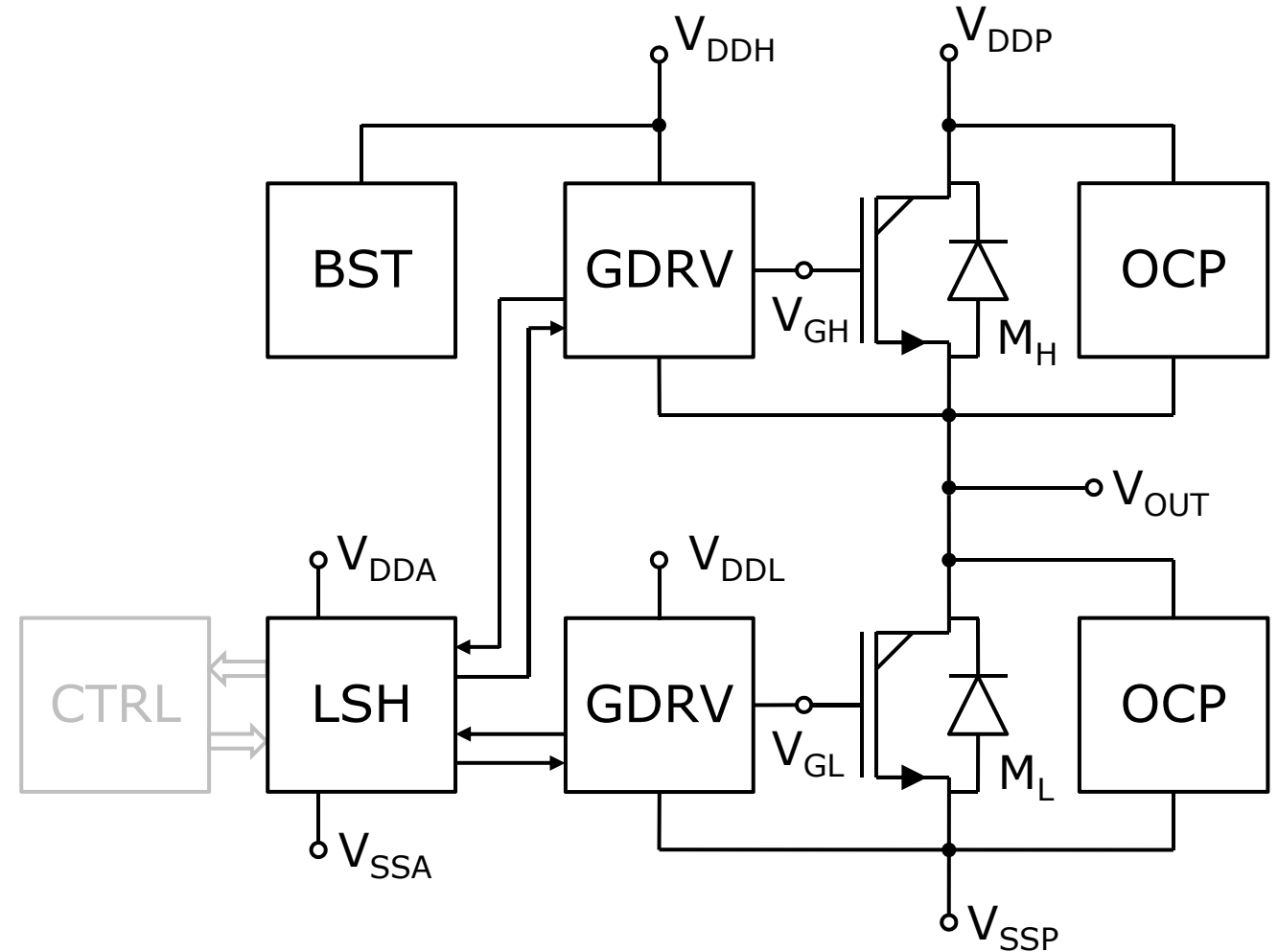
Challenges of High Voltage

- limit voltage swings
 - clamps => inactive during normal operation
 - beware overdrive cases



Half Bridge Essentials

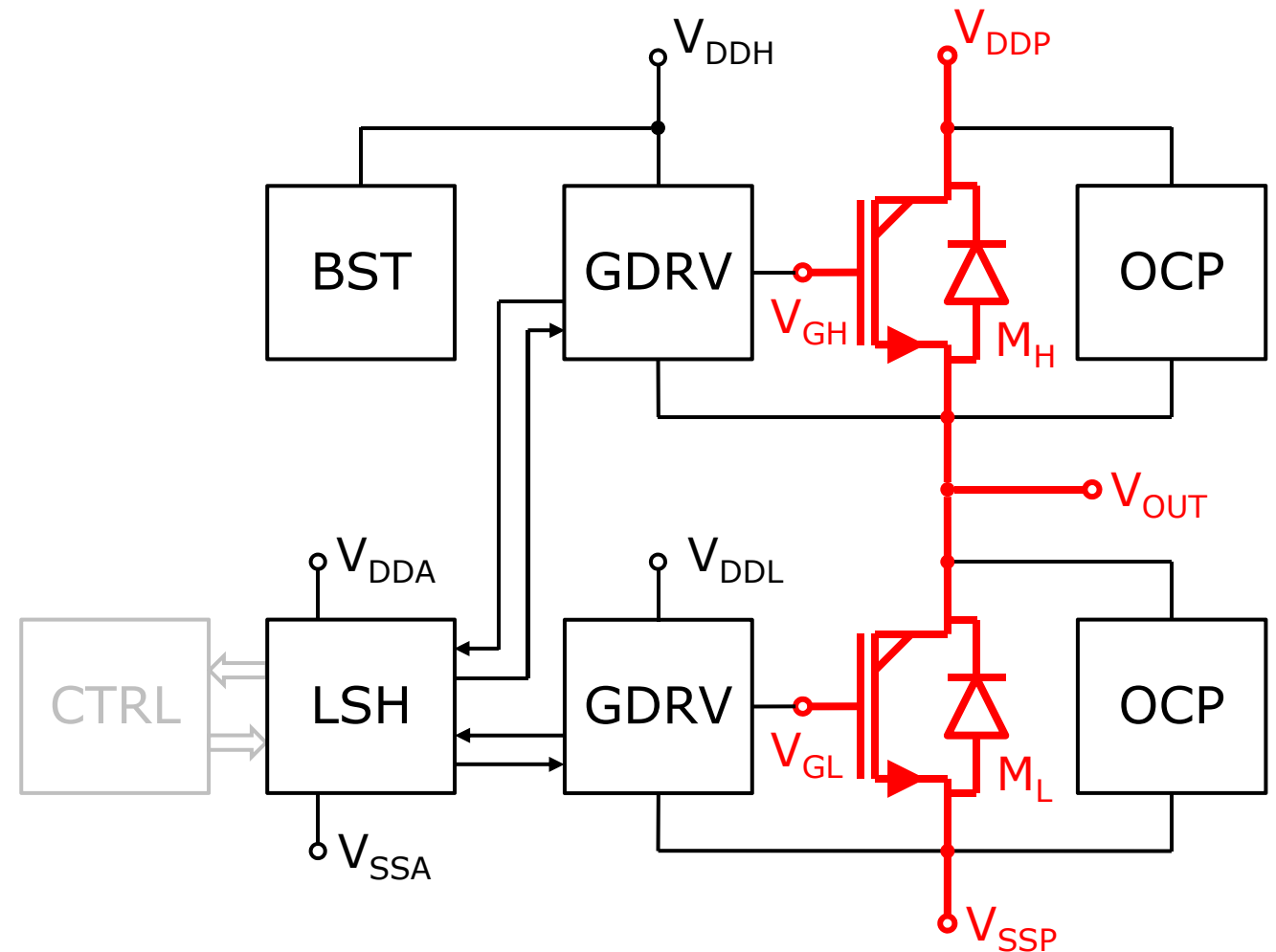
- PowerFETs
- BST
 - highside gatedrive voltage
- GDRV
 - powerFET gate driver
- LSH
 - levelshifters
- OCP
 - overcurrent protection



Half Bridge Essentials

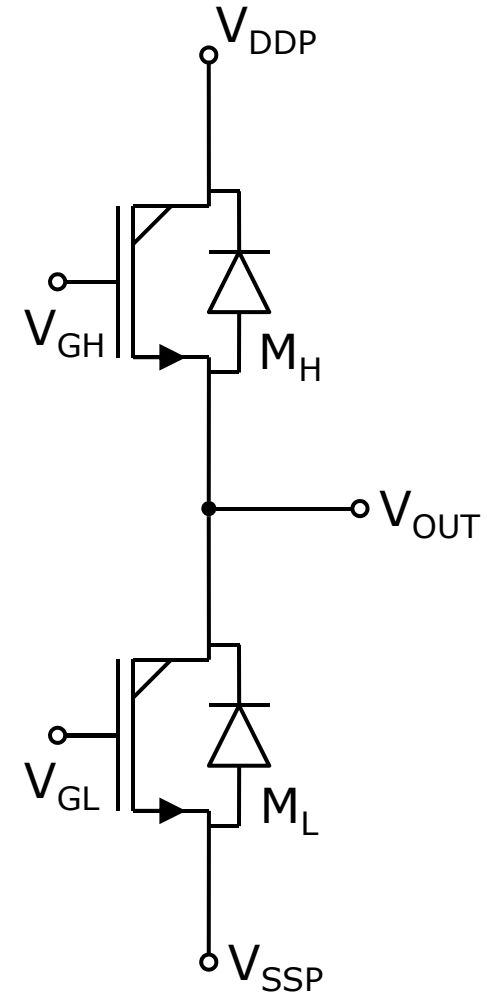
□ PowerFETs

- BST
 - highside gatedrive voltage
- GDRV
 - powerFET gate driver
- LSH
 - levelshifters
- OCP
 - overcurrent protection



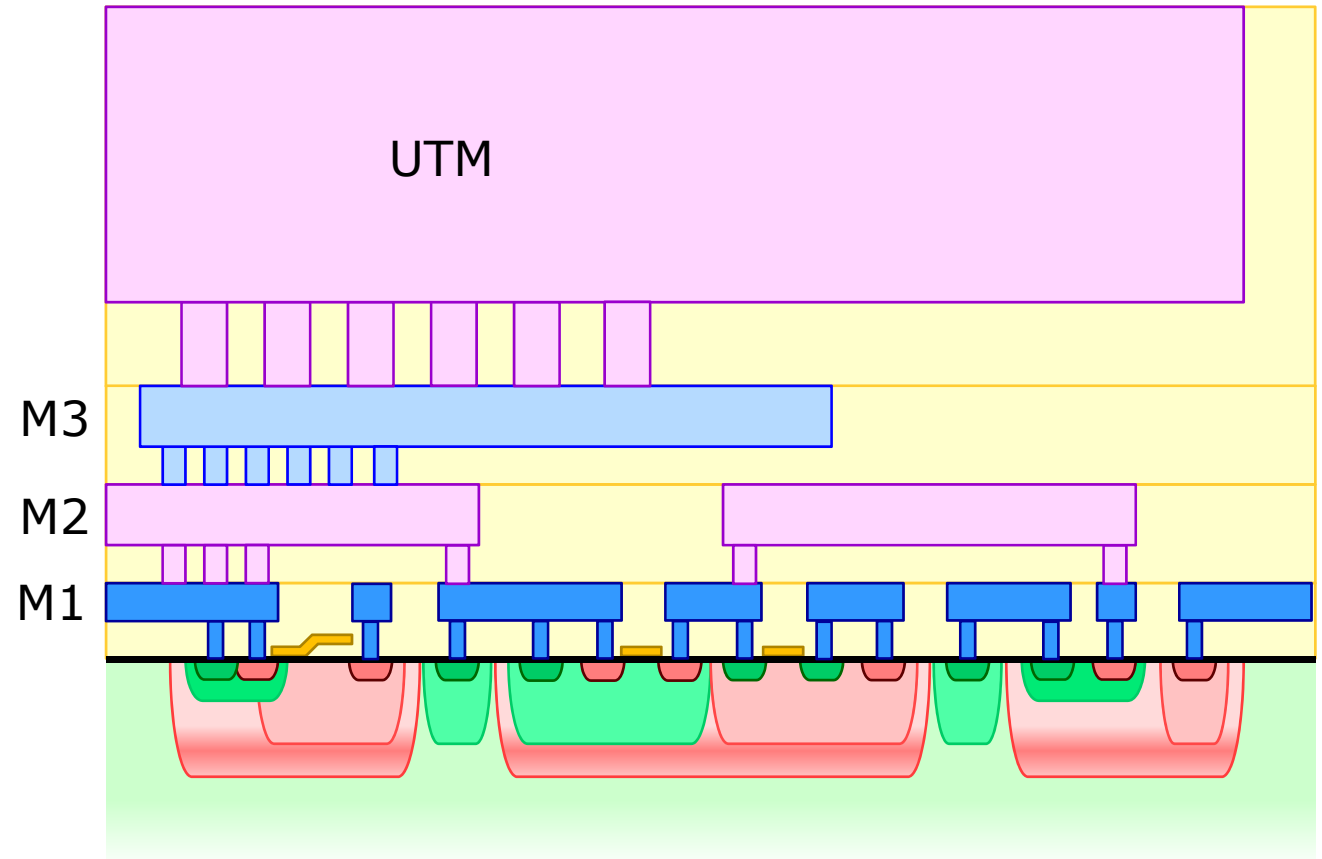
Design Considerations on PowerFETs

- powerFET design target: resistance in ON state
 - conduction loss
- intrinsic R_{DSon}
 - determine size using $R_{DSon} * Area$
- account for interconnect resistance
- account for temperature dependence



PowerFET Interconnect

- typical BCD technology
 - 4 to 6 interconnect layers
- Alumin(i)um (Al) or Copper (Cu)
 - Cu has 40% lower resistivity
- ultra thick top metal (UTM)
 - $3\mu\text{m}$ Cu: $6\text{m}\Omega/\square$
 - $3\mu\text{m}$ Al: $10\text{m}\Omega/\square$

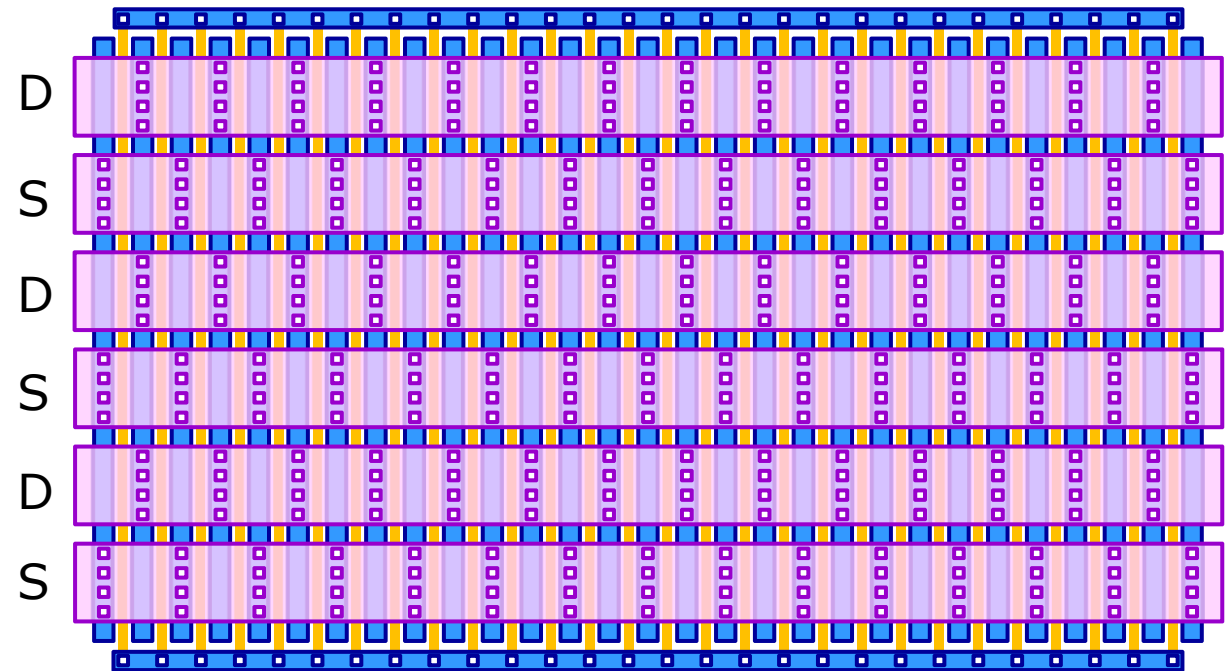


[illegible]

- Downloaded from <http://www.jstor.org/stable/2346129> by University of Cambridge on Tue, 20 Jun 2017 12:01:04 UTC

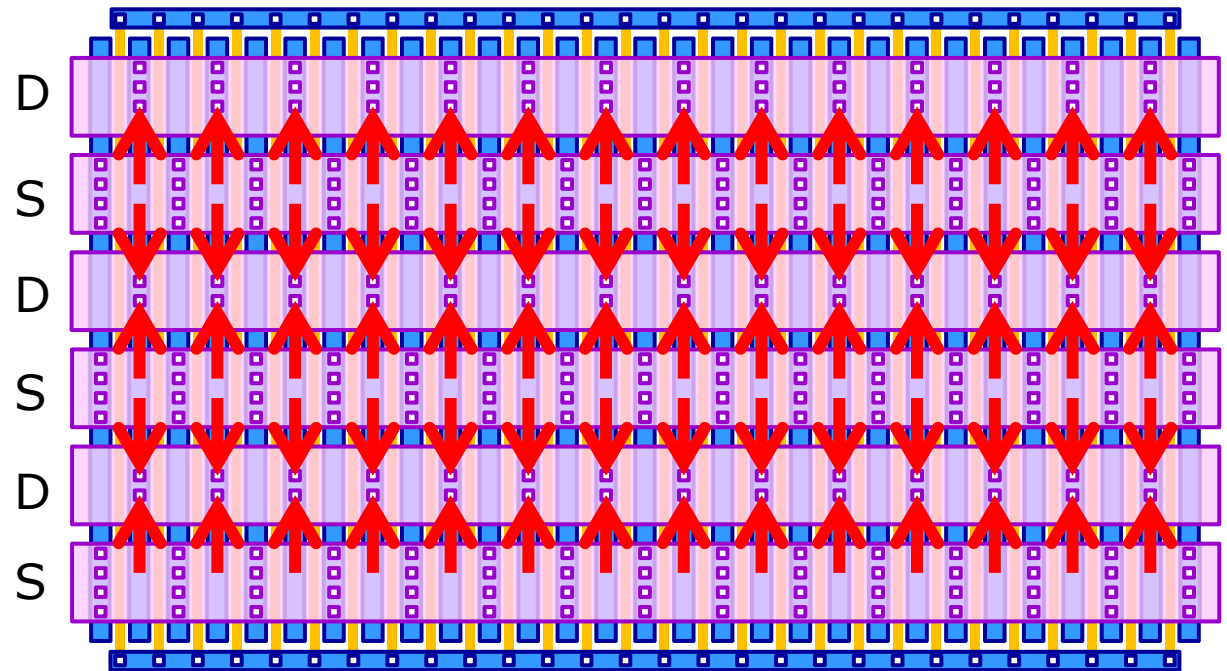
PowerFET Interconnect

- second metal: M2
 - perpendicular to M1
 - maximize vias



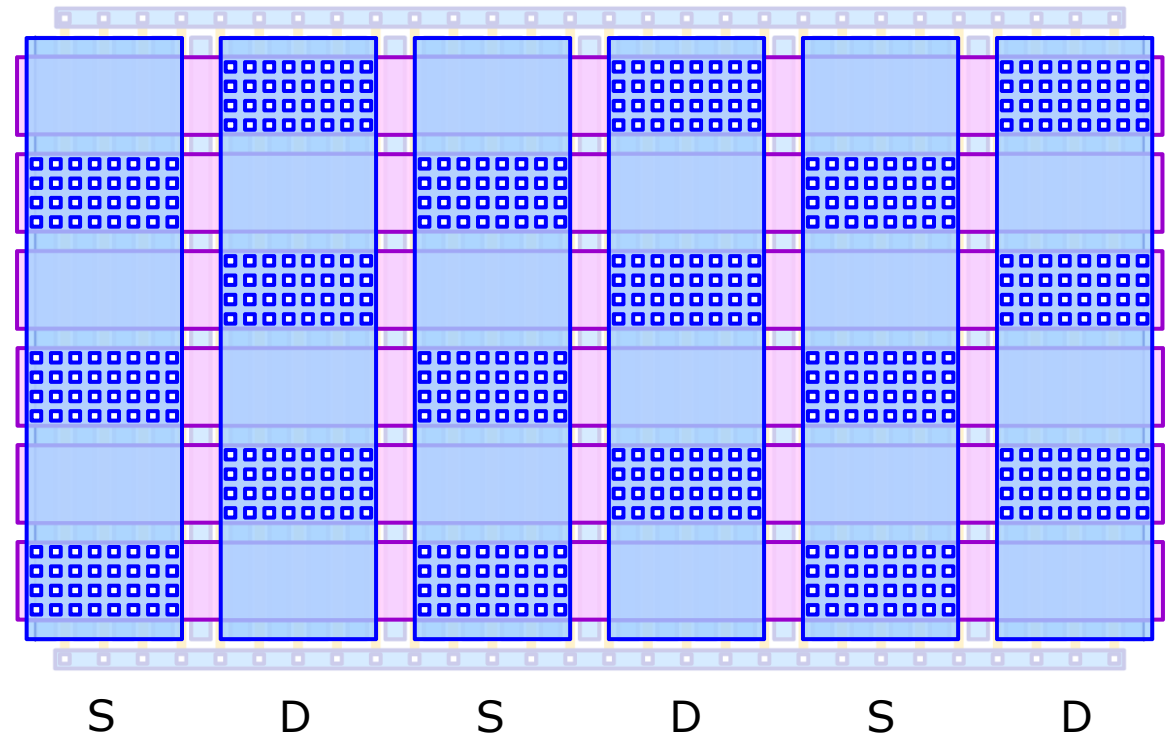
PowerFET Interconnect

- second metal: M2
 - perpendicular to M1
 - maximize vias
- M1 stripes connect in parallel
 - negligible contribution M1 to total resistance



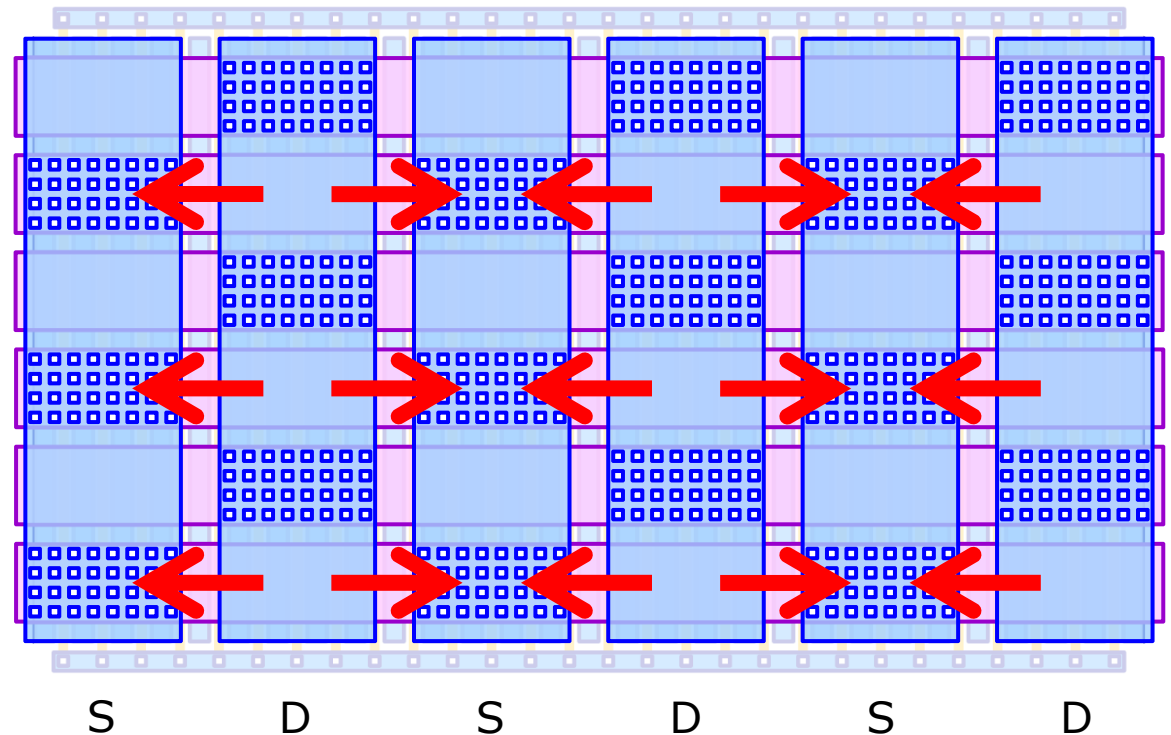
PowerFET Interconnect

- third metal: M3
 - perpendicular to M3
 - maximize vias



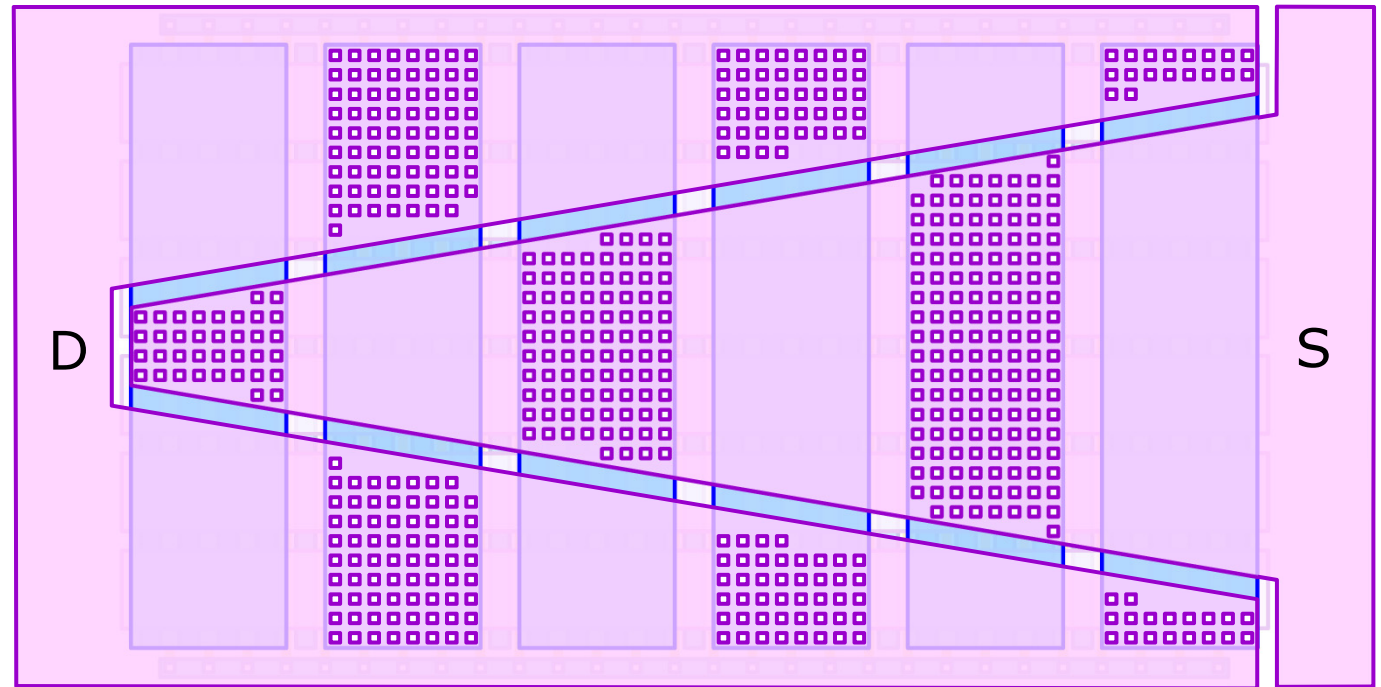
PowerFET Interconnect

- third metal: M3
 - perpendicular to M3
 - maximize vias
- M2 stripes connect in parallel
 - negligible contribution M2 to total resistance



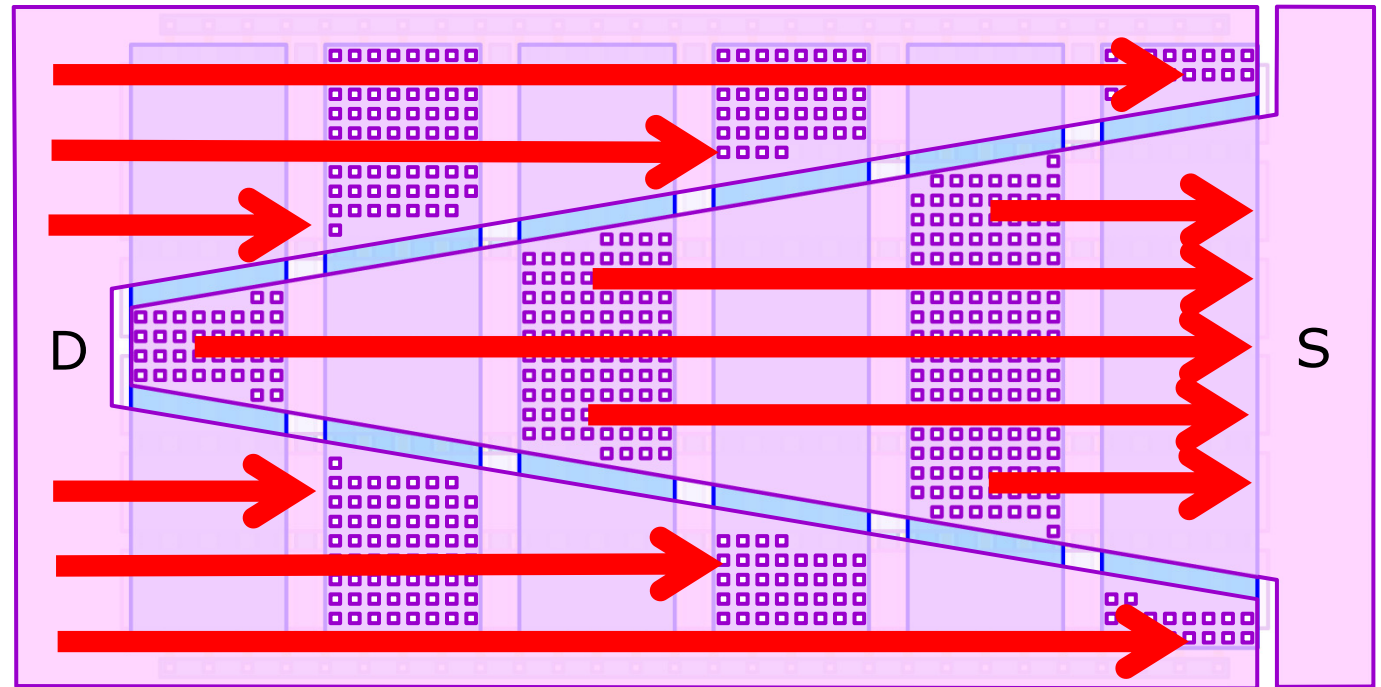
PowerFET Interconnect

- top metal UTM
 - ultra thick
 - maximize vias
 - tapered fingers



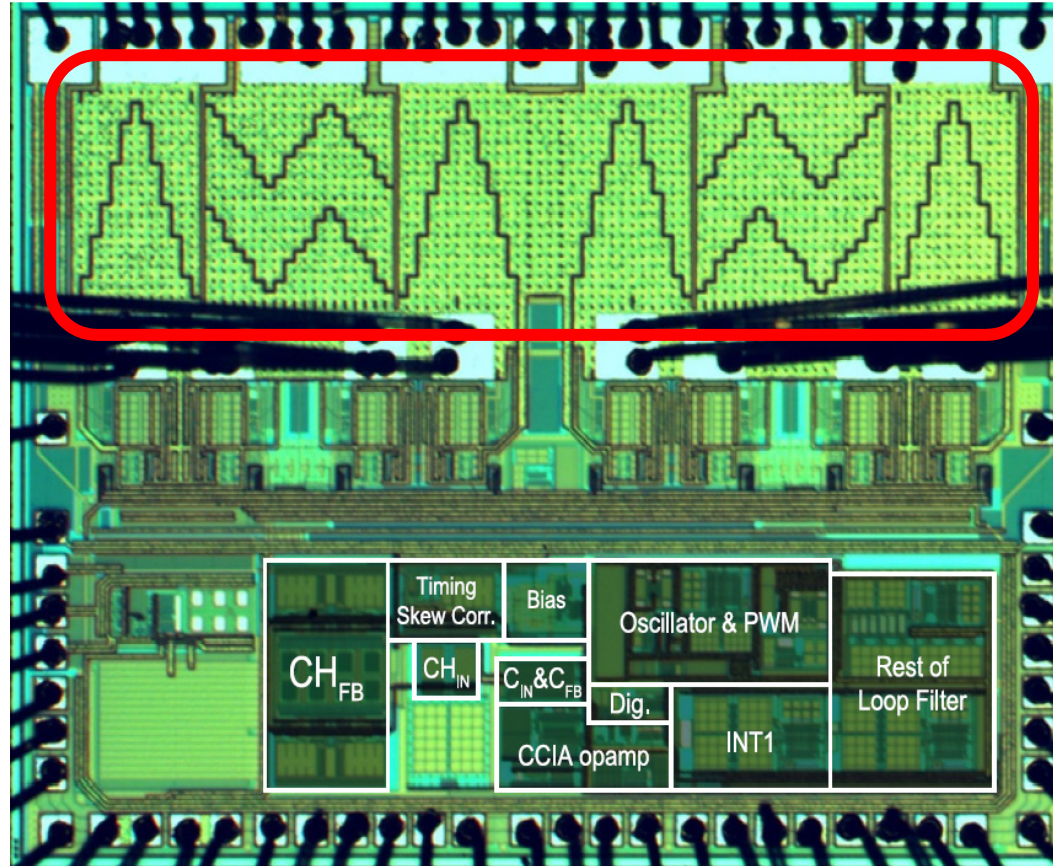
PowerFET Interconnect

- top metal UTM
 - ultra thick
 - maximize vias
 - tapered fingers
- uniform current density
 - minimal resistance
- approximate resistance
 - $R_{\square \text{UTM}} \times \text{Aspect Ratio}$
- more accurate
 - 3D resistive extraction tools

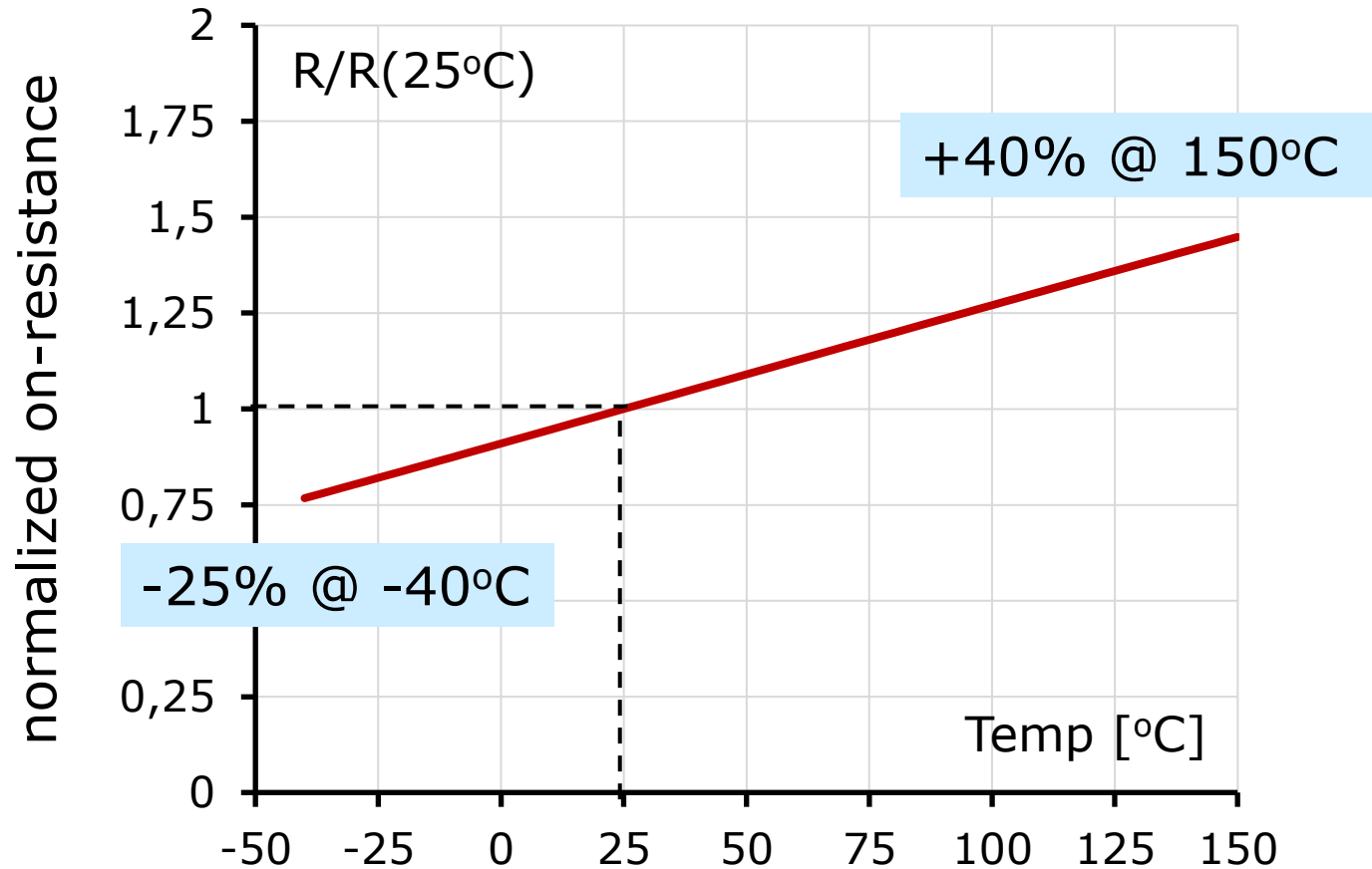


PowerFET Interconnect

- example: class-D audio amplifier (Paper 31.2)

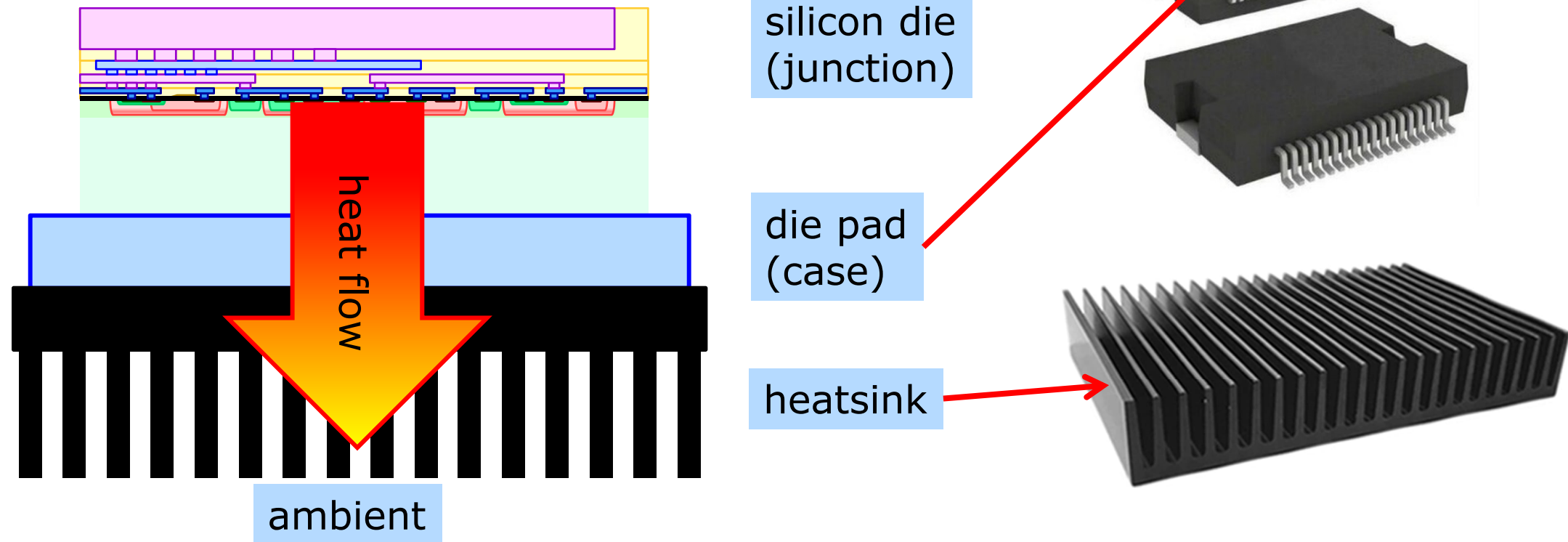


Temperature Dependence On-Resistance



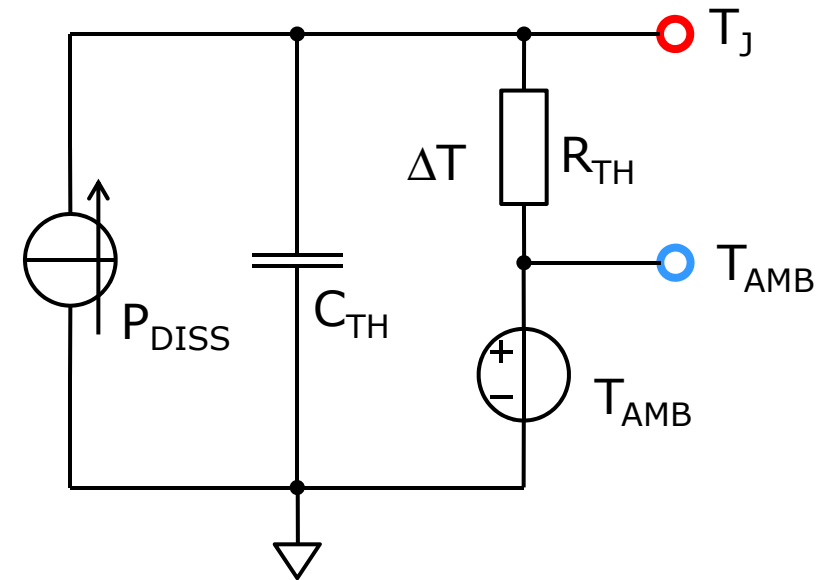
Dissipation & Heat Flow

- dissipation
 - conversion of (electrical) power into heat



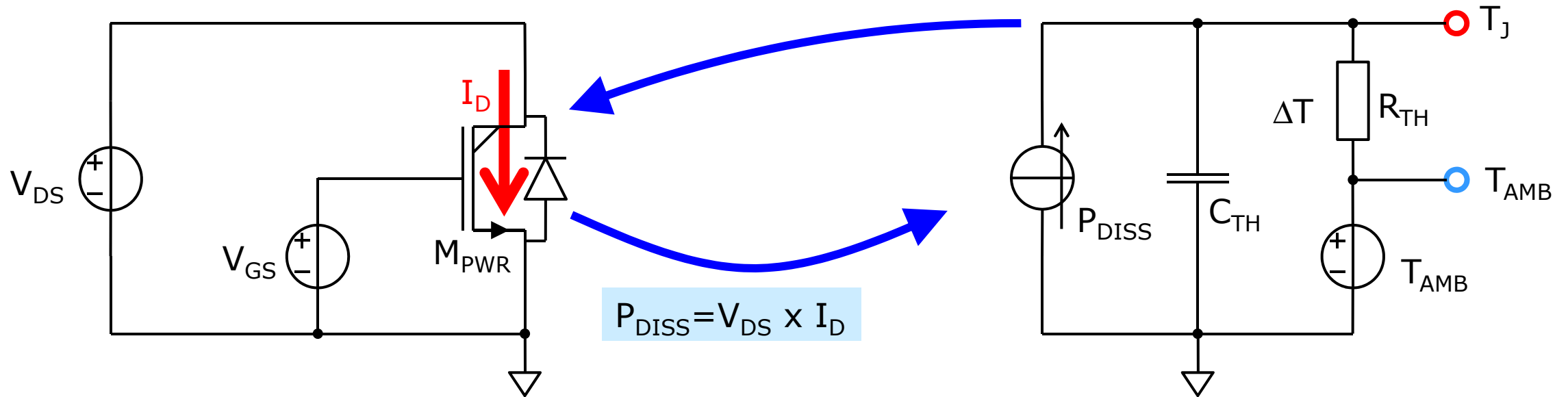
Thermal Modeling

- electrical equivalent
 - current $\longleftrightarrow$ power dissipation
 - voltage $\longleftrightarrow$ temperature
- thermal circuit
 - thermal resistance R_{TH} [K/W]
 - thermal capacitance C_{TH} [K/J]
- ambient temperature T_{AMB}
- junction temperature T_J



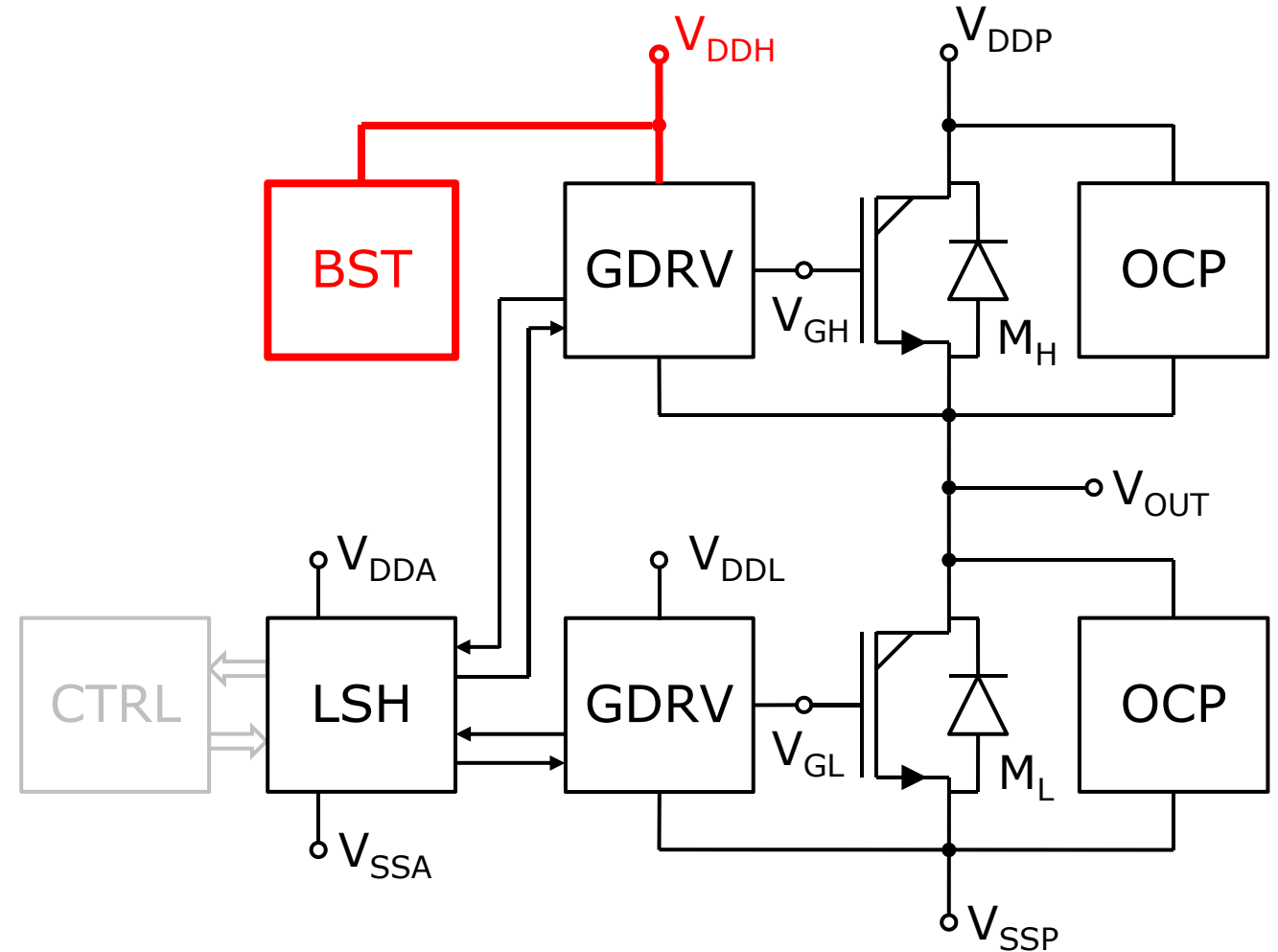
Thermal Modeling

- electrical-thermal co-simulation
- simultaneous
 - not all circuit simulators can do this
- sequential iterations
 - electrical simulation yields P_{DISS}
 - thermal simulation yields T_J
 - rerun simulations until convergence



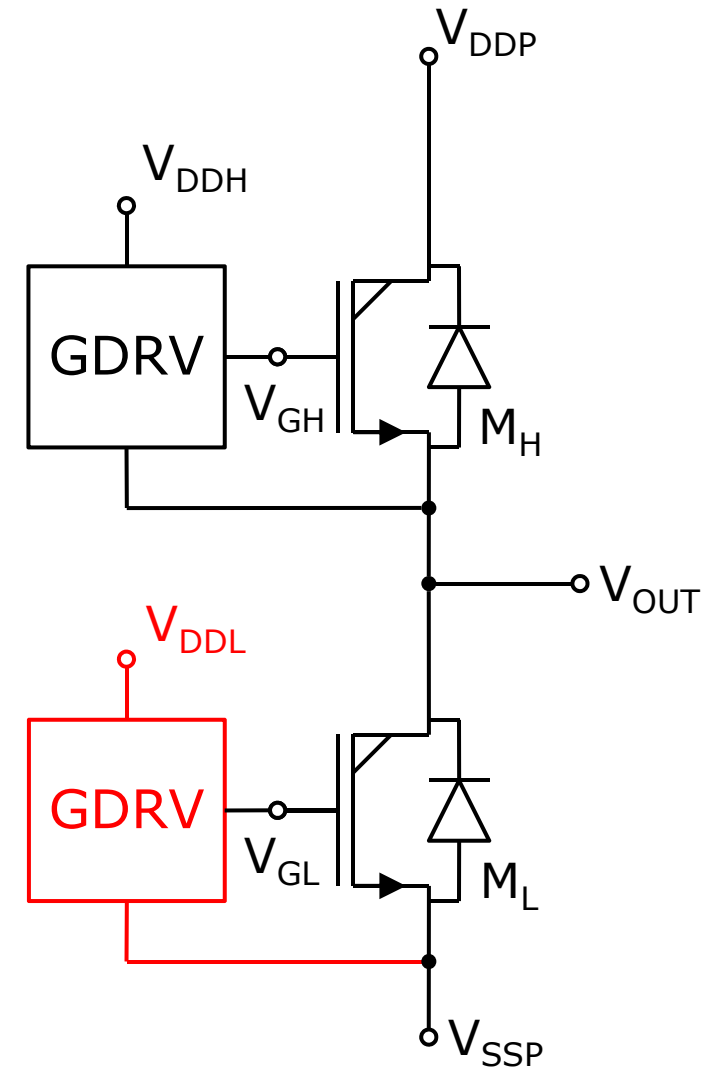
Half Bridge Essentials

- PowerFETs
- BST
 - highside gatedrive voltage
- GDRV
 - powerFET gate driver
- LSH
 - levelshifters
- OCP
 - overcurrent protection



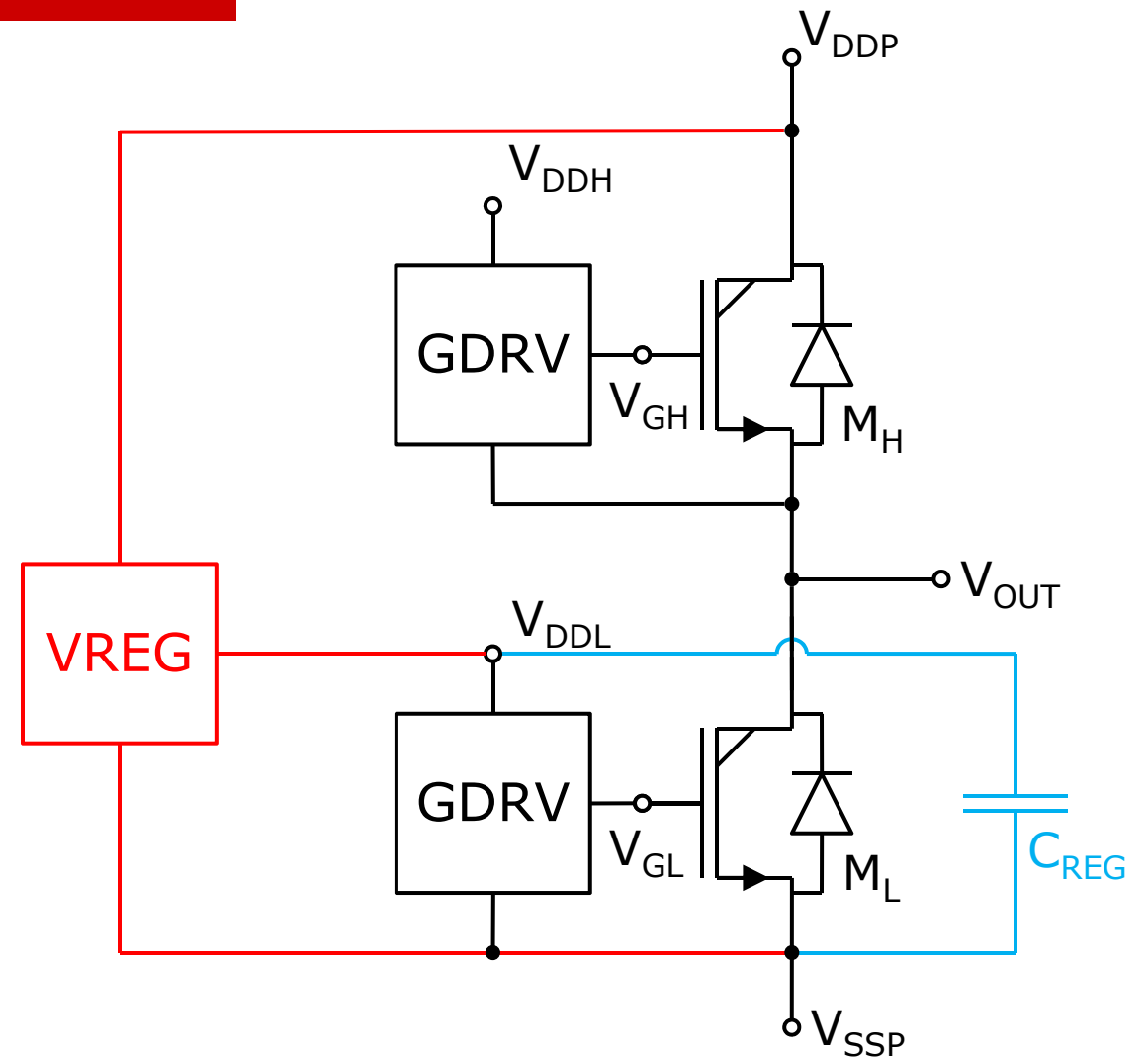
Bootstrap

- lowside gatedriver supply V_{DDL}
 - limit V_{GS} of M_L



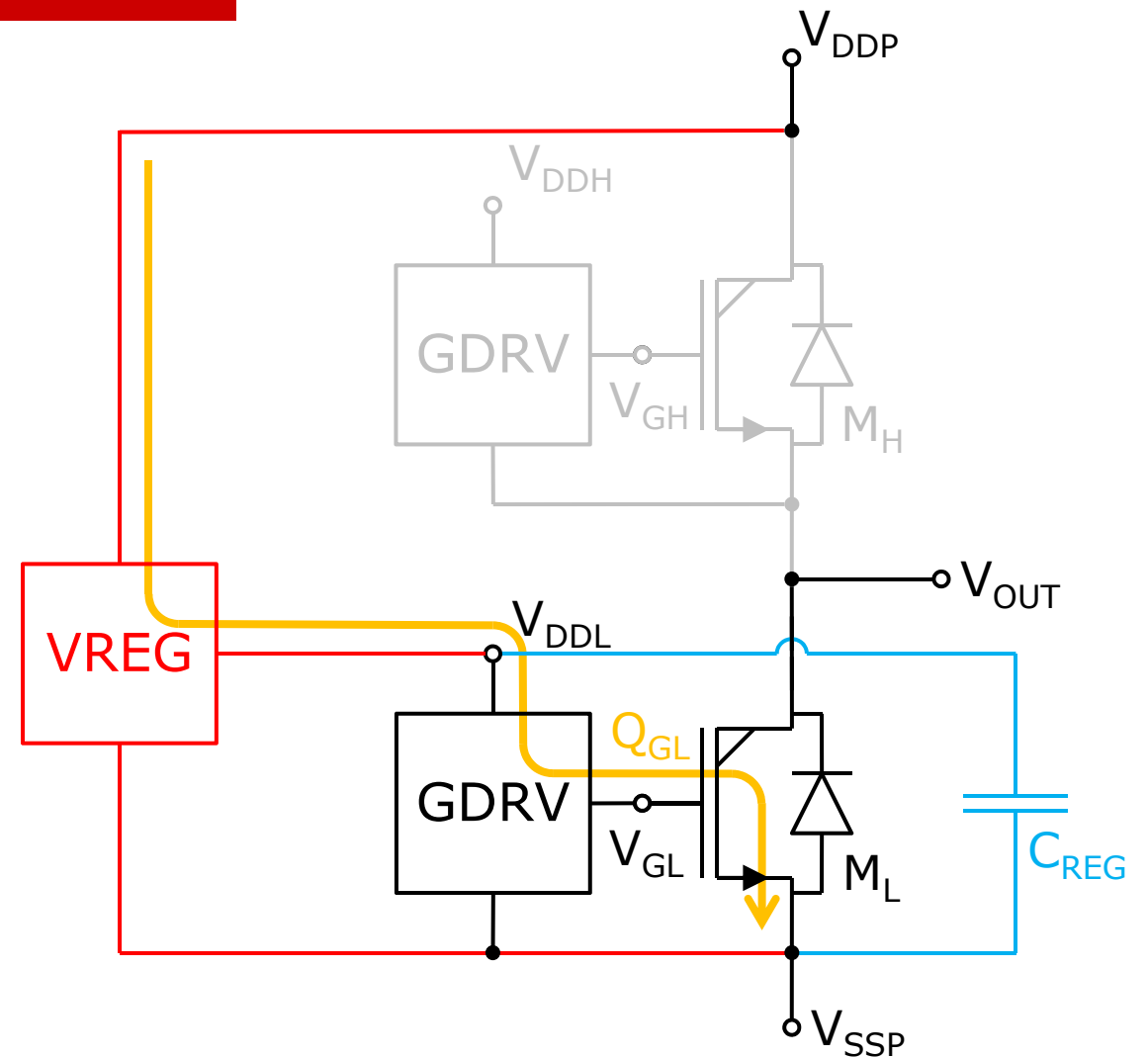
Bootstrap

- lowside gatedriver supply V_{DDL}
 - limit V_{GS} of M_L
- linear regulator
 - external decoupling C_{REG}



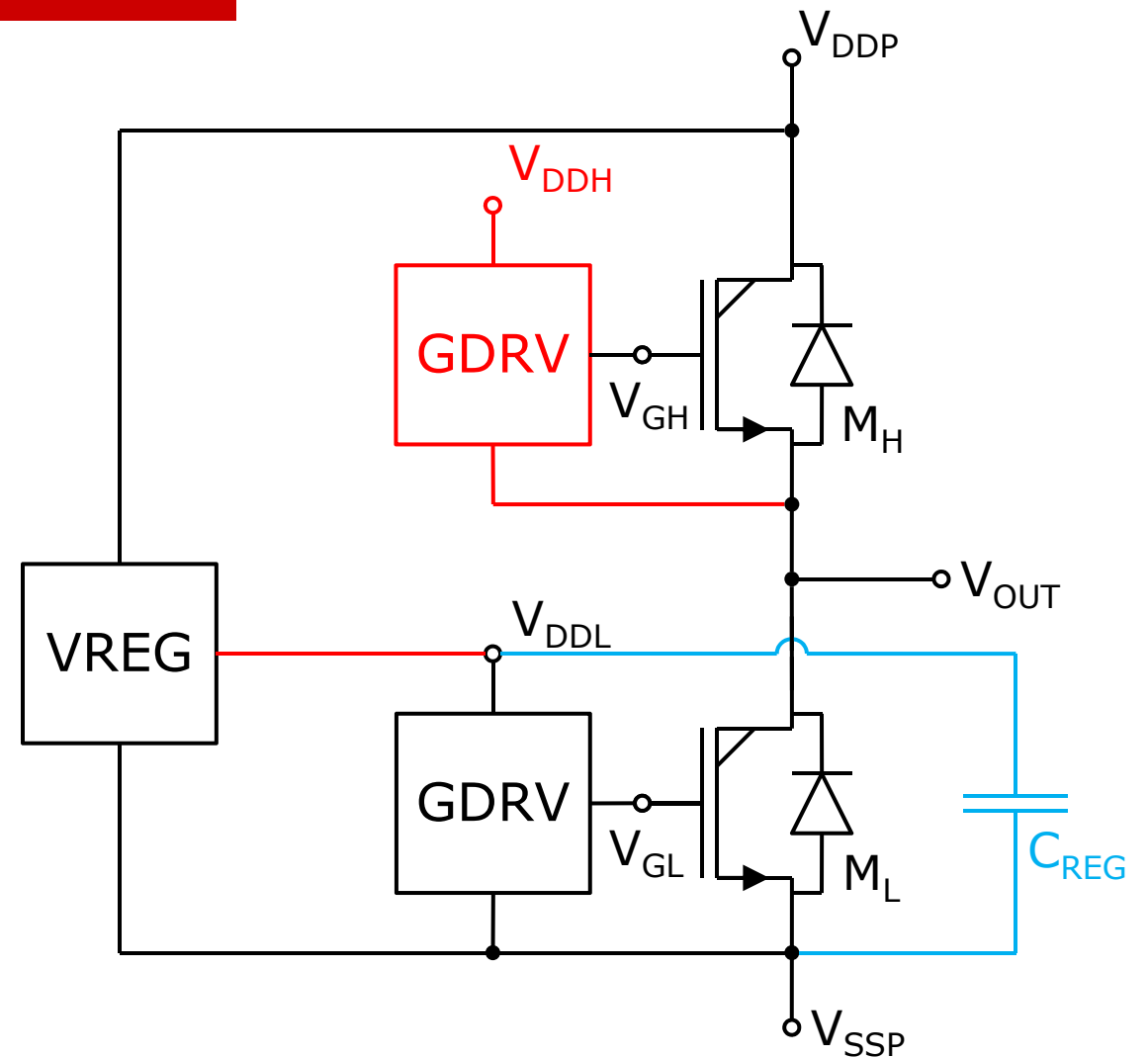
Bootstrap

- lowside gatedriver supply V_{DDL}
 - limit V_{GS} of M_L
- linear regulator
 - external decoupling C_{REG}
- gate charge Q_{GL} drawn from V_{DDP}
 - via regulator



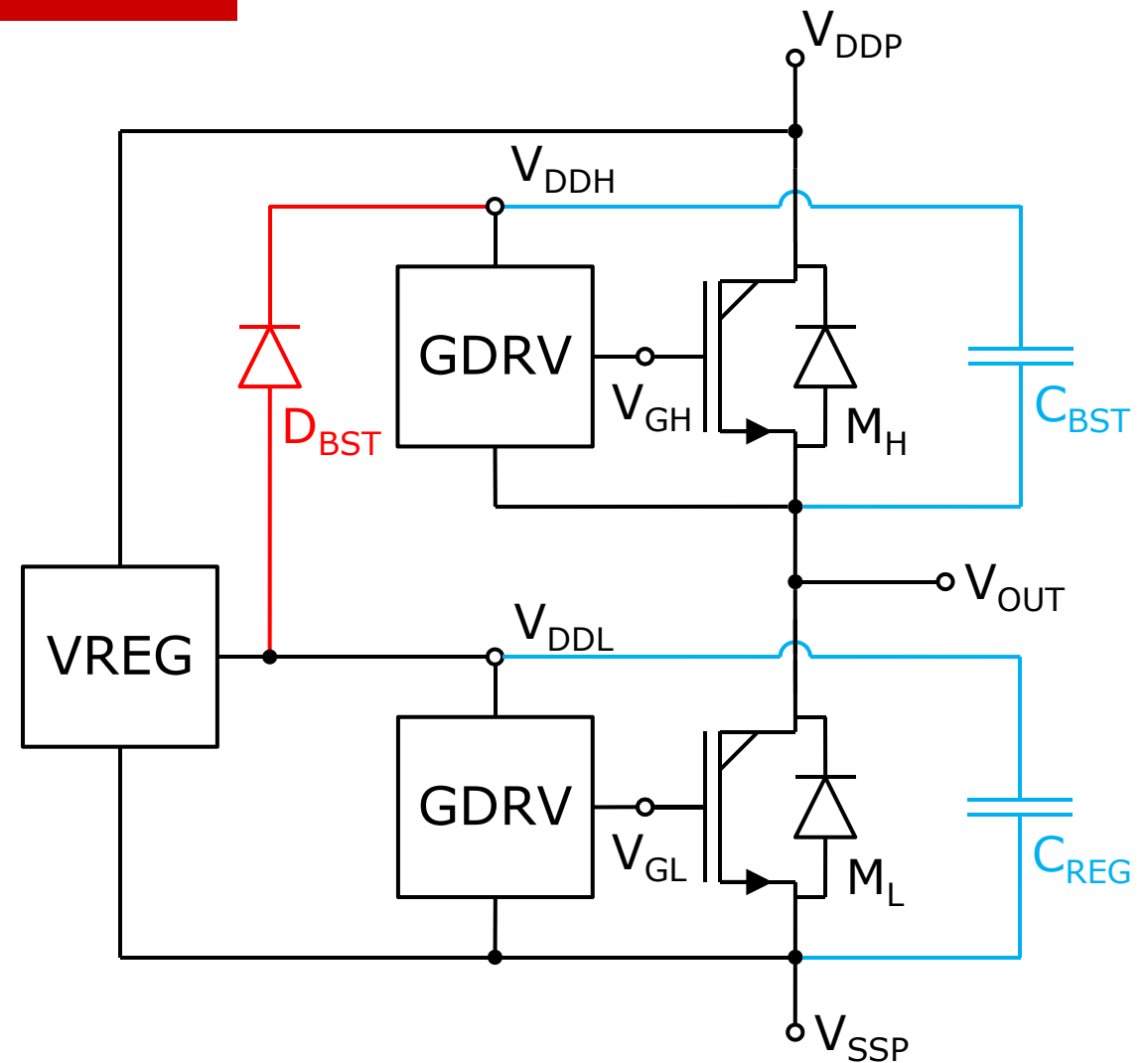
Bootstrap

- highside gatedriver supply V_{DDH}
 - limit V_{GS} of M_H



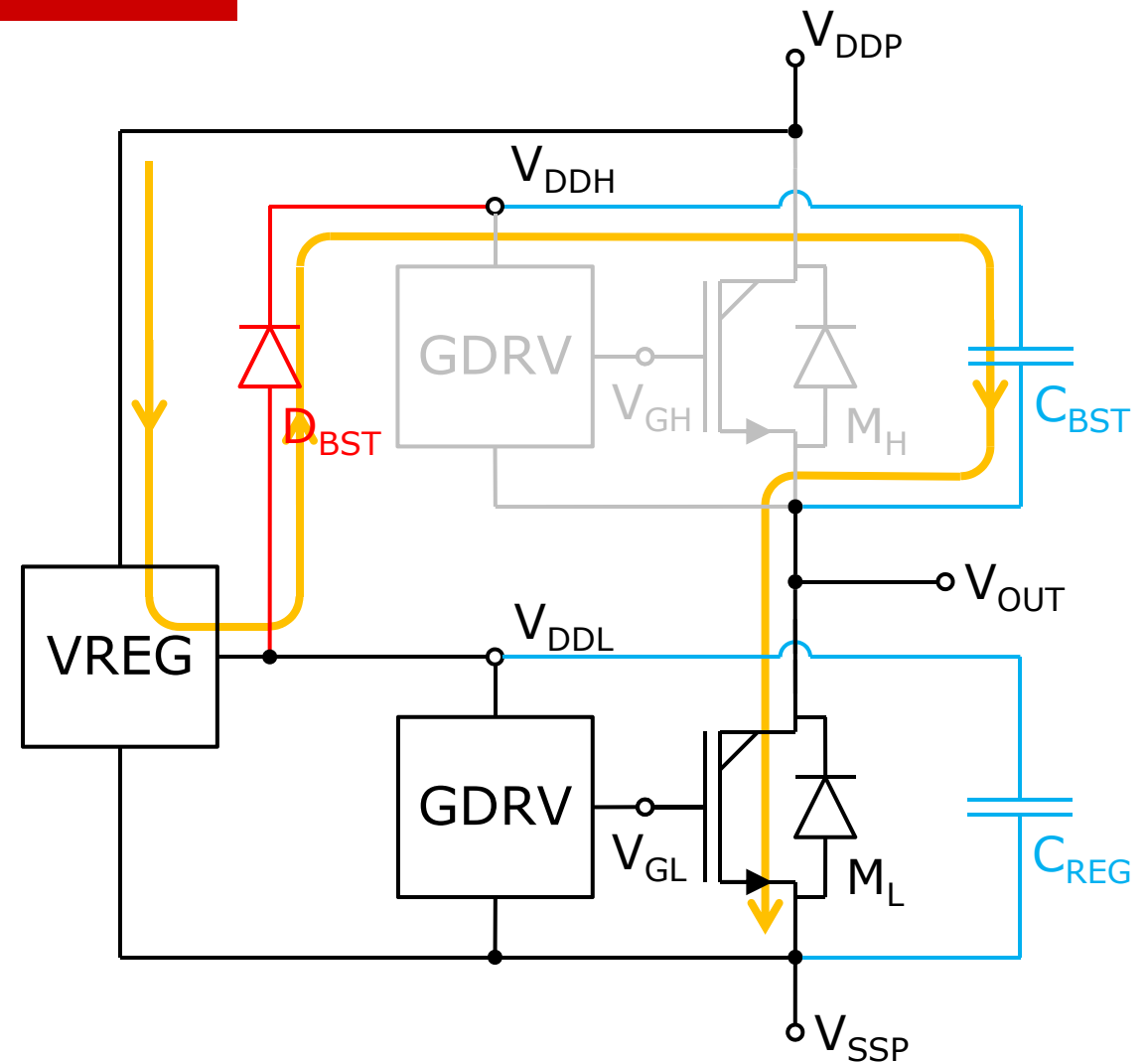
Bootstrap

- highside gatedriver supply V_{DDH}
 - limit V_{GS} of M_H
- external cap C_{BST}
 - diode D_{BST} connection to V_{DDL}



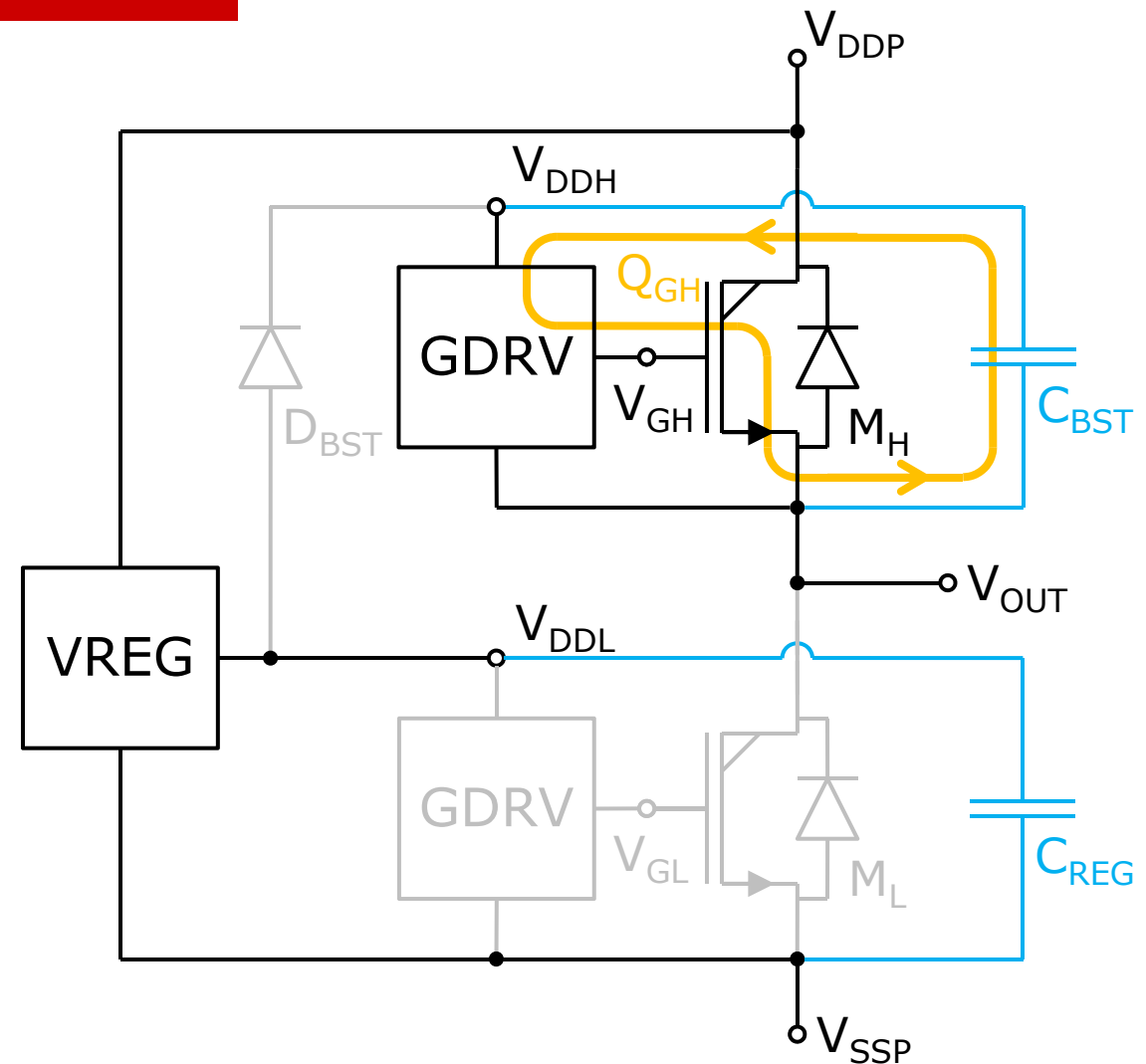
Bootstrap

- highside gatedriver supply V_{DDH}
 - limit V_{GS} of M_H
- external cap C_{BST}
 - diode D_{BST} connection to V_{DDL}
- turn on lowside M_L
 - C_{BST} charged via D_{BST} from V_{DDP}



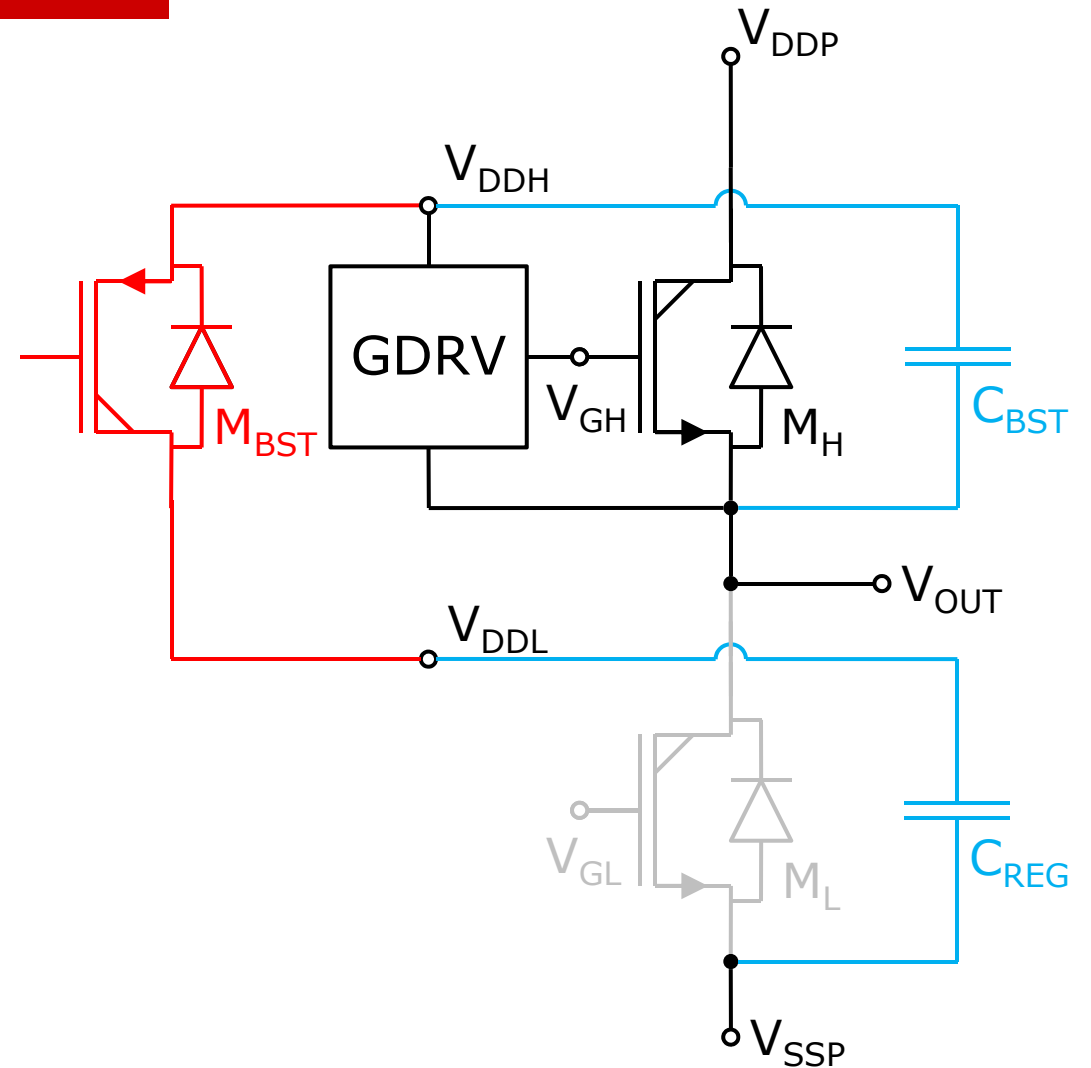
Bootstrap

- highside gatedriver supply V_{DDH}
 - limit V_{GS} of M_H
- external cap C_{BST}
 - diode D_{BST} connection to V_{DDL}
- turn on lowside M_L
 - C_{BST} charged via D_{BST} from V_{DDP}
- turn on highside M_H
 - D_{BST} reverse biased
 - gate charge Q_{GH} drawn from C_{BST}

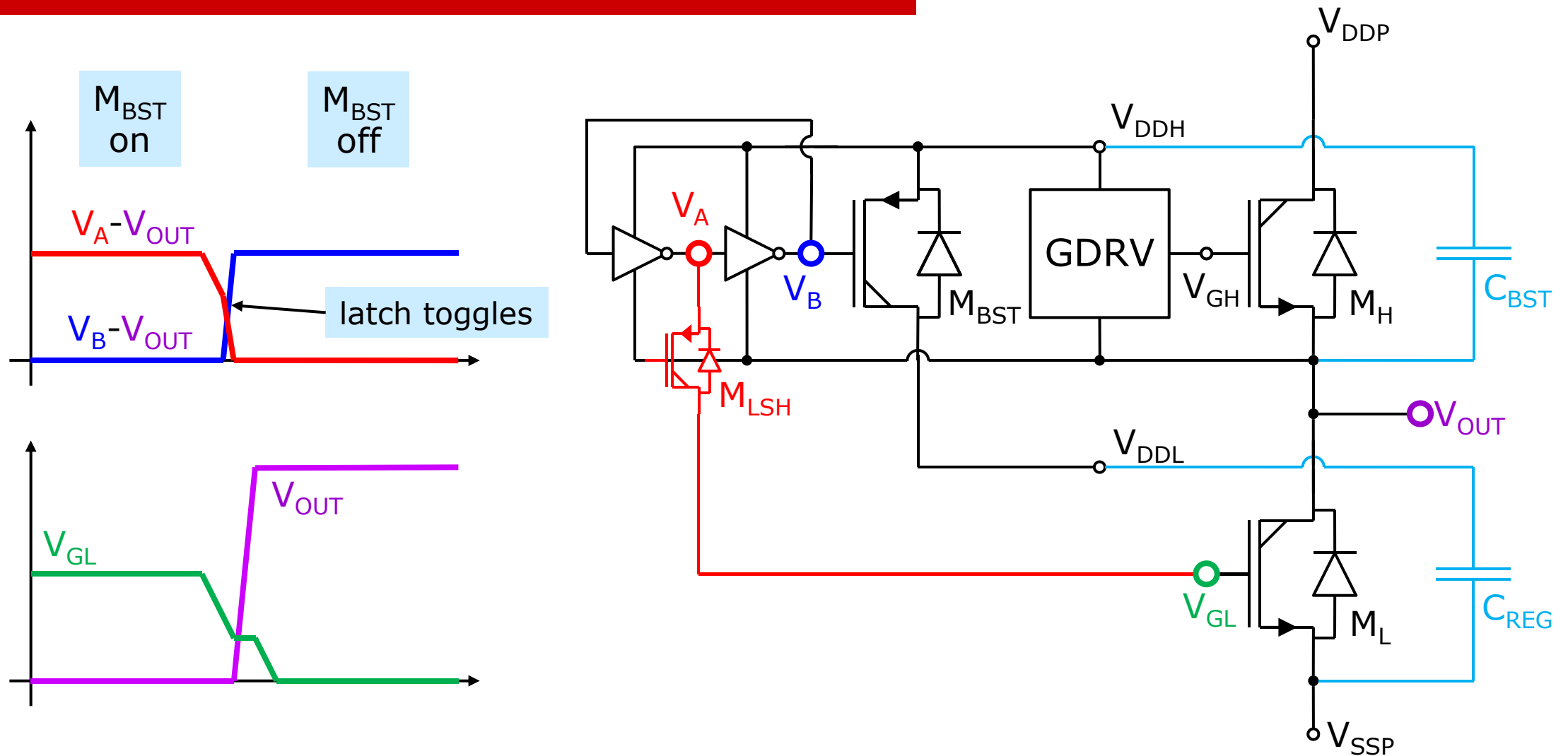


Bootstrap Switch

- turn off M_{BST} when V_{OUT} is high
 - body diode reverse biased
- turn on M_{BST} when V_{OUT} is low
 - body diode bypassed
 - no forward diode voltage drop

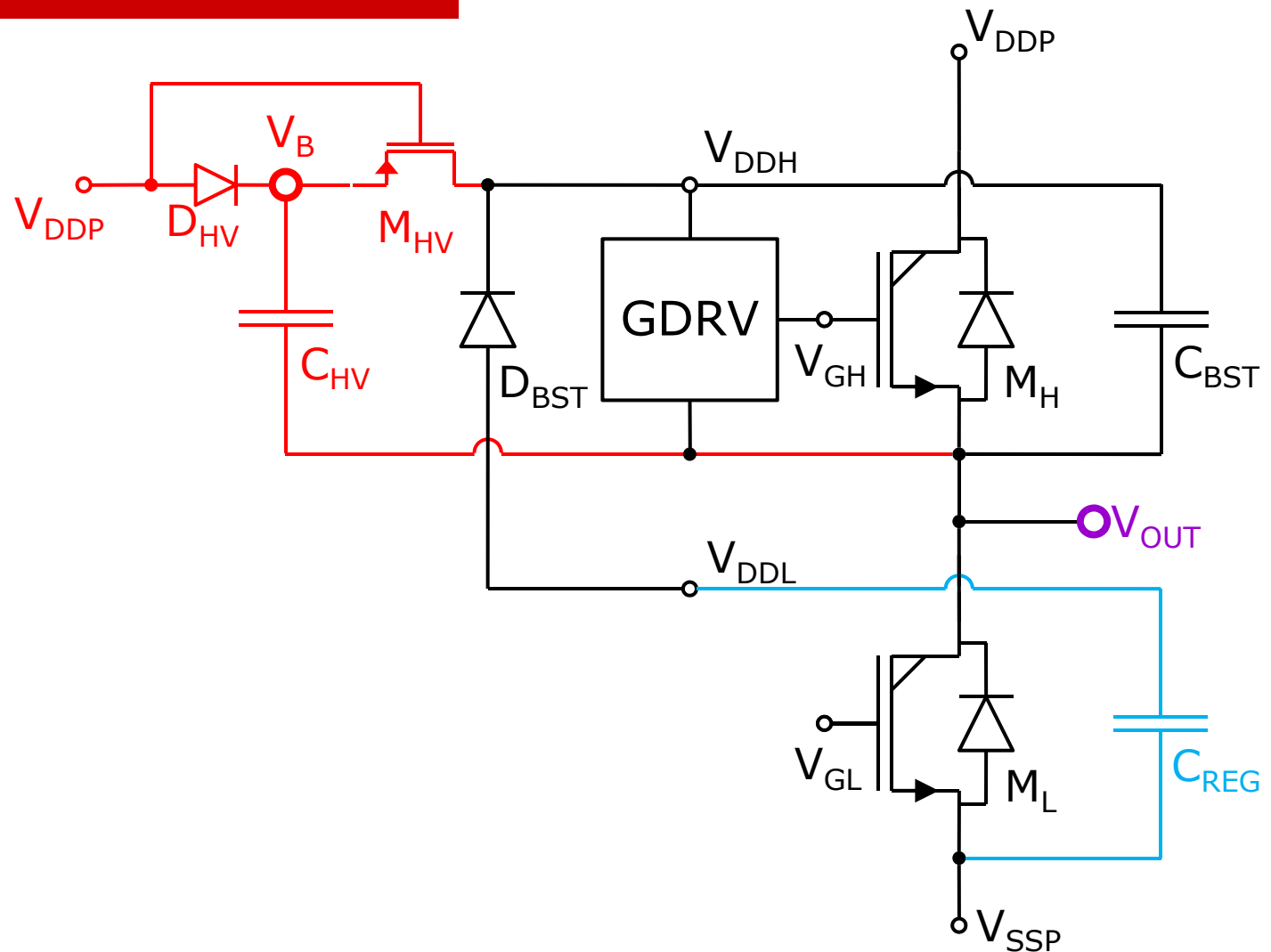


Bootstrap Switch



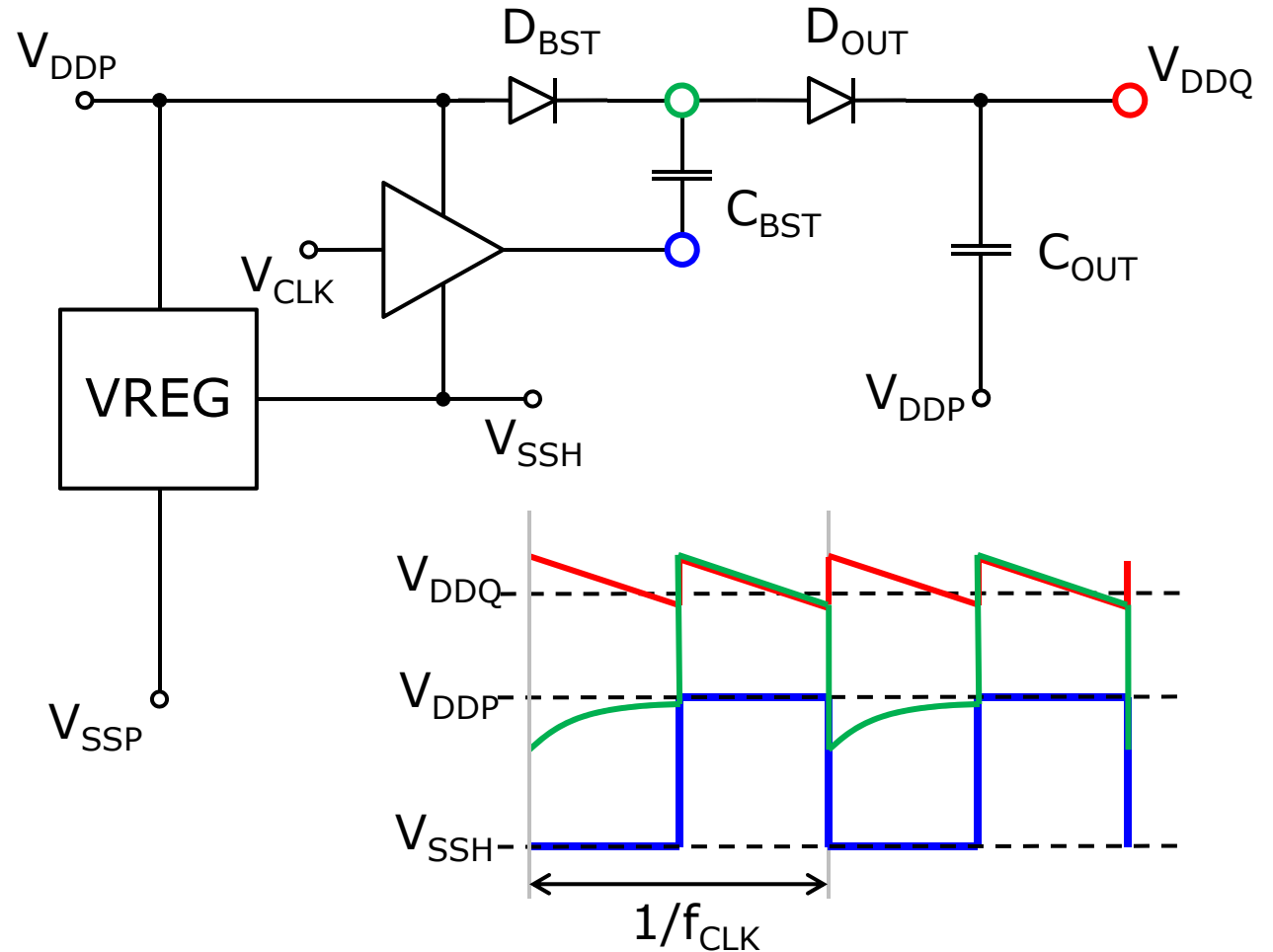
Integrated Bootstrap [20]

- two stage bootstrap
 - small HV capacitor C_{HV}
- output V_{OUT} low
 - switch M_{HV} off
 - C_{BST} charged to V_{DDL}
 - C_{HV} charged to V_{DDP}
- output V_{OUT} high
 - switch M_{HV} on
 - C_{HV} discharges to C_{BST}
 - $Q_{GH} = C_{HV} \times (V_{DDP} - V_{DDL})$



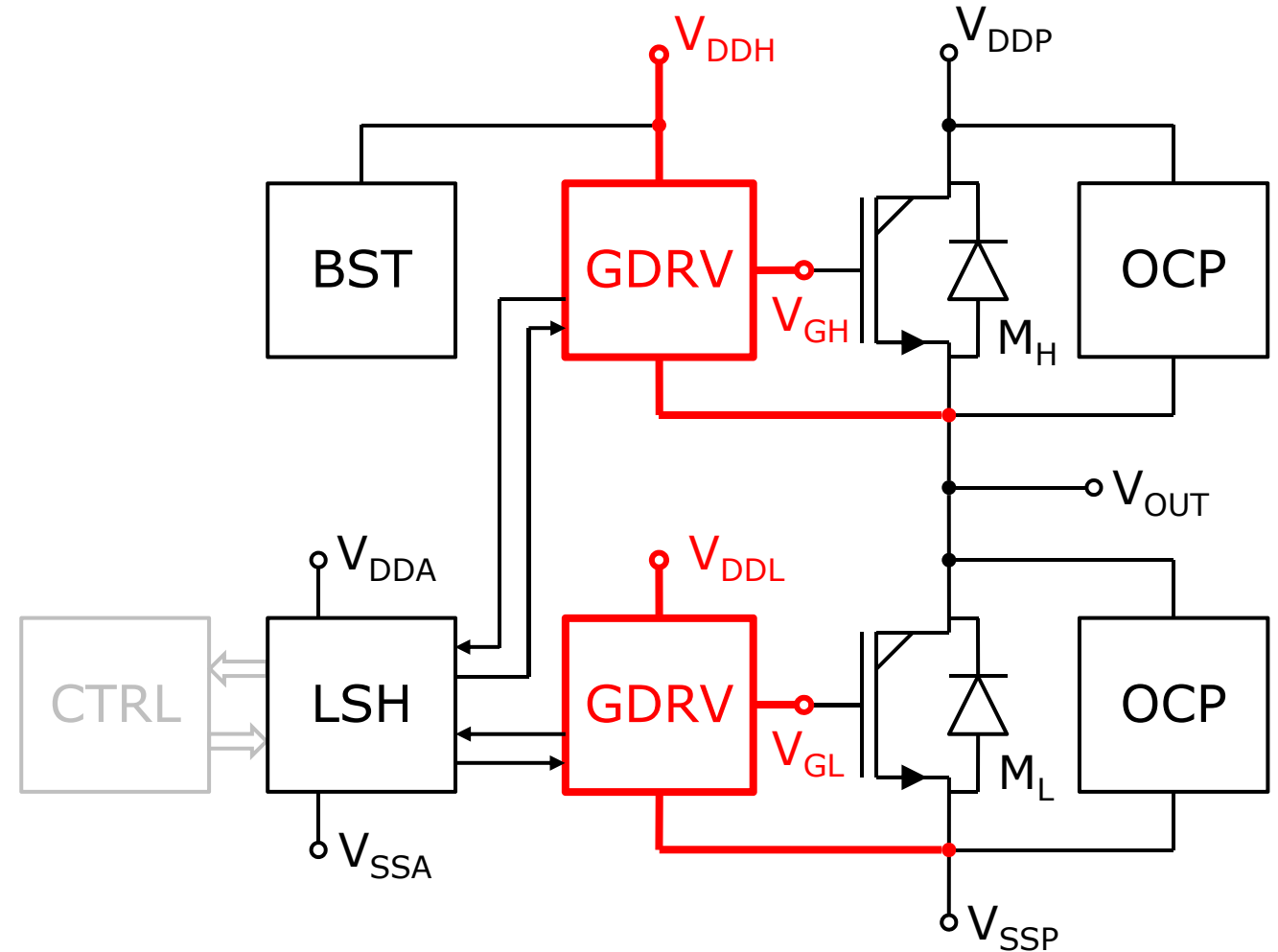
Chargepump^[21] + Regulator

- input voltage
 - $V_{IN} = V_{DDP} - V_{SSH}$
- output voltage
 - $V_{DDQ} = V_{DDP} + V_{IN}$
- attractive in applications with multiple half bridges
 - e.g. 4-channel class-D audio amplifier would need 8 bootstrap caps



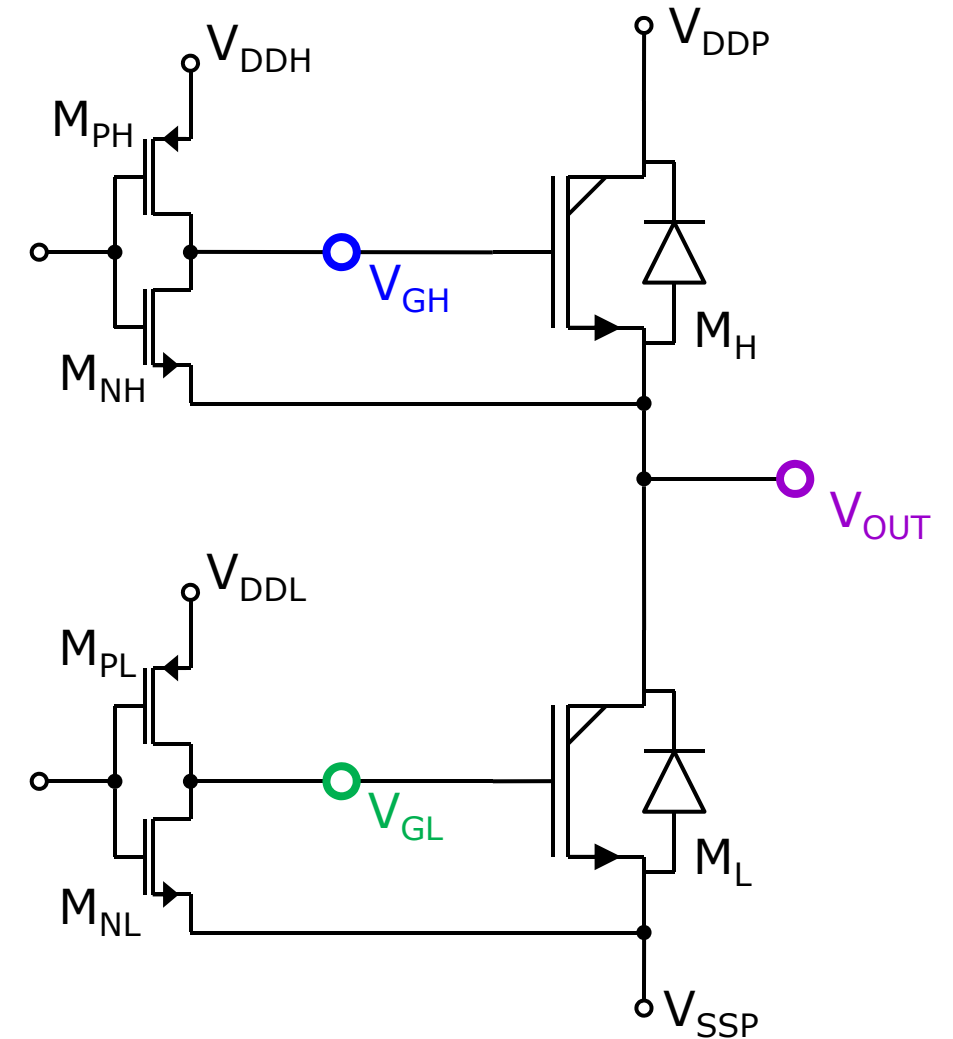
Half Bridge Essentials

- PowerFETs
- BST
 - highside gatedrive voltage
- GDRV
 - powerFET gate driver
- LSH
 - levelshifters
- OCP
 - overcurrent protection



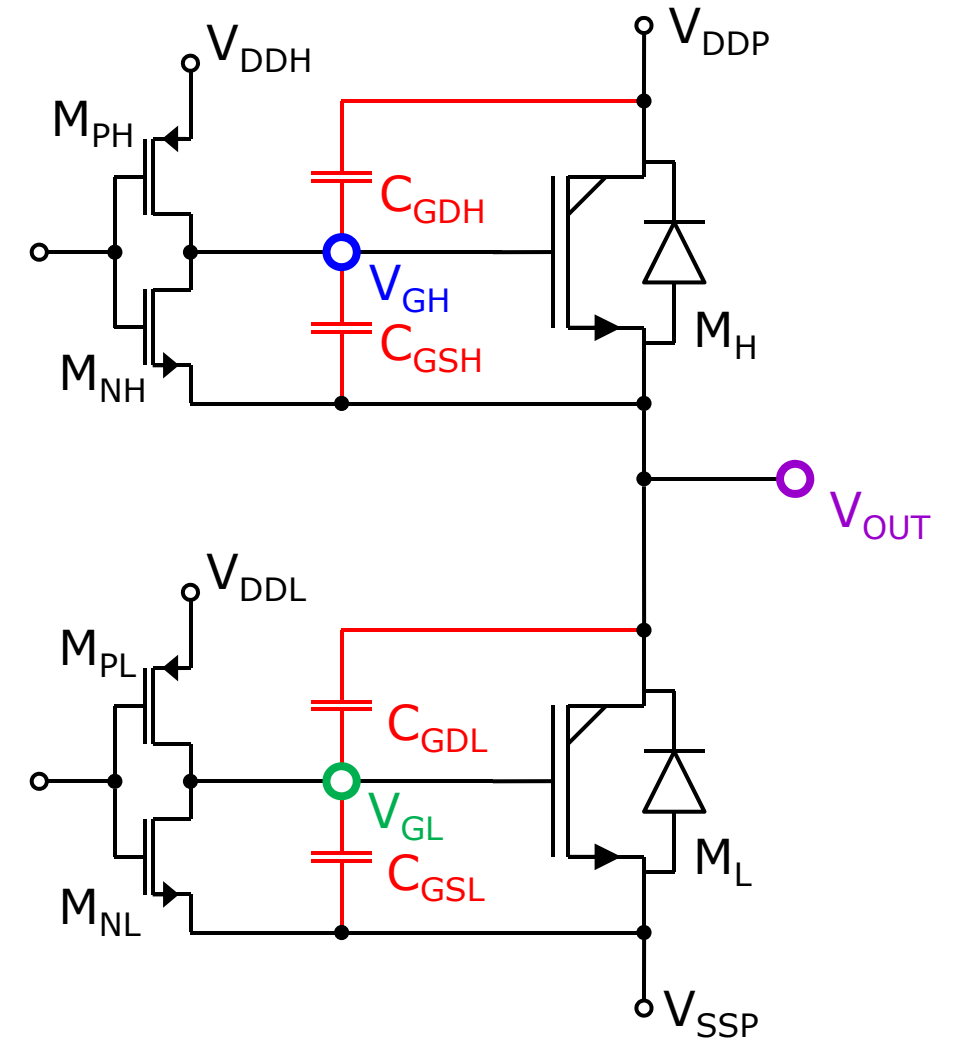
Inverter Gate Drivers

- lowside gatedriver
 - supply V_{DDL} (regulator)
 - (local) ground V_{SSP}
- highside gatedriver
 - supply V_{DDH} (bootstrap)
 - (local) ground V_{OUT}



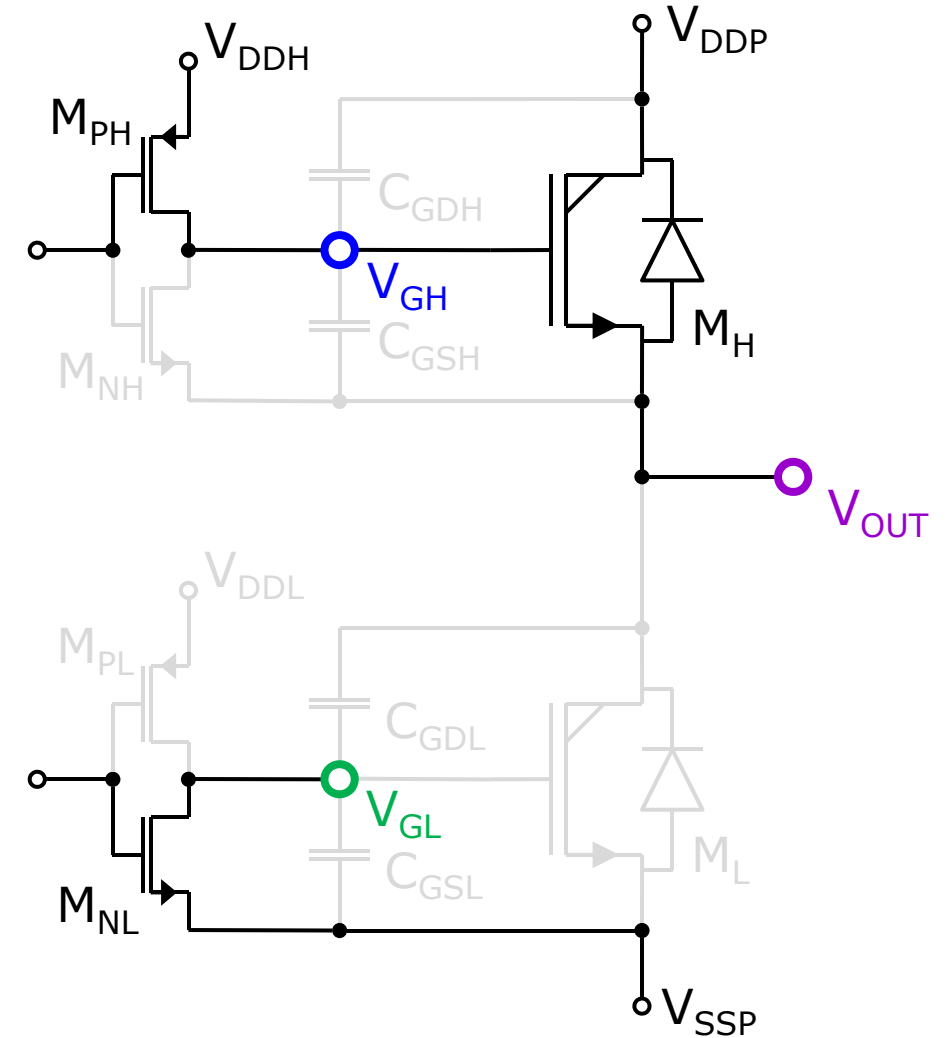
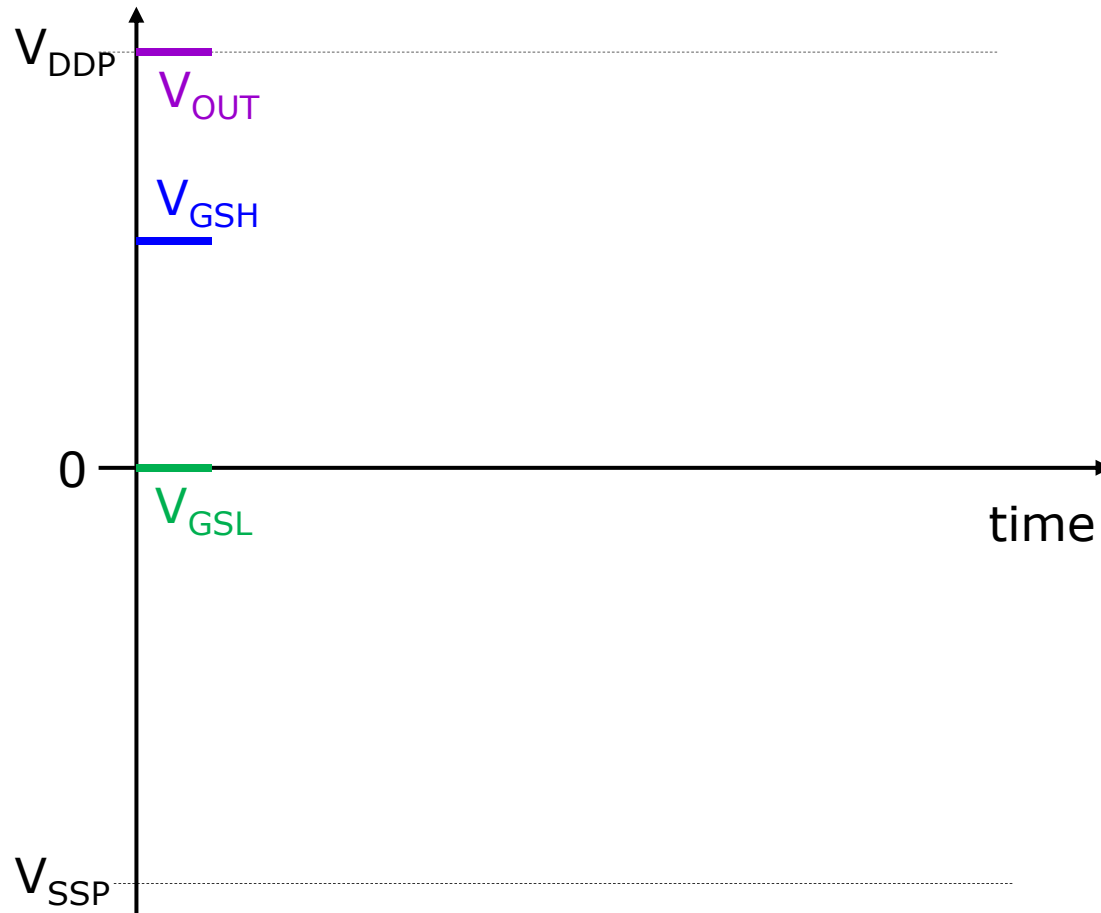
Parasitic Capacitances

- lowside gatedriver
 - supply V_{DDL} (regulator)
 - (local) ground V_{SSP}
- highside gatedriver
 - supply V_{DDH} (bootstrap)
 - (local) ground V_{OUT}
- switching dynamics
 - gate driver size
 - parasitic capacitances powerFETs

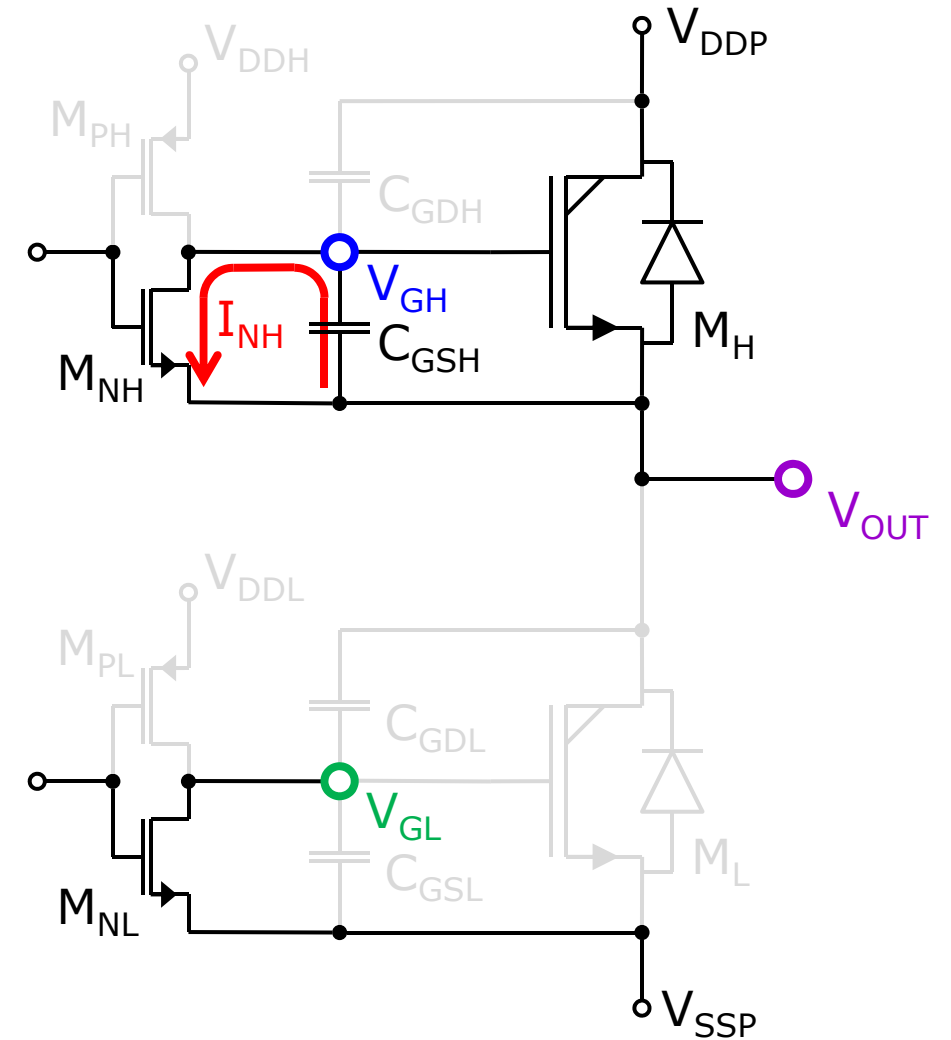
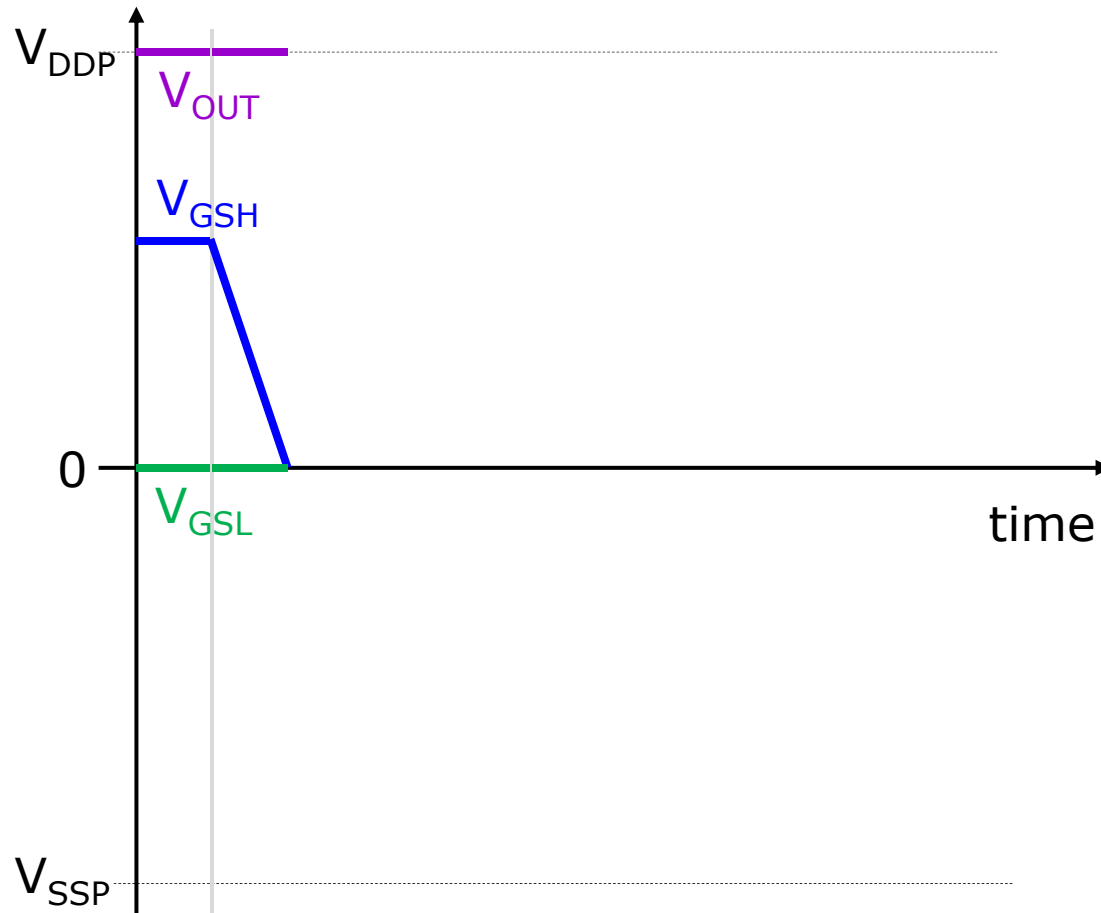


Zero Current Switching

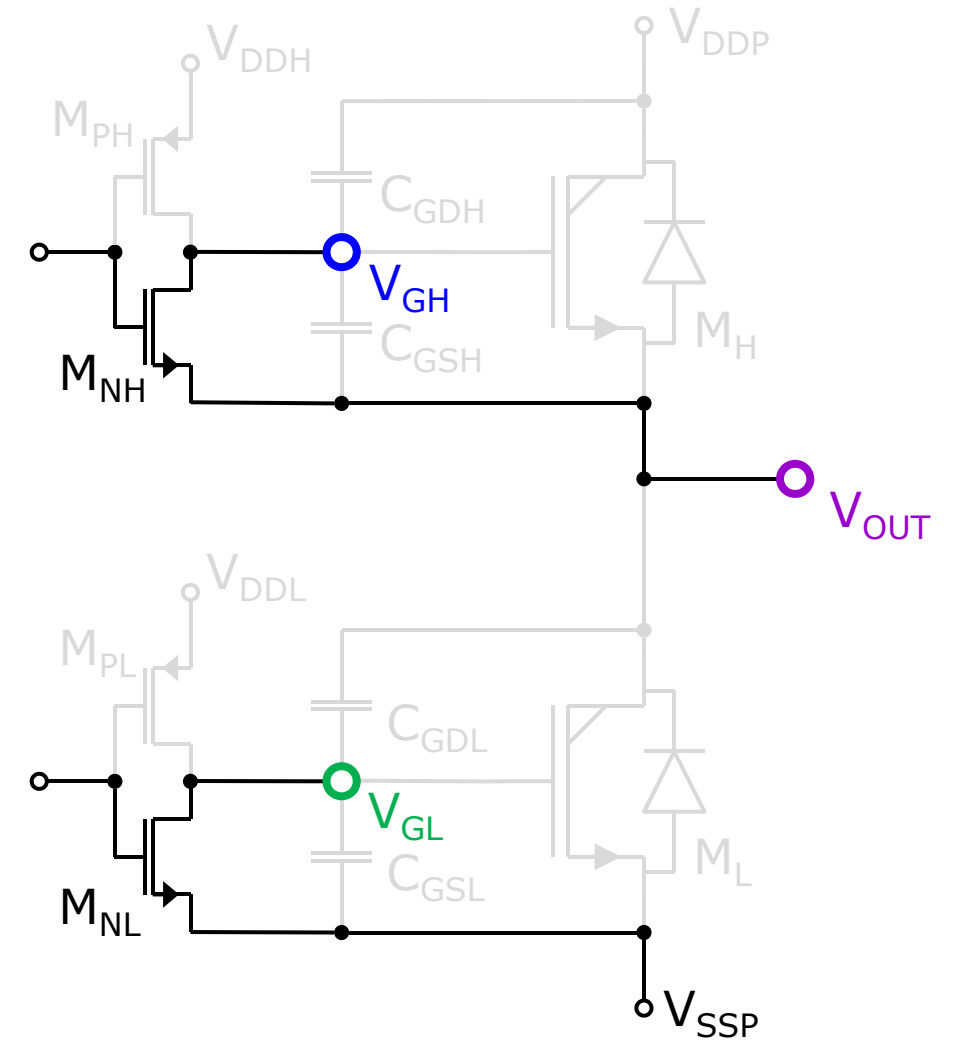
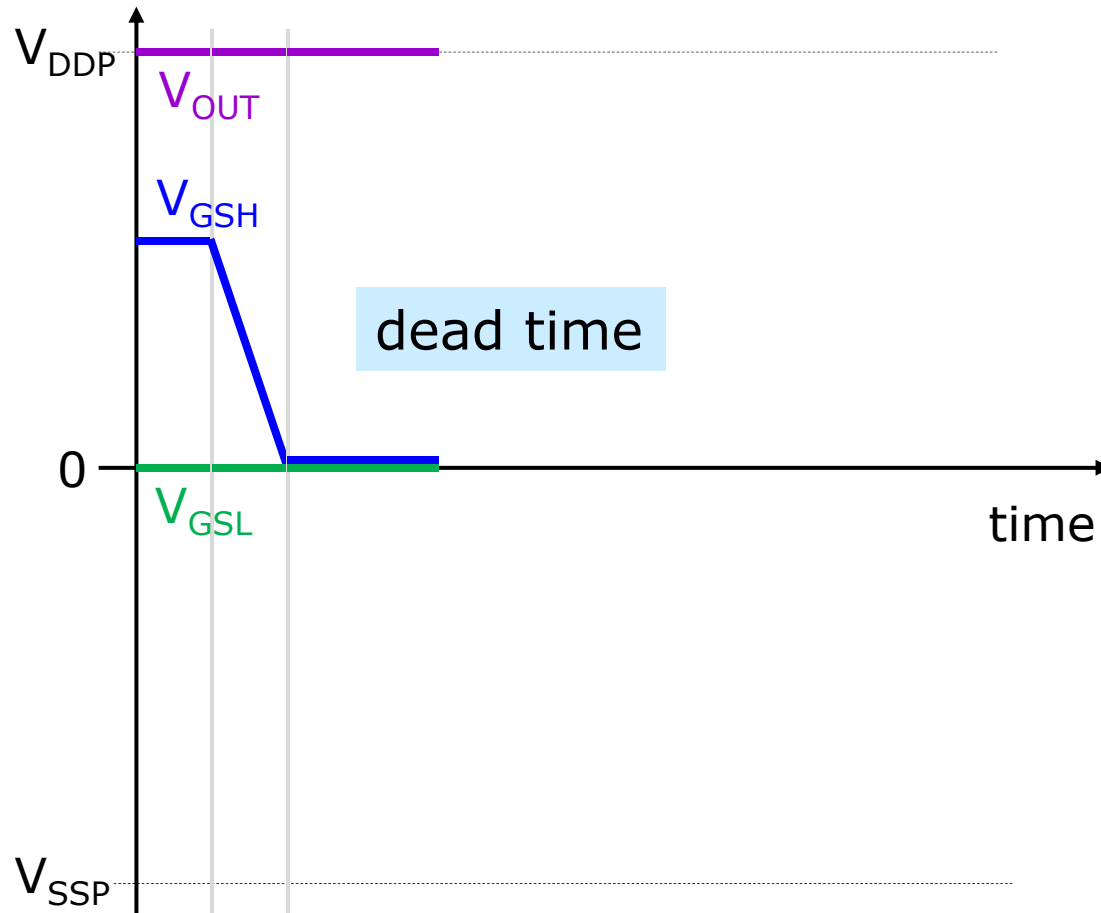
Zero Current Switching



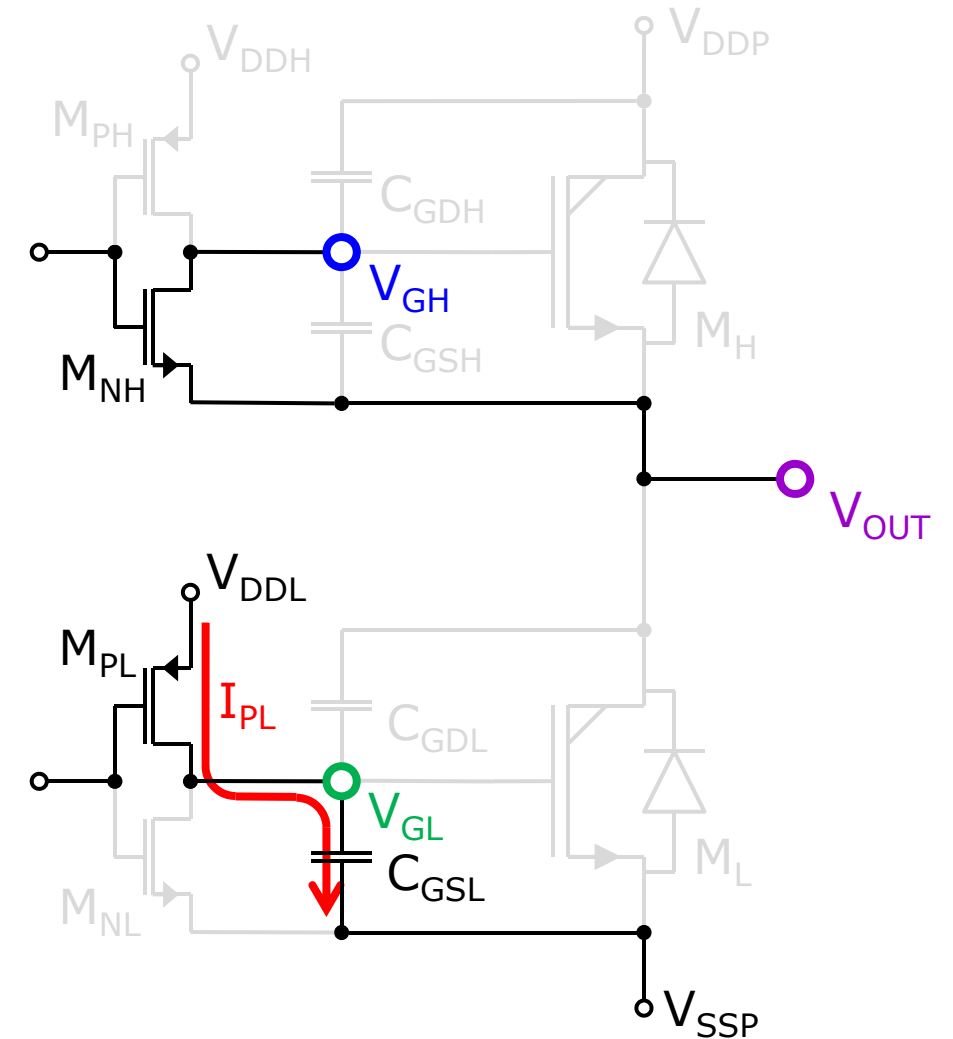
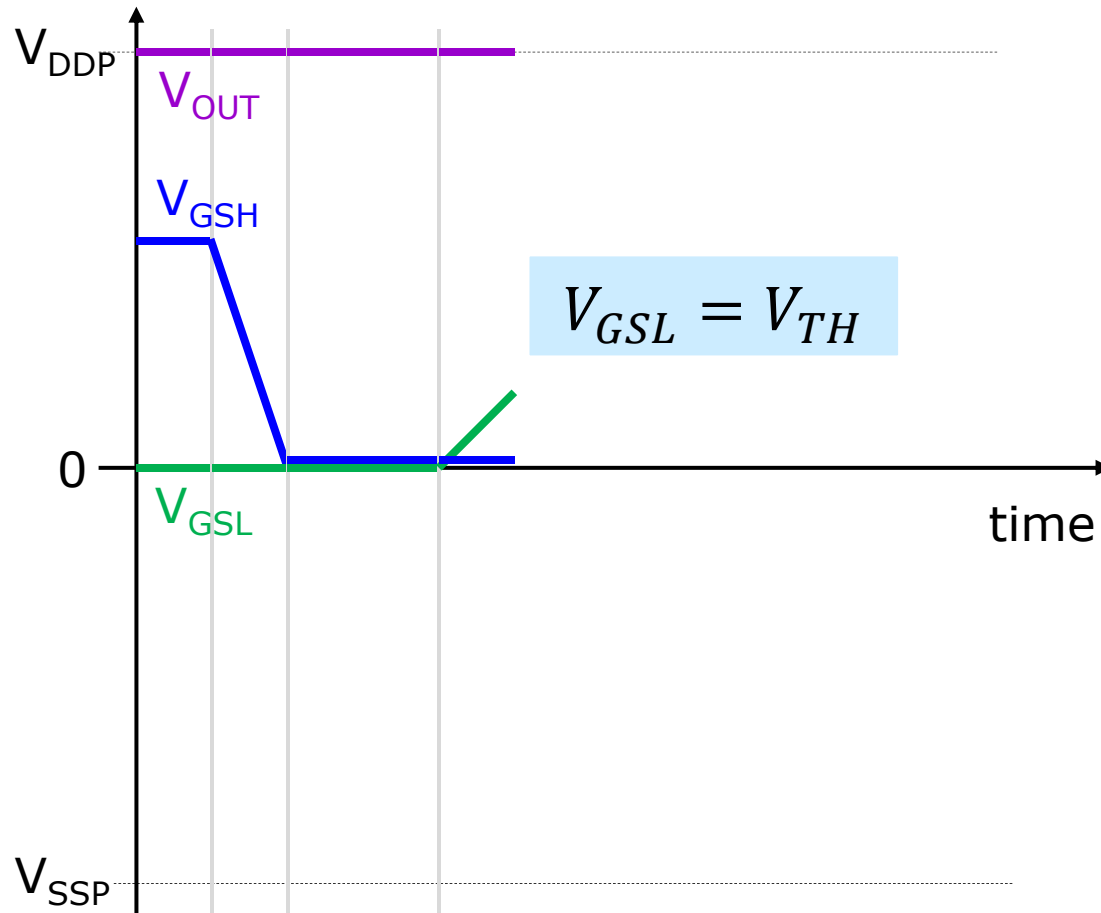
Zero Current Switching



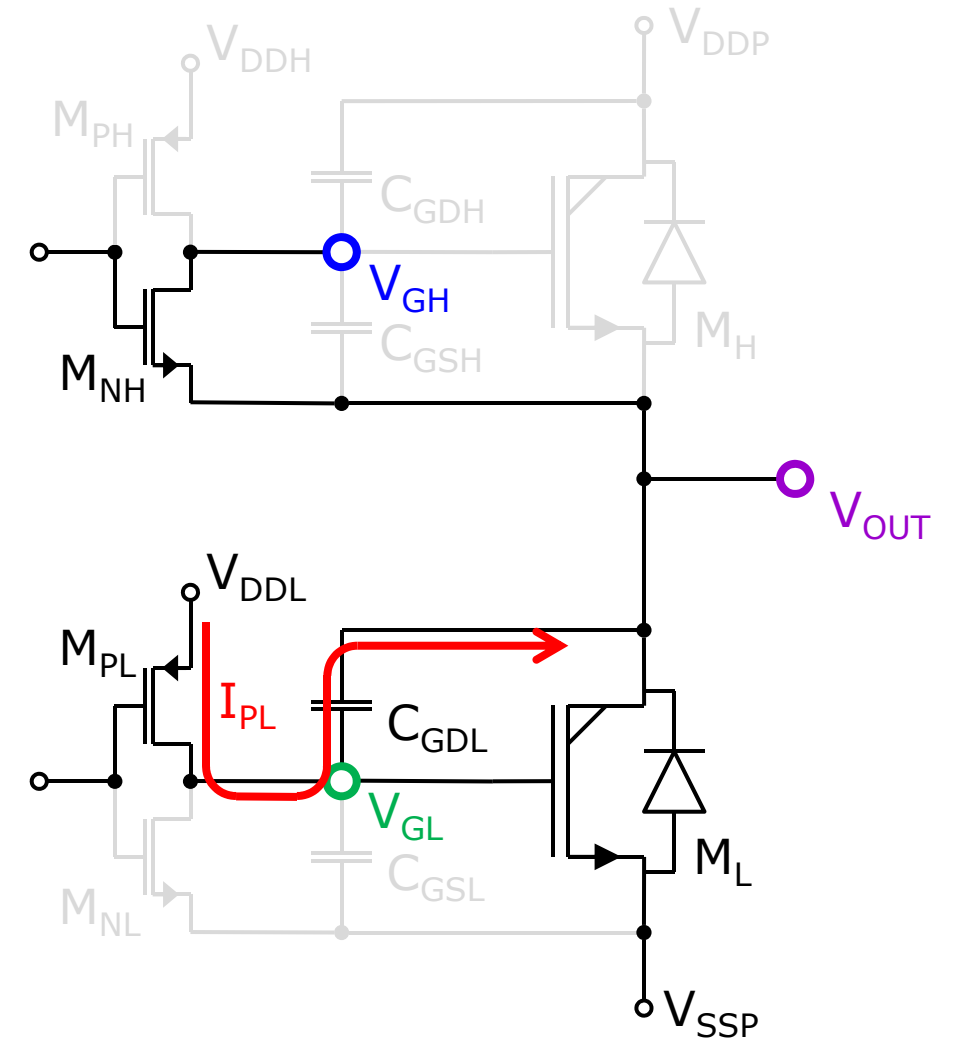
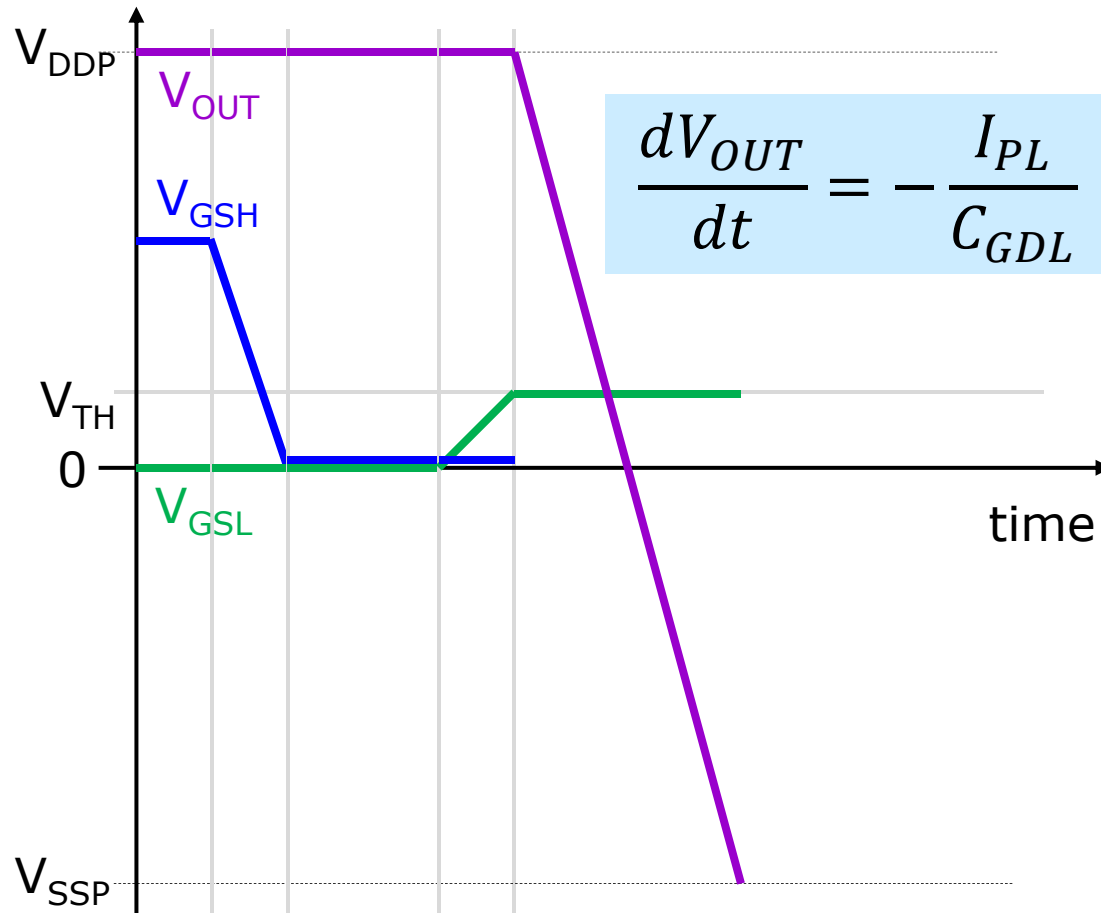
Zero Current Switching



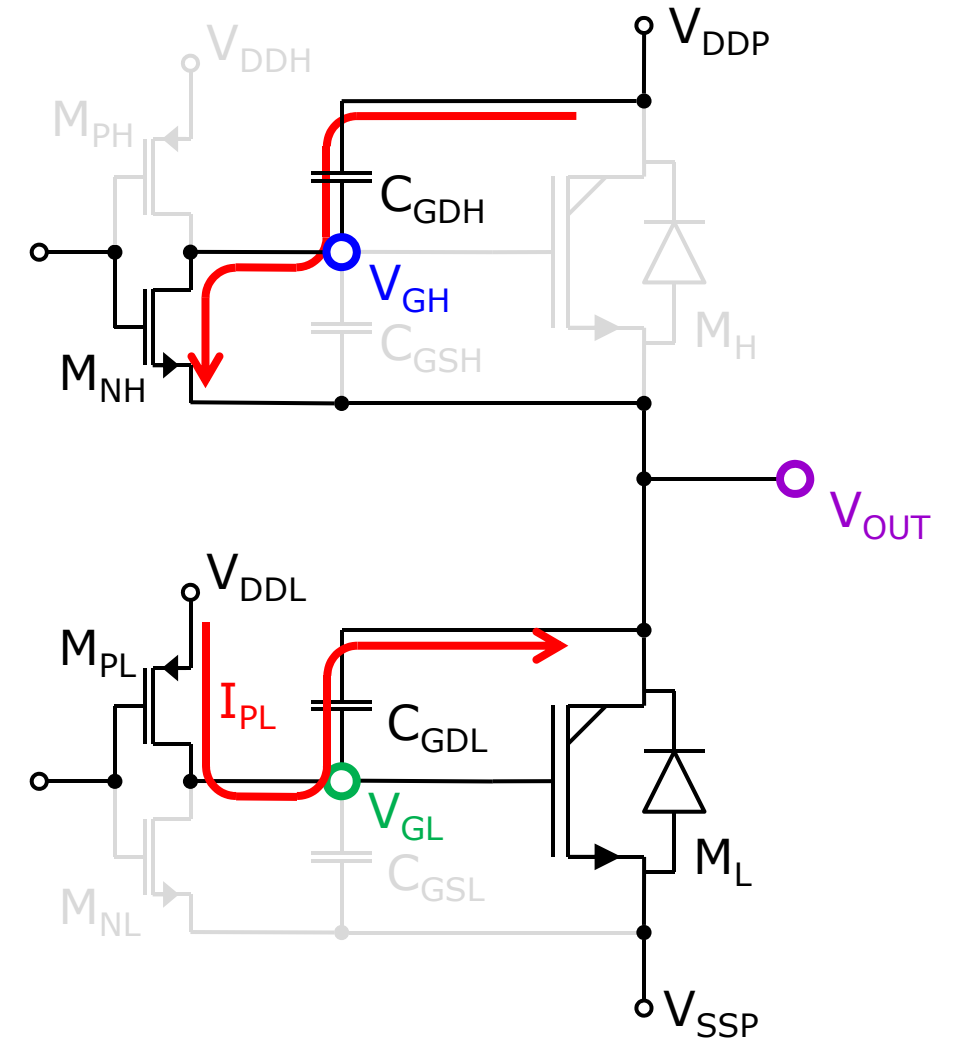
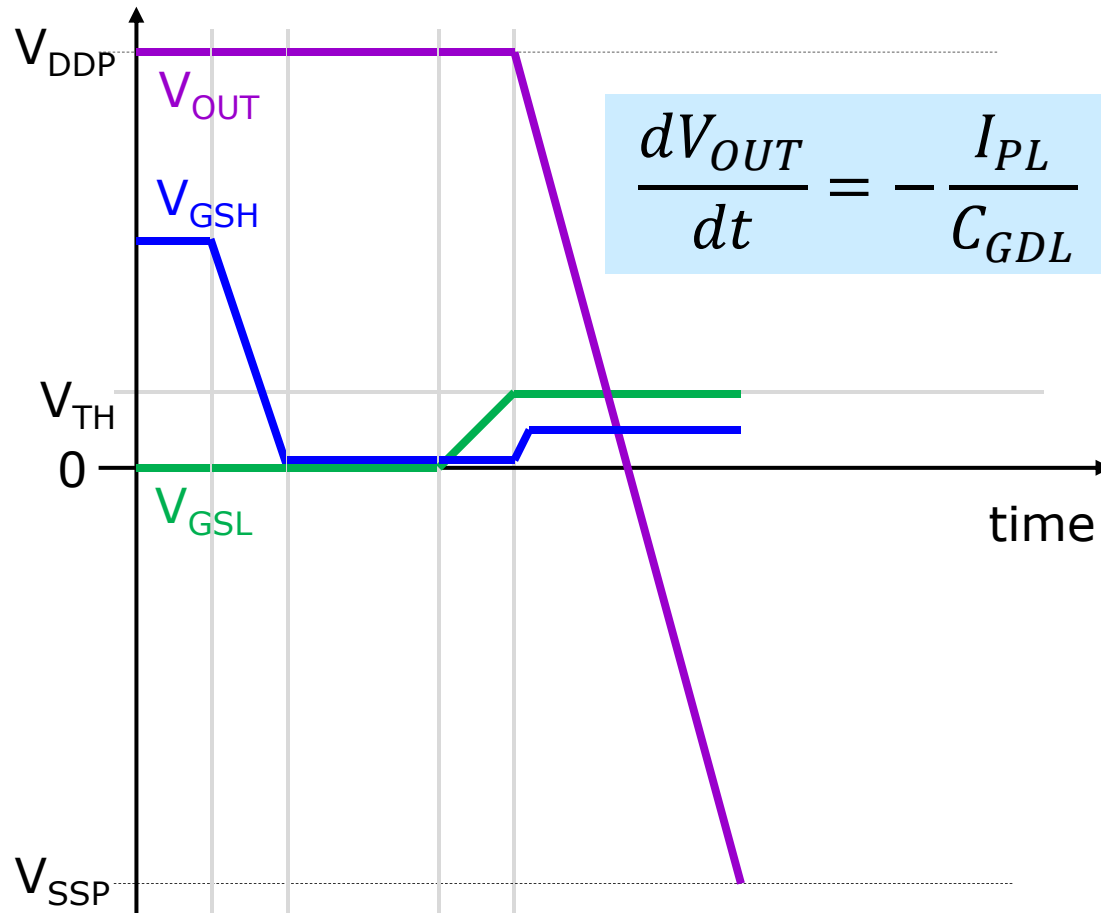
Zero Current Switching



Zero Current Switching



Zero Current Switching



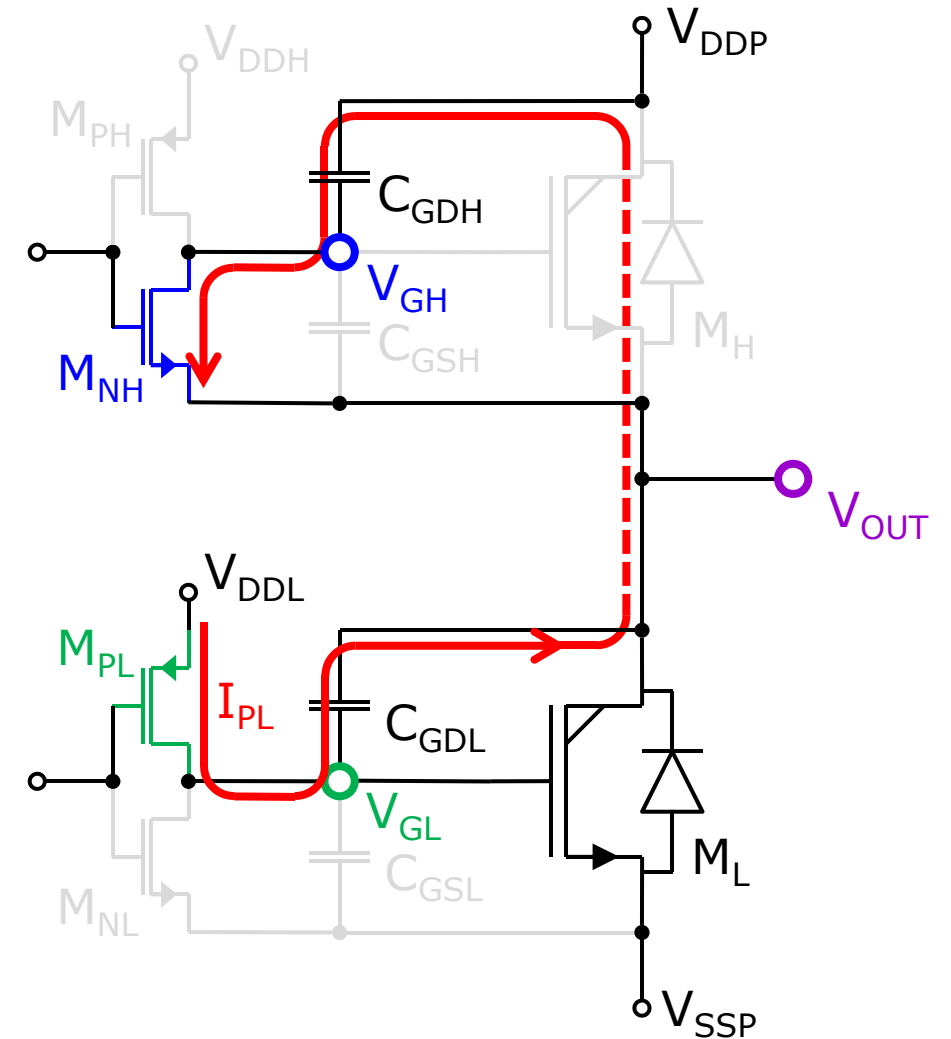
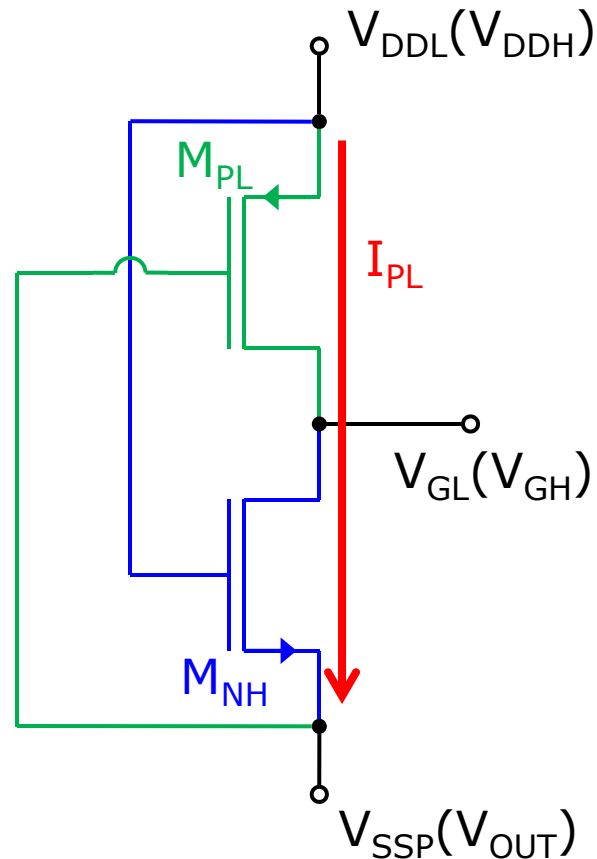
Zero Current Switching

prevent shoot-through:

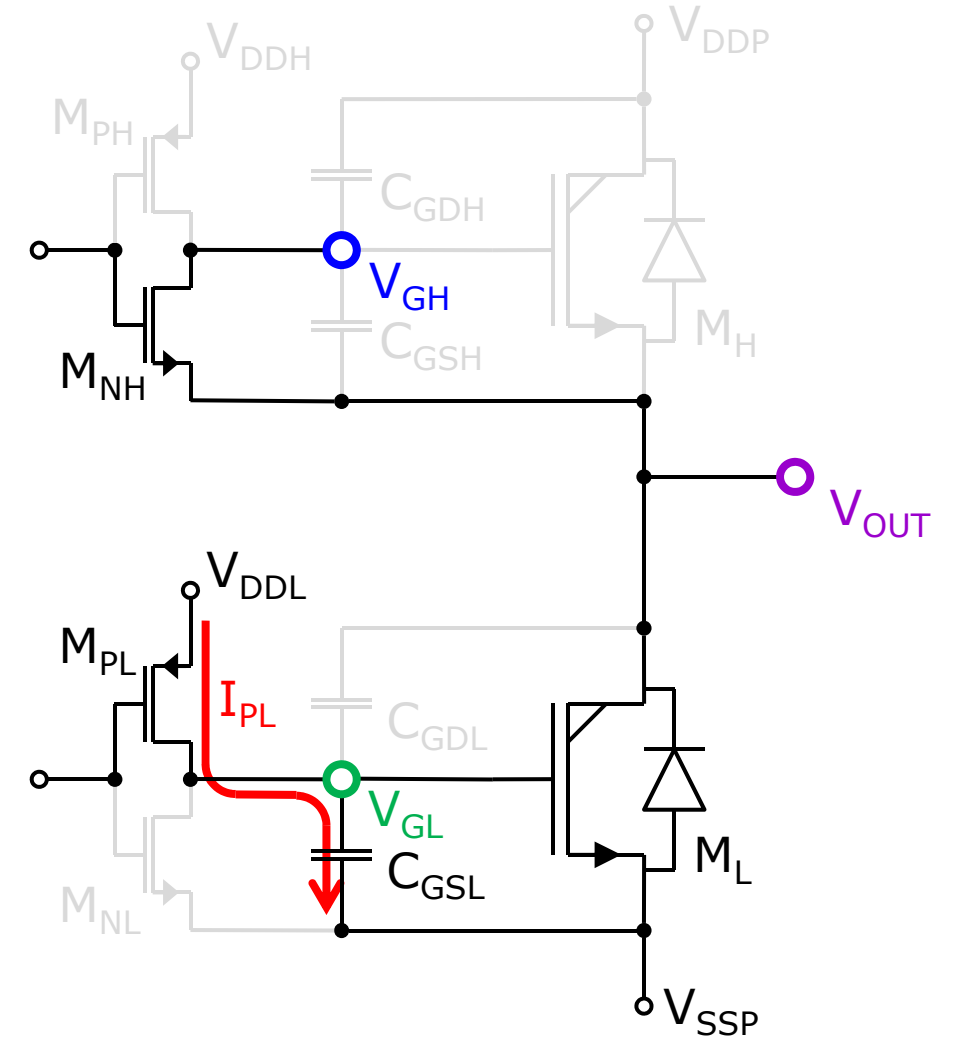
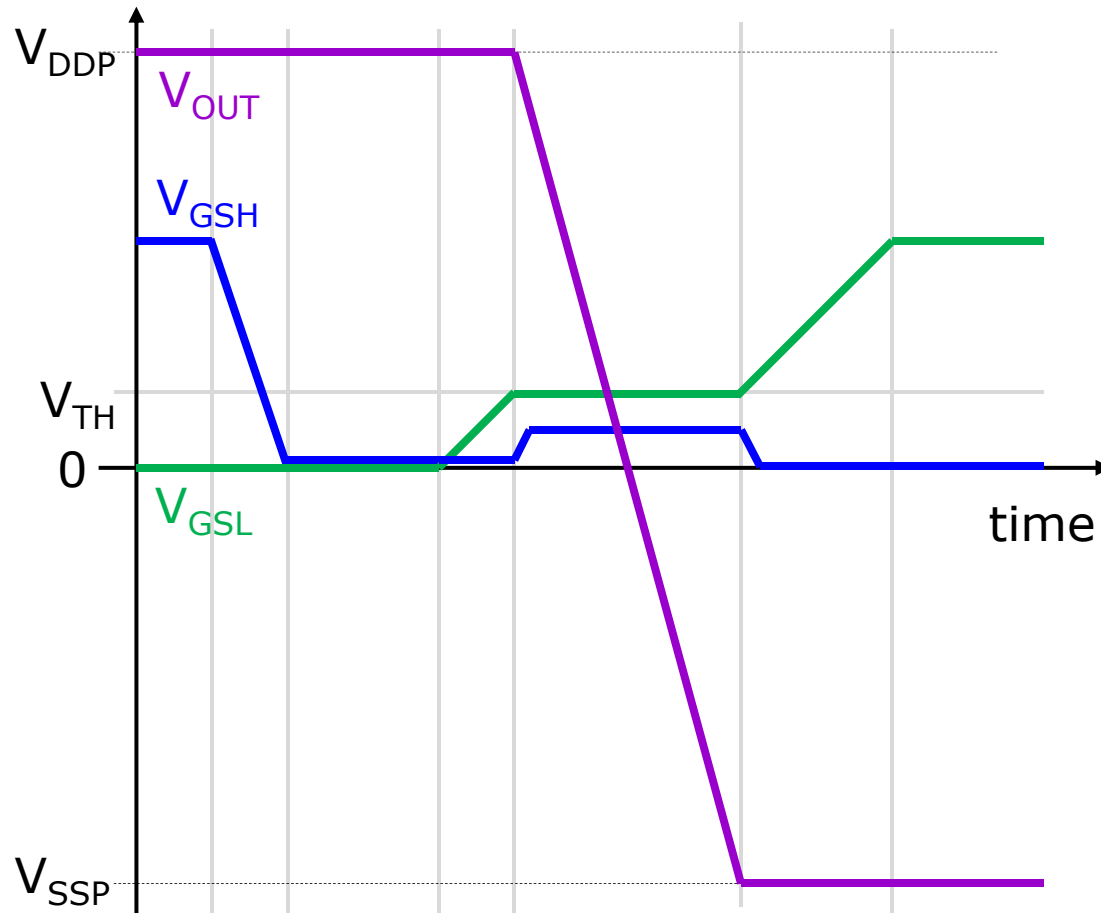
$$(W/L)_{NH} : (W/L)_{PL}$$



$$V_{GH} < V_{TH}$$

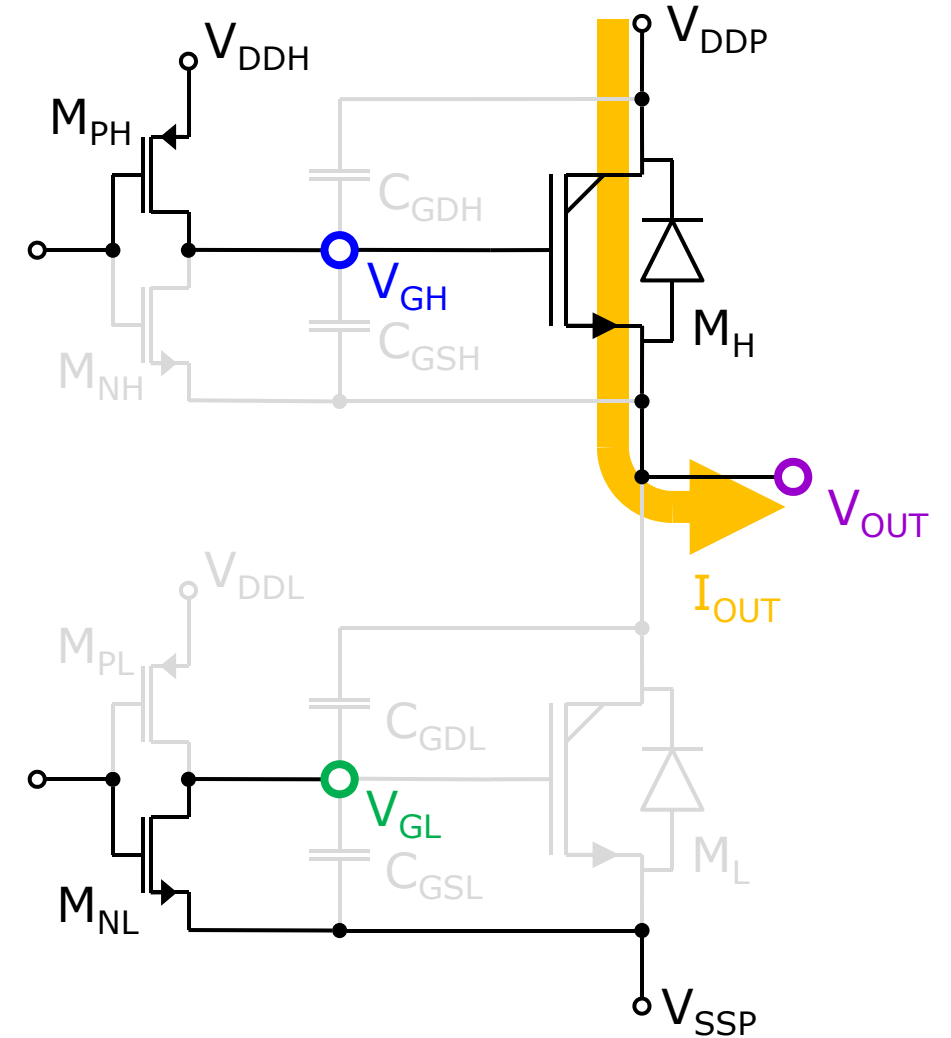
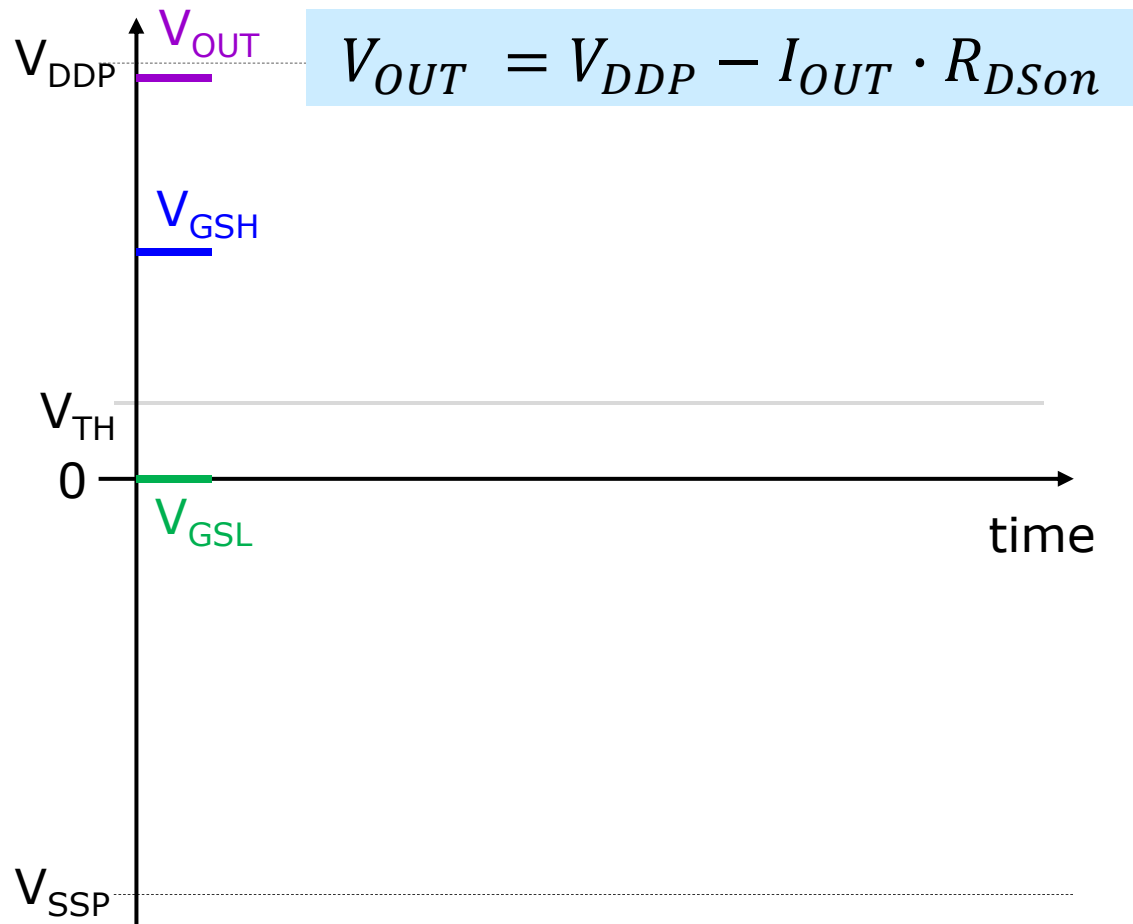


Zero Current Switching

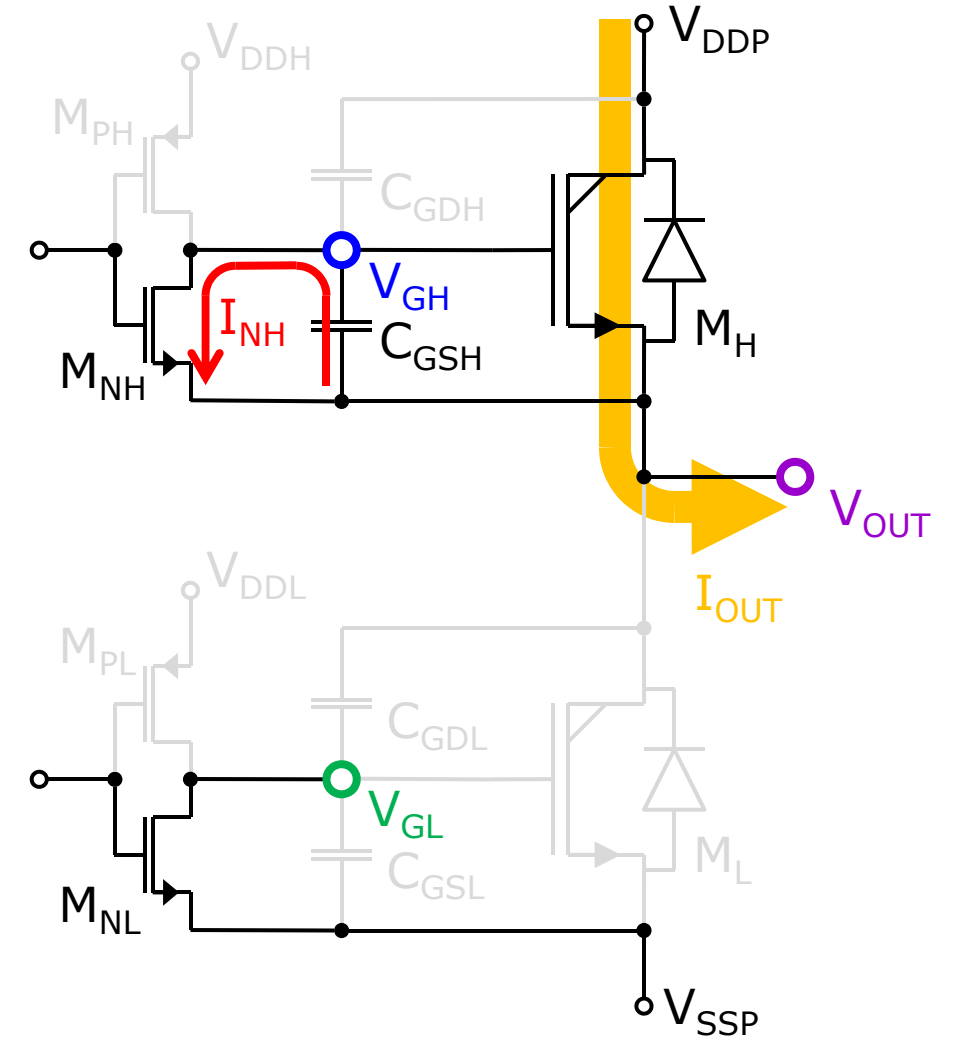
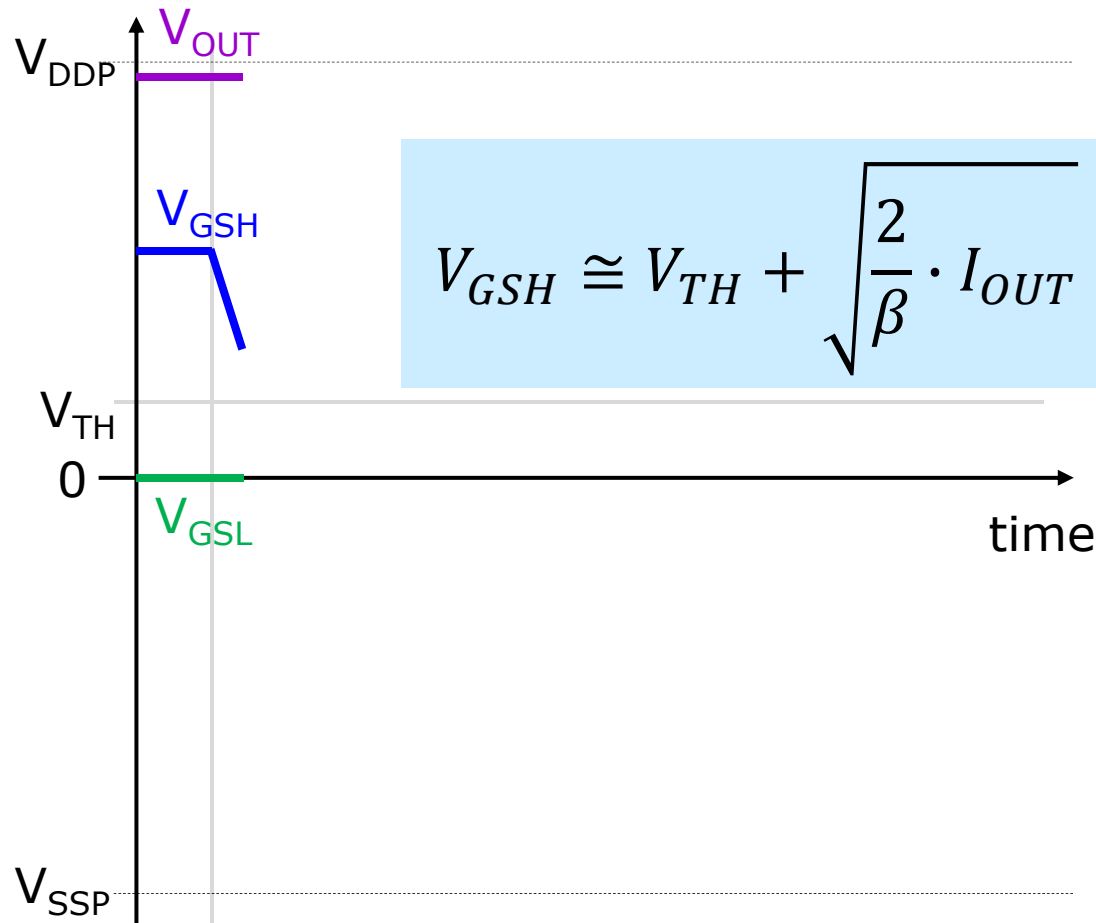


Soft Switching

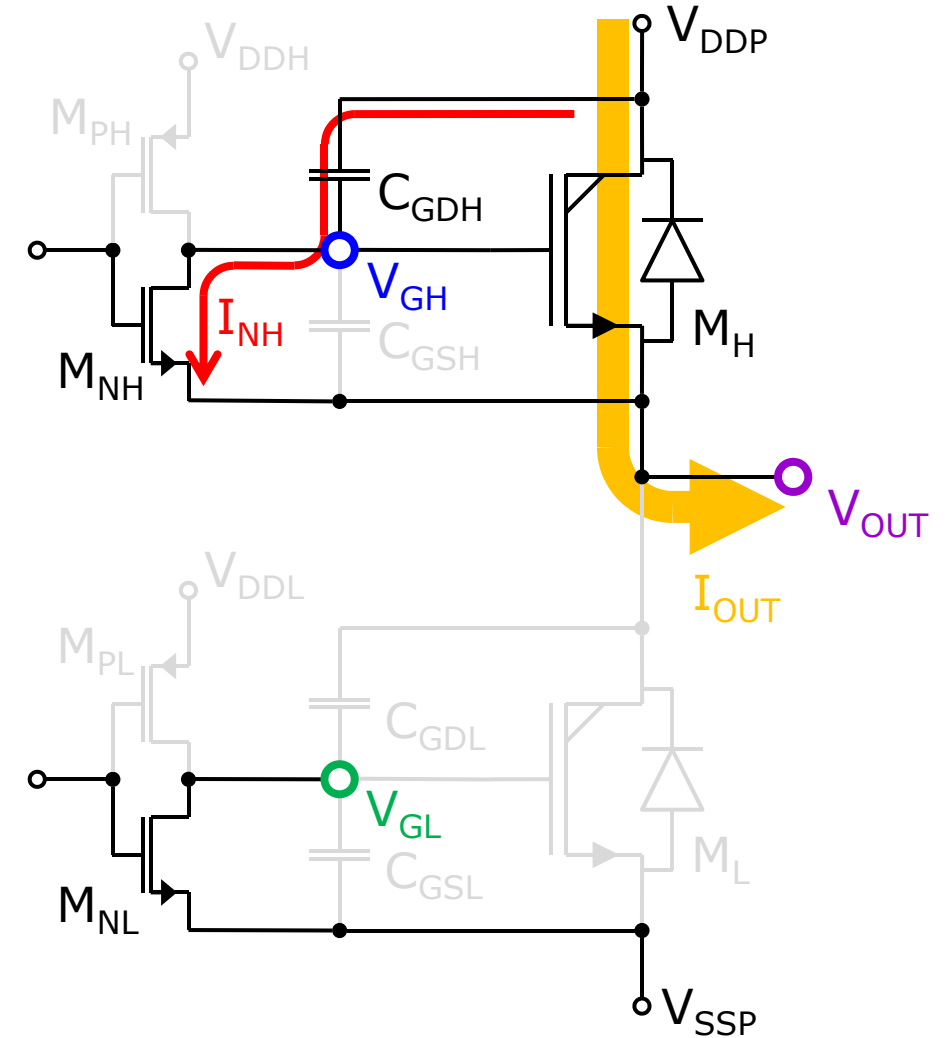
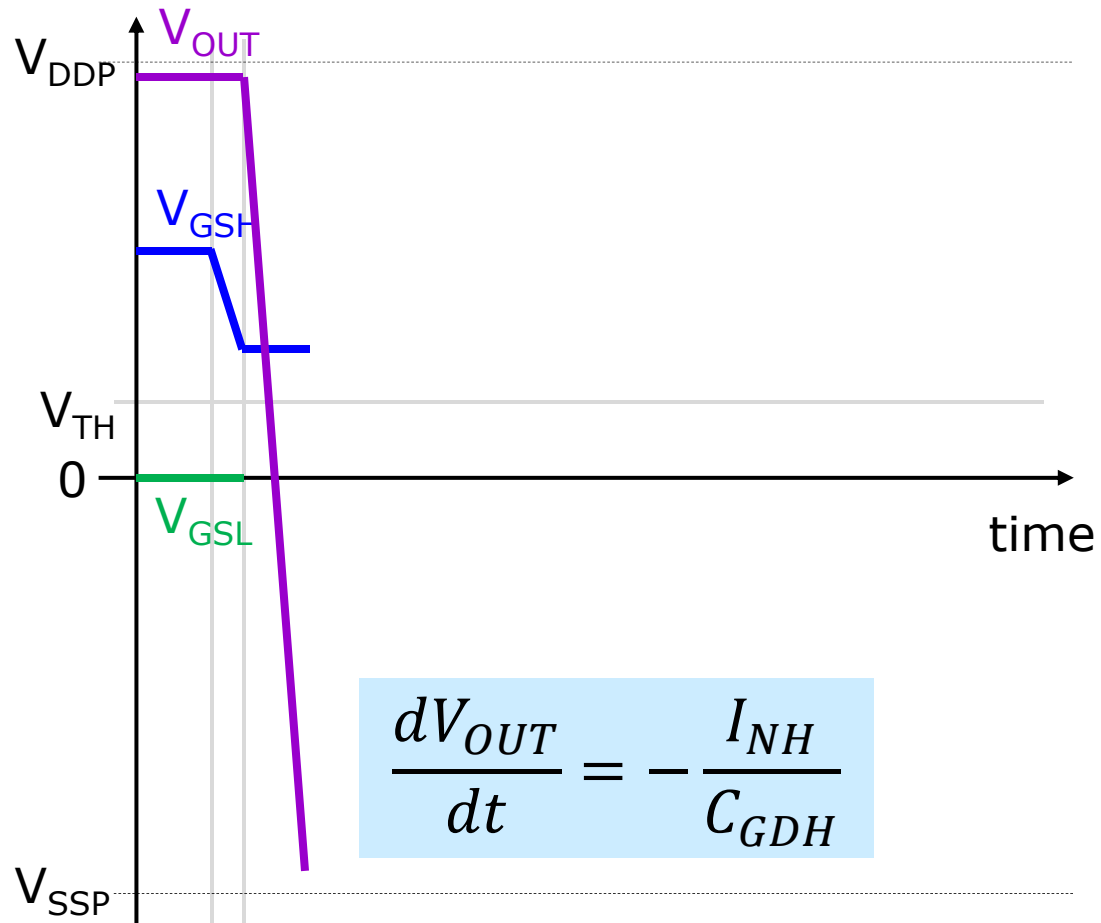
Soft Switching



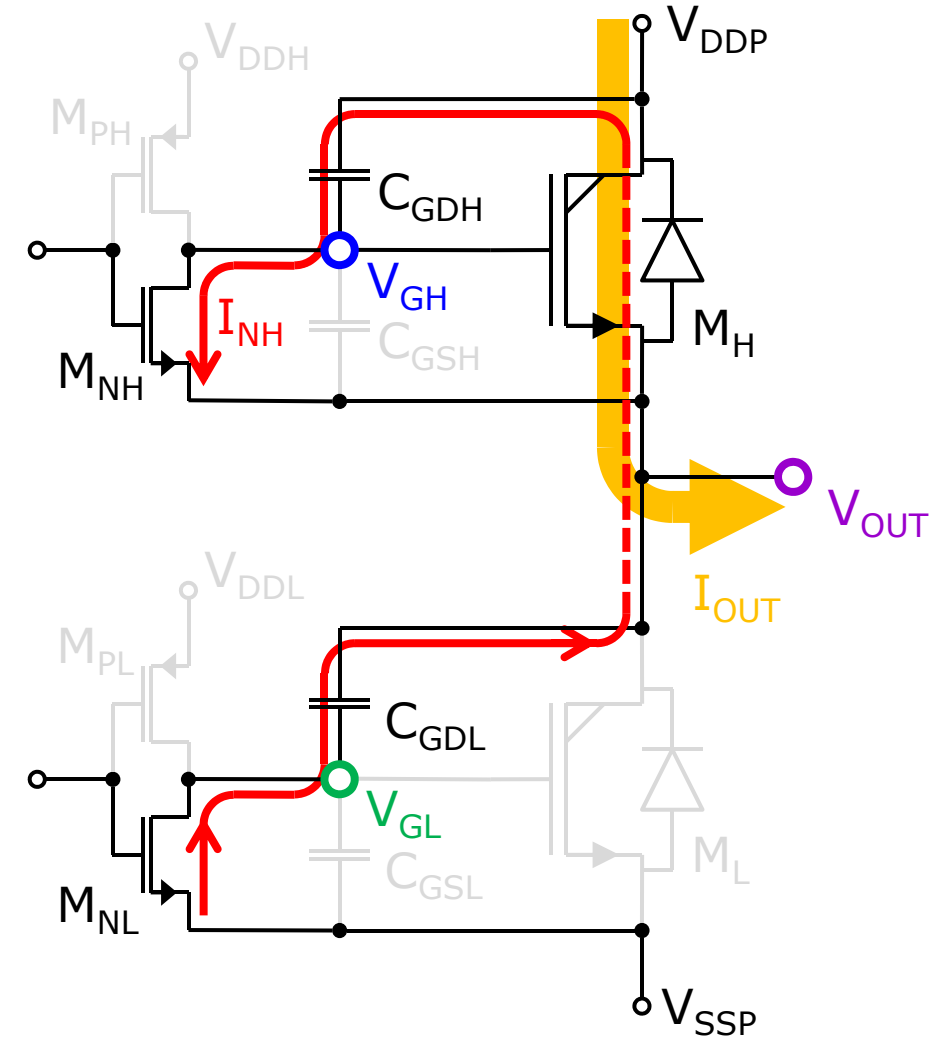
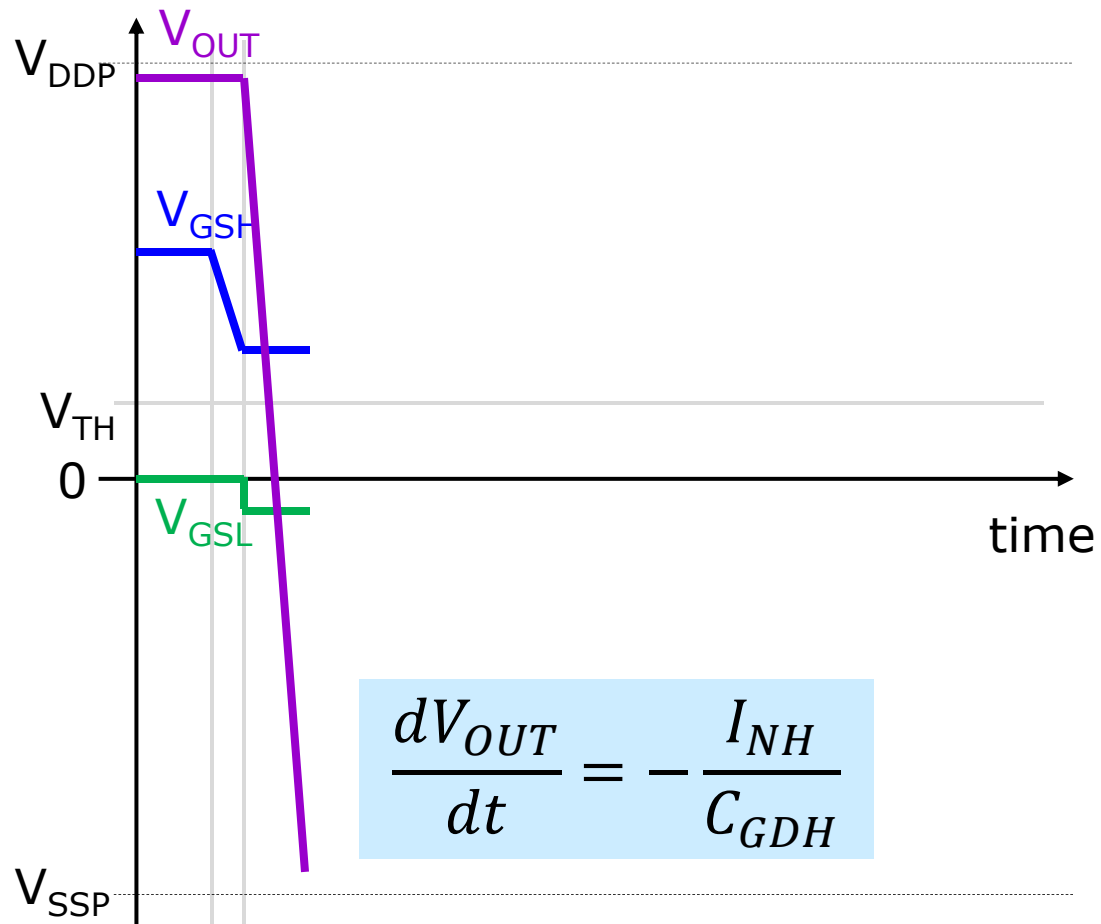
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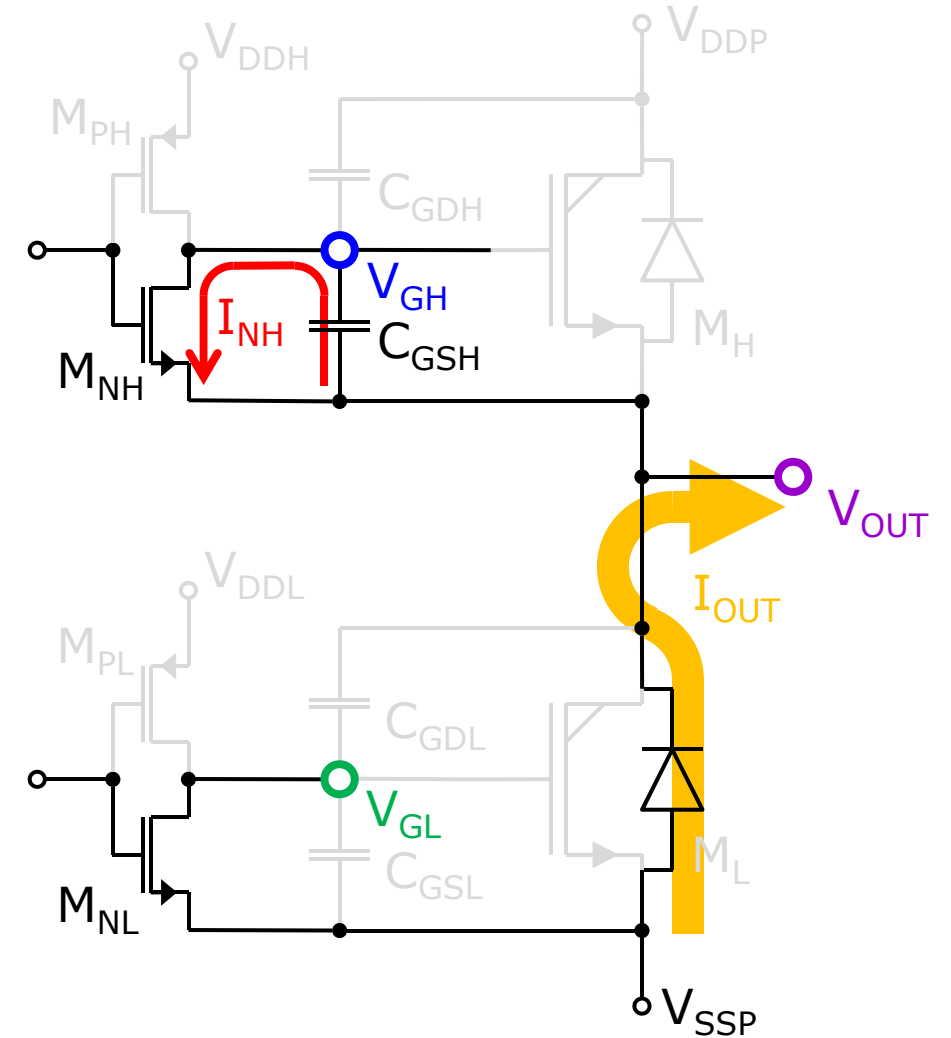
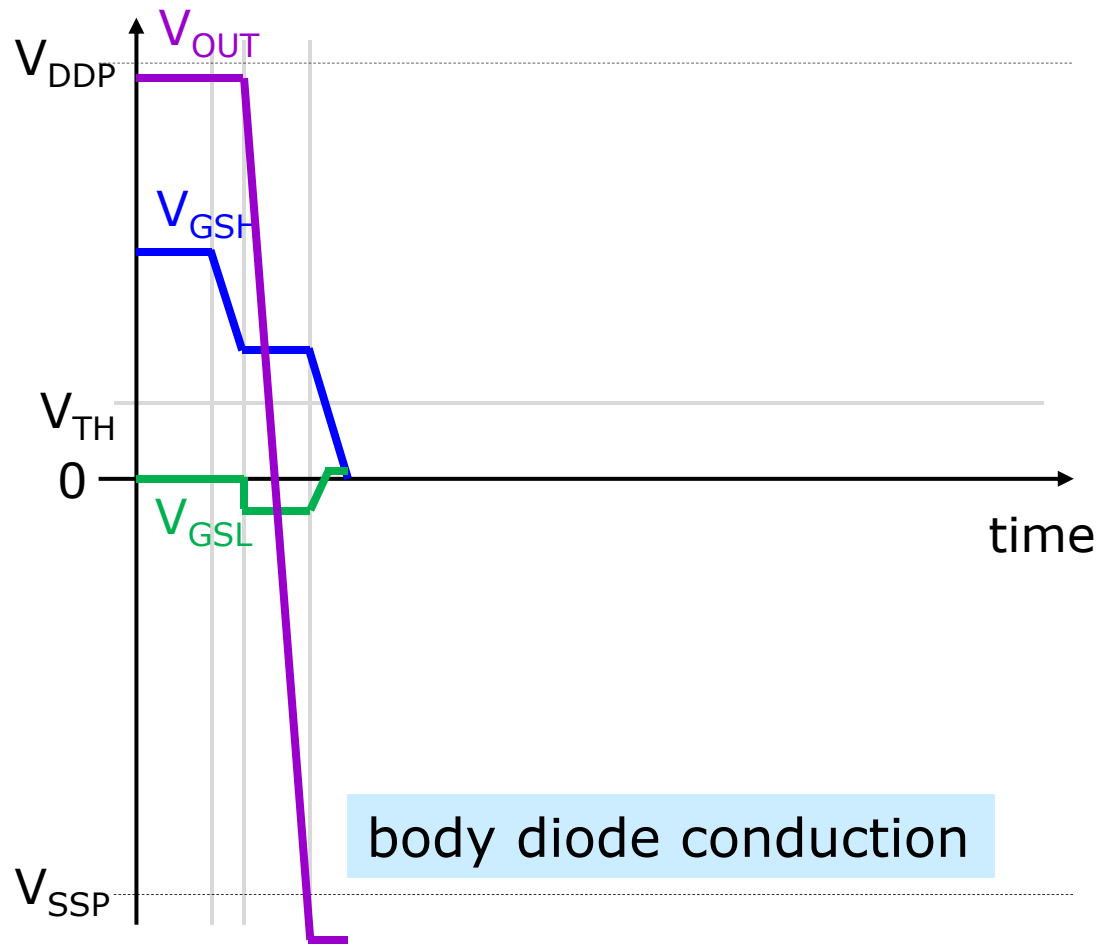
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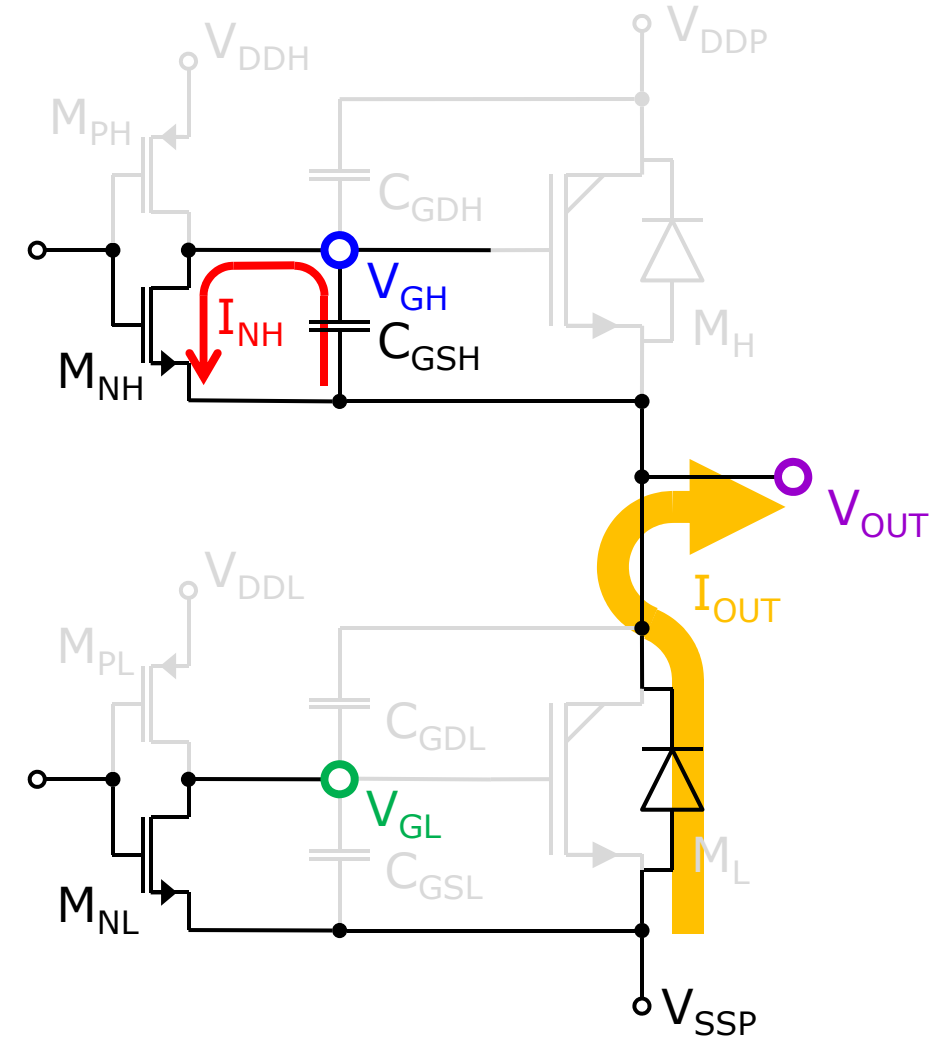
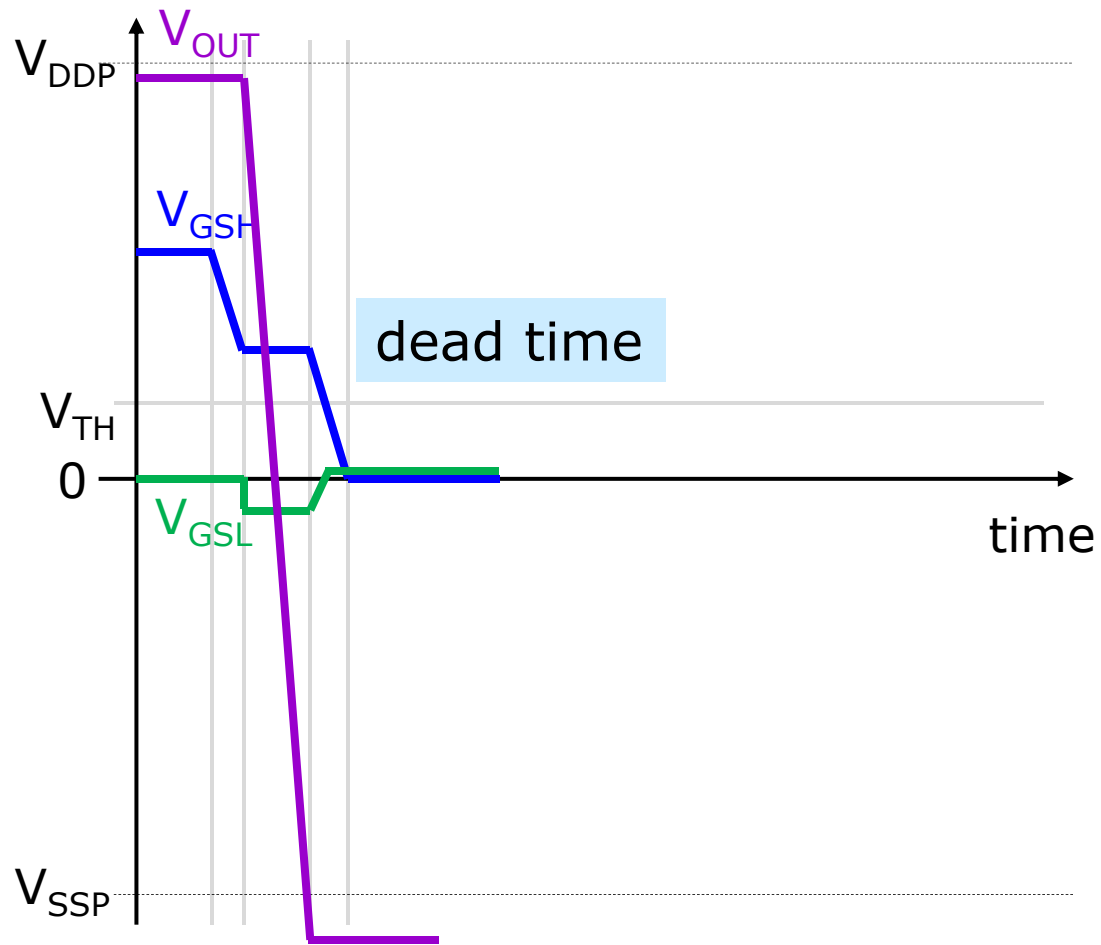
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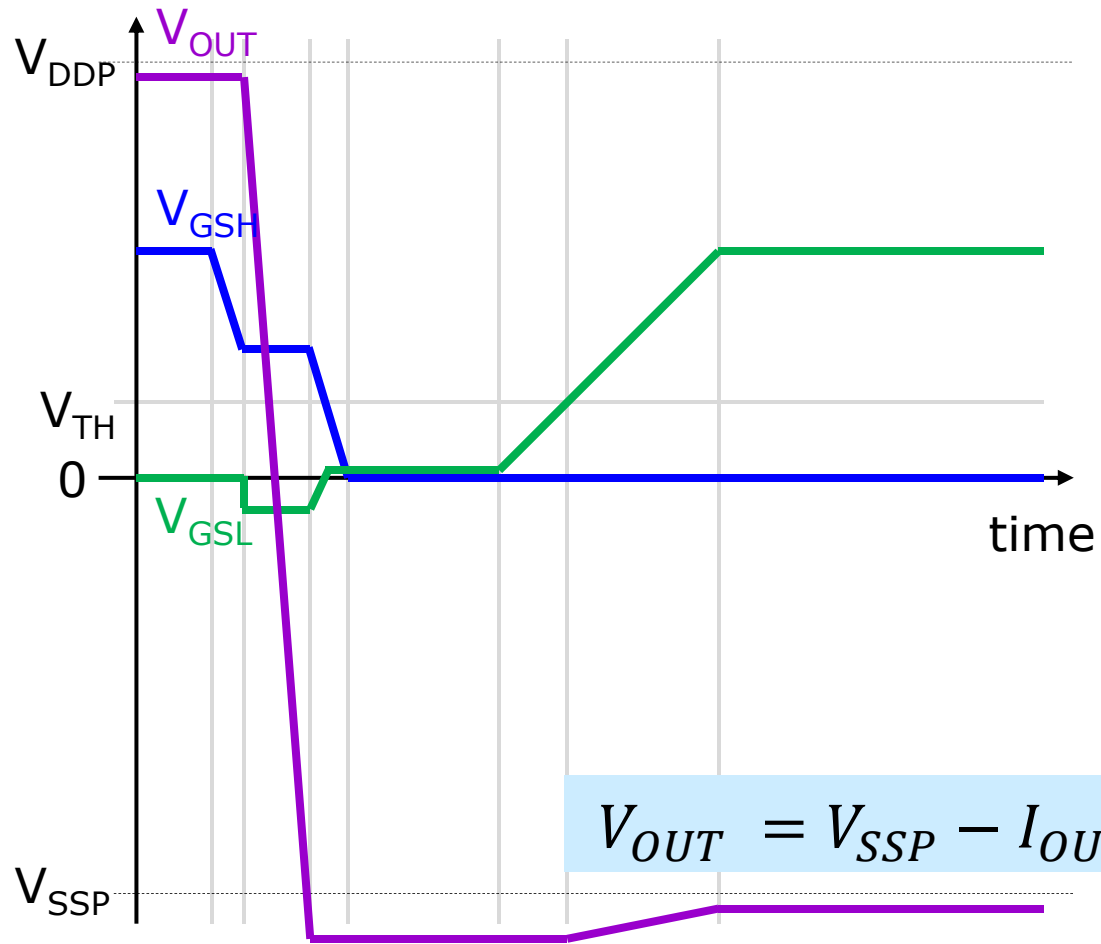
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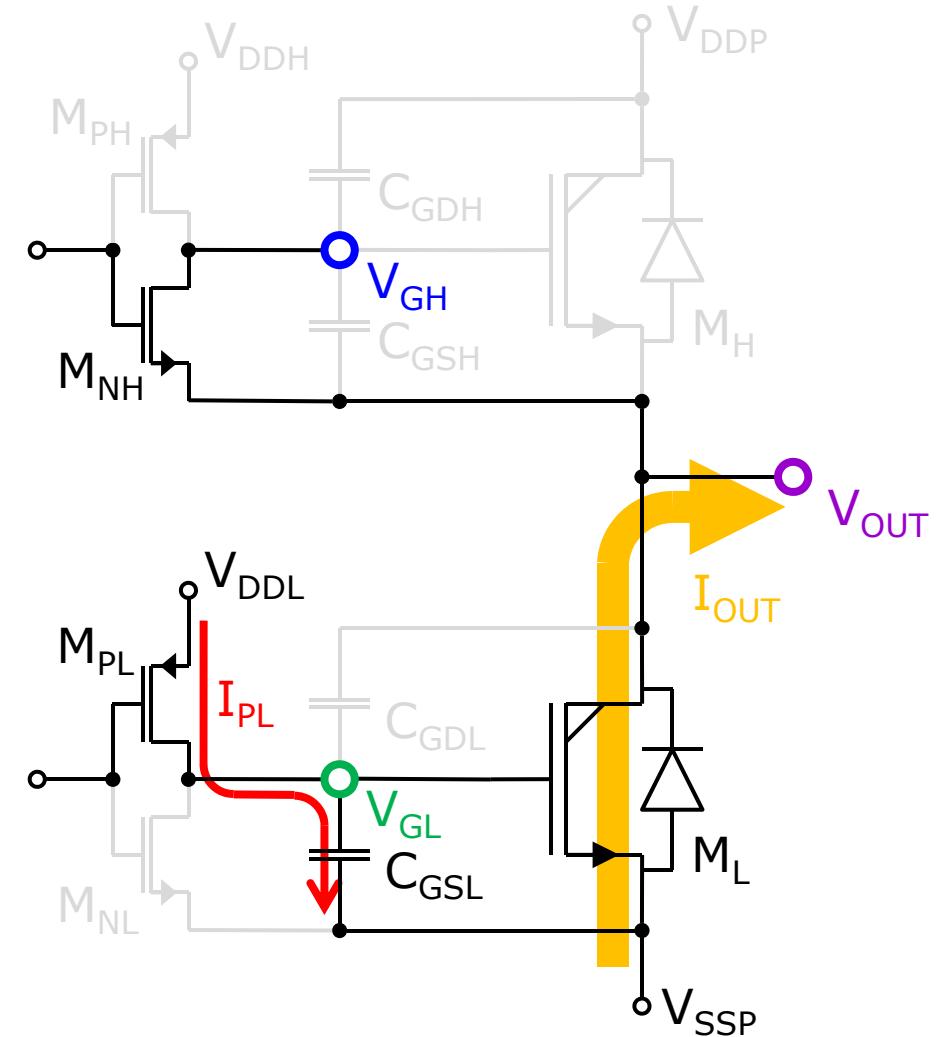
Soft Switching



Soft Switching

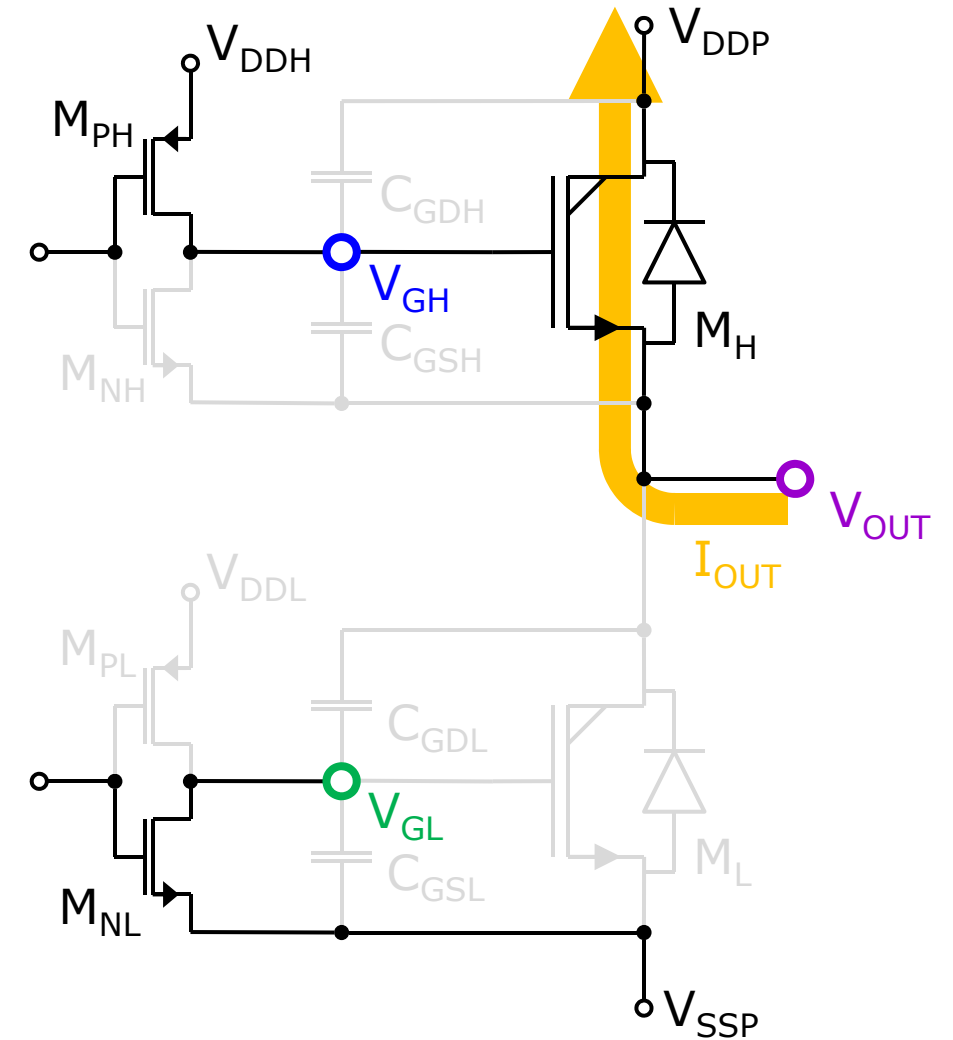
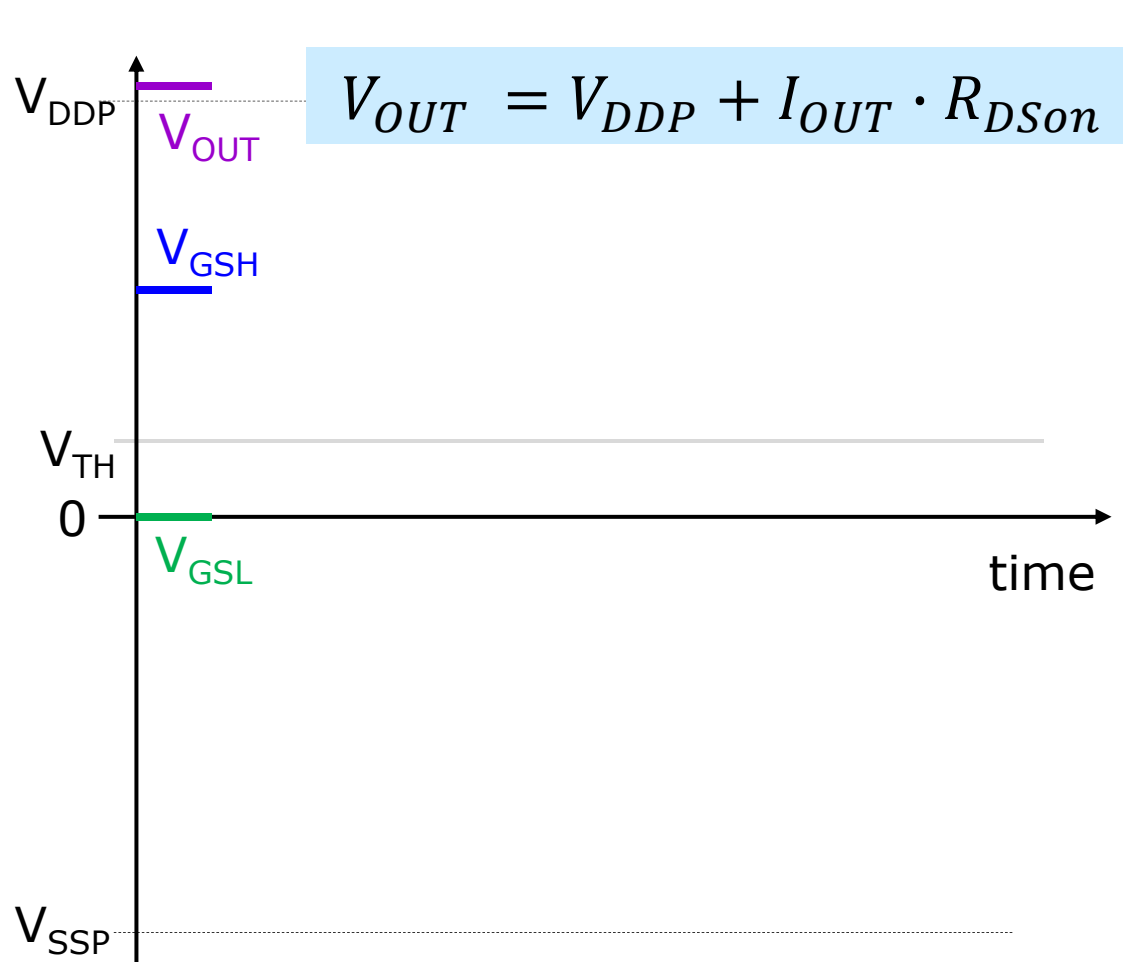


$$V_{OUT} = V_{SSP} - I_{OUT} \cdot R_{Dson}$$

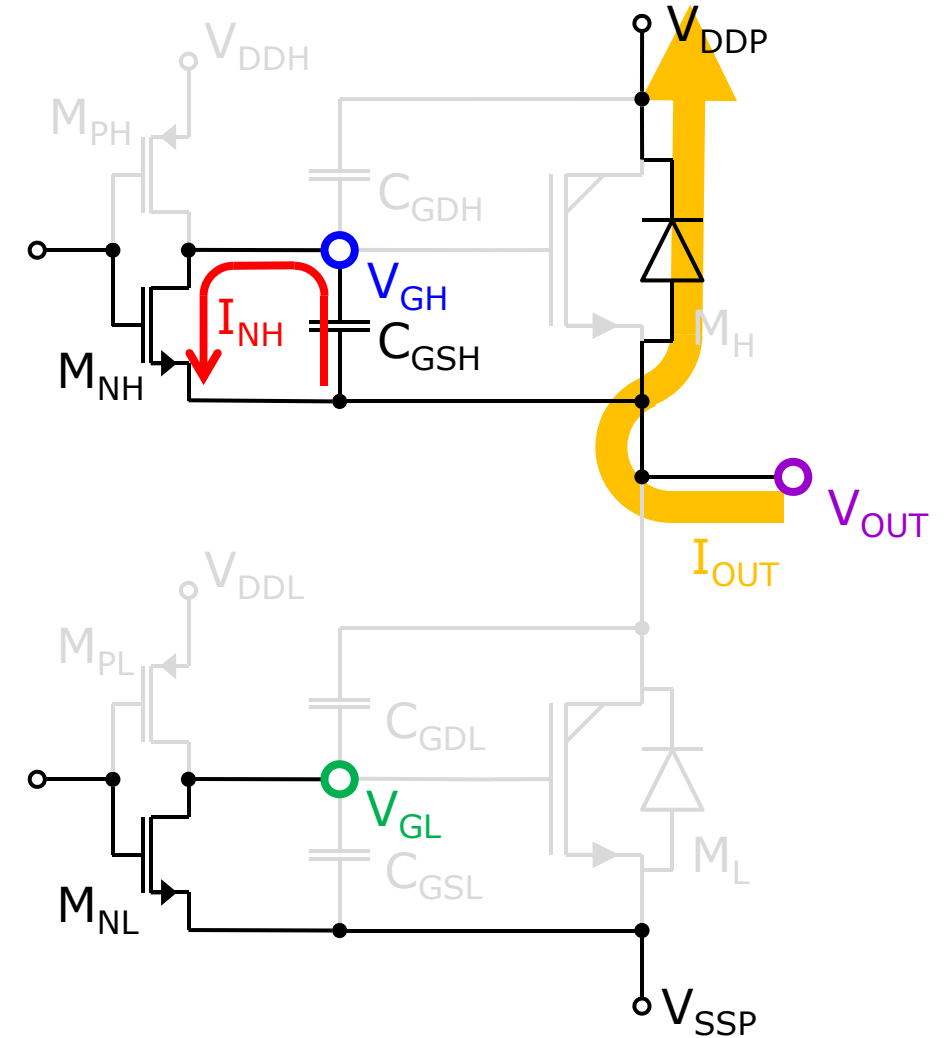
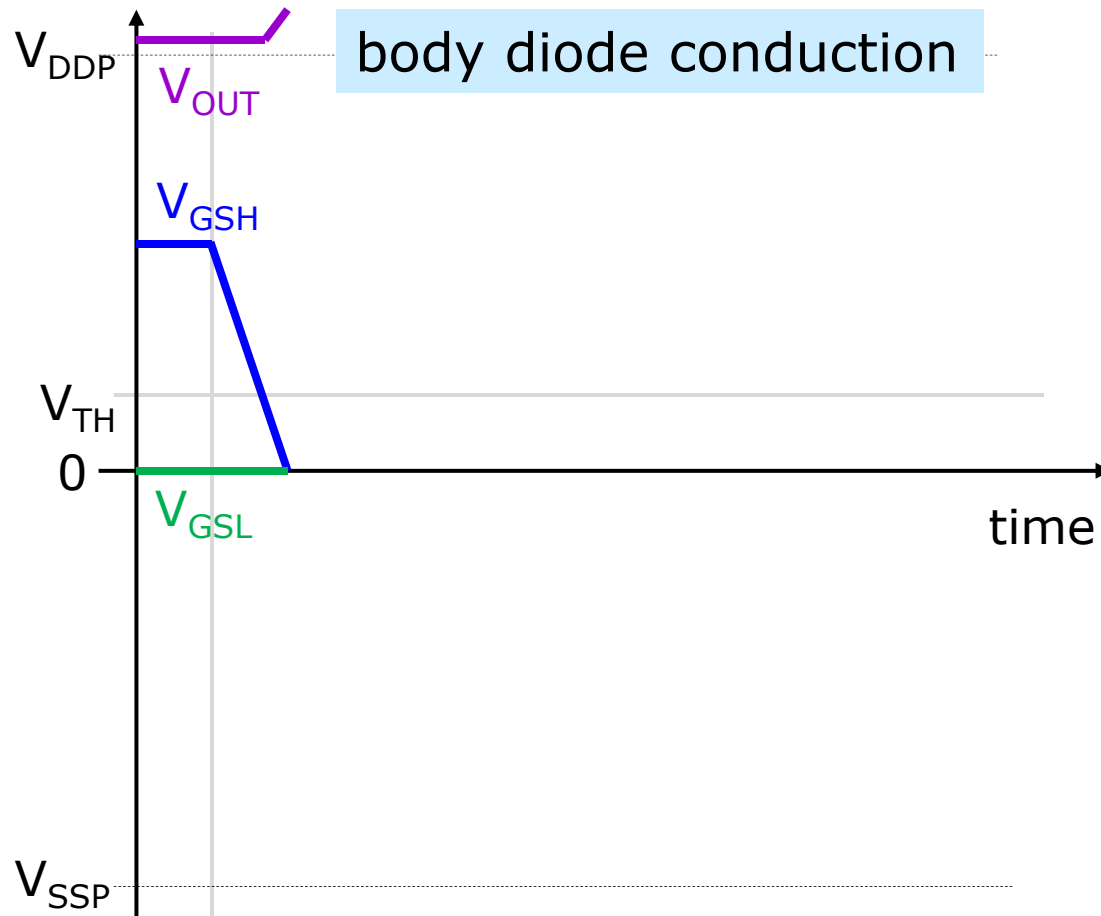


Hard Switching

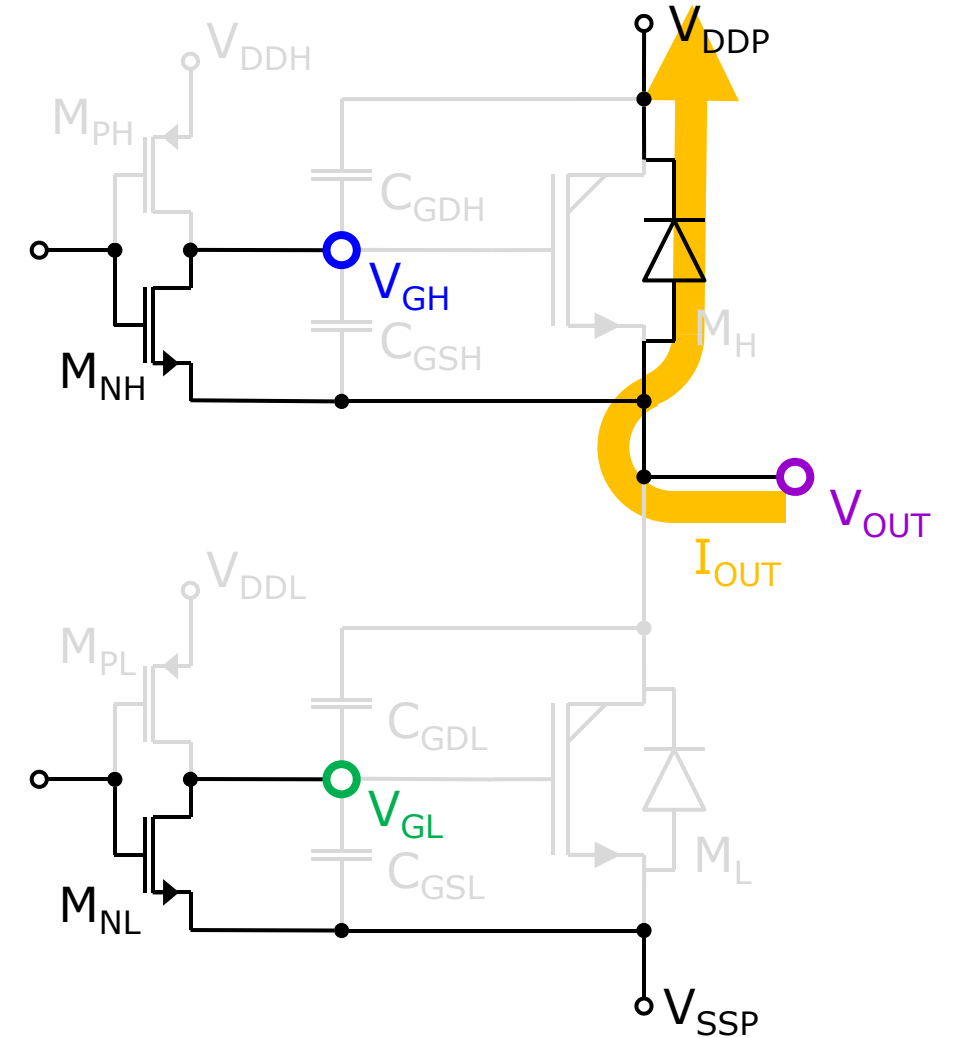
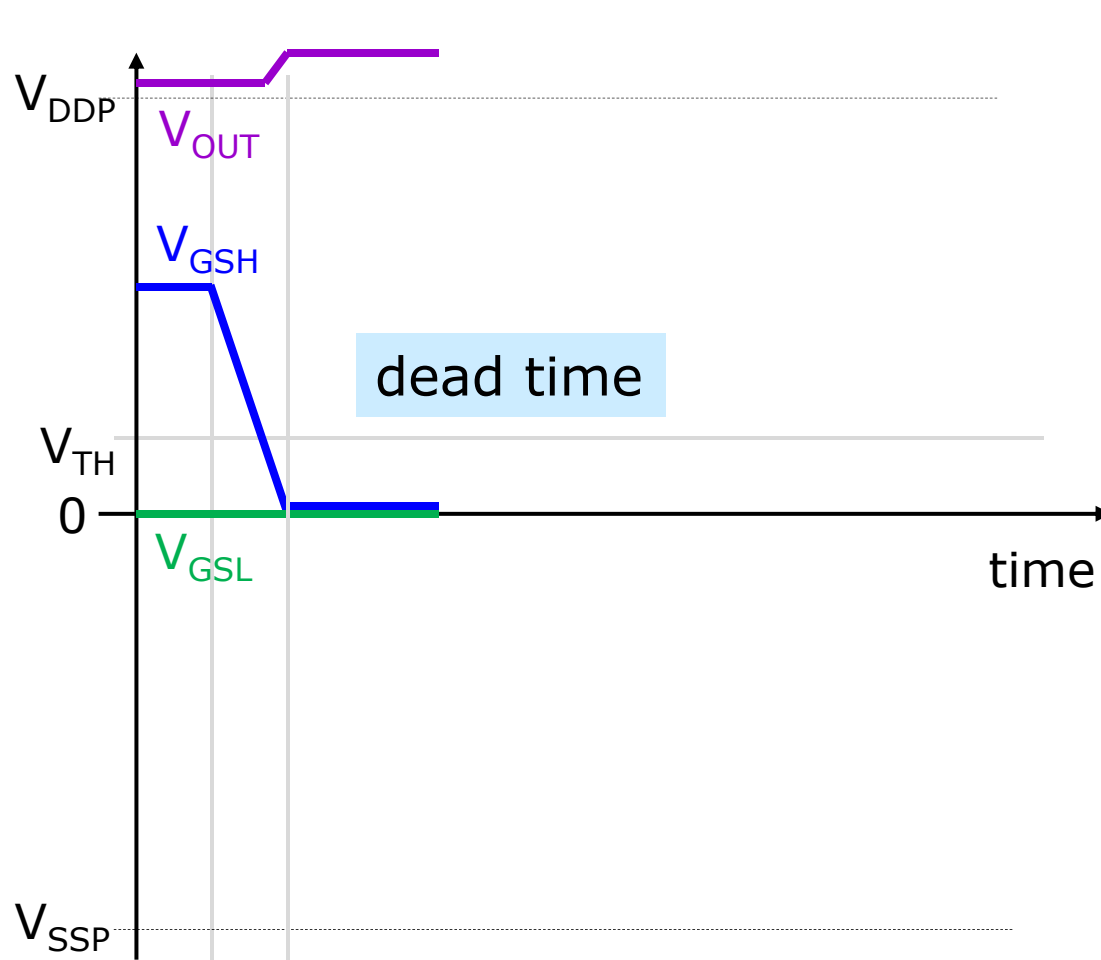
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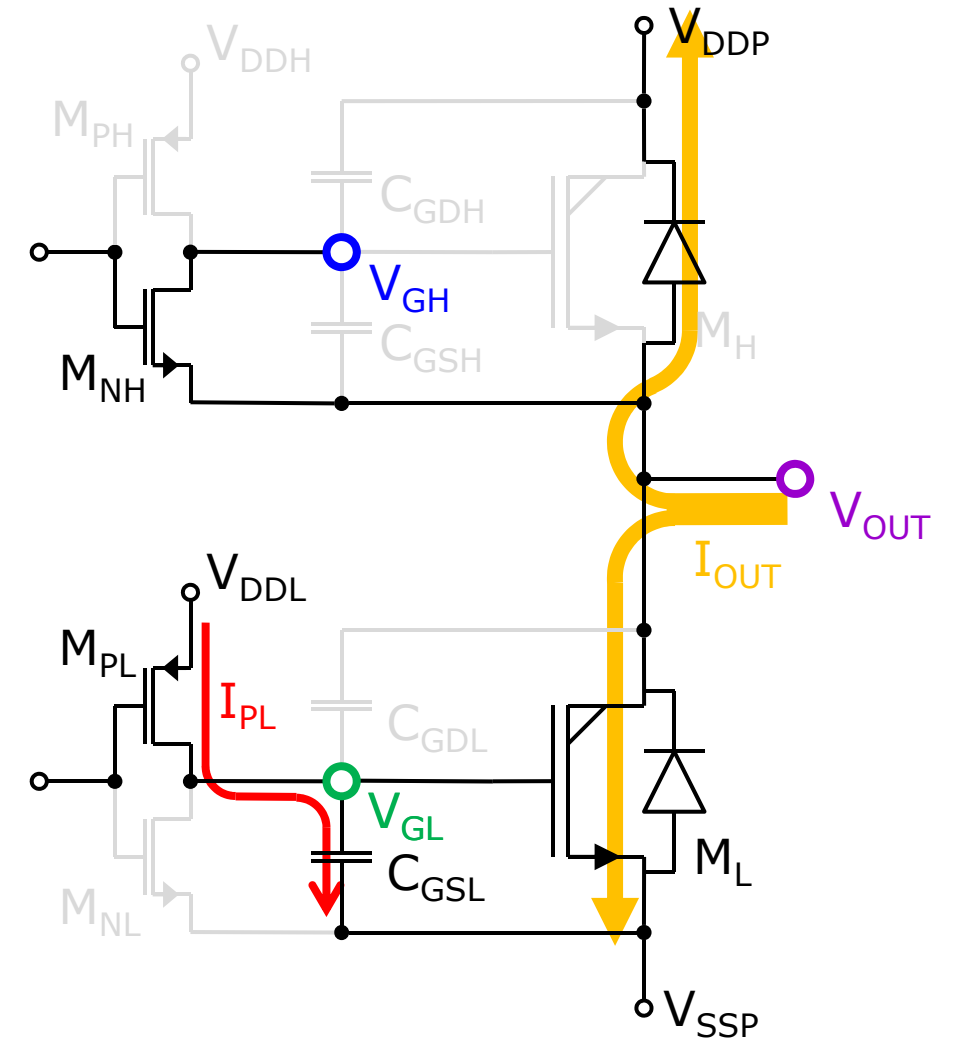
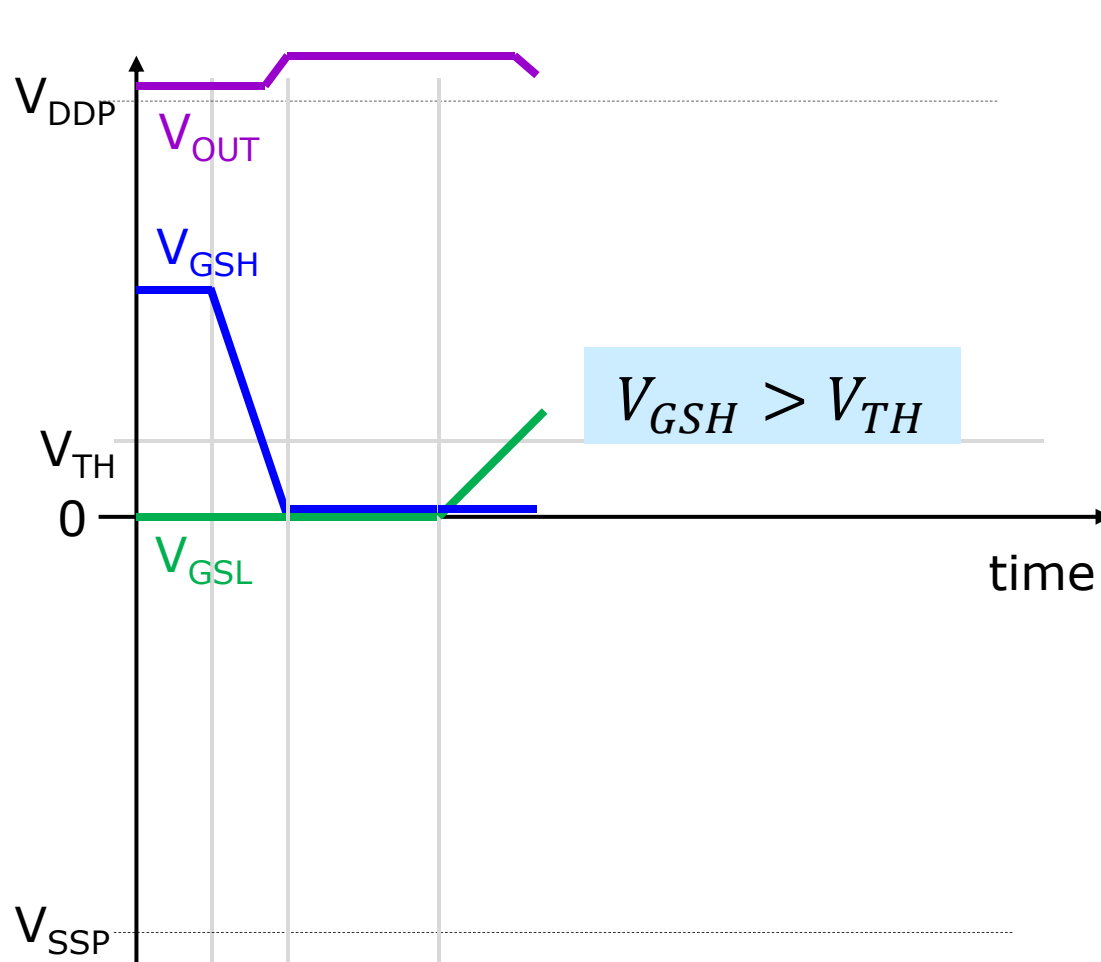
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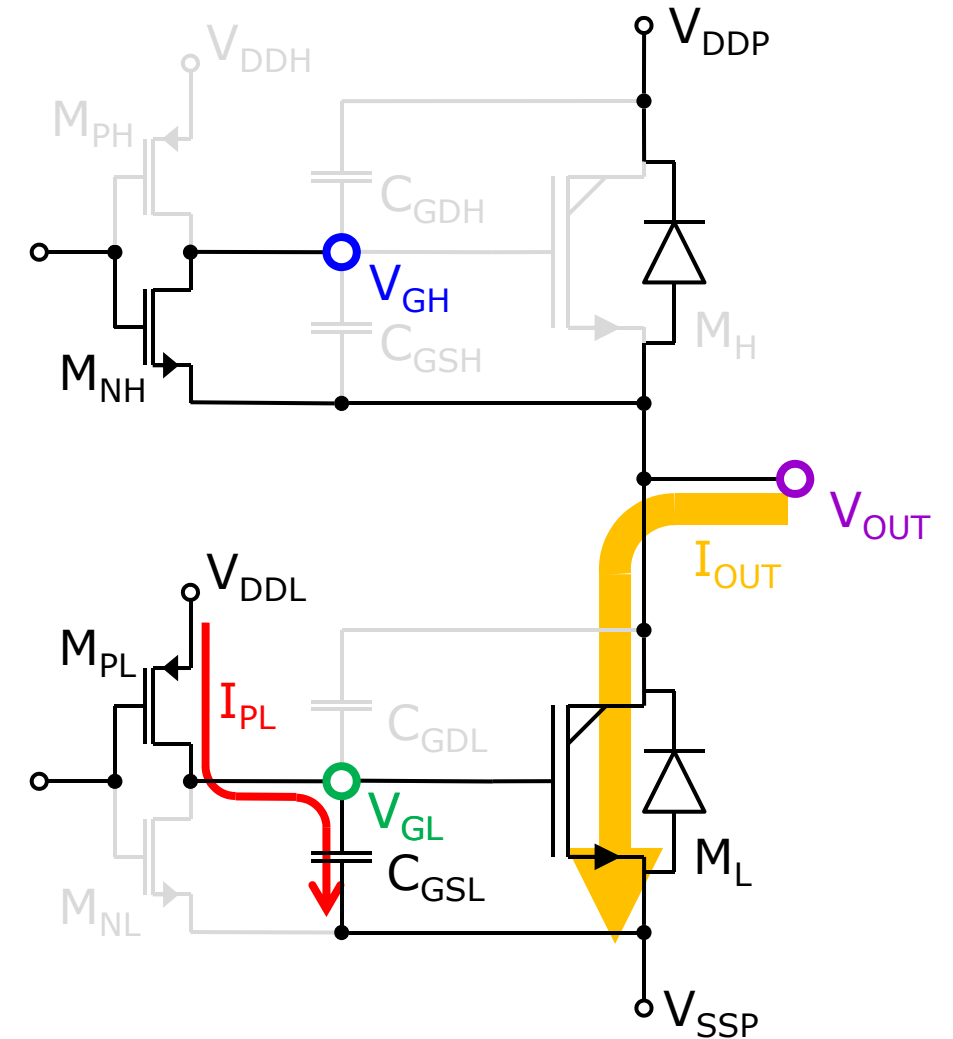
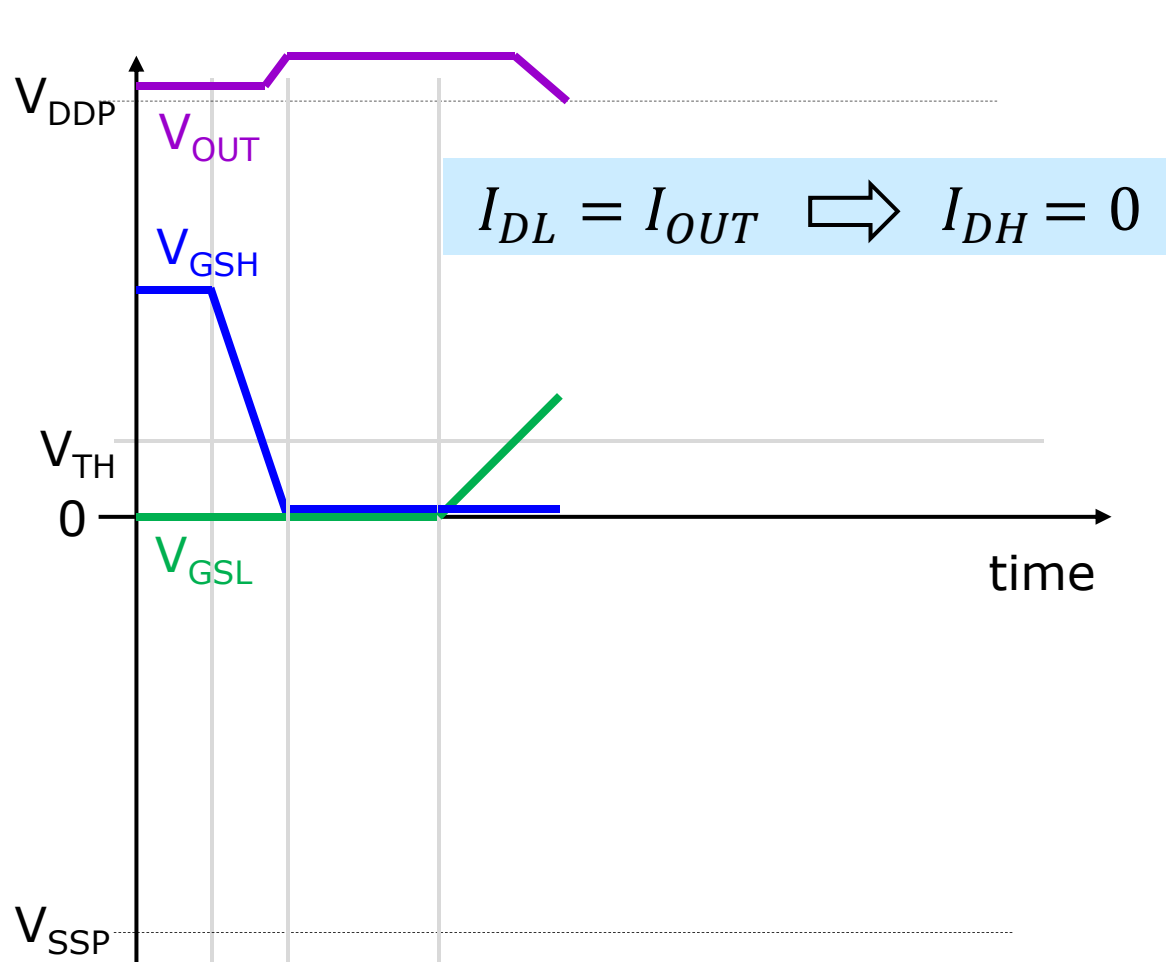
Hard Switching



Hard Switching

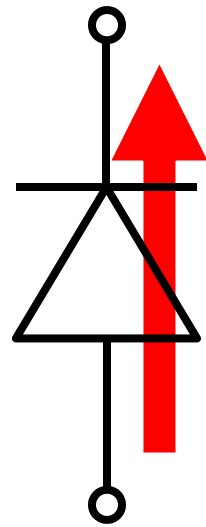


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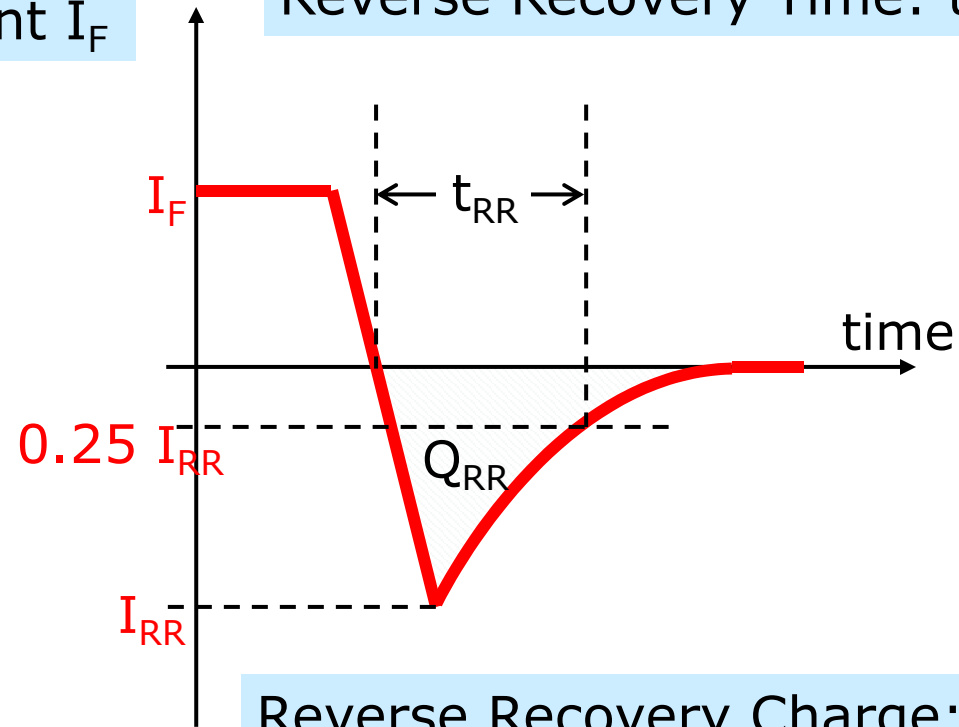


Reverse Recovery

Forward Current I_F

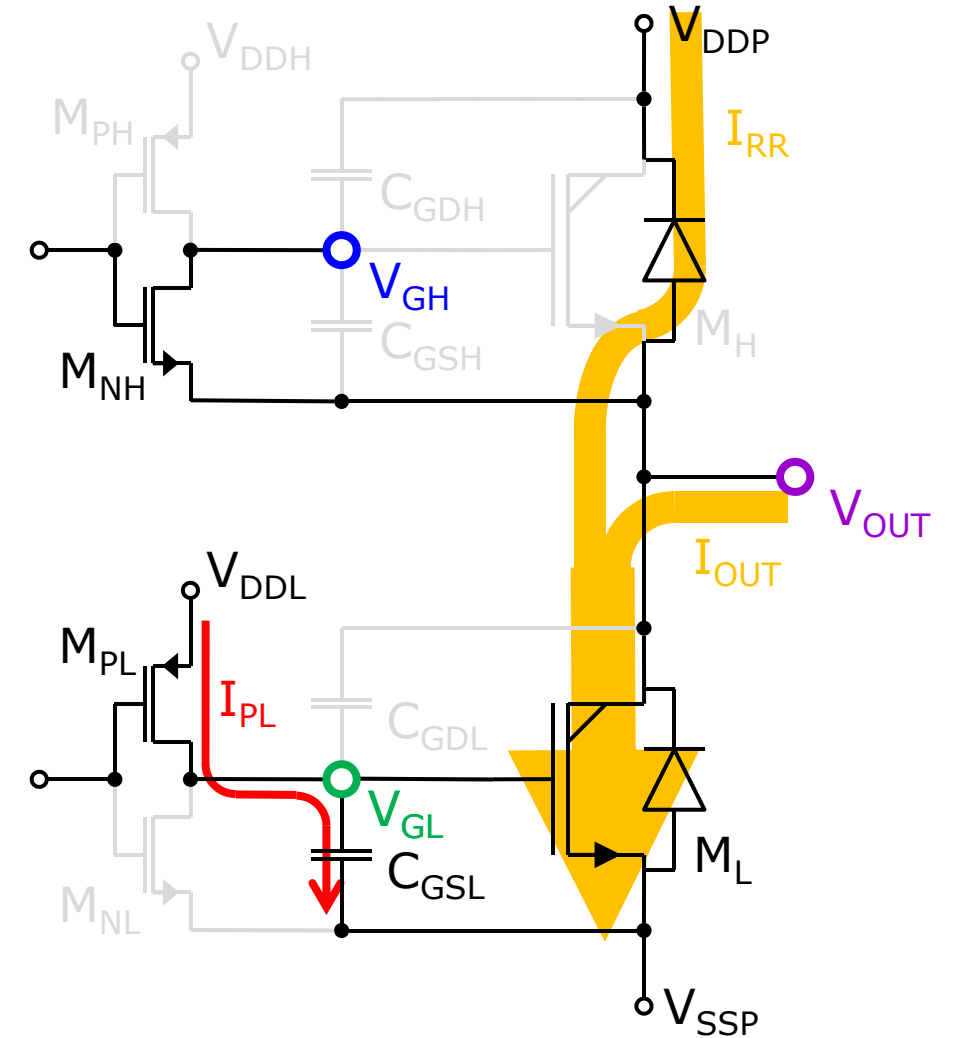
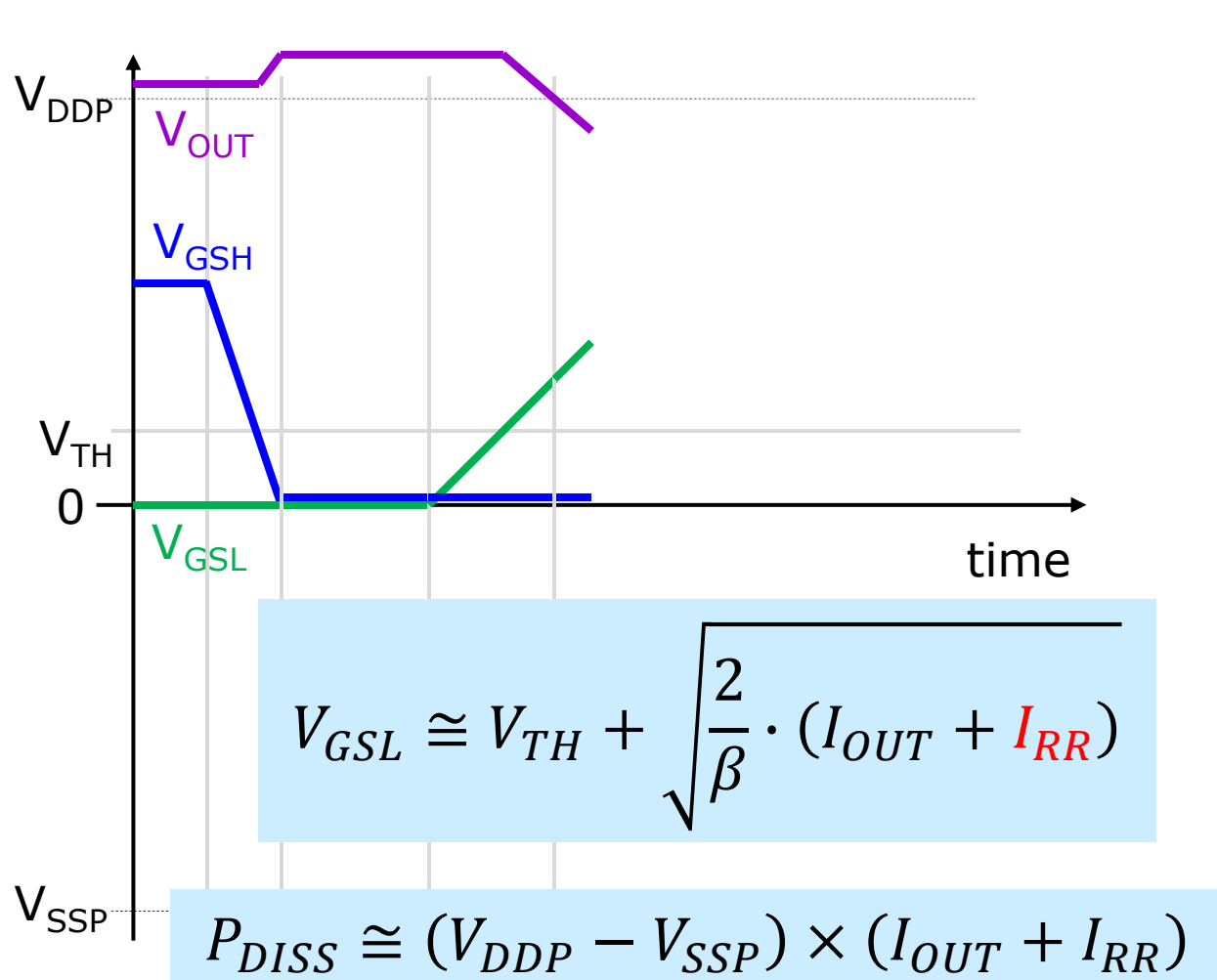


Reverse Recovery Time: t_{RR}

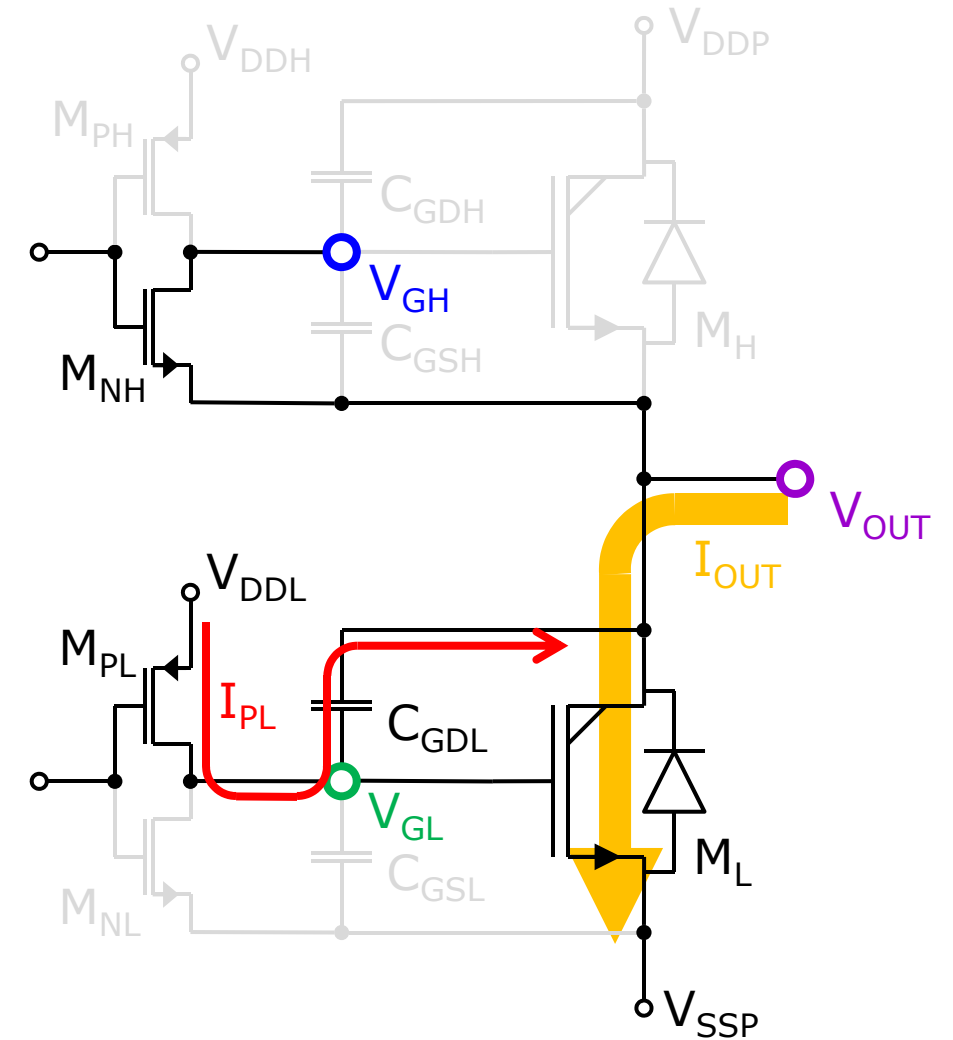
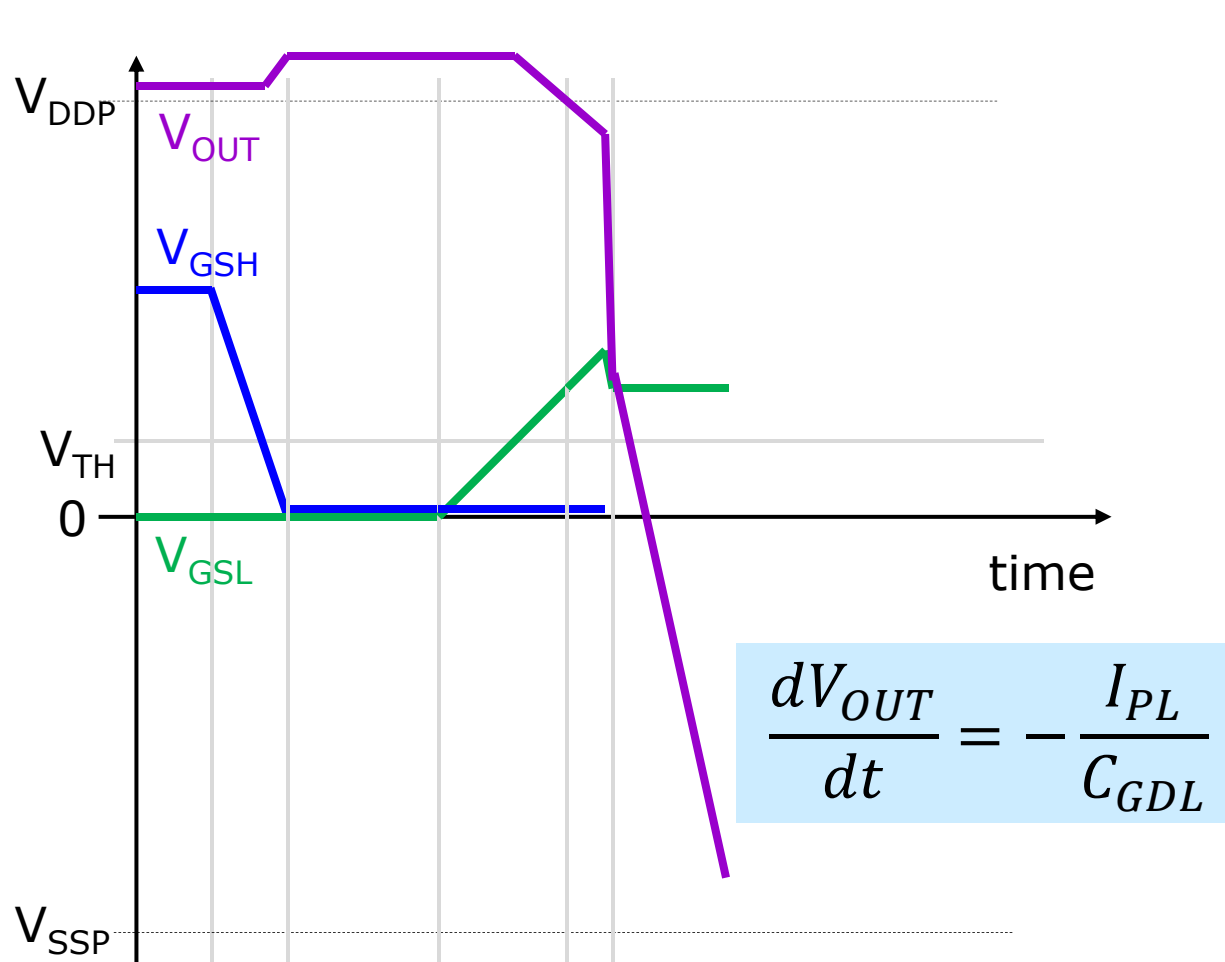


Reverse Recovery Charge: Q_{RR}

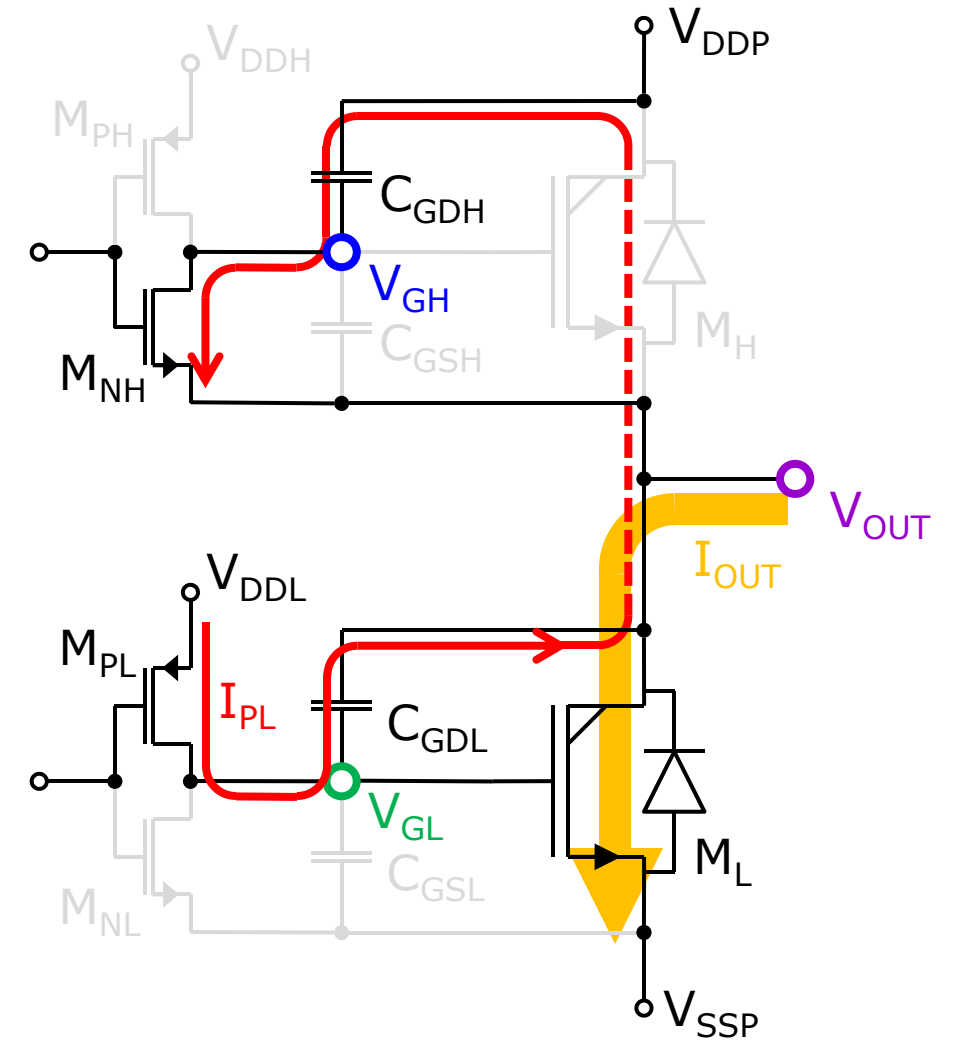
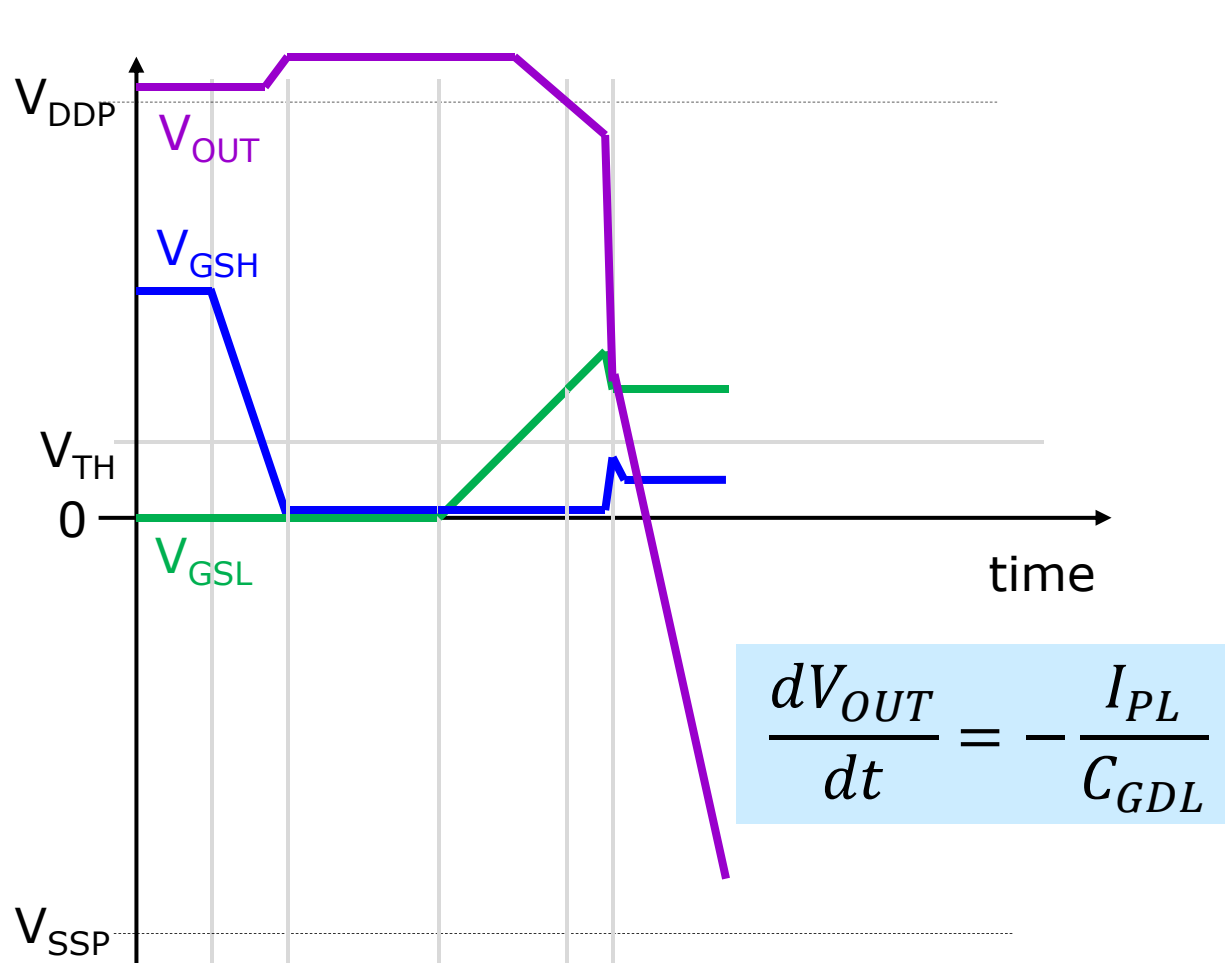
Hard Switching



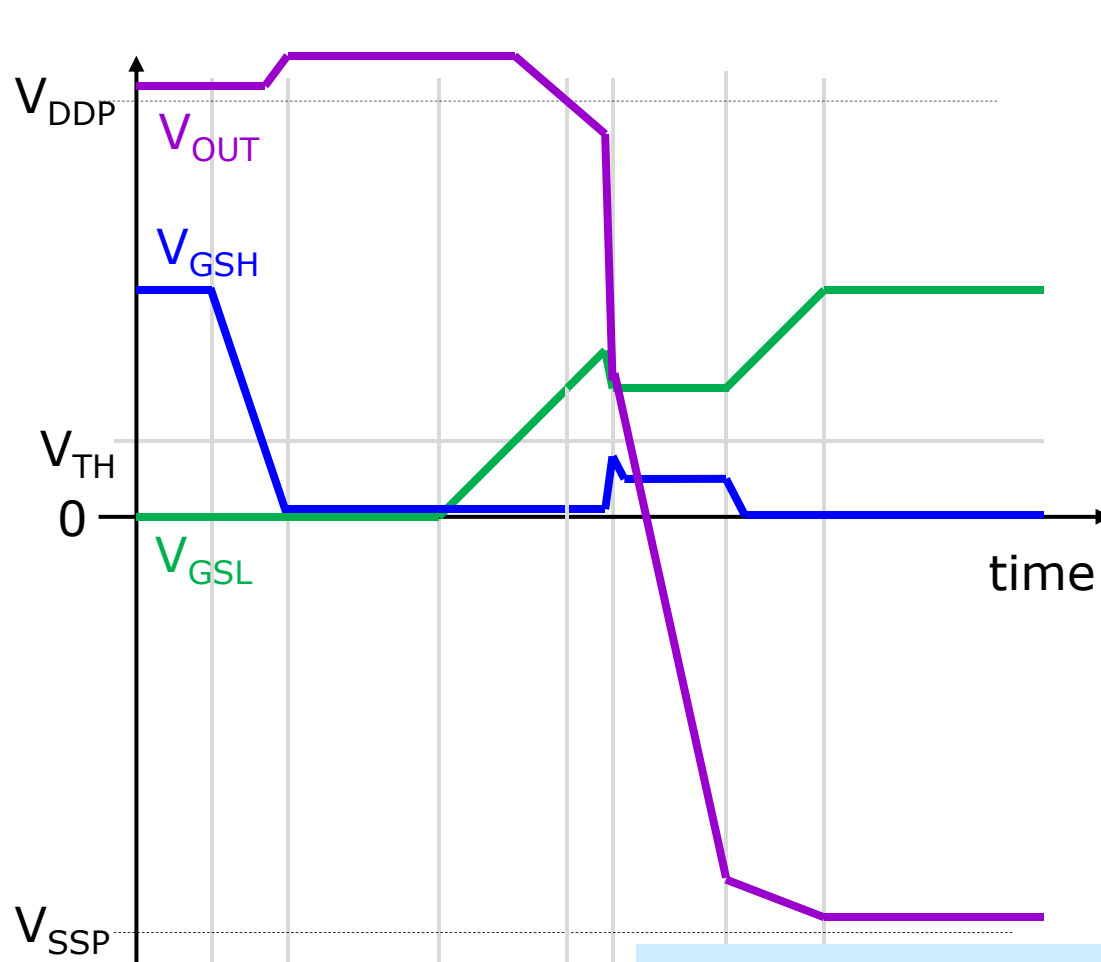
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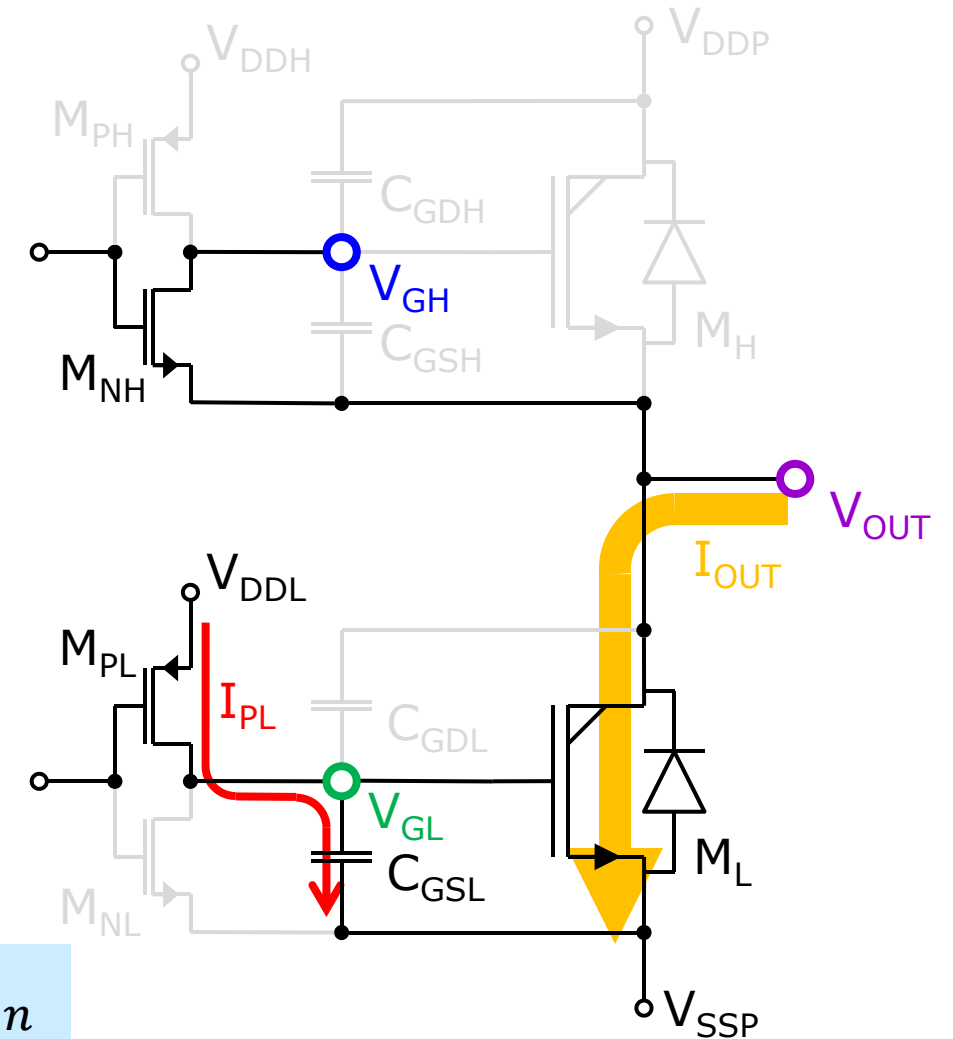
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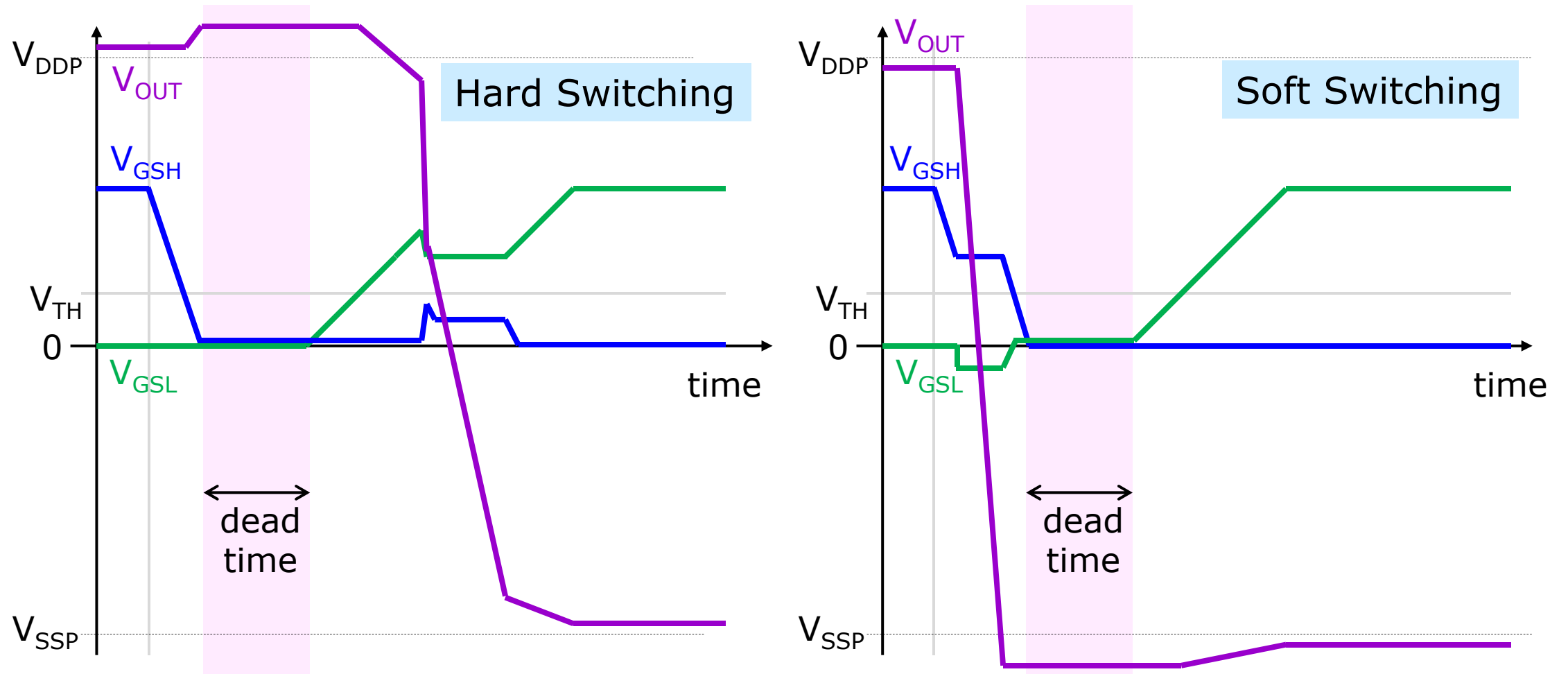
Hard Switching



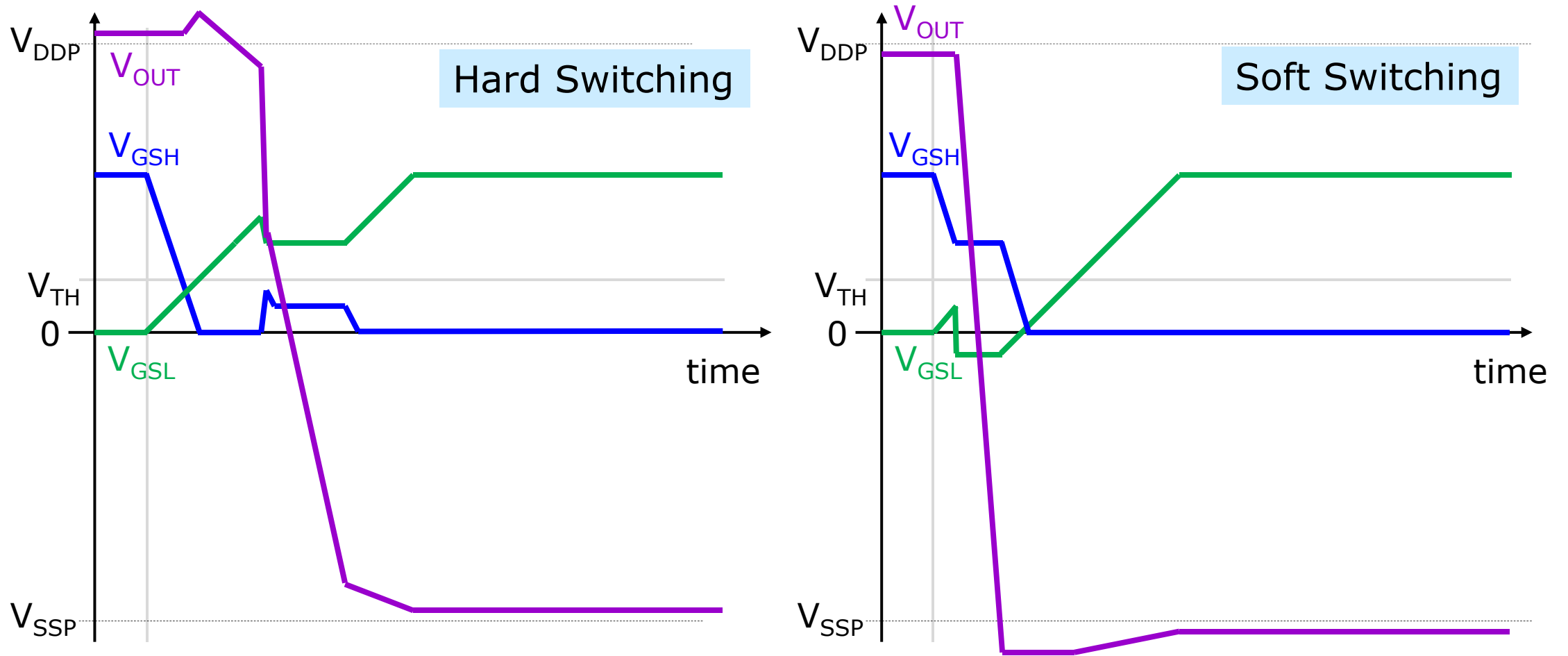
$$V_{OUT} = V_{SSP} + I_{OUT} \cdot R_{DSon}$$



Dead Time

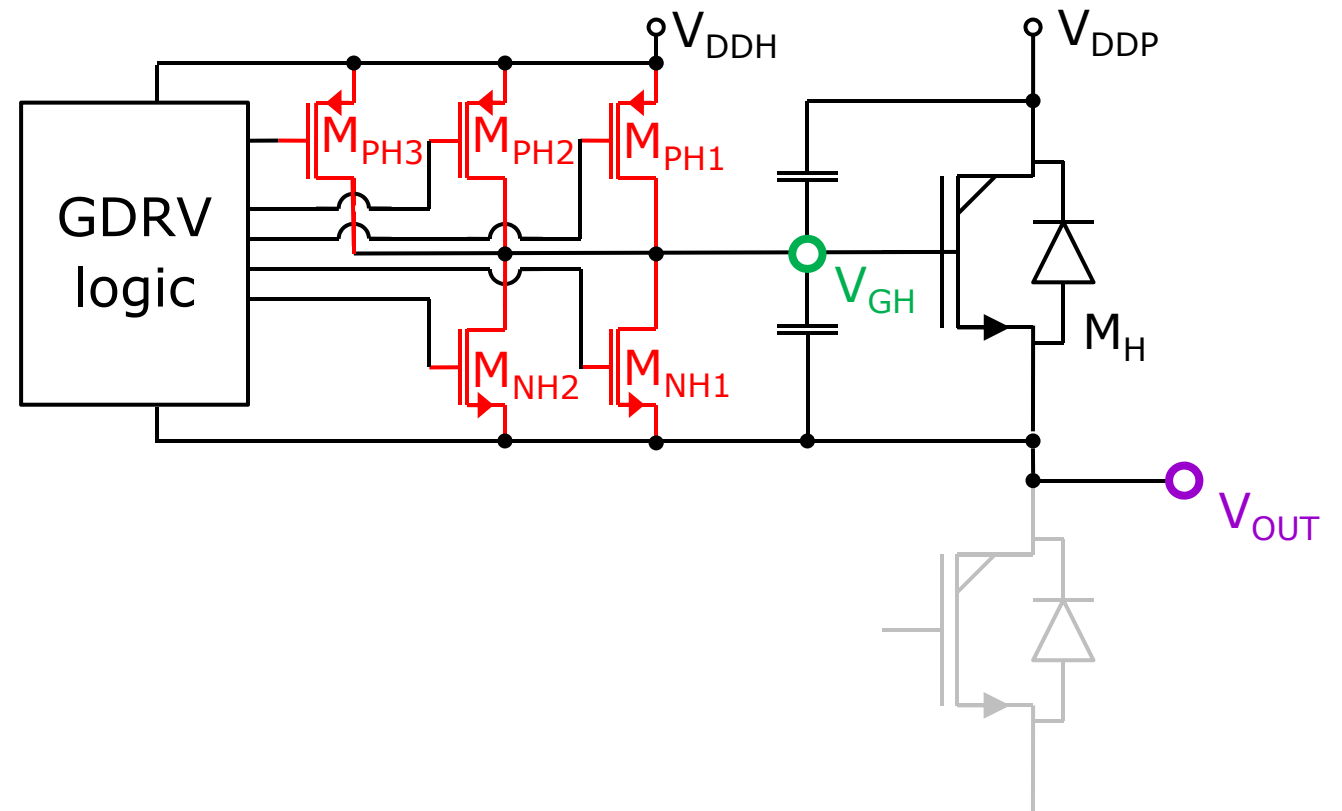


Dead Time^[22]



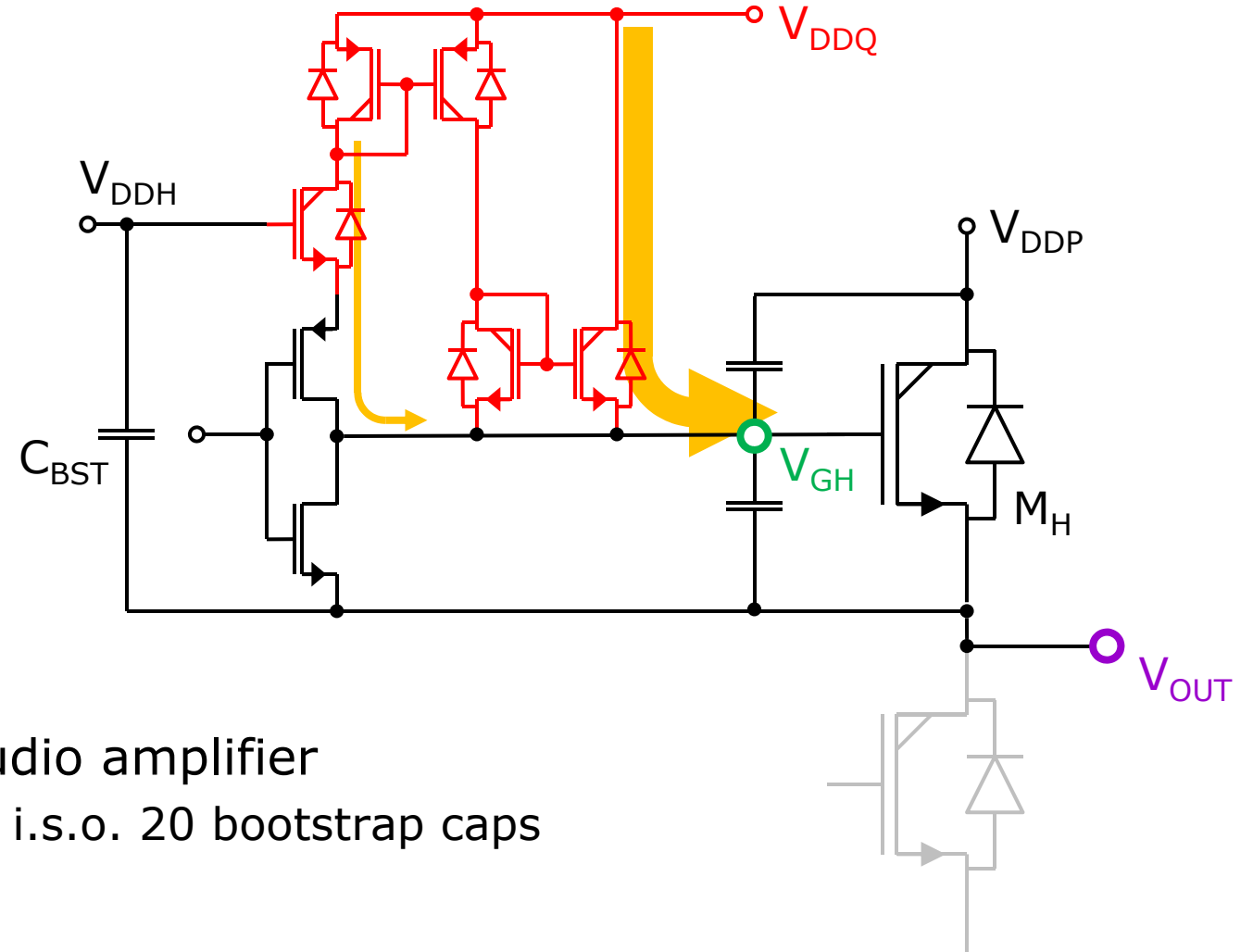
Adaptive Gatedriver^[10]

- adjustable driver sizes
 - optimize size per scenario
- negative dead-time
 - eliminate reverse recovery



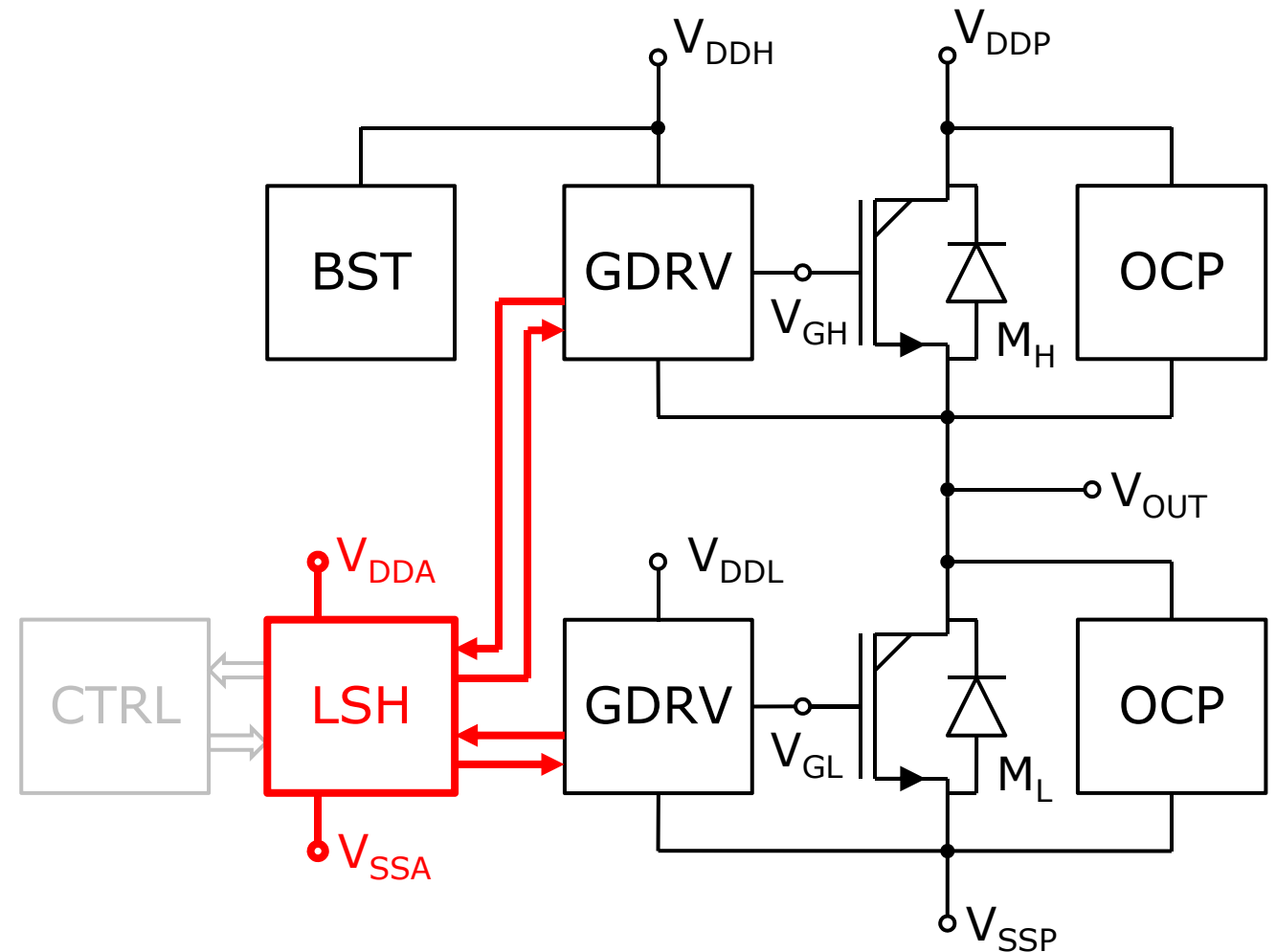
Chargepump Gatedriver^[19]

- constant chargepump voltage
 - $V_{DDQ} = V_{DDP} + 5V$
- integrated bootstrap capacitor
 - $V_{DDH} = V_{OUT} + 3.3V + V_{THn}$
- charge current multiplier
- 5-channel multiphase class-D audio amplifier
 - 2 external caps for chargepump i.s.o. 20 bootstrap caps



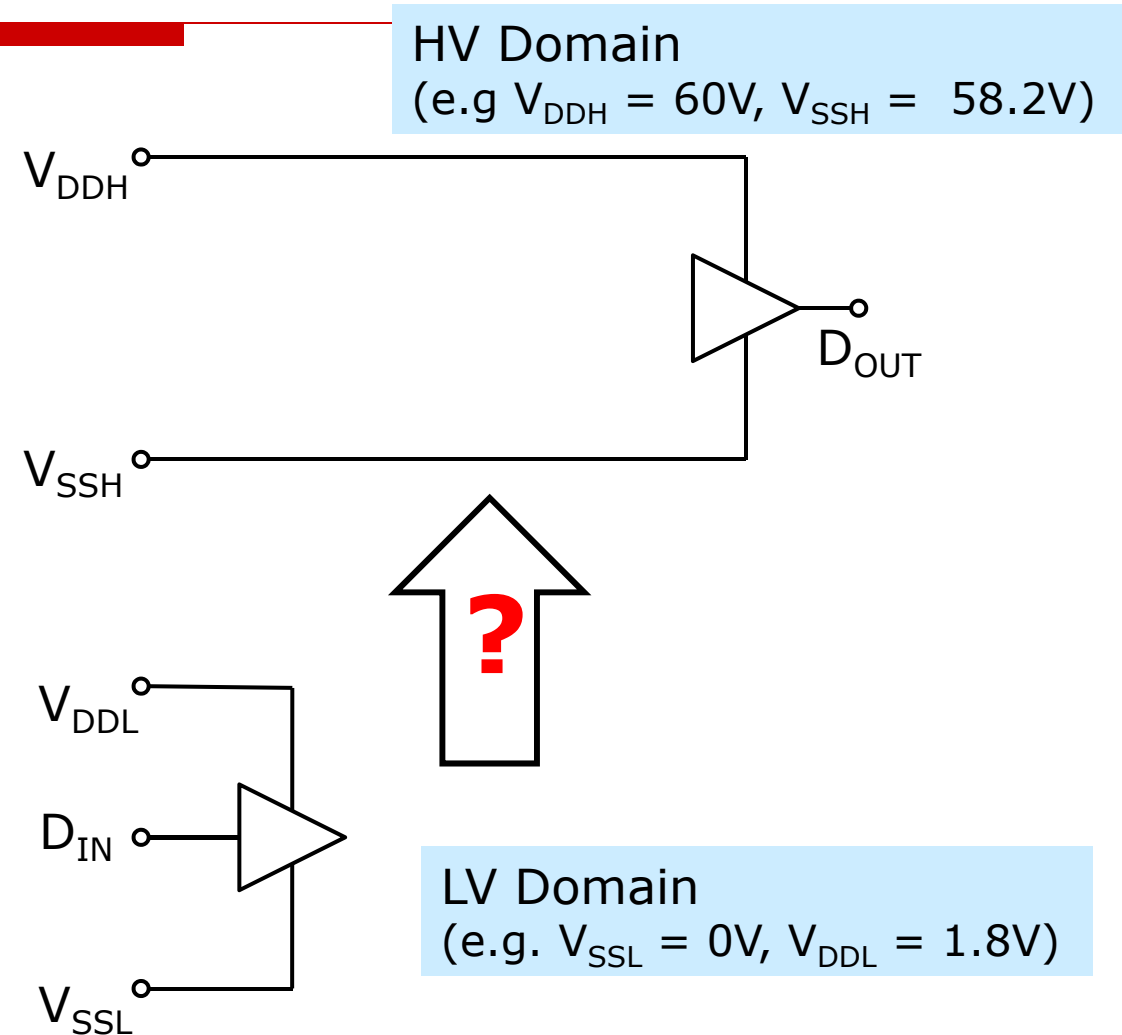
Half Bridge Essentials

- PowerFETs
- BST
 - highside gatedrive voltage
- GDRV
 - powerFET gate driver
- LSH
 - levelshifters
- OCP
 - overcurrent protection



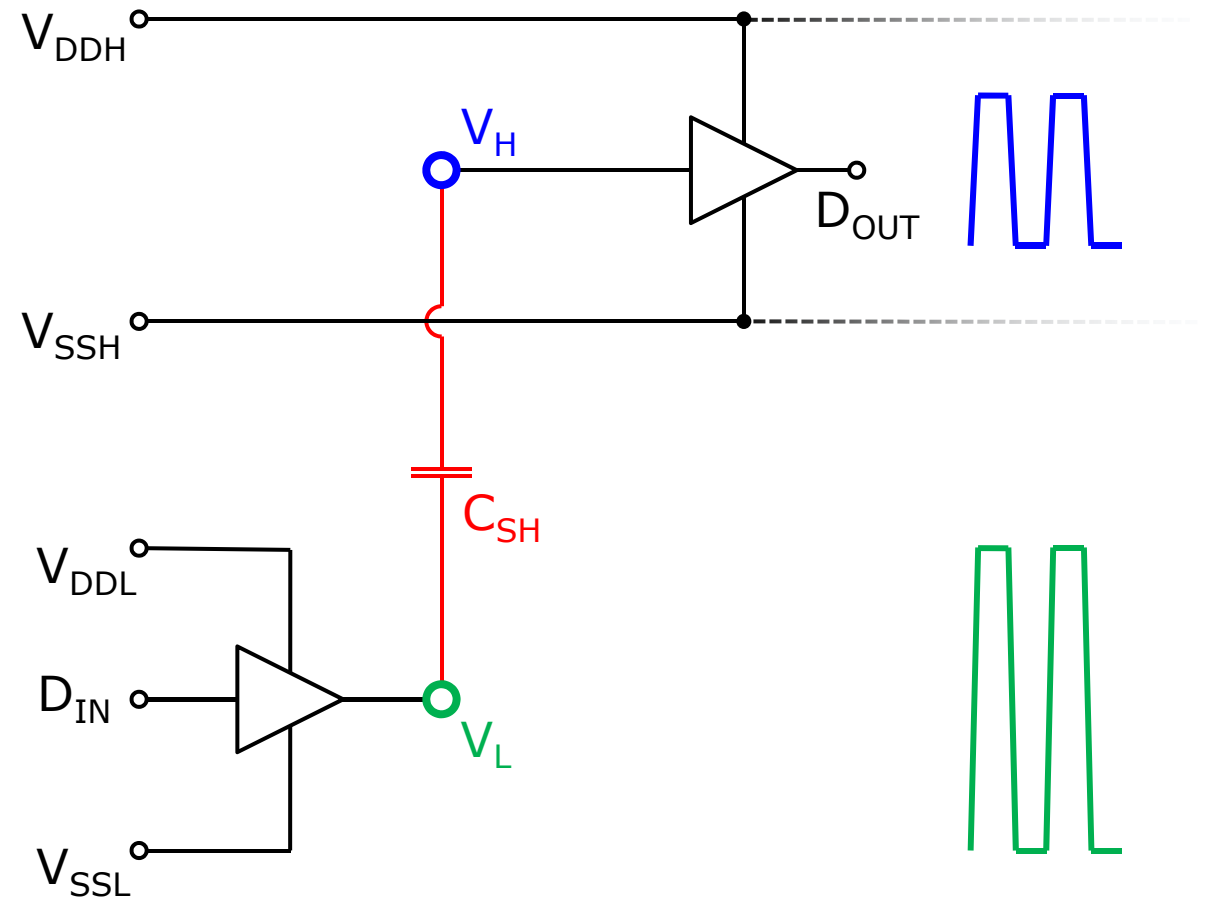
Levelshifters

- circuit that transfers (logic) signals between voltage domains
- output voltage domain can
 - positive / negative
 - static / dynamic
- robustness / sensitivity



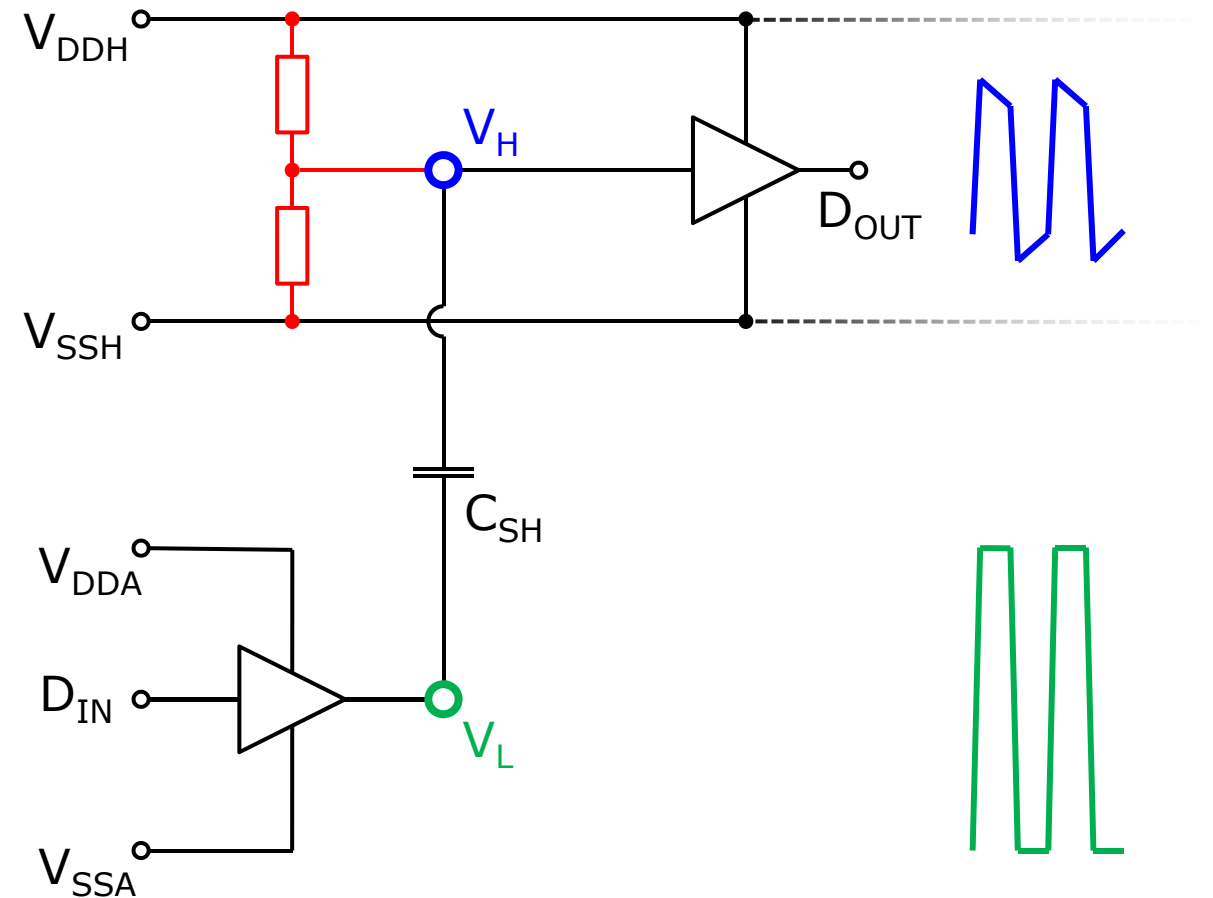
Levelshifters

- capacitive
- very fast
- HV domain can be both positive or negative
- no HV devices required
 - other than C_{SH}



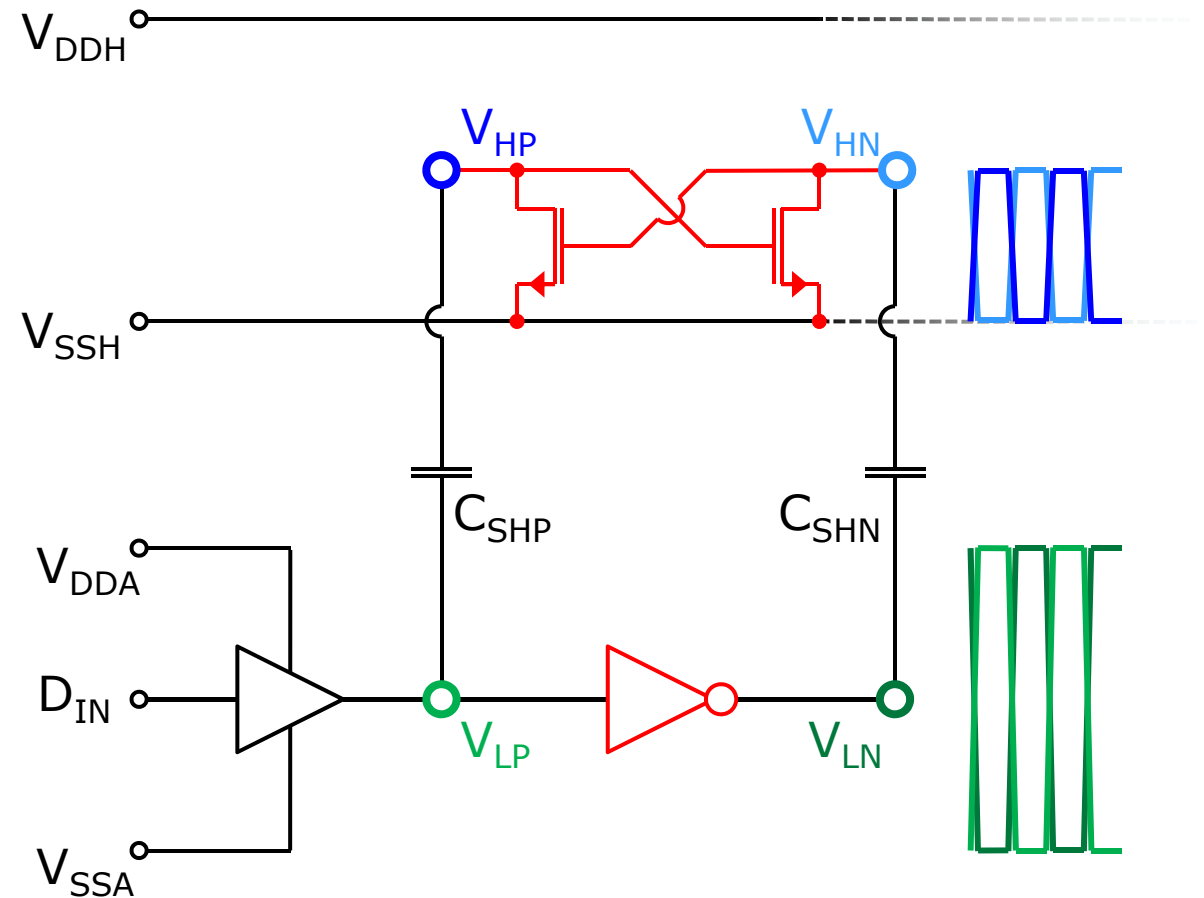
Levelshifters

- no signal gain
 - V_H swing $<$ V_L swing
 - capacitive load C_{SH} top plate
- top plate voltage drift
 - DC biasing
 - requires regular activity
 - not suitable for static signals
- sensitive to V_{SSH} transients
 - cannot distinguish between transient on V_L and V_{SSH}

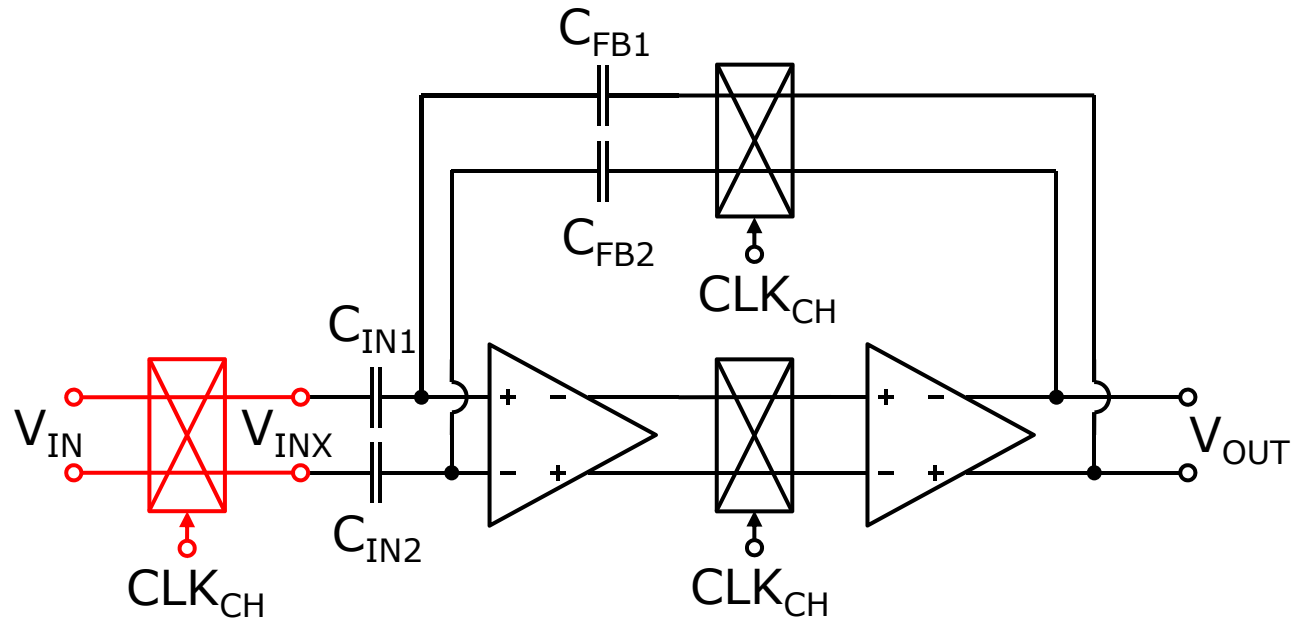


Levelshifters

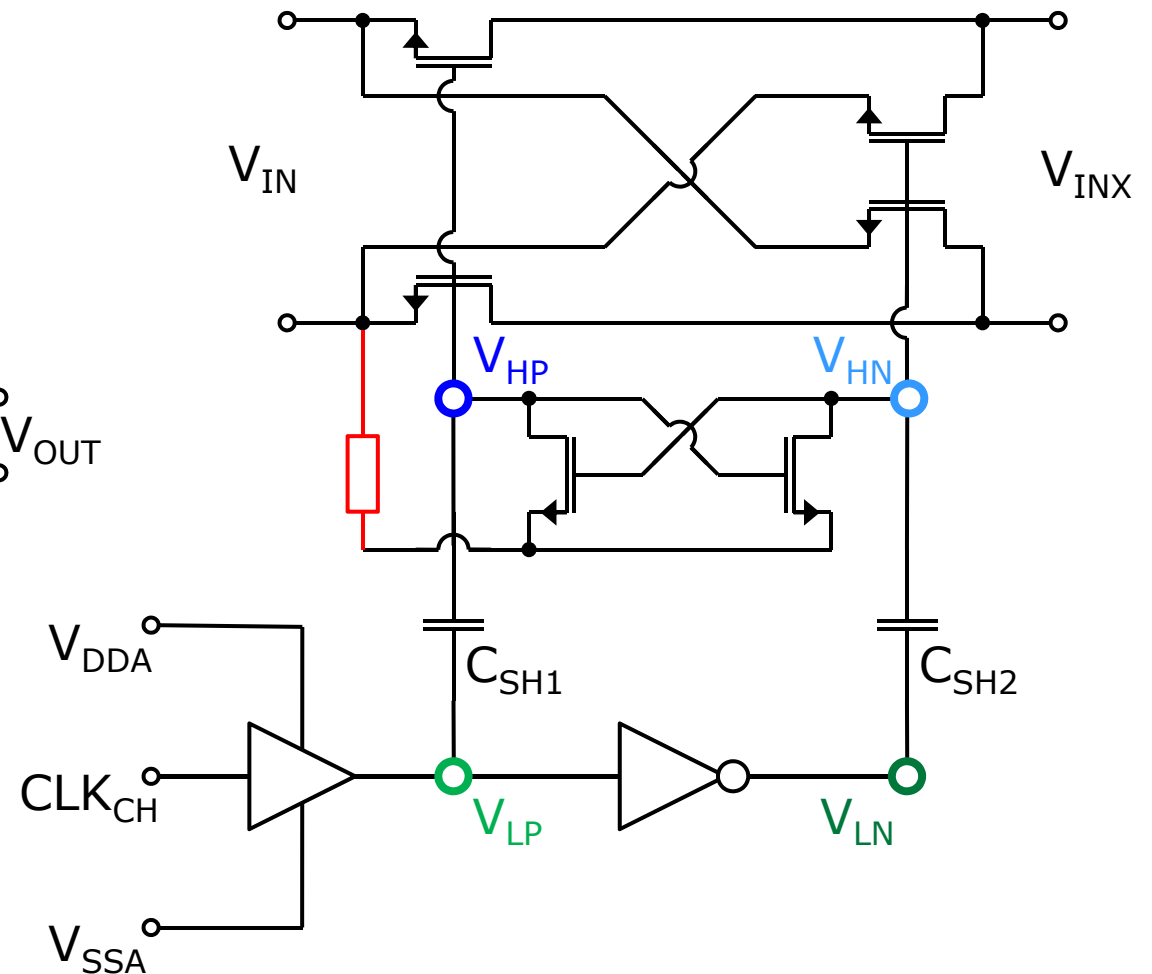
- differential operation
 - shift capacitors driven in anti-phase
 - cross-coupled latch fixes low level
- $V_{HP/N}$ to V_{SSH}



Levelshifters^[23]

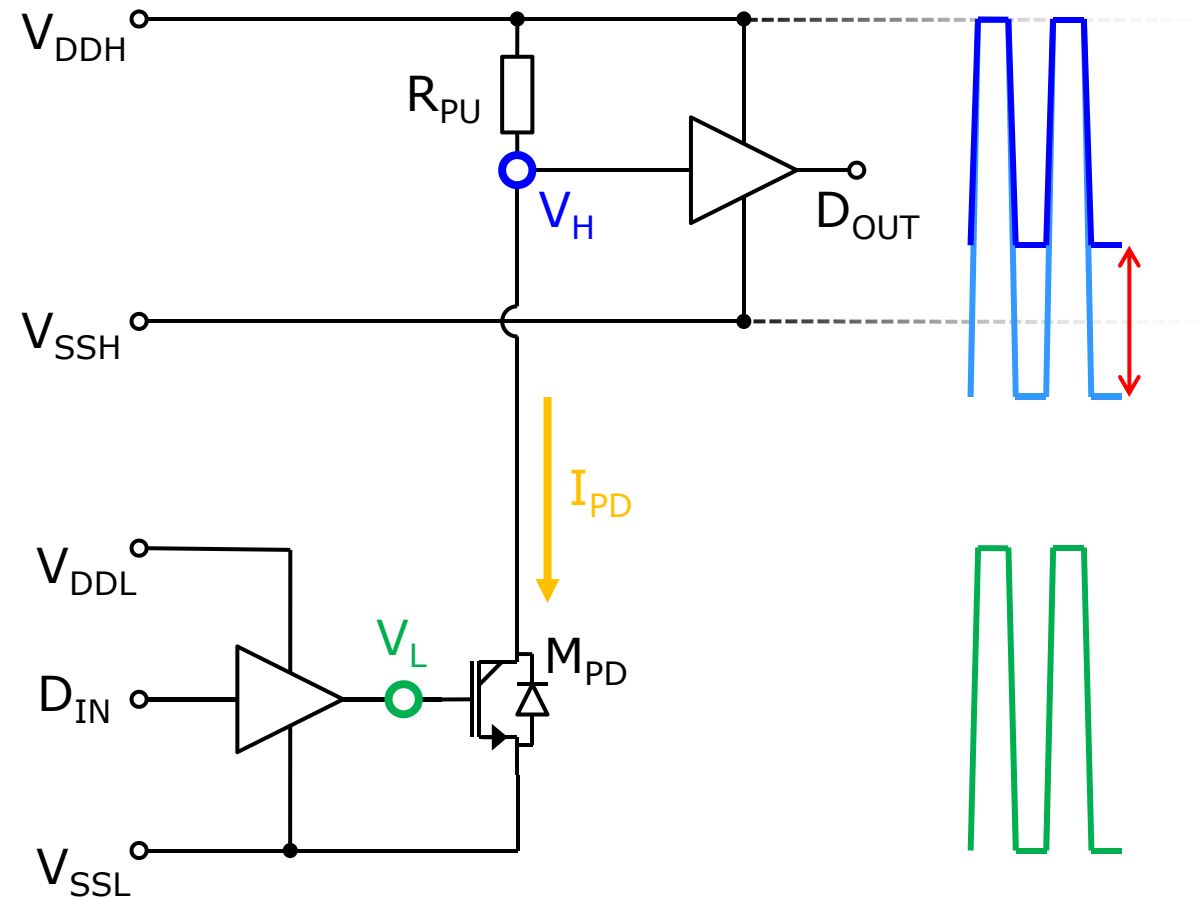


- Example: HV input chopper
 - resistor fixes low level $V_{HP/N}$ to V_{IN}
 - $\pm 30V$ input CM range



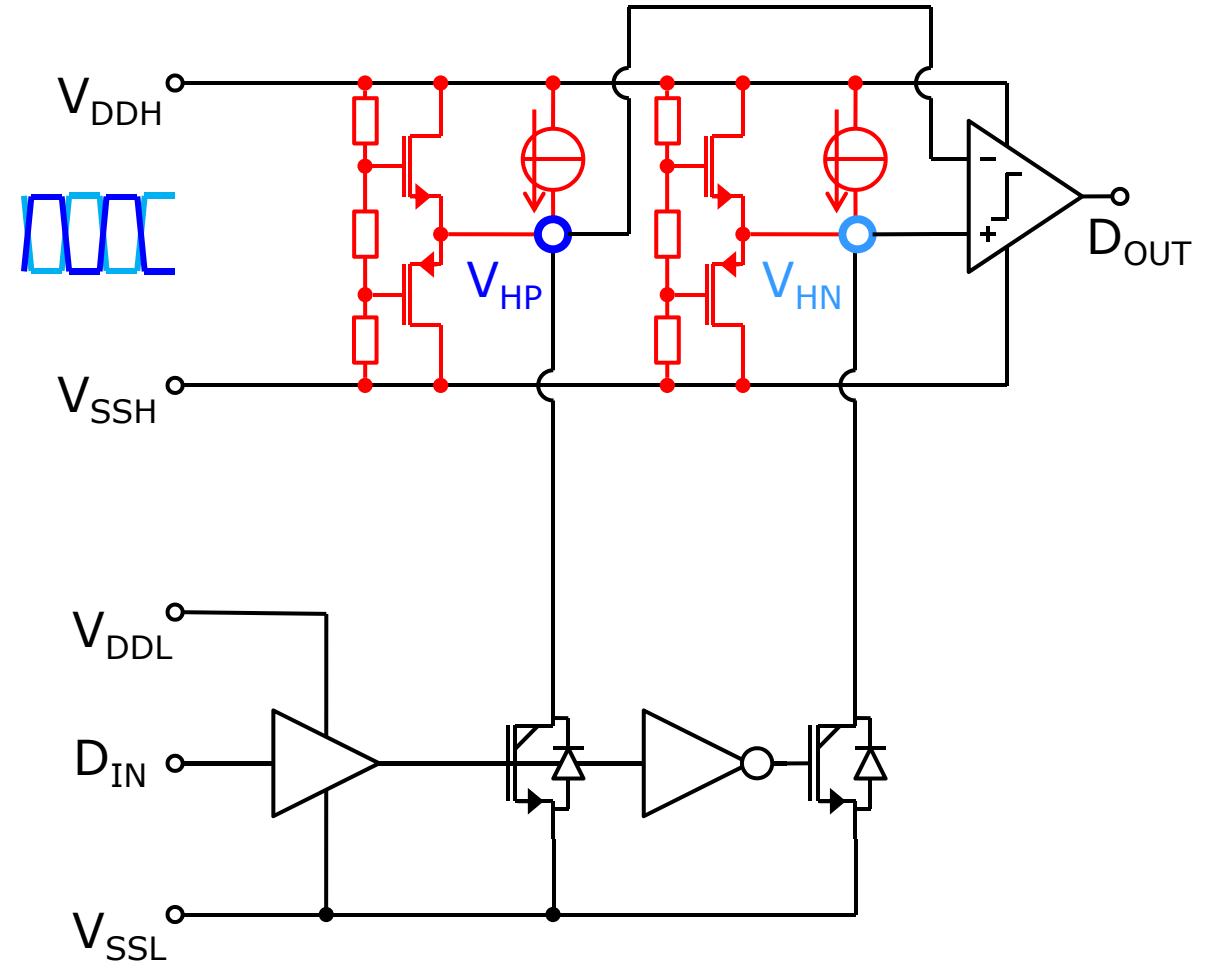
Levelshifters

- static
 - only one HV pulldown device
 - positive HV domain only
 - complementary version for negative HV domains
- no floating nodes
 - static & dynamic signals
 - V_H swing PVT sensitive
 - current consumption when active
- speed / power trade-off



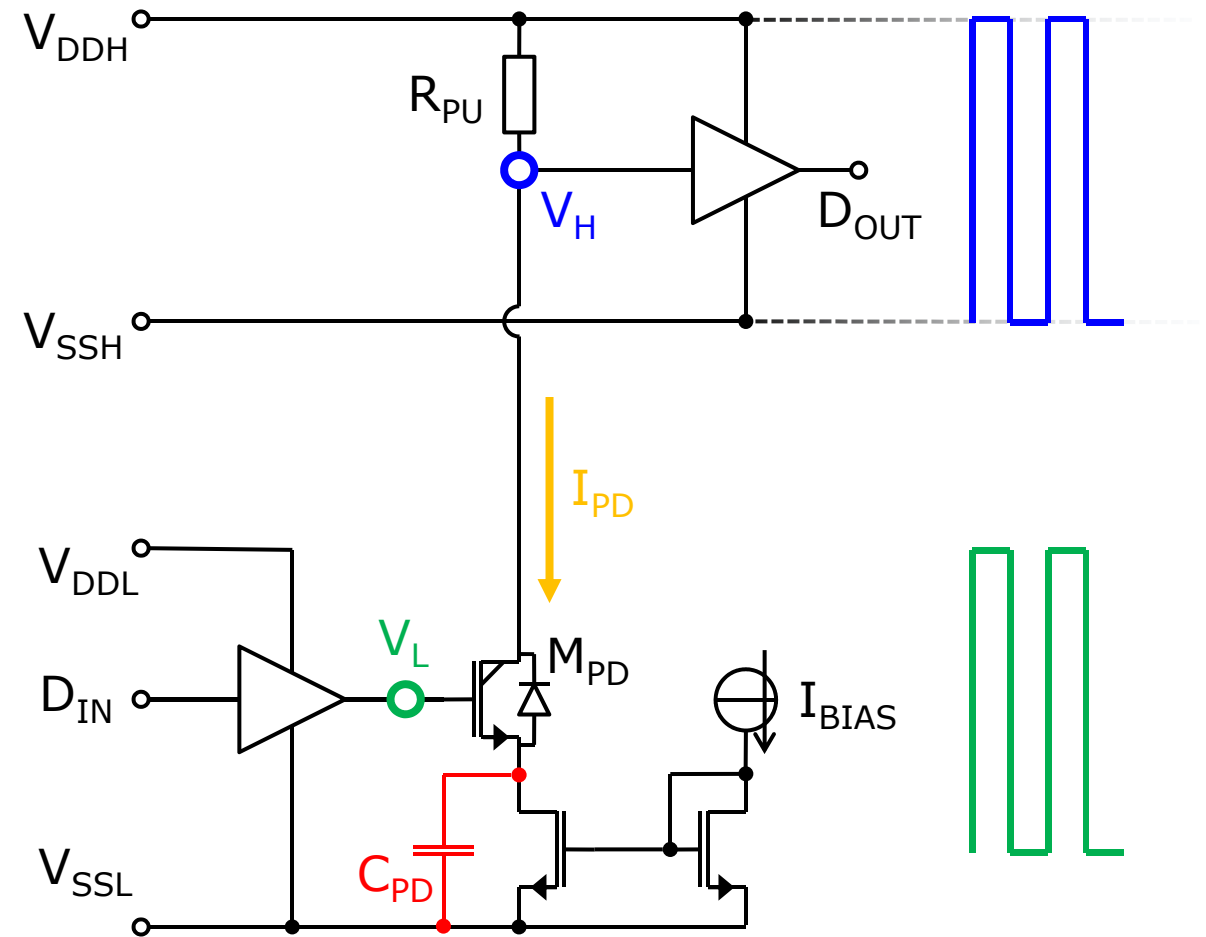
Levelshifters^[24]

- Example: clamping
 - differential operation
 - non-overlapping NMOS/PMOS clamps
 - insensitive to V_{SSH} transients
- V_{SSH} swing 0 – 50V
- propagation delay < 5ns



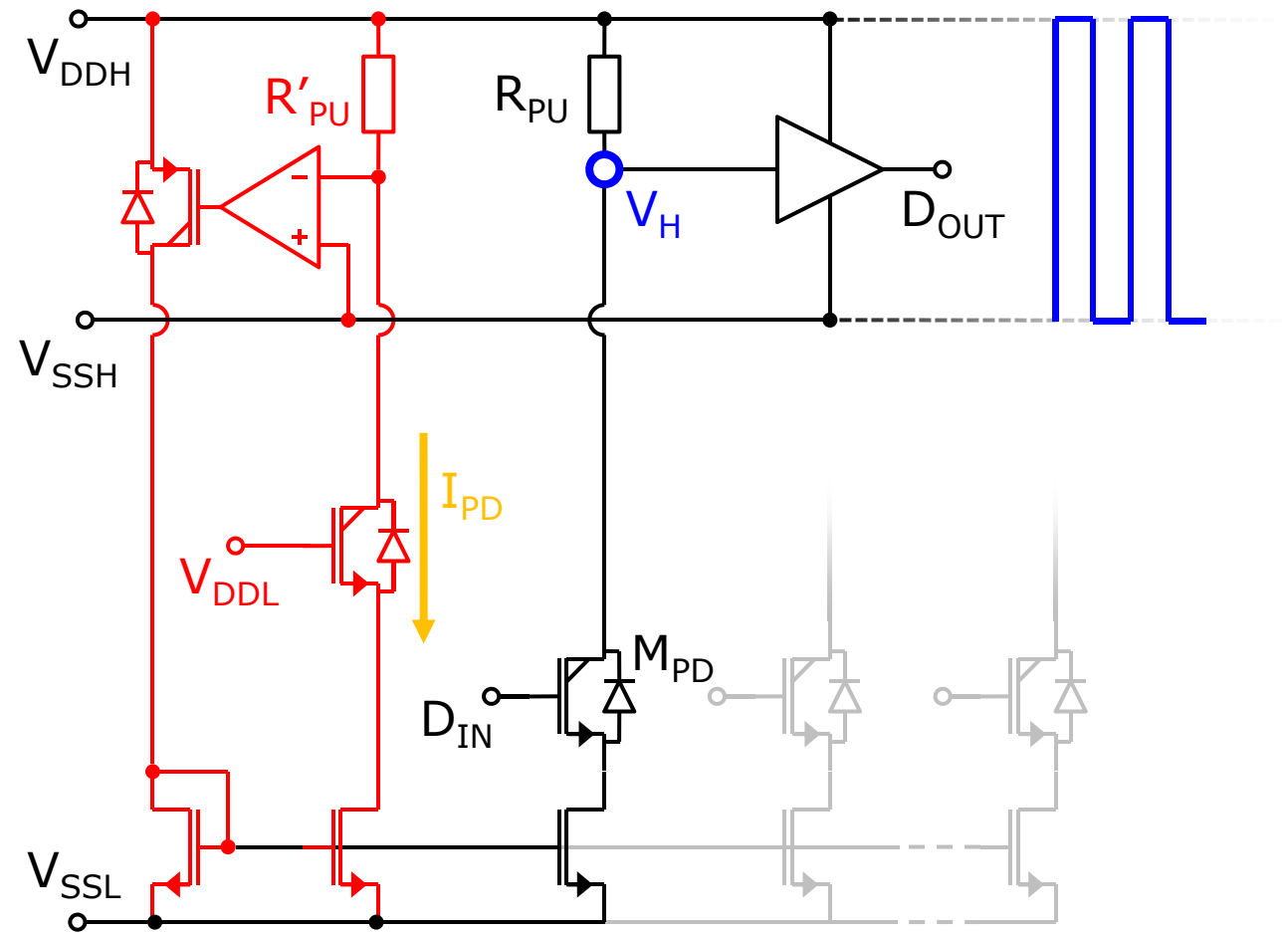
Levelshifters

- fix V_H swing
 - match I_{BIAS} to R_{PU}
- speed-up capacitor C_{PD}
 - current spike at rising edge V_L



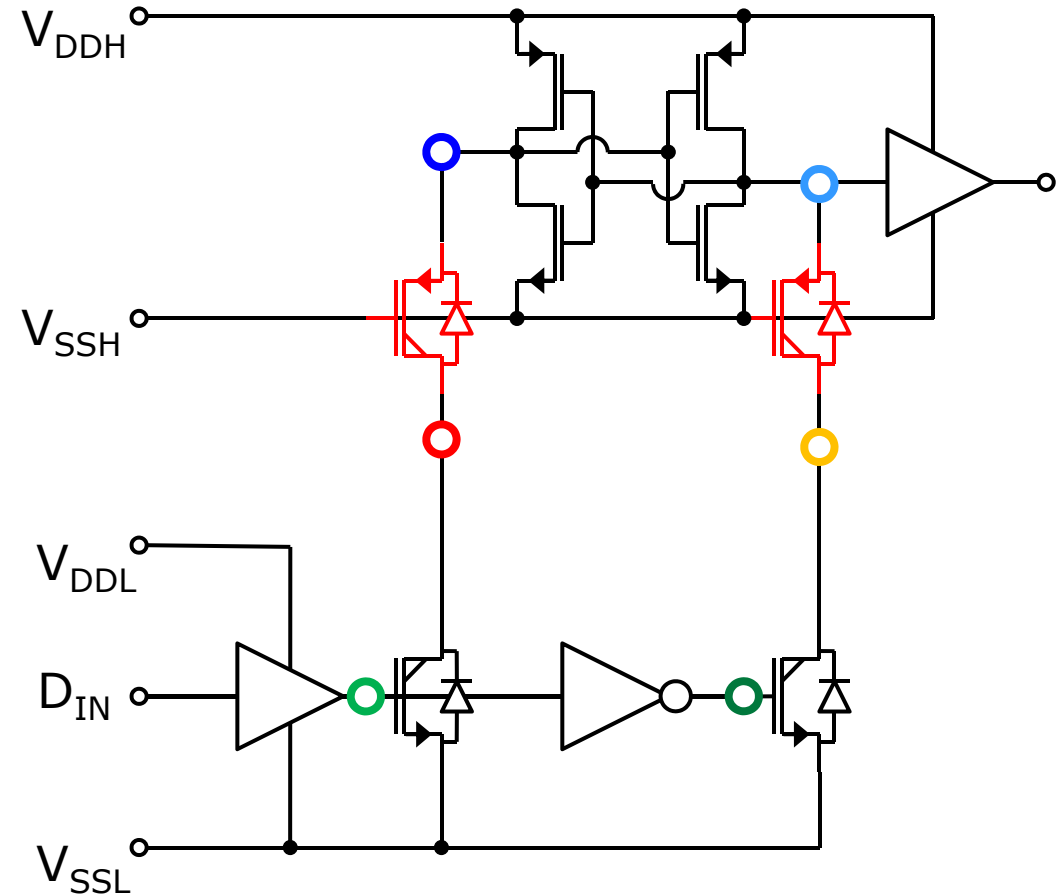
Levelshifters^[25]

- Example: multichannel
 - feedback loop regulates V_H swing to match $V_{DDH} - V_{SSH}$
 - biasing efficiently shared by 32 channels
 - $V_{SSL}/V_{DDL} = 0V/3V$
 - $V_{SSH}/V_{DDH} = 6V/9V$



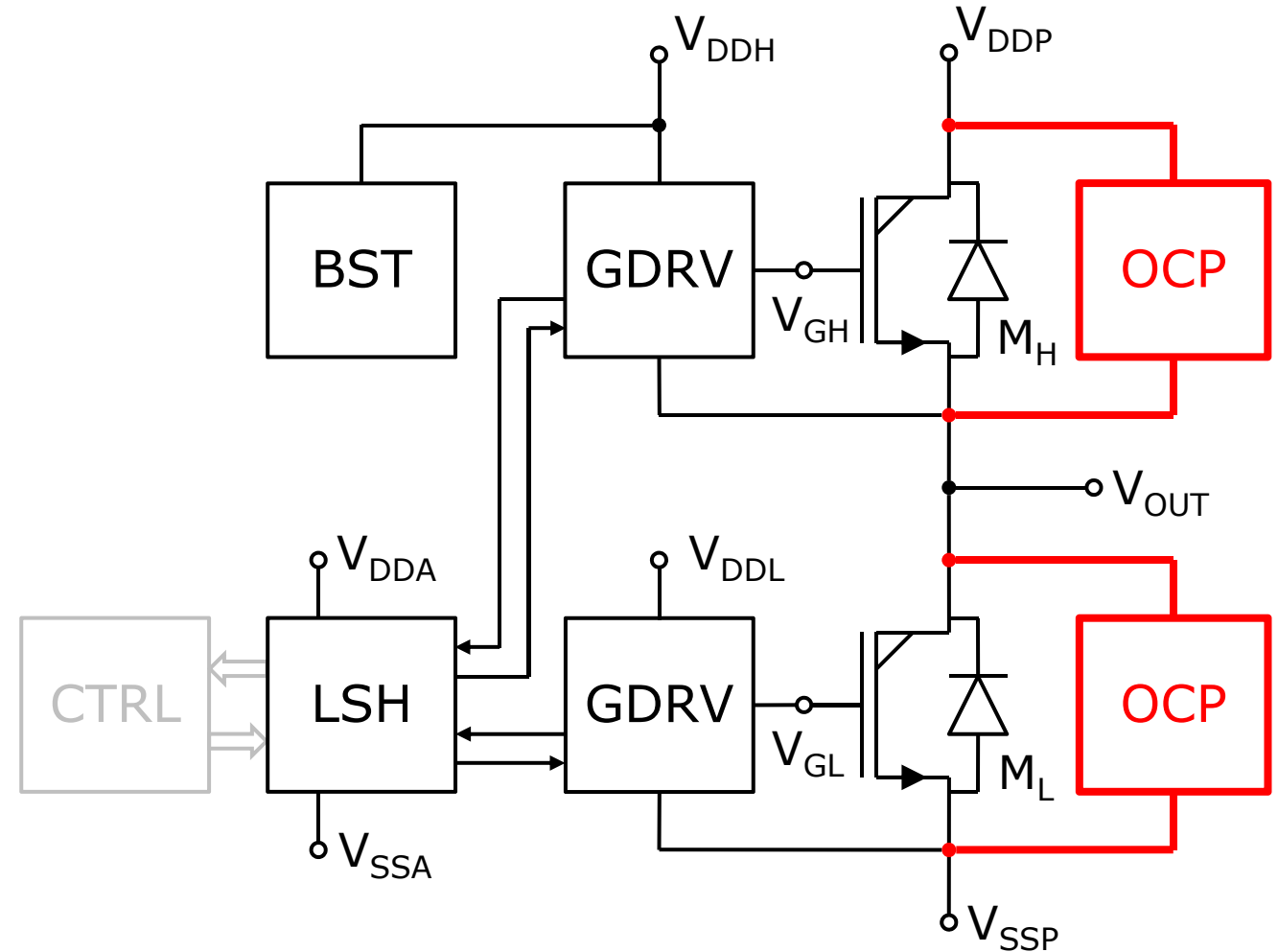
Levelshifters^[26]

- highside latch
 - cross coupled inverters
- HV PMOS devices terminate pulldown currents
 - no DC current after transition
 - good for bootstrapped highside drivers
 - requires careful dimensioning to cover PVT corners
- expensive
 - 4 HV devices required for 1 signal



Half Bridge Essentials

- PowerFETs
- BST
 - highside gatedrive voltage
- GDRV
 - powerFET gate driver
- LSH
 - levelshifters
- OCP
 - overcurrent protection

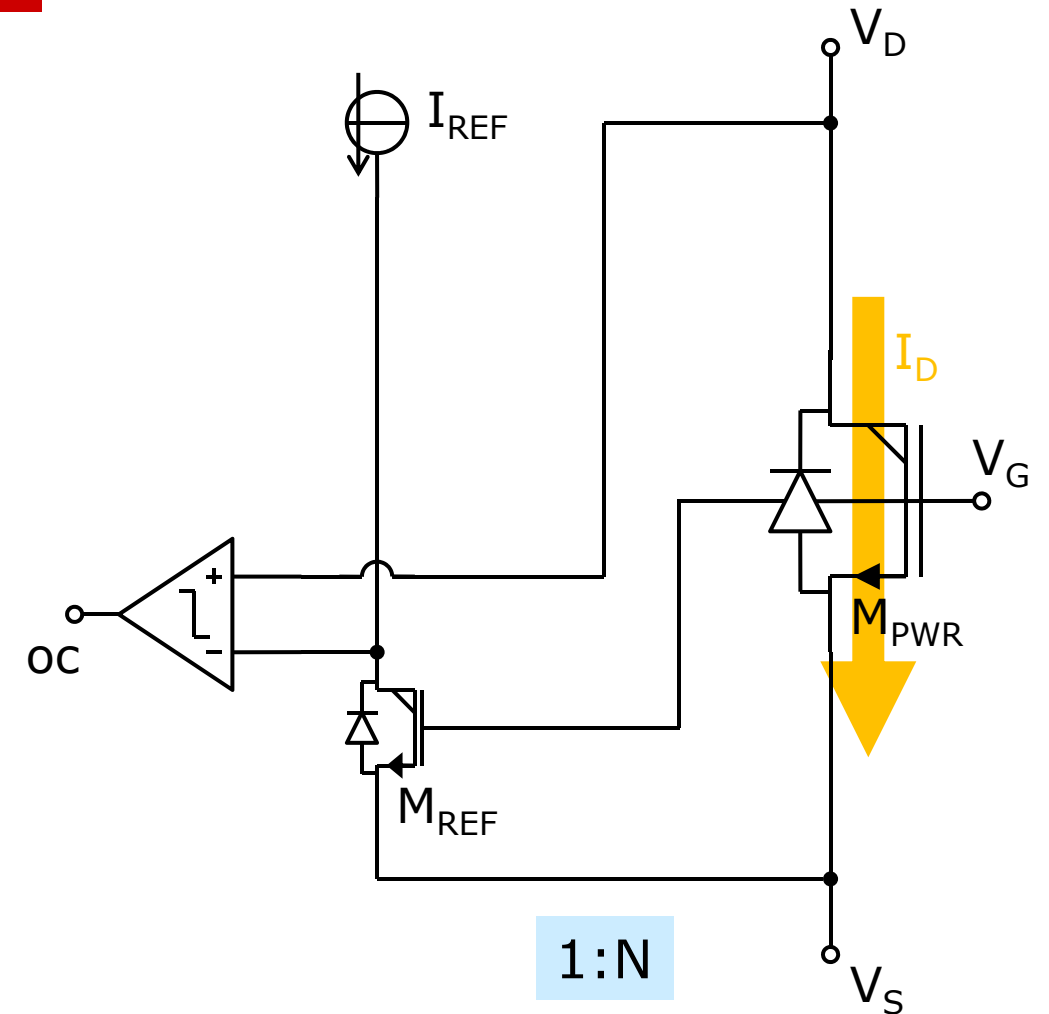


Overcurrent Protection

- latch-off
 - permanent shutdown
- hiccup mode
 - periodic restart
- cycle-by-cycle current limiting
 - continue operation
 - prevent 'audio-holes'
- series current sensing
 - (external) sense resistors
 - accurate
 - power loss
- parallel current sensing
 - replica FETs
 - matching
 - lossless

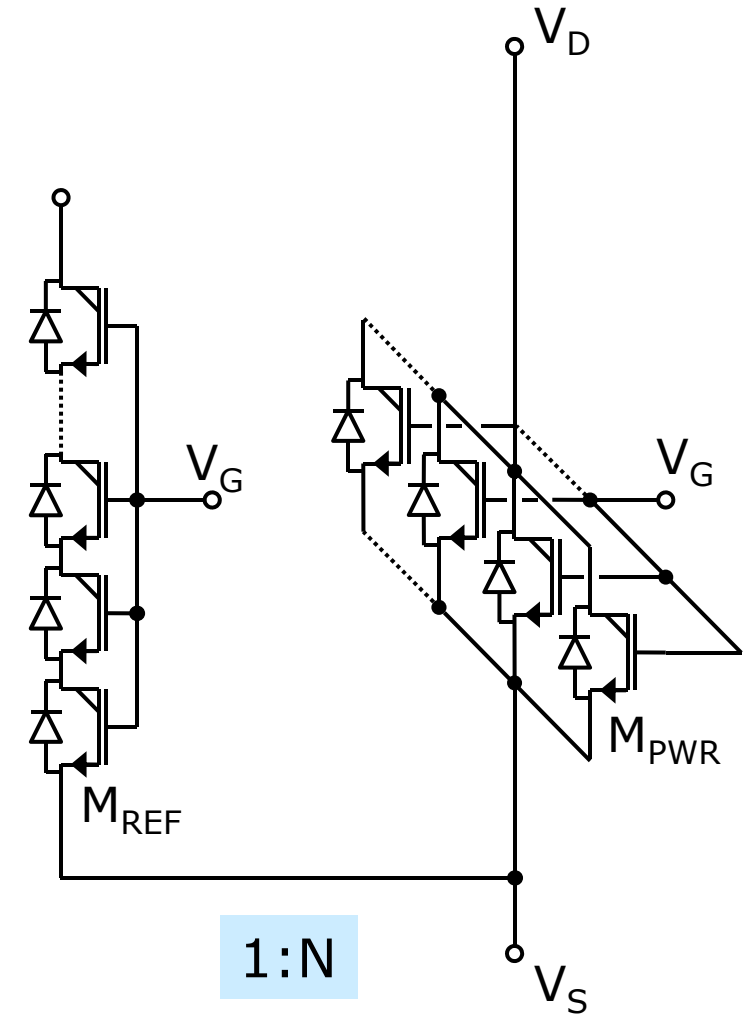
Replica FET^[27]

- replica M_{REF}
 - ratio 1:N (e.g. 1:10000)
 - gates & sources connected
 - matching (geometric & thermal)
- overcurrent comparator
 - M_{REF} biased at constant current I_{REF}
 - overcurrent flag **OC** if
$$I_D > N \times I_{\text{REF}}$$
- only accurate for positive currents
 - diode conduction not homogeneous



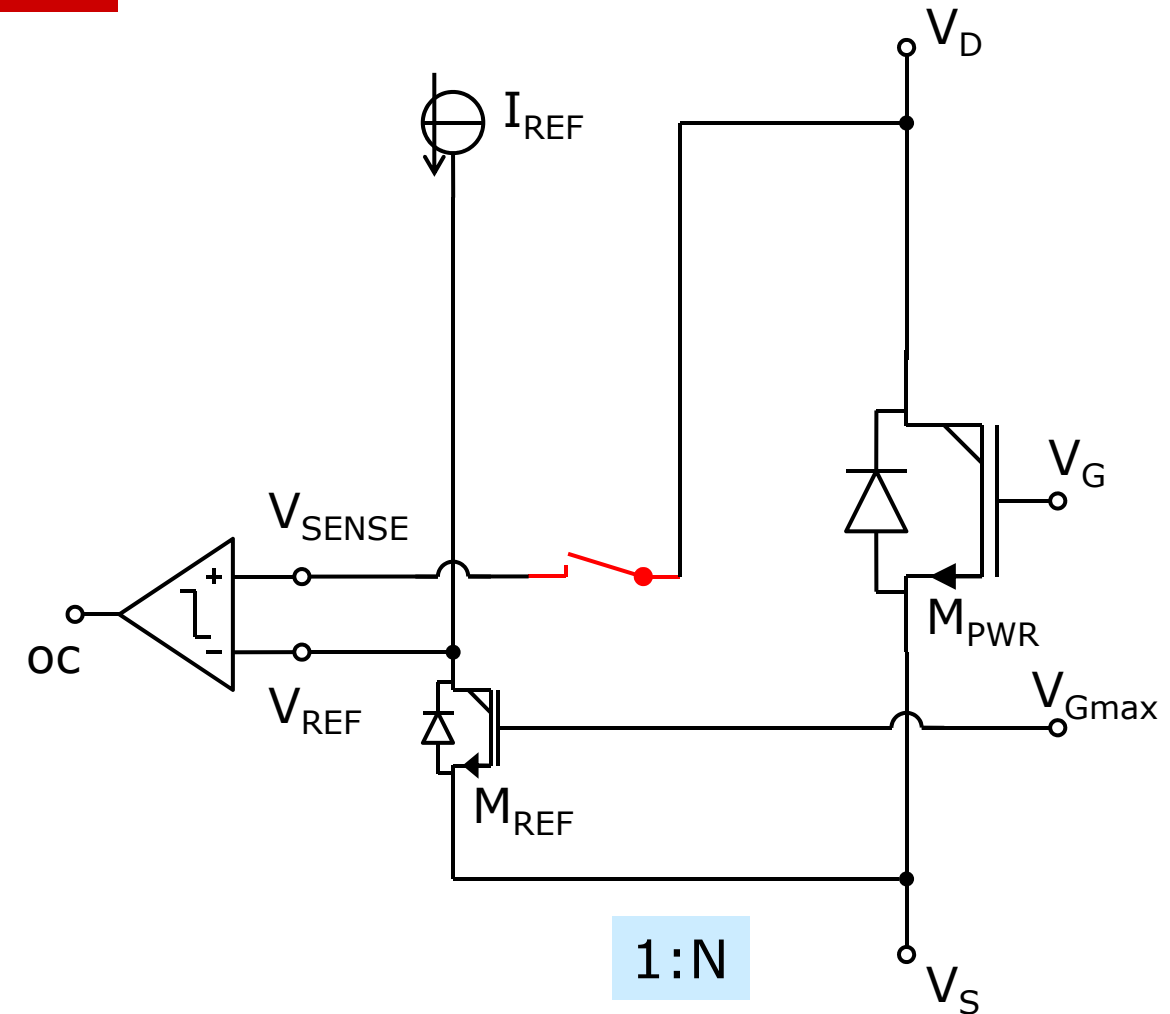
Matching

- DMOS has fixed channel length
 - only width scaling
- powerFET
 - very big (e.g. 10cm gate width)
 - many (folded) fingers in parallel
- replica stacking
 - increase ratio 1:N
 - increase matching (larger area)



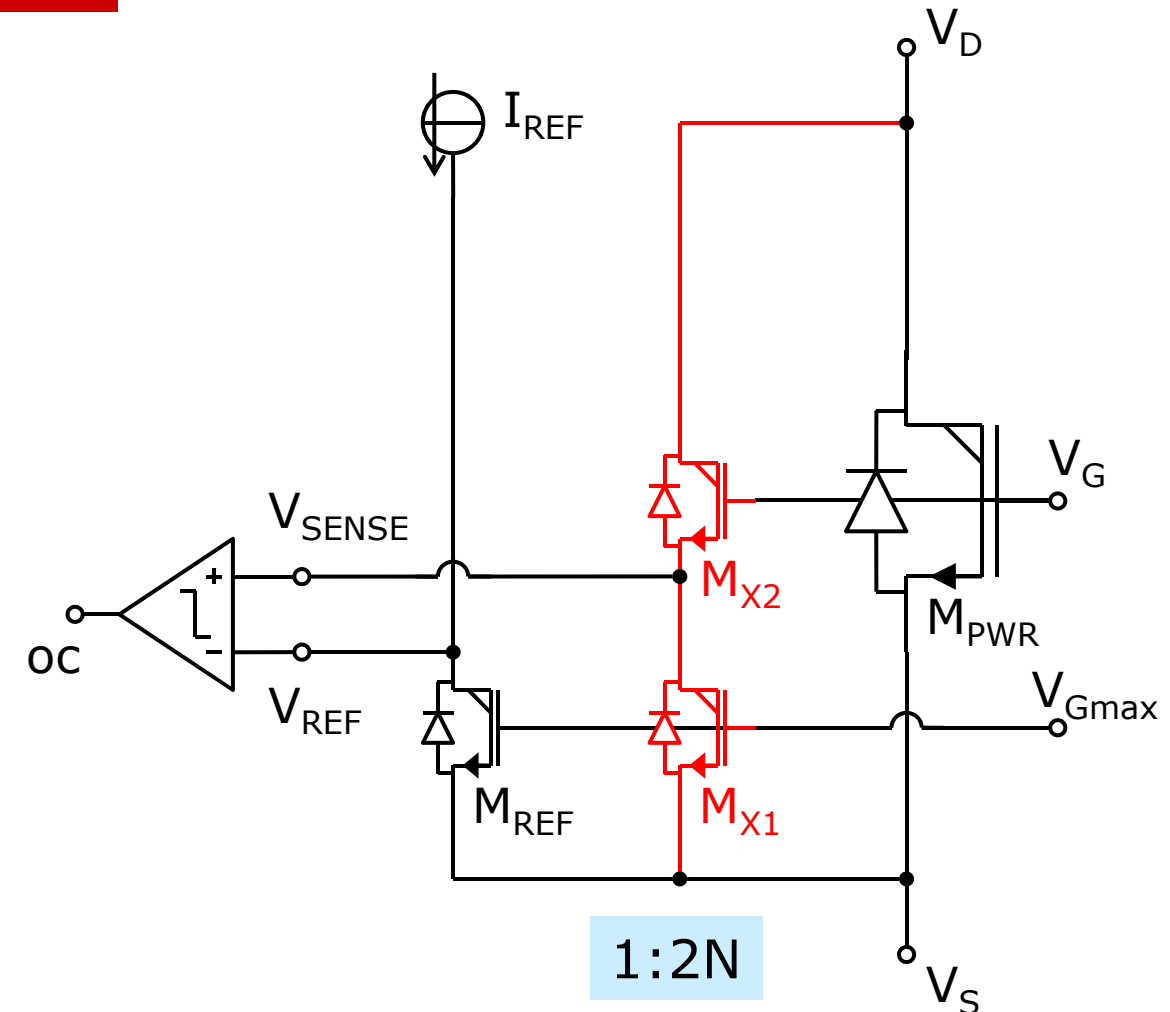
Overcurrent Comparator

- current ratio only valid in on-state
 - $V_{GS} = V_{GSmax}$: FETs in linear region
 - V_{DS} small
- off-state V_{DS} can be very high
 - damage comparator inputs
 - disconnect switch

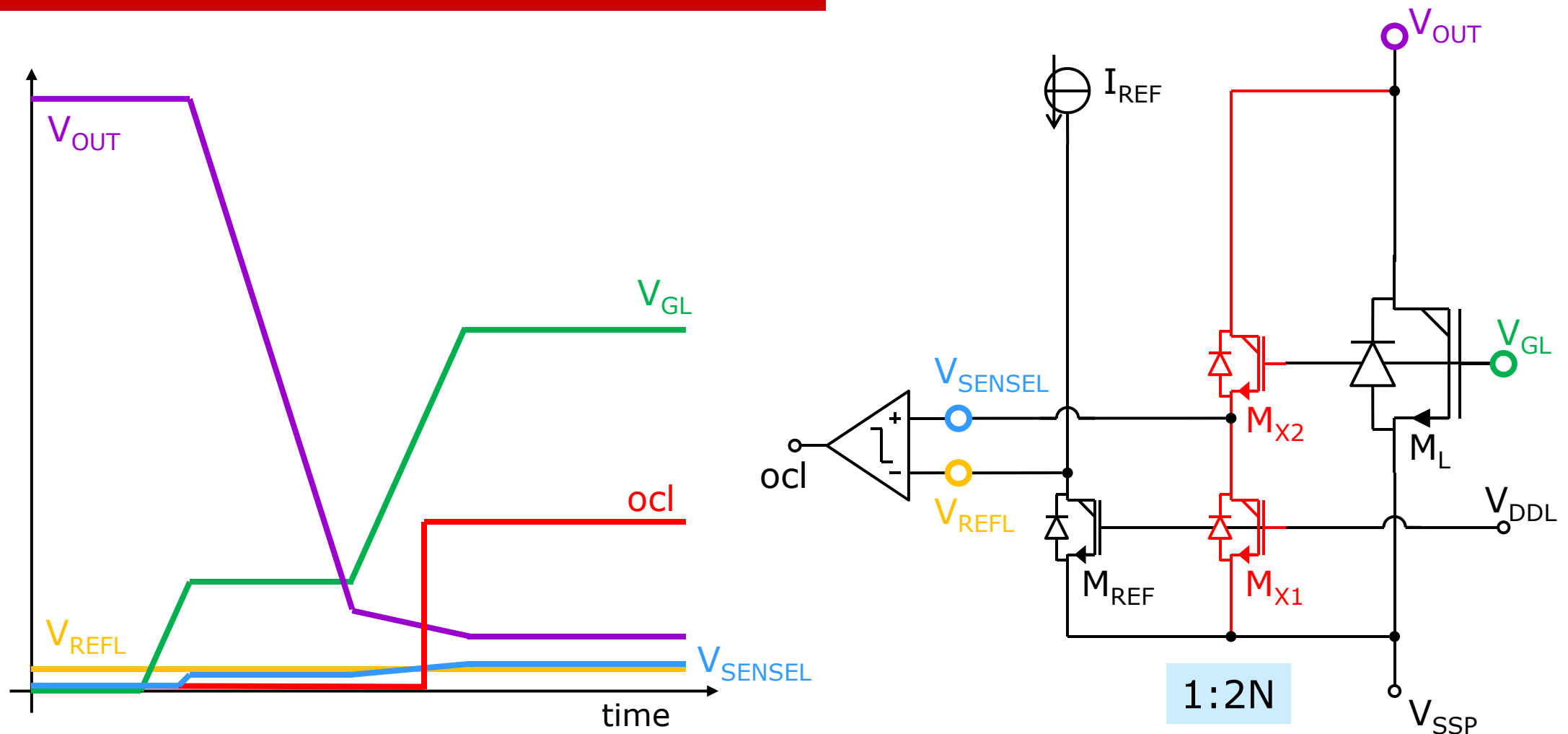


Overcurrent Comparator

- current ratio only valid in on-state
 - $V_{GS} = V_{GSmax}$: FETs in linear region
 - V_{DS} small
- off-state V_{DS} can be very high
 - damage comparator inputs
 - disconnect switch
- divider switch M_{X1}/M_{X2}
 - M_R and M_{X1} always on
 - off-state: pulldown V_X
 - on-state: linear division of V_{DS}
 - increase ratio 1:N

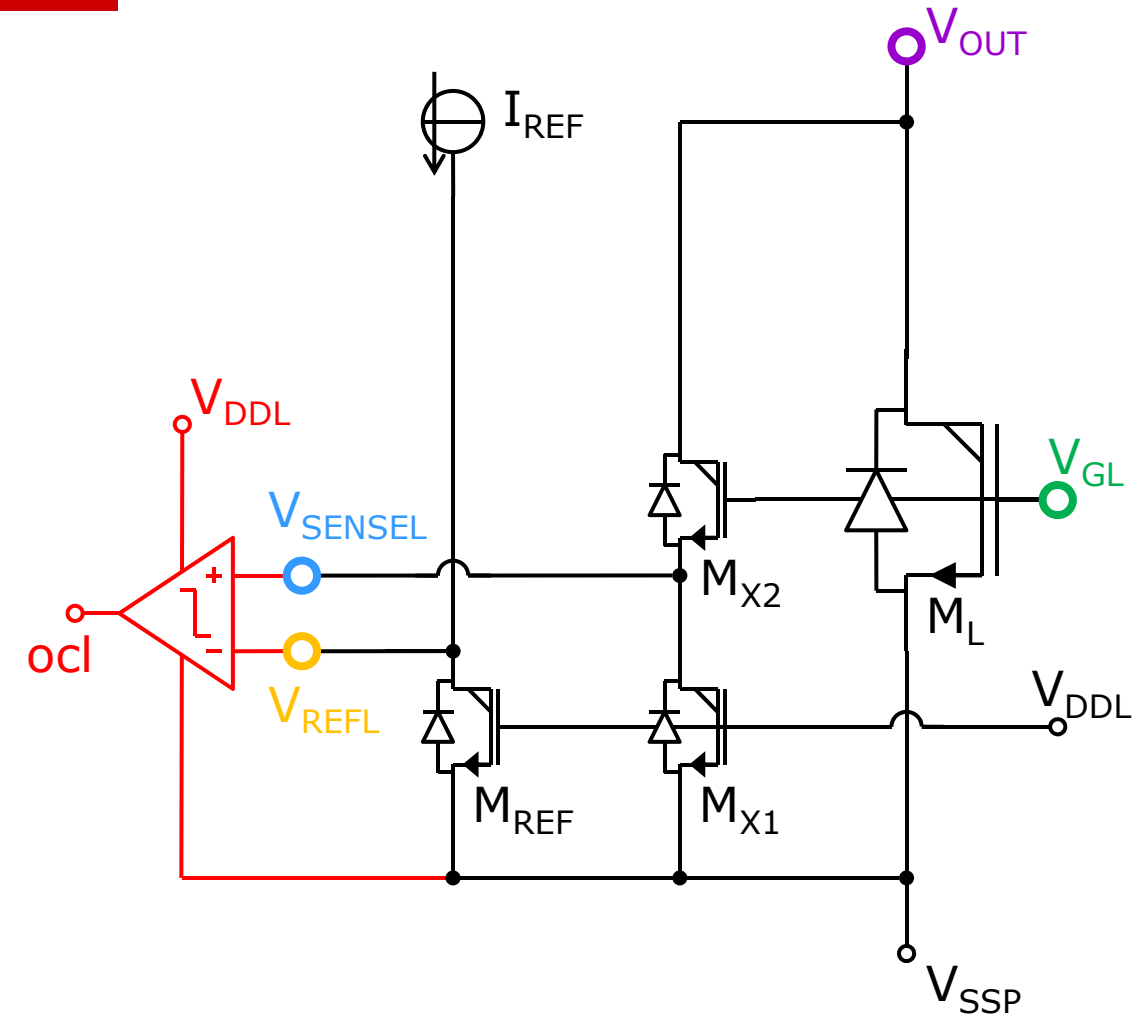


Lowside Overcurrent Comparator



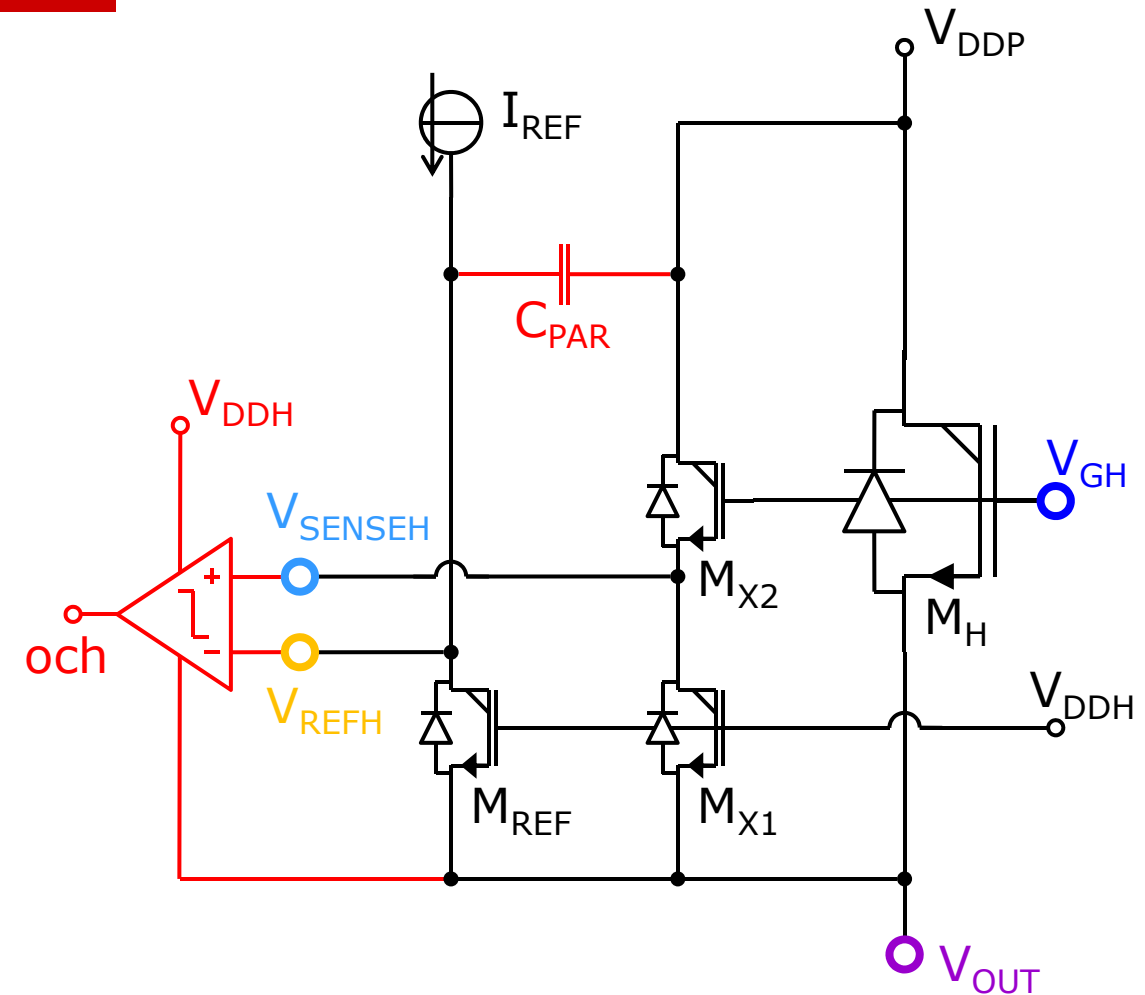
Lowside Overcurrent Comparator

- ❑ lowside comparator inputs
 - V_{REFL} & V_{SENSEL} low voltage w.r.t. V_{SSP}
- ❑ straightforward LV comparator design
 - low voltage domain [$V_{\text{SSP}}, V_{\text{DDL}}$]

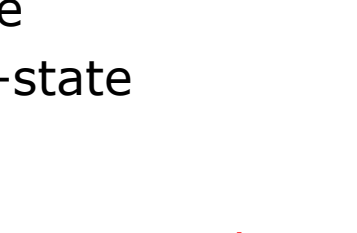


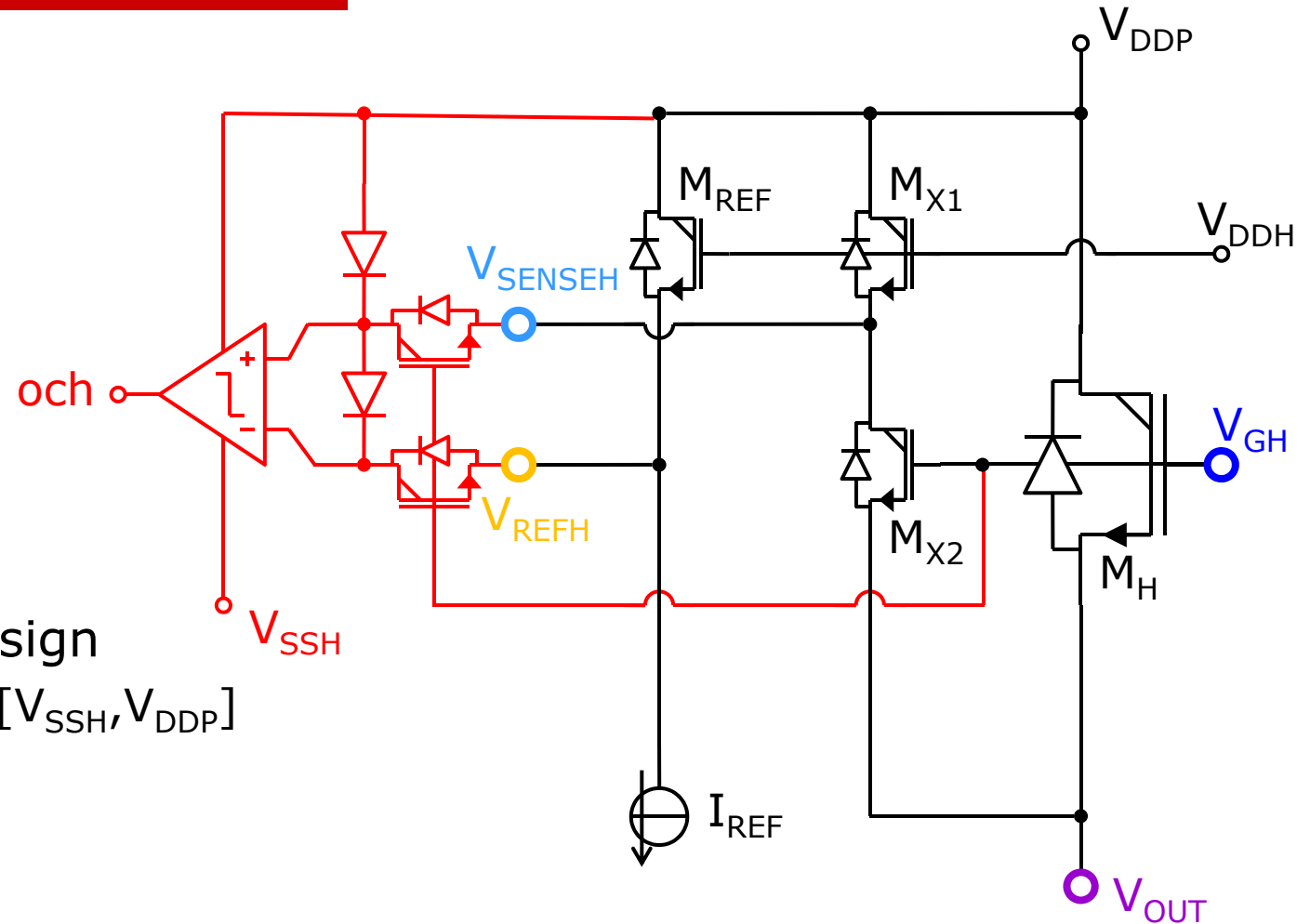
Highside Overcurrent Comparator

- highside comparator inputs
 - V_{REFH} & V_{SENSEH} low voltage w.r.t. V_{OUT}
- straightforward LV comparator design
 - switching voltage domain [V_{OUT} , V_{DDH}]
- HV levelshifter
 - **och** w.r.t switching node V_{OUT}
 - fast & dV/dt insensitive
- capacitive coupling to static nodes
 - e.g. I_{REF} to supply V_{DDP}



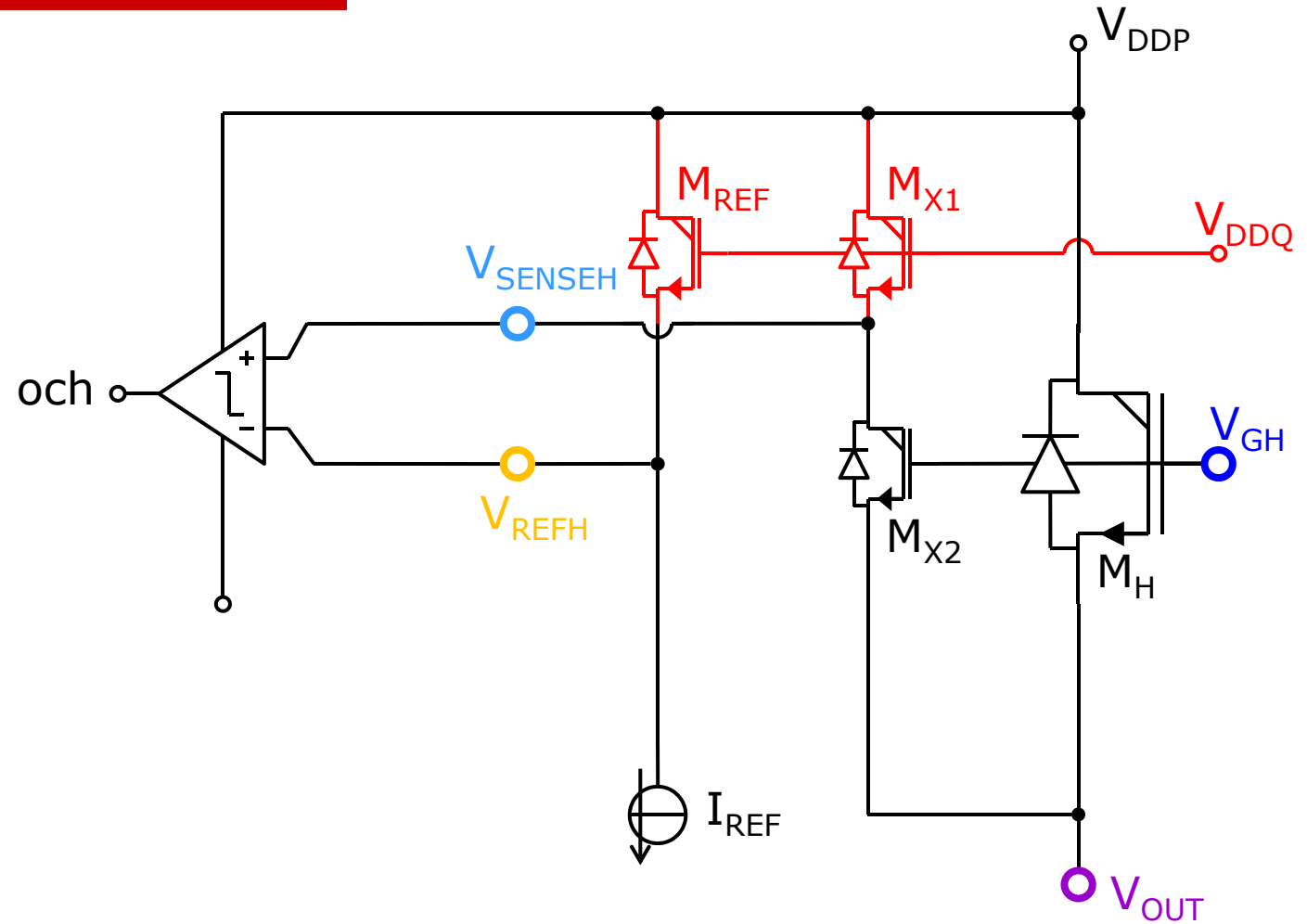
Highside Overcurrent Comparator (2)

- input switches
 - connect to highside gate
 - disconnect inputs in off-state
 - inputs clamped to V_{DDP}
 - limited CM swing
 - straightforward comparator design
 - output in highside LV domain [V_{SSH}, V_{DDP}]
 - HV levelshifter
 - no dV/dt requirement
- 



Highside Overcurrent Comparator (3)

- chargepump voltage V_{DDQ}
 - M_{REF} and M_{X1} always on
 - V_{SENSEH} and V_{REFH} close to V_{DDP}
- inputs close to V_{DDP}
 - limited CM swing
 - no clamps needed



Papers to see at ISSCC2022

- Session 14: GaN, High-Voltage and Wireless Power
 - 14.4 “A 2.5–5MHz 87% Peak Efficiency 48V-to-1V Integrated Hybrid DC-DC Converter Adopting Ladder SC Network with Capacitor Assisted Dual Inductor Filtering”
 - 14.6 “A 27W D2D Wireless Power Transfer System with Compact Single-Stage Regulated Class-E Architecture and Adaptive ZVS Control”
 - 14.7 “A 1.2W 51%-Peak-Efficiency Isolated DC-DC Converter with a Cross-Coupled Shoot-Through-Free Class-D Oscillator Meeting the CISPR-32 Class-B EMI Standard”

- Session 18: DC-DC Converters
 - 18.2 “A 12V/24V-to-1V DSD Power Converter with 56mV Droop and 0.9 μ s 1% Settling Time for a 3A/20ns Load Transient”
 - 18.7 “A 2–5MHz Multiple DC Output Hybrid Boost Converter with Scalable CR Boosting Scheme Achieving 91% Efficiency at a Conversion Ratio of 12”

Papers to see at ISSCC2022 (2)

□ Session 30: Power Management Techniques

- 30.2 "A 130V Triboelectric Energy-Harvesting Interface in 180nm BCD with Scalable Multi-Chip-Stacked Bias-Flip and Daisy-Chained Synchronous Signaling Technique"
- 30.3 "A Reconfigurable Series-Parallel Charger for Dual-Battery Applications with 89W 97.7% Efficiency in Direct Charging Mode"

□ Session 31: Audio Amplifiers

- 31.2 "A 121.4 dB DR -109.8 dB THD+N Capacitively-Coupled Chopper Class-D Audio Amplifier"
- 31.4 "A -91 dB THD+N, Resistor-Less Class-D Piezoelectric Speaker Driver using a Dual Voltage/ Current Feedback for LC Resonance Damping"

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13. M. Haas, M.Ortmanns, "A floating high-voltage level-shifter with high area efficiency for biomedical implants", PRIME 2016
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15. Y. Gao, et.al., "An AC-Input Inductorless LED Driver for Visible-Light-Communication Applications with 8Mb/s Data-Rate and 6.4% Low-Frequency Flicker", ISSCC 2017

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17. S. Stanzione, et.al., "A Self-Biased 5-to-60V Input Voltage and 25-to-1600 μ W Integrated DC-DC Buck Converter with Fully Analog MPPT Algorithm Reaching up to 88% End-to-End Efficiency", ISSCC 2013
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Live Q&A Session:

February 20, 2022
8:40am-9:00am, PST